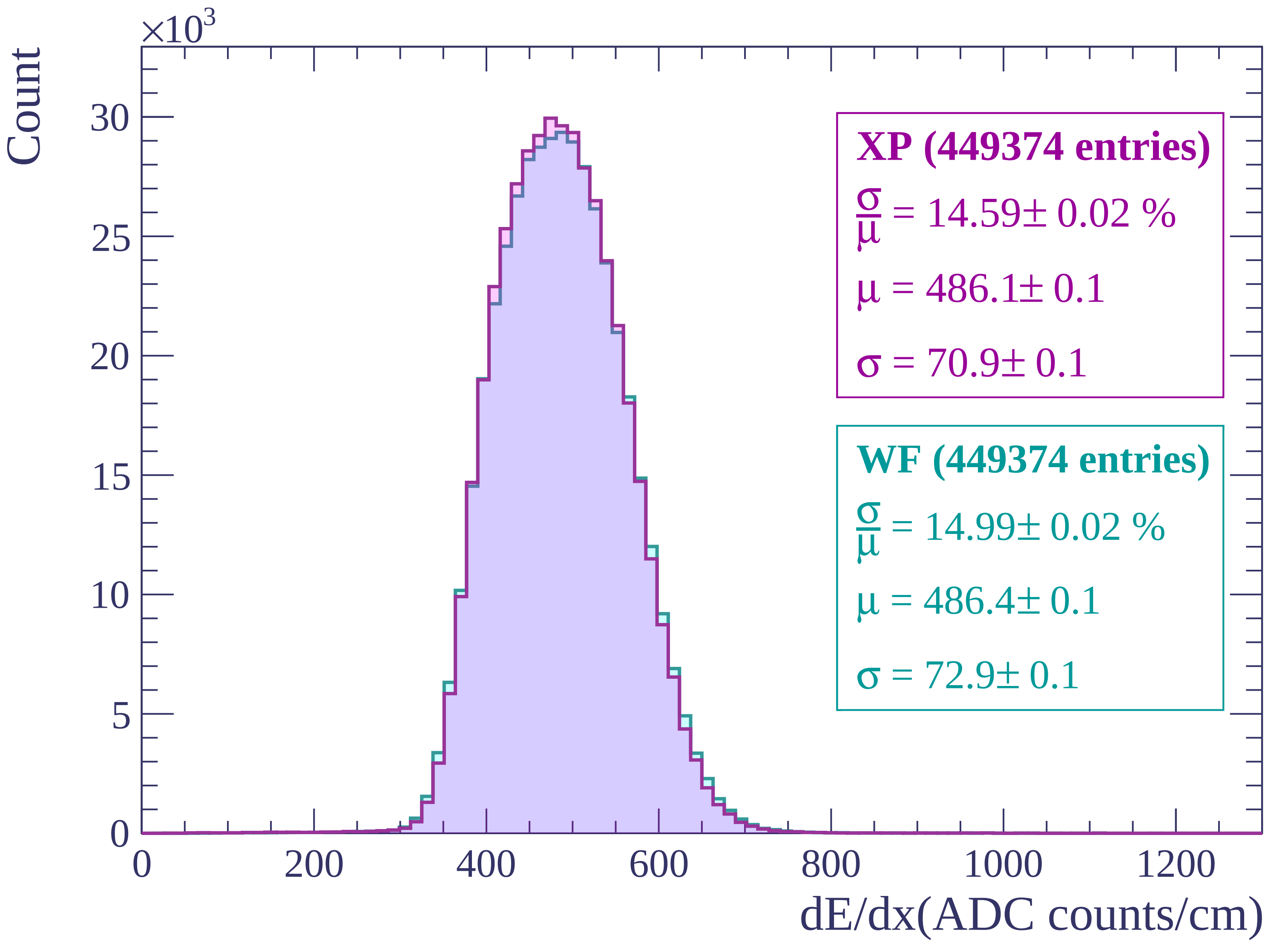
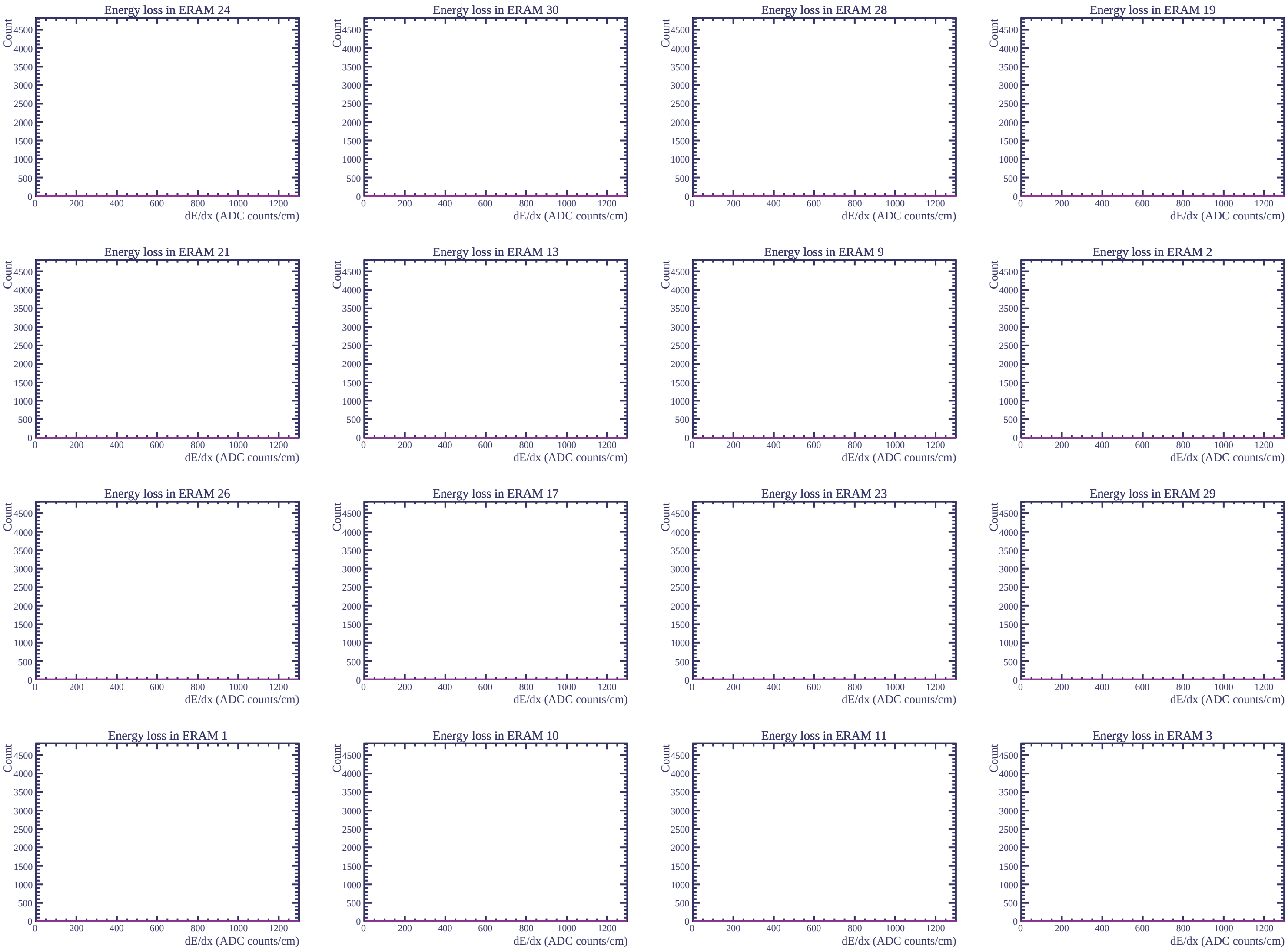
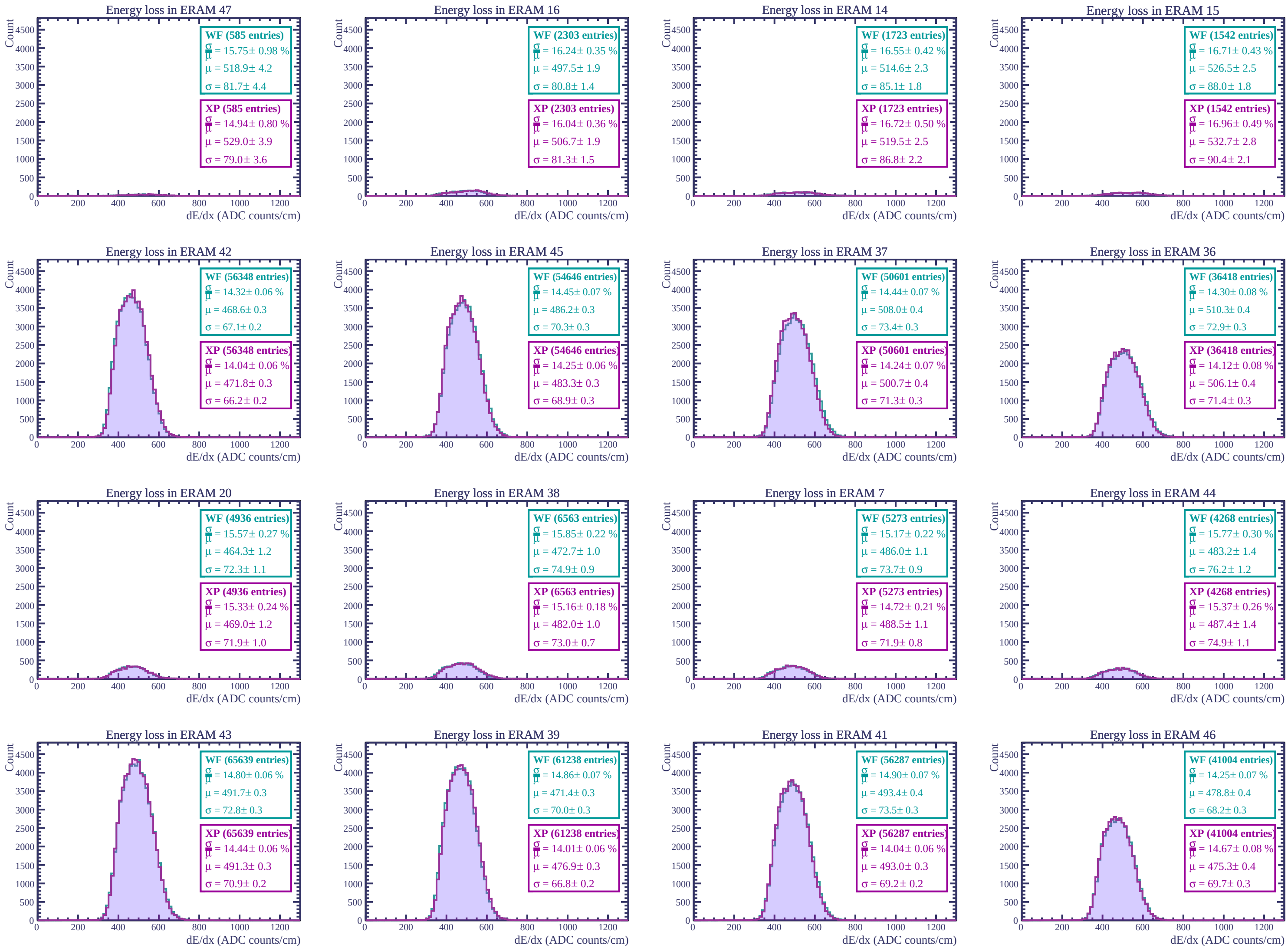
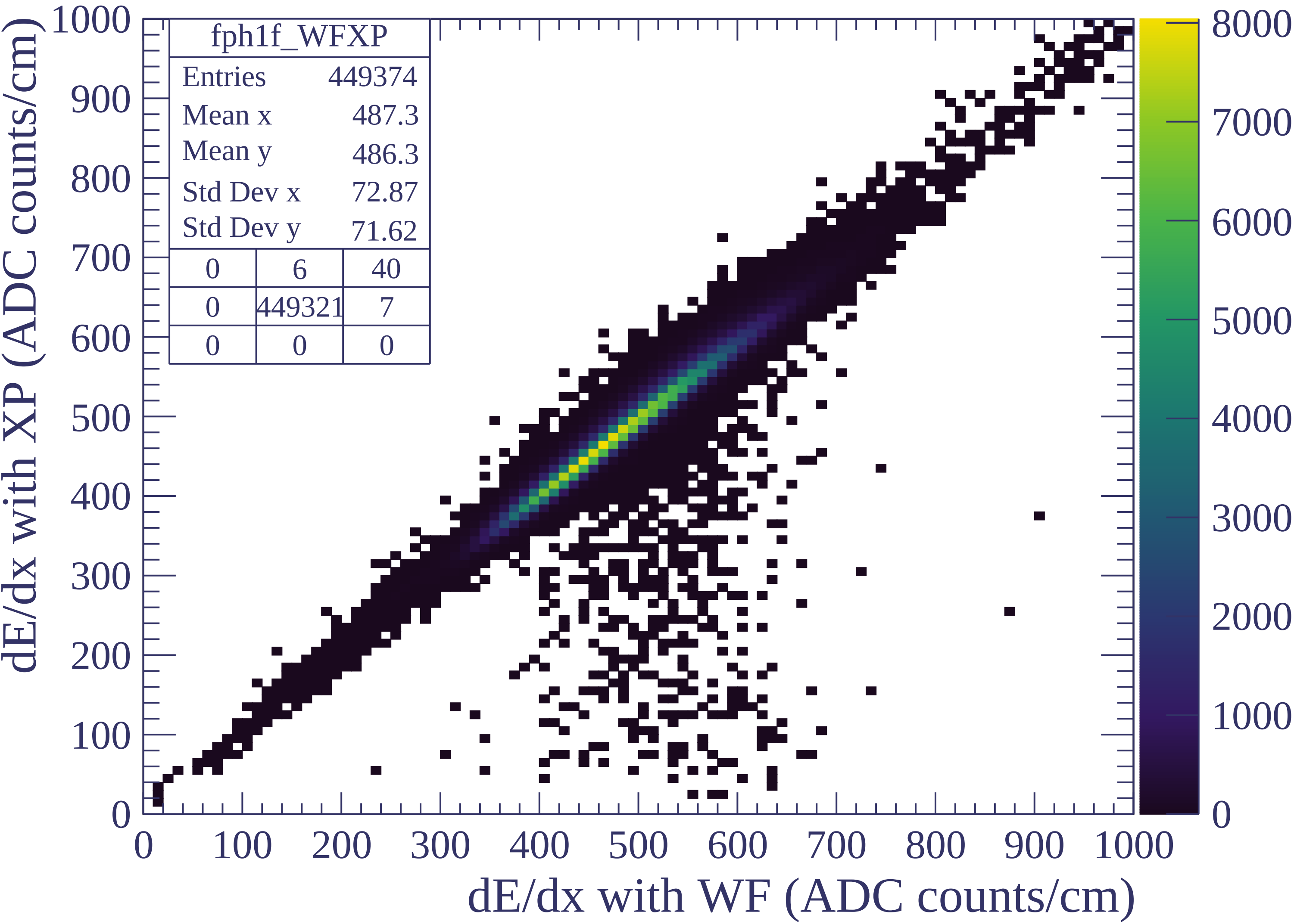
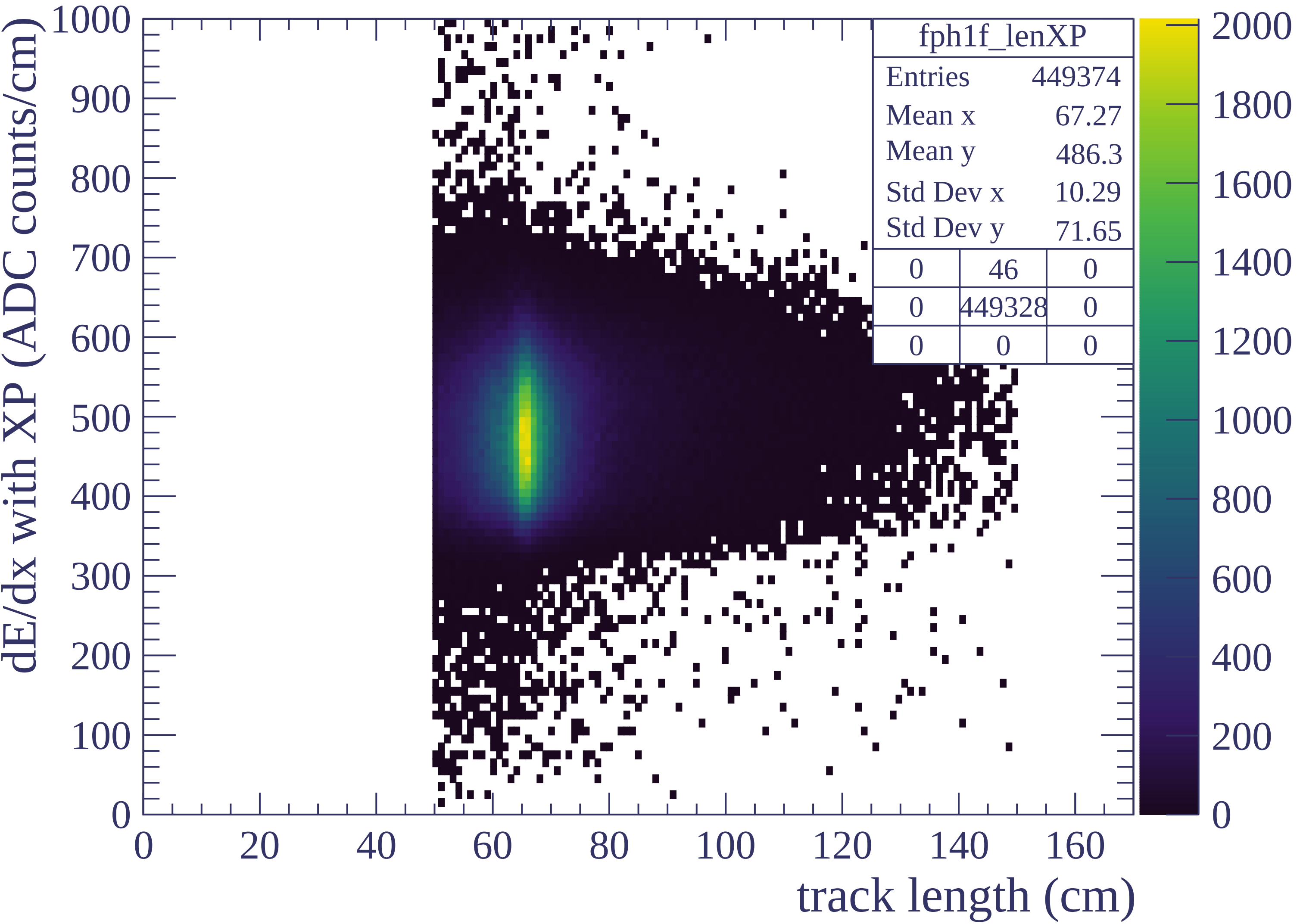

×10^3
Count
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP (449374 entries)
σ/μ = 14.59± 0.02 %
μ = 486.1± 0.1
σ = 70.9± 0.1
WF (449374 entries)
σ/μ = 14.99± 0.02 %
μ = 486.4± 0.1
σ = 72.9± 0.1

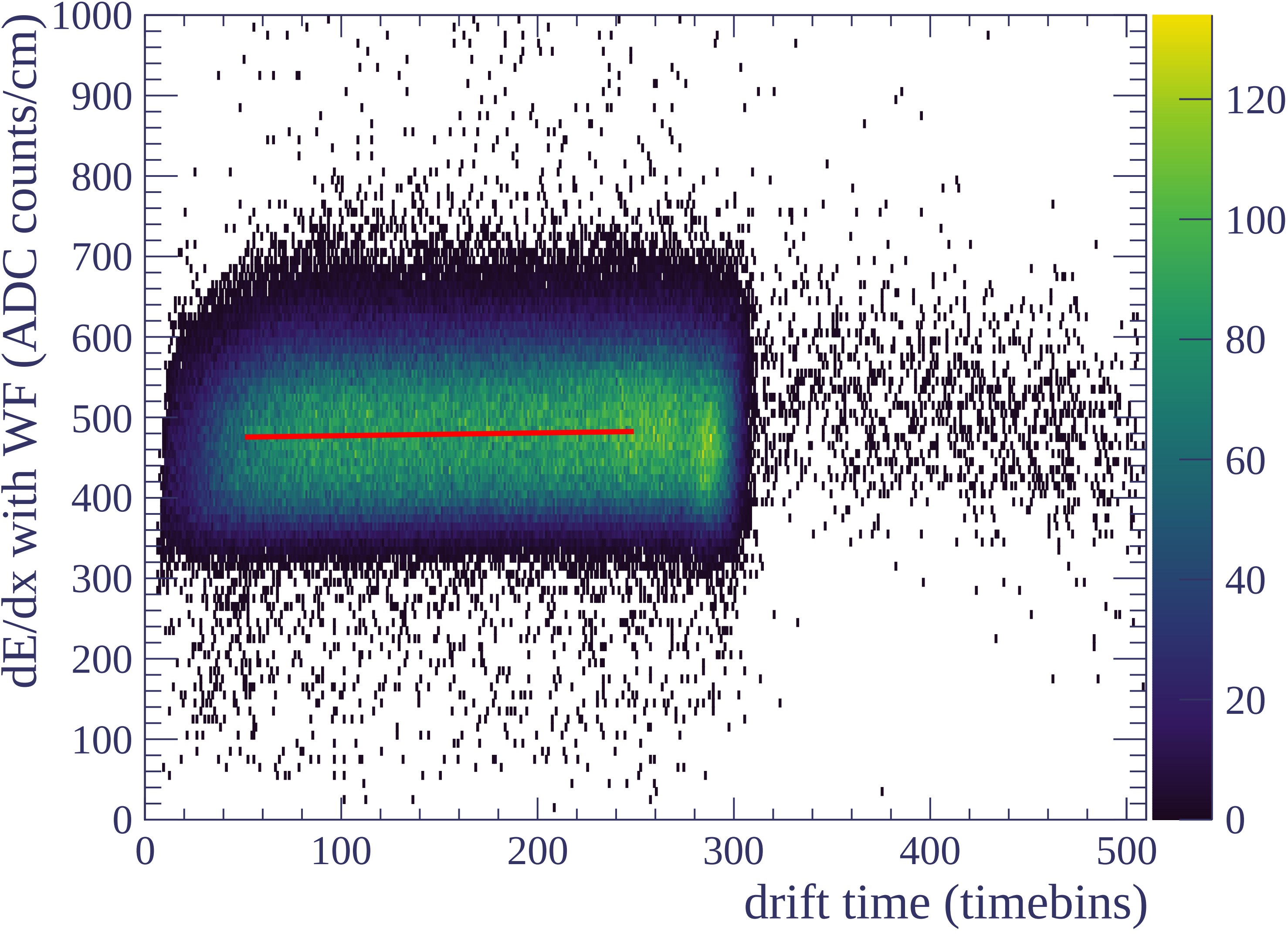

Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
Energy loss in ERAM 21
Energy loss in ERAM 13
Energy loss in ERAM 9
Energy loss in ERAM 2
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
Energy loss in ERAM 1
Energy loss in ERAM 10
Energy loss in ERAM 11
Energy loss in ERAM 3
Count
dE/dx (ADC counts/cm)



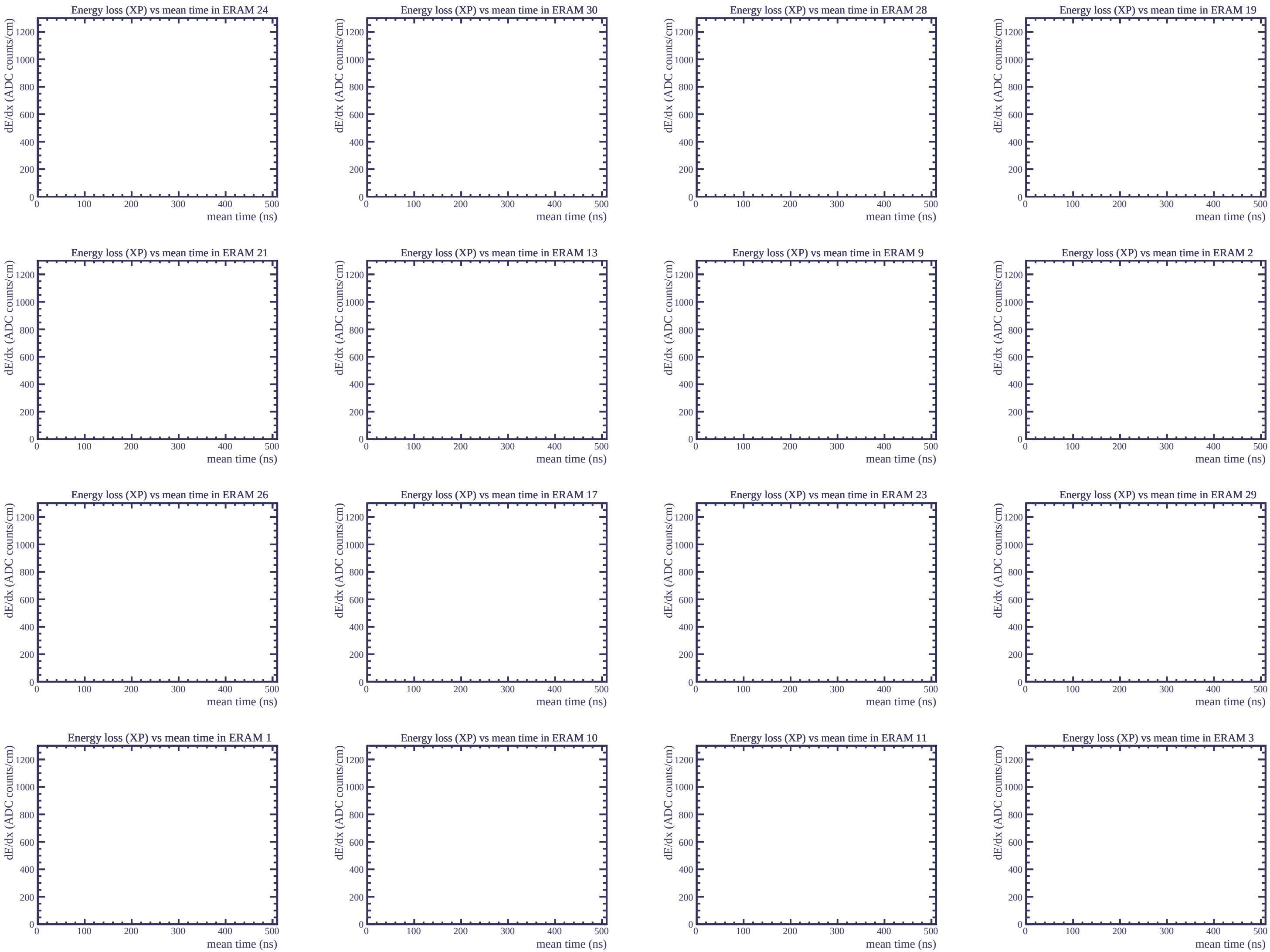
Energy loss in ERAM 47
WF (585 entries)
σ/μ = 15.75± 0.98 %
μ = 518.9± 4.2
σ = 81.7± 4.4
XP (585 entries)
σ/μ = 14.94± 0.80 %
μ = 529.0± 3.9
σ = 79.0± 3.6
Energy loss in ERAM 16
WF (2303 entries)
σ/μ = 16.24± 0.35 %
μ = 497.5± 1.9
σ = 80.8± 1.4
XP (2303 entries)
σ/μ = 16.04± 0.36 %
μ = 506.7± 1.9
σ = 81.3± 1.5
Energy loss in ERAM 14
WF (1723 entries)
σ/μ = 16.55± 0.42 %
μ = 514.6± 2.3
σ = 85.1± 1.8
XP (1723 entries)
σ/μ = 16.72± 0.50 %
μ = 519.5± 2.5
σ = 86.8± 2.2
Energy loss in ERAM 15
WF (1542 entries)
σ/μ = 16.71± 0.43 %
μ = 526.5± 2.5
σ = 88.0± 1.8
XP (1542 entries)
σ/μ = 16.96± 0.49 %
μ = 532.7± 2.8
σ = 90.4± 2.1
Energy loss in ERAM 42
WF (56348 entries)
σ/μ = 14.32± 0.06 %
μ = 468.6± 0.3
σ = 67.1± 0.2
XP (56348 entries)
σ/μ = 14.04± 0.06 %
μ = 471.8± 0.3
σ = 66.2± 0.2
Energy loss in ERAM 45
WF (54646 entries)
σ/μ = 14.45± 0.07 %
μ = 486.2± 0.3
σ = 70.3± 0.3
XP (54646 entries)
σ/μ = 14.25± 0.06 %
μ = 483.3± 0.3
σ = 68.9± 0.3
Energy loss in ERAM 37
WF (50601 entries)
σ/μ = 14.44± 0.07 %
μ = 508.0± 0.4
σ = 73.4± 0.3
XP (50601 entries)
σ/μ = 14.24± 0.07 %
μ = 500.7± 0.4
σ = 71.3± 0.3
Energy loss in ERAM 36
WF (36418 entries)
σ/μ = 14.30± 0.08 %
μ = 510.3± 0.4
σ = 72.9± 0.3
XP (36418 entries)
σ/μ = 14.12± 0.08 %
μ = 506.1± 0.4
σ = 71.4± 0.3
Energy loss in ERAM 20
WF (4936 entries)
σ/μ = 15.57± 0.27 %
μ = 464.3± 1.2
σ = 72.3± 1.1
XP (4936 entries)
σ/μ = 15.33± 0.24 %
μ = 469.0± 1.2
σ = 71.9± 1.0
Energy loss in ERAM 38
WF (6563 entries)
σ/μ = 15.85± 0.22 %
μ = 472.7± 1.0
σ = 74.9± 0.9
XP (6563 entries)
σ/μ = 15.16± 0.18 %
μ = 482.0± 1.0
σ = 73.0± 0.7
Energy loss in ERAM 7
WF (5273 entries)
σ/μ = 15.17± 0.22 %
μ = 486.0± 1.1
σ = 73.7± 0.9
XP (5273 entries)
σ/μ = 14.72± 0.21 %
μ = 488.5± 1.1
σ = 71.9± 0.8
Energy loss in ERAM 44
WF (4268 entries)
σ/μ = 15.77± 0.30 %
μ = 483.2± 1.4
σ = 76.2± 1.2
XP (4268 entries)
σ/μ = 15.37± 0.26 %
μ = 487.4± 1.4
σ = 74.9± 1.1
Energy loss in ERAM 43
WF (65639 entries)
σ/μ = 14.80± 0.06 %
μ = 491.7± 0.3
σ = 72.8± 0.3
XP (65639 entries)
σ/μ = 14.44± 0.06 %
μ = 491.3± 0.3
σ = 70.9± 0.2
Energy loss in ERAM 39
WF (61238 entries)
σ/μ = 14.86± 0.07 %
μ = 471.4± 0.3
σ = 70.0± 0.3
XP (61238 entries)
σ/μ = 14.01± 0.06 %
μ = 476.9± 0.3
σ = 66.8± 0.2
Energy loss in ERAM 41
WF (56287 entries)
σ/μ = 14.90± 0.07 %
μ = 493.4± 0.4
σ = 73.5± 0.3
XP (56287 entries)
σ/μ = 14.04± 0.06 %
μ = 493.0± 0.3
σ = 69.2± 0.2
Energy loss in ERAM 46
WF (41004 entries)
σ/μ = 14.25± 0.07 %
μ = 478.8± 0.4
σ = 68.2± 0.3
XP (41004 entries)
σ/μ = 14.67± 0.08 %
μ = 475.3± 0.4
σ = 69.7± 0.3
Count
dE/dx (ADC counts/cm)

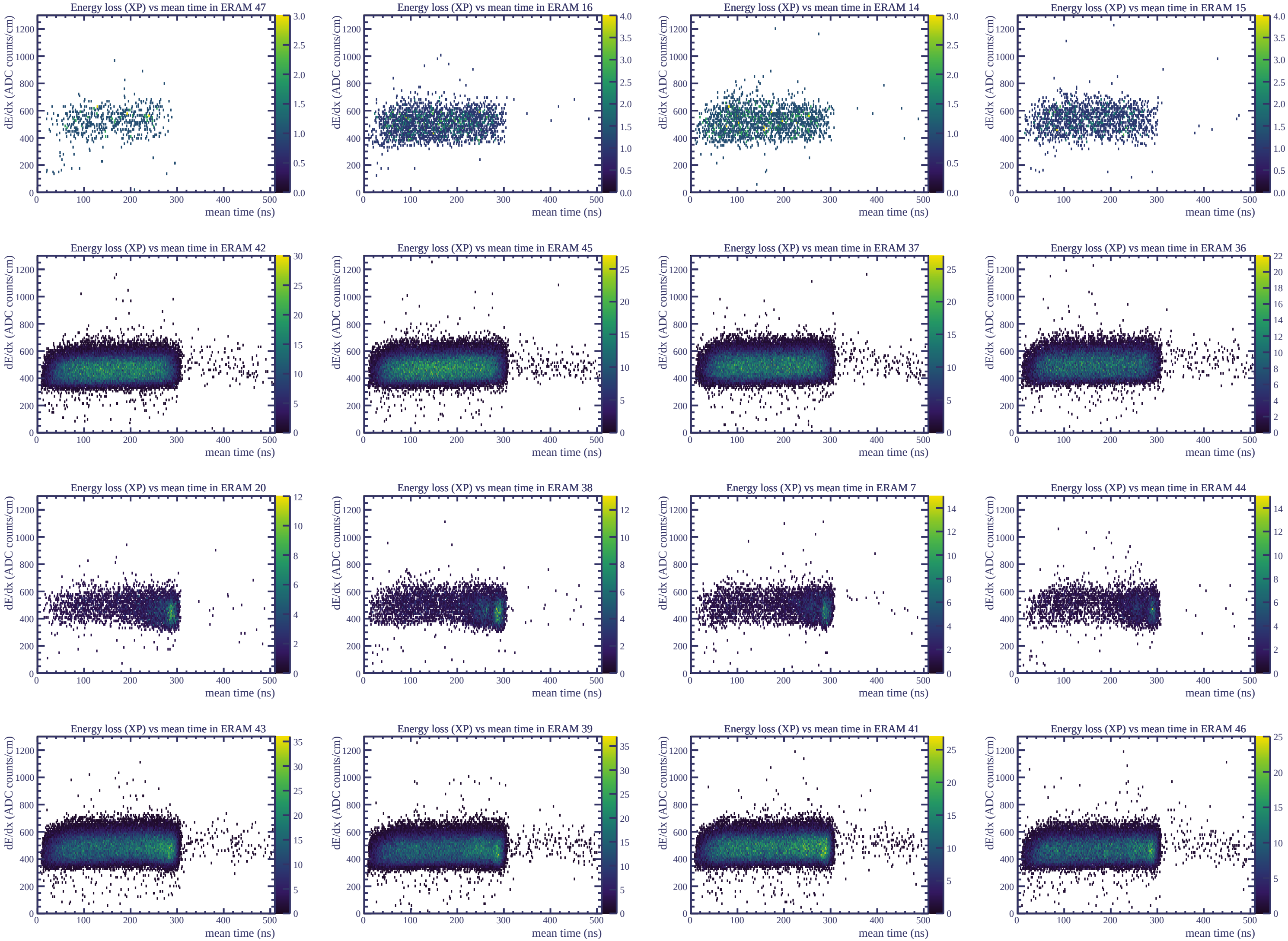

fph1f_WFXP
Entries 449374
Mean x 487.3
Mean y 486.3
Std Dev x 72.87
Std Dev y 71.62
0 6 40
0 449321 7
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
1000
2000
3000
4000
5000
6000
7000
8000

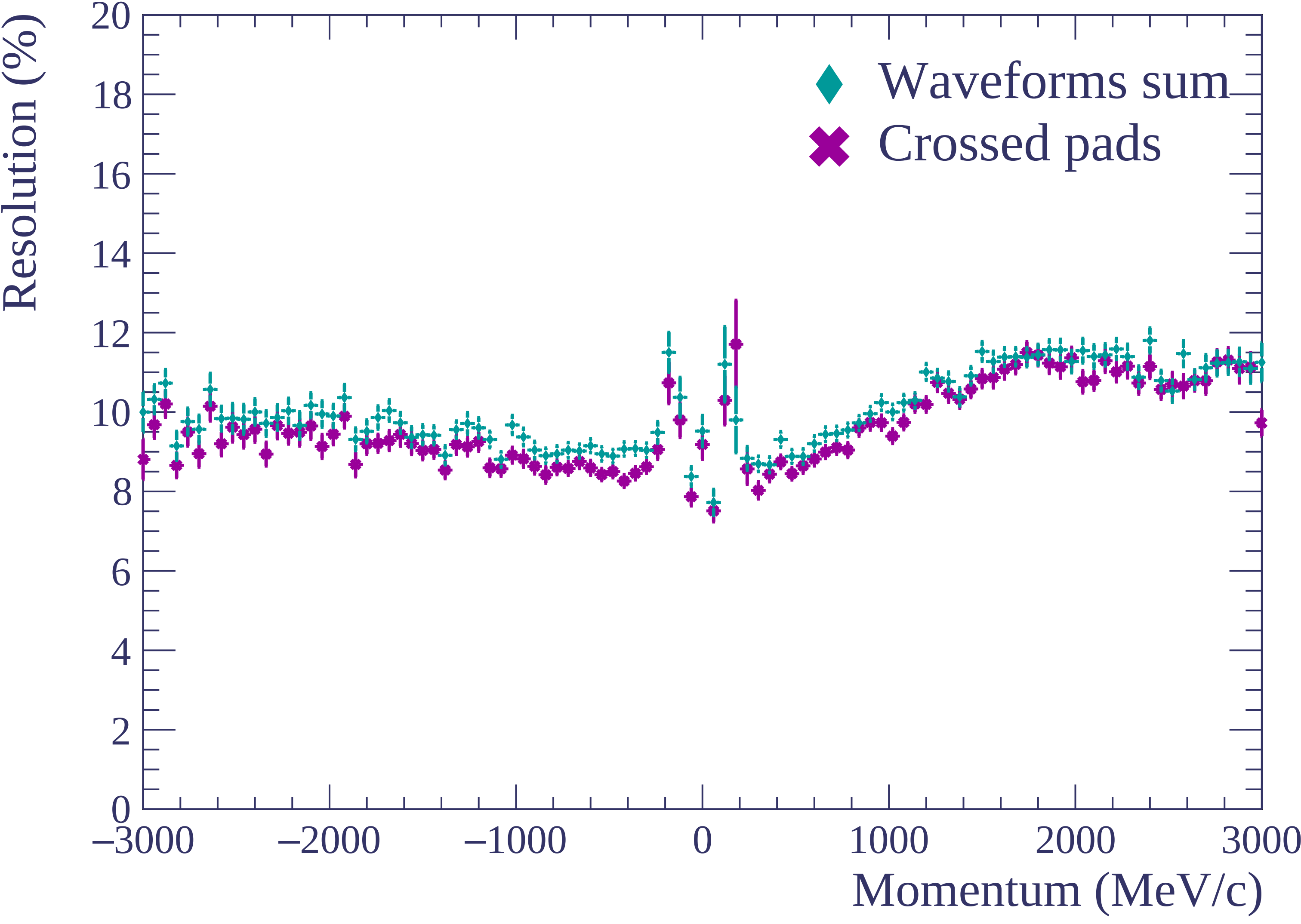

fph1f_lenXP
Entries 449374
Mean x 67.27
Mean y 486.3
Std Dev x 10.29
Std Dev y 71.65
0 46 0
0 449328 0
0 0 0
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000

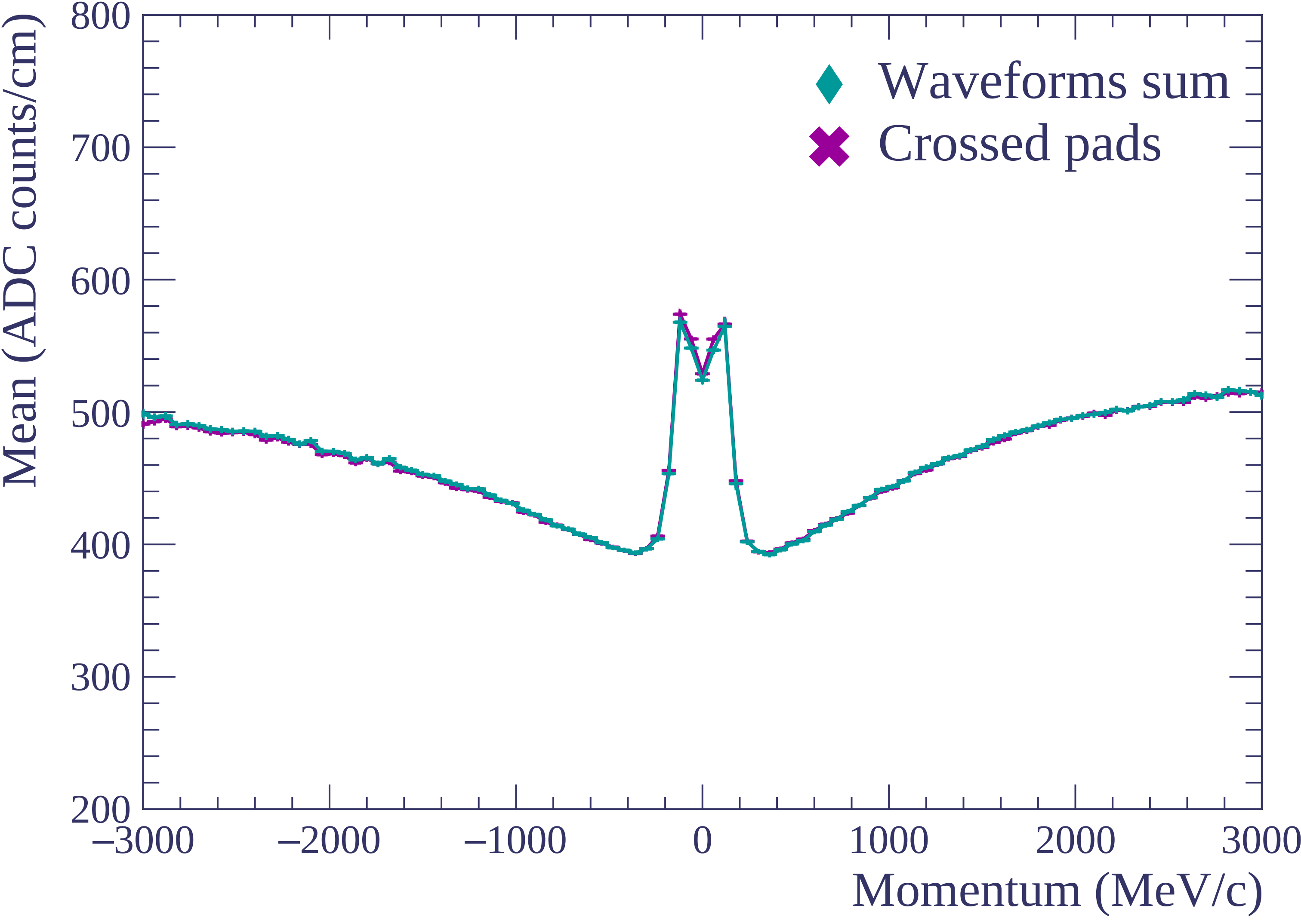


dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
20
40
60
80
100
120

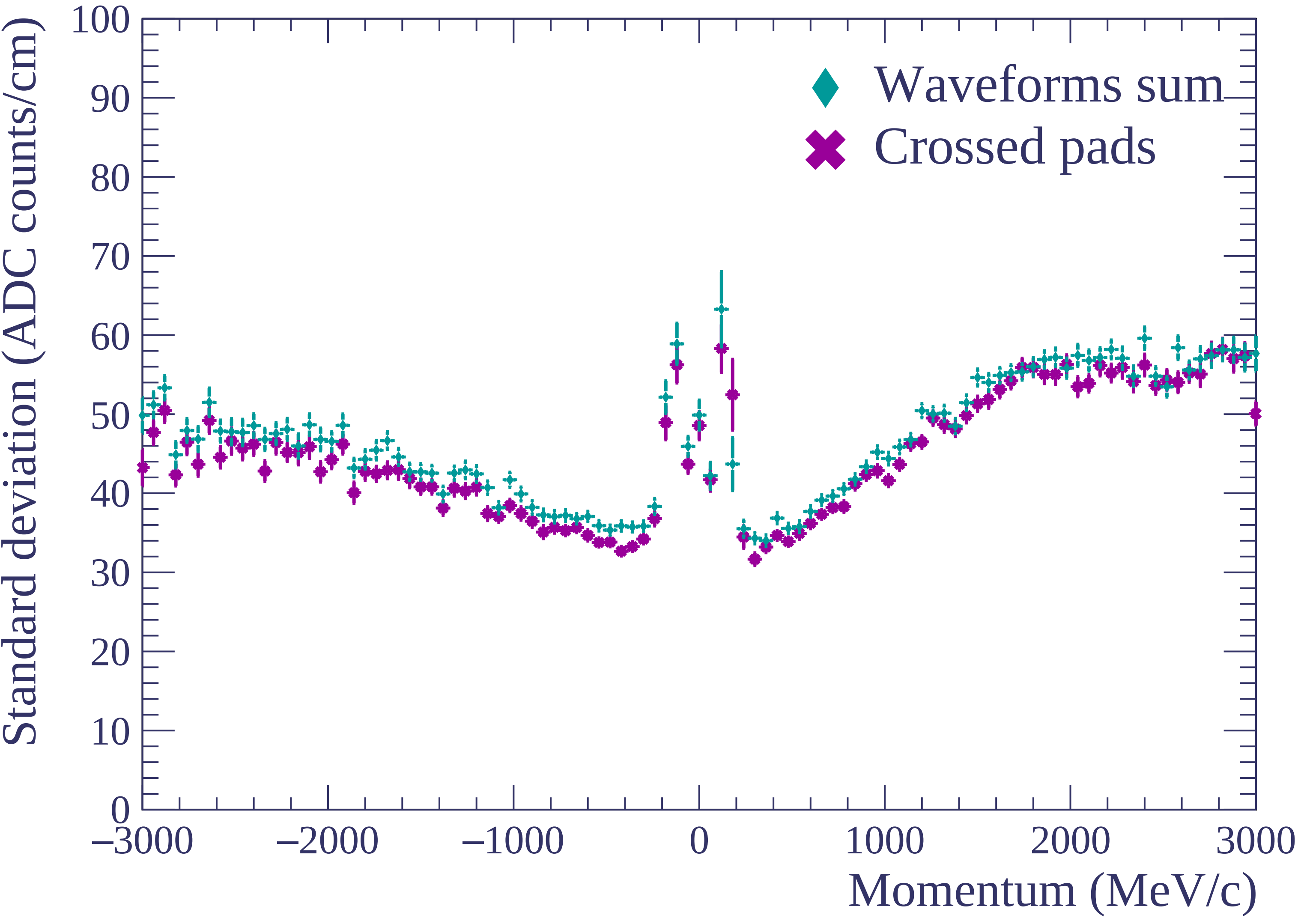

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)

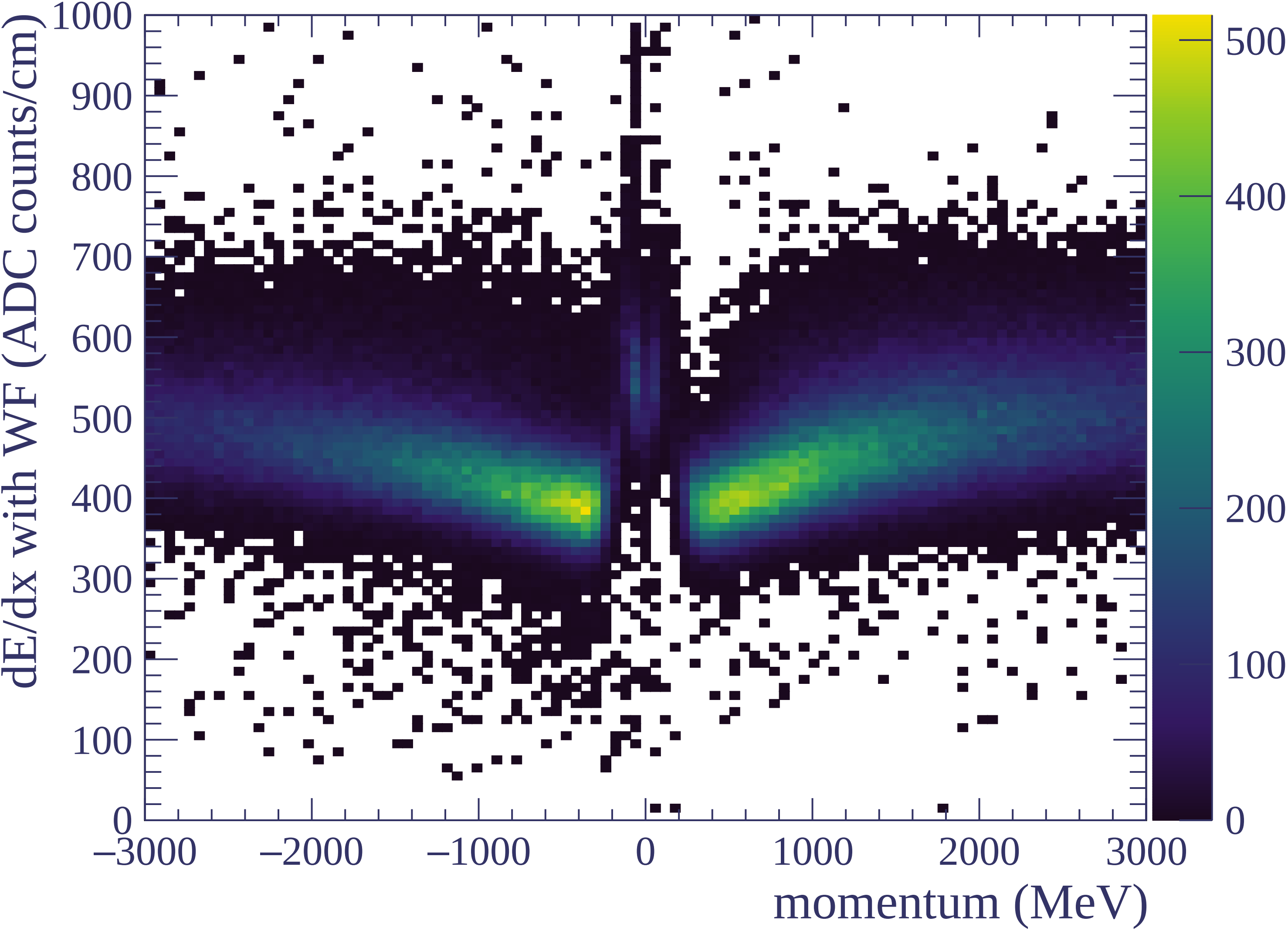

Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)

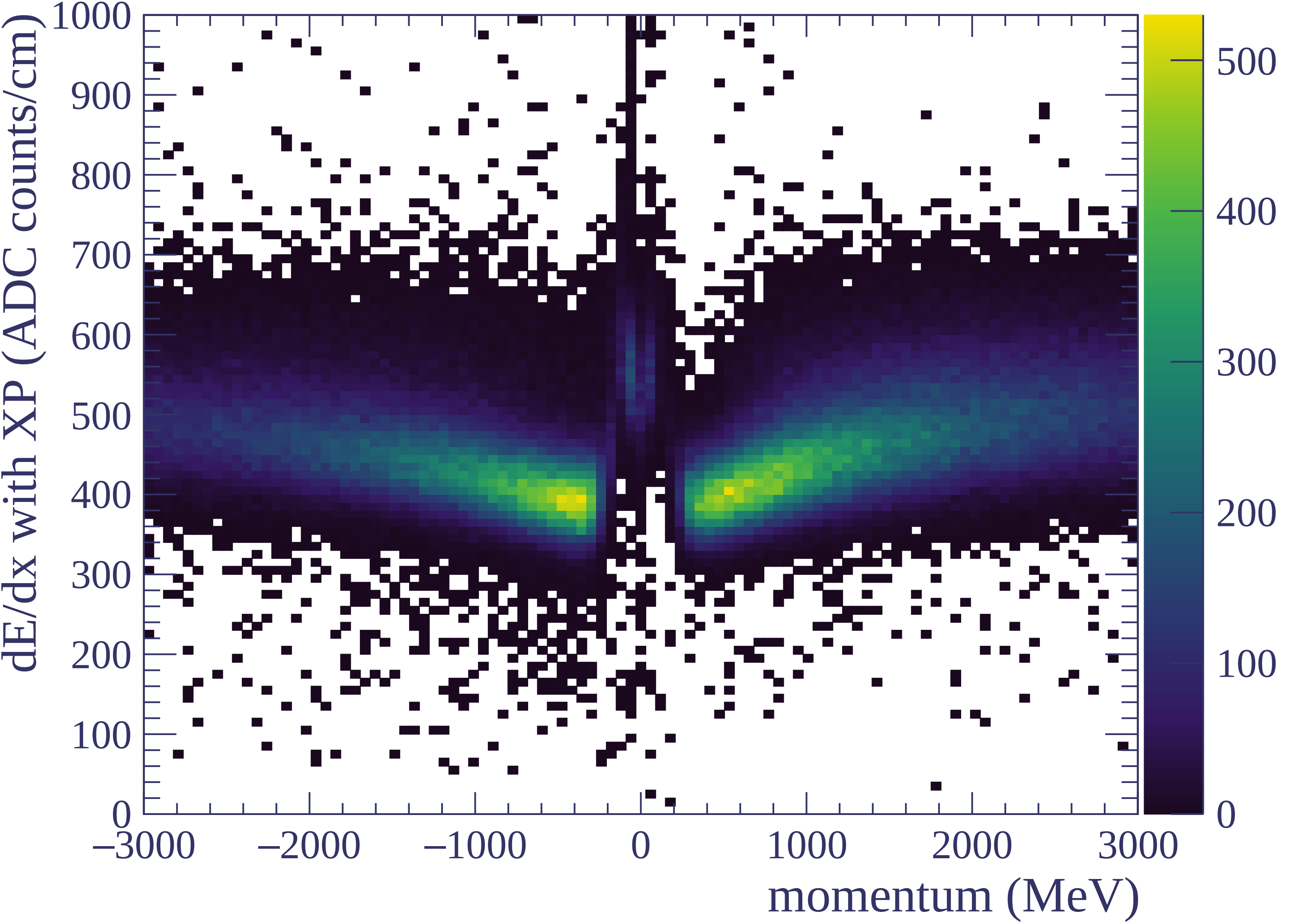

Waveforms sum
Crossed pads
Resolution (%)
Momentum (MeV/c)
0
2
4
6
8
10
12
14
16
18
20
−3000
−2000
−1000
0
1000
2000
3000



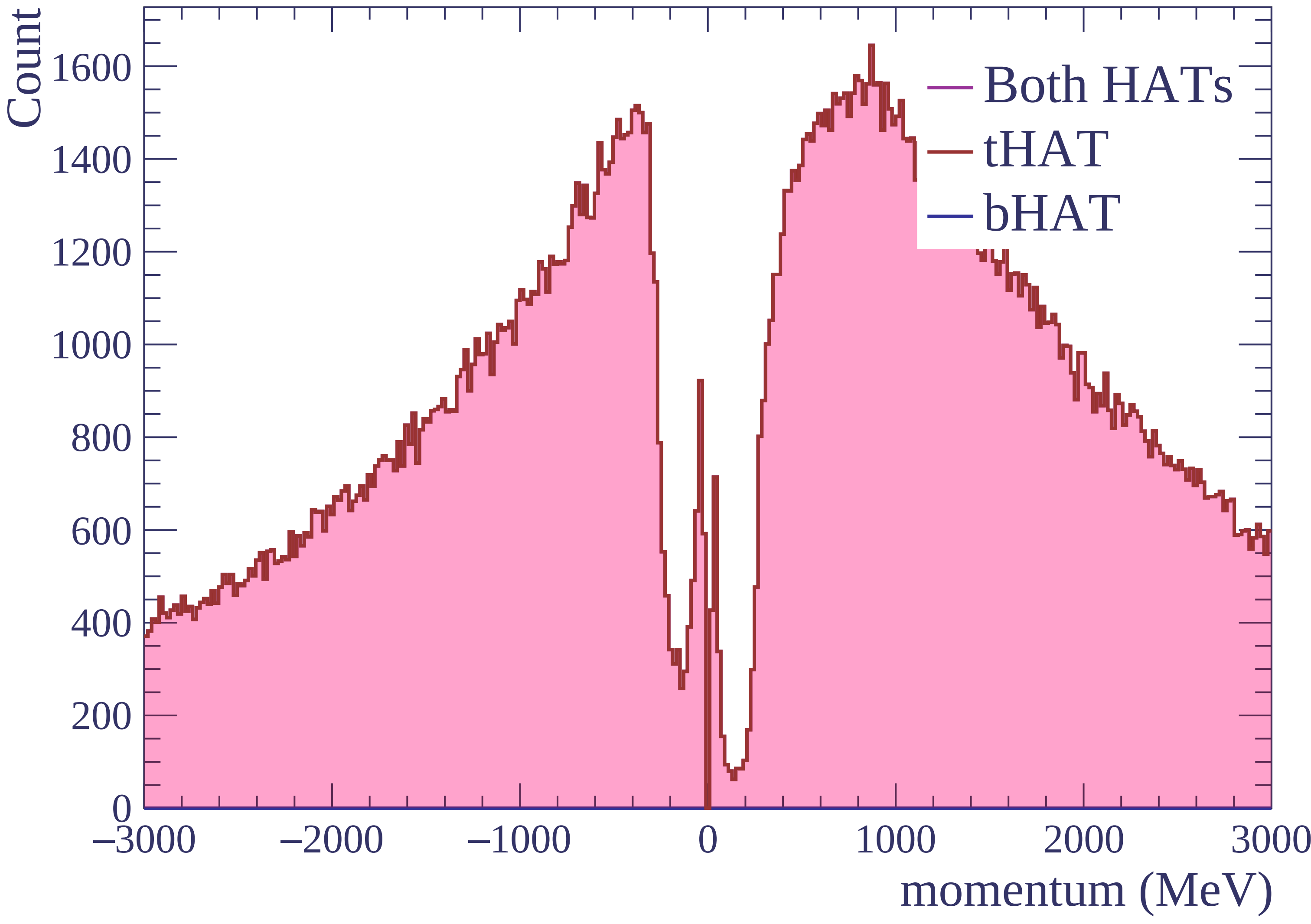
Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000


Waveforms sum
Crossed pads
Standard deviation (ADC counts/cm)
Momentum (MeV/c)
100
90
80
70
60
50
40
30
20
10
0
−3000
−2000
−1000
0
1000
2000
3000

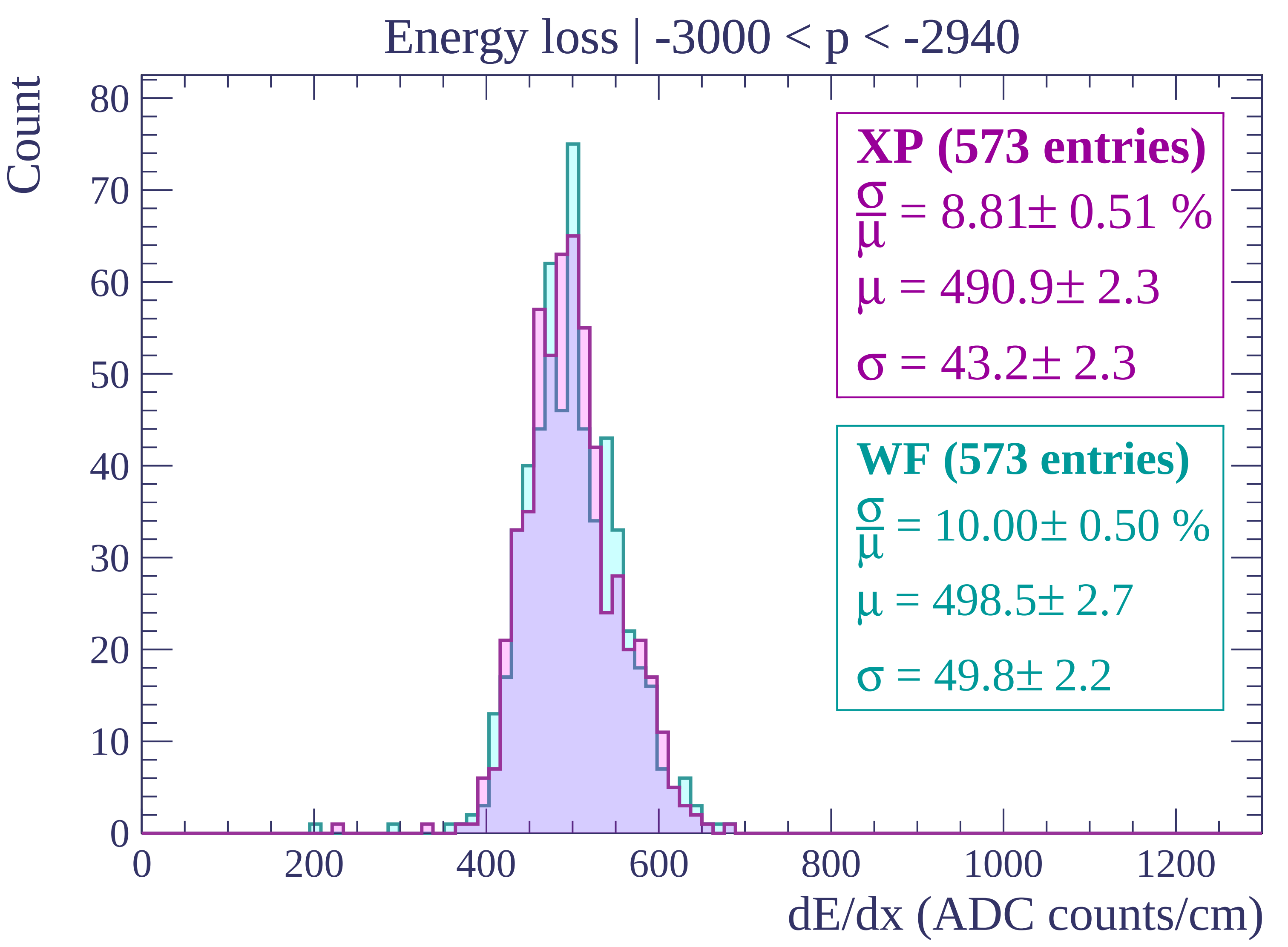

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
500
400
300
200
100
0



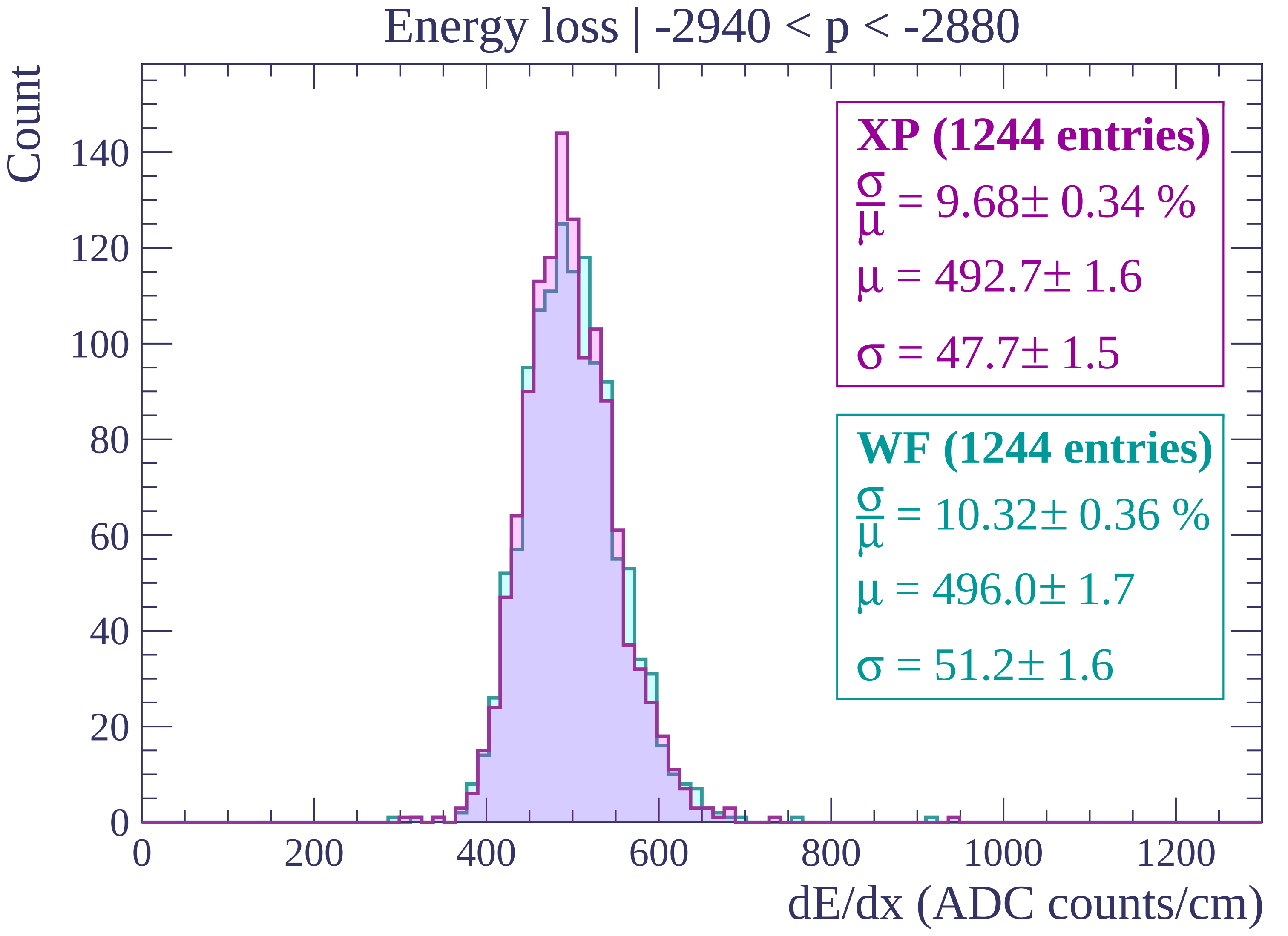
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
500
400
300
200
100
0

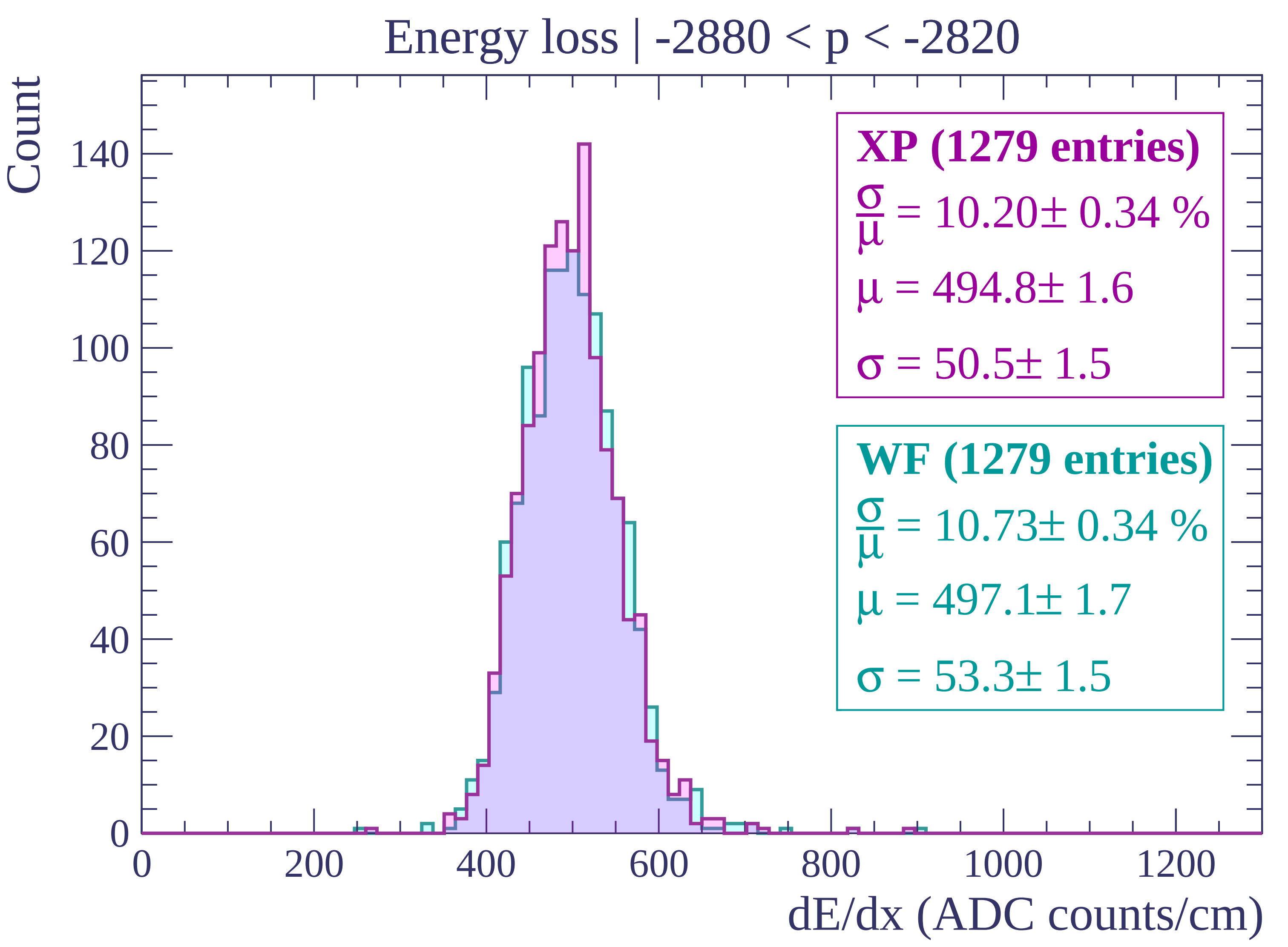


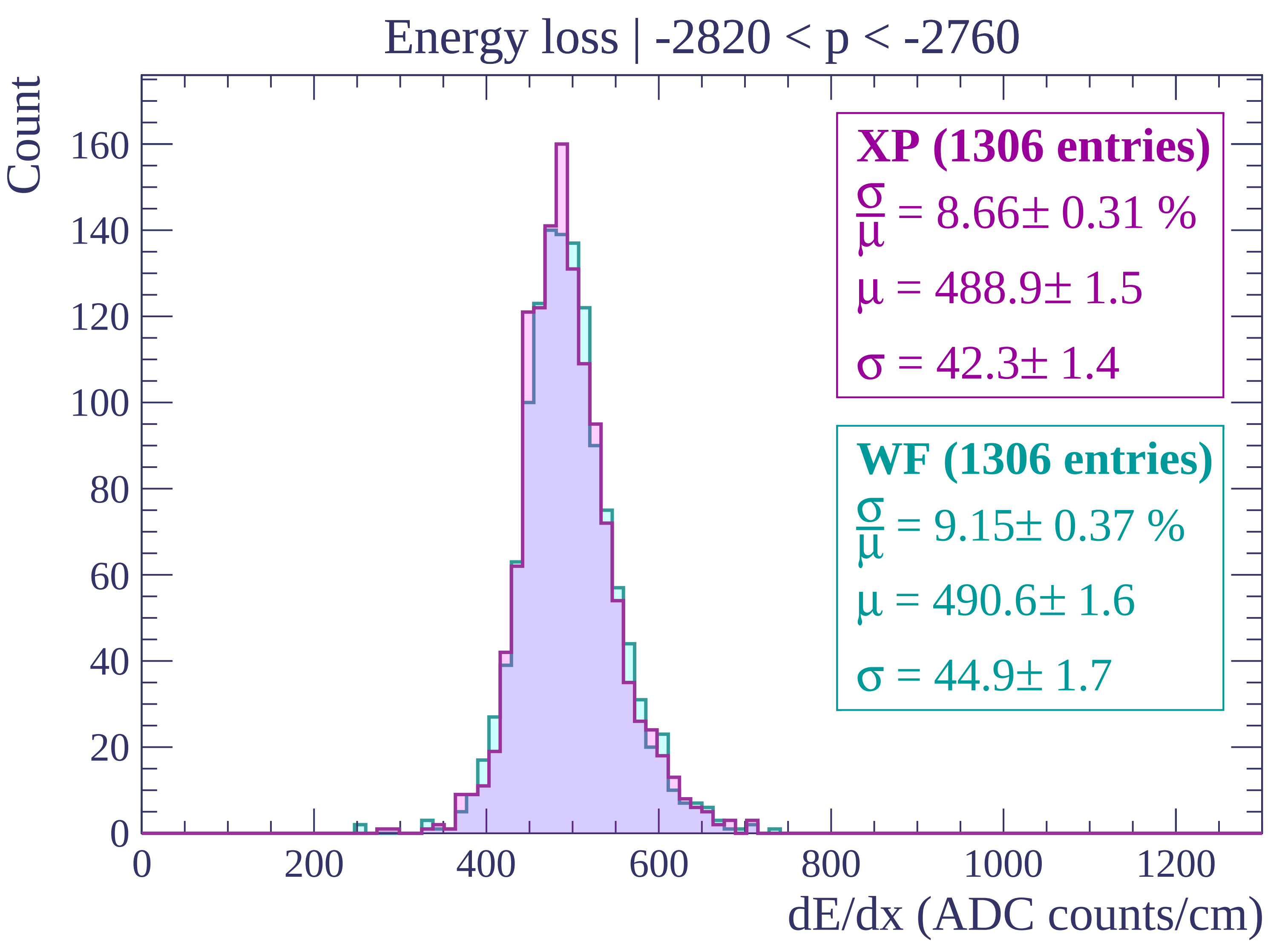
Count
1600
1400
1200
1000
800
600
400
200
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



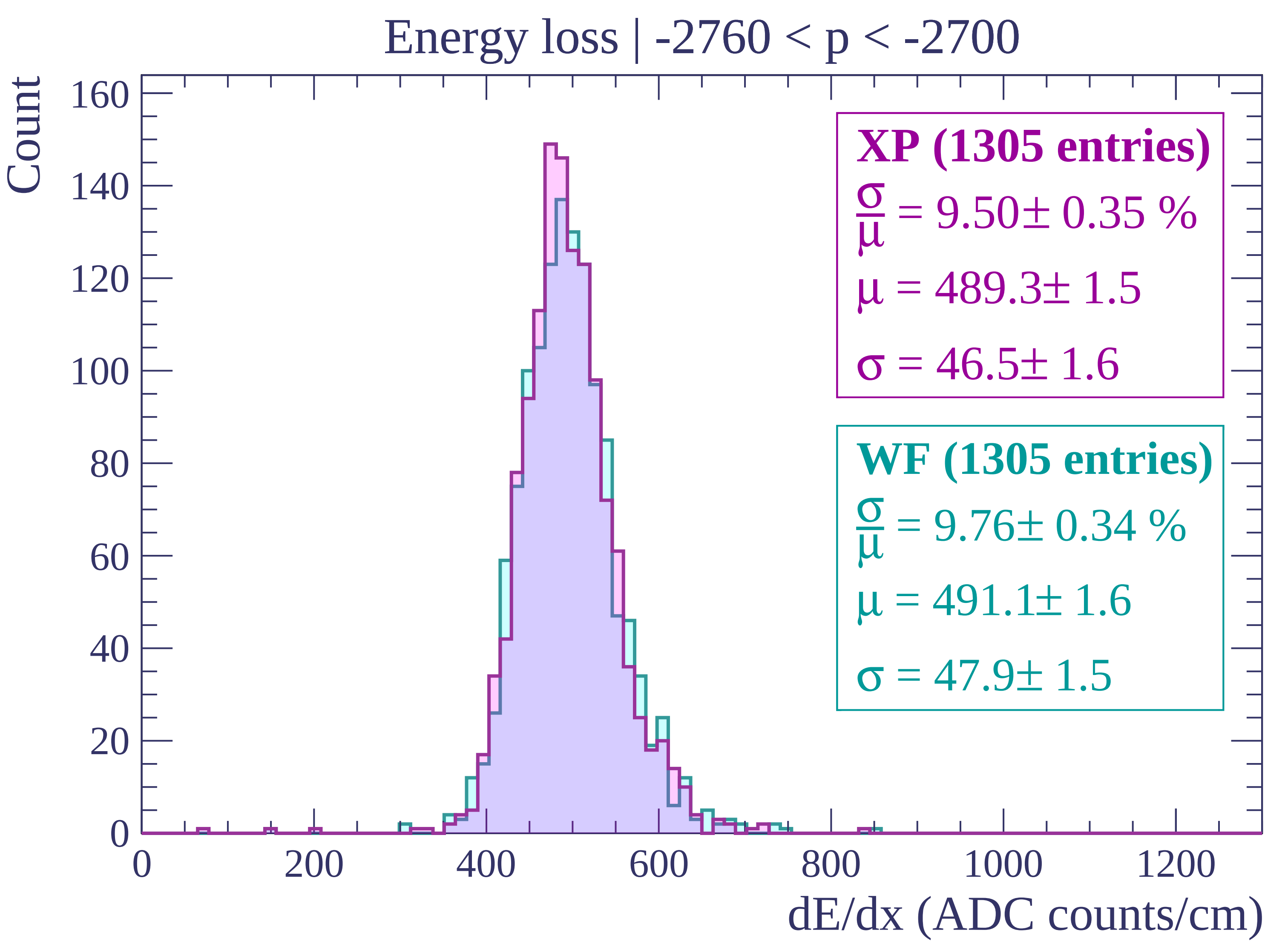
Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200
XP (573 entries)
σ/μ = 8.81± 0.51 %
μ = 490.9± 2.3
σ = 43.2± 2.3
WF (573 entries)
σ/μ = 10.00± 0.50 %
μ = 498.5± 2.7
σ = 49.8± 2.2

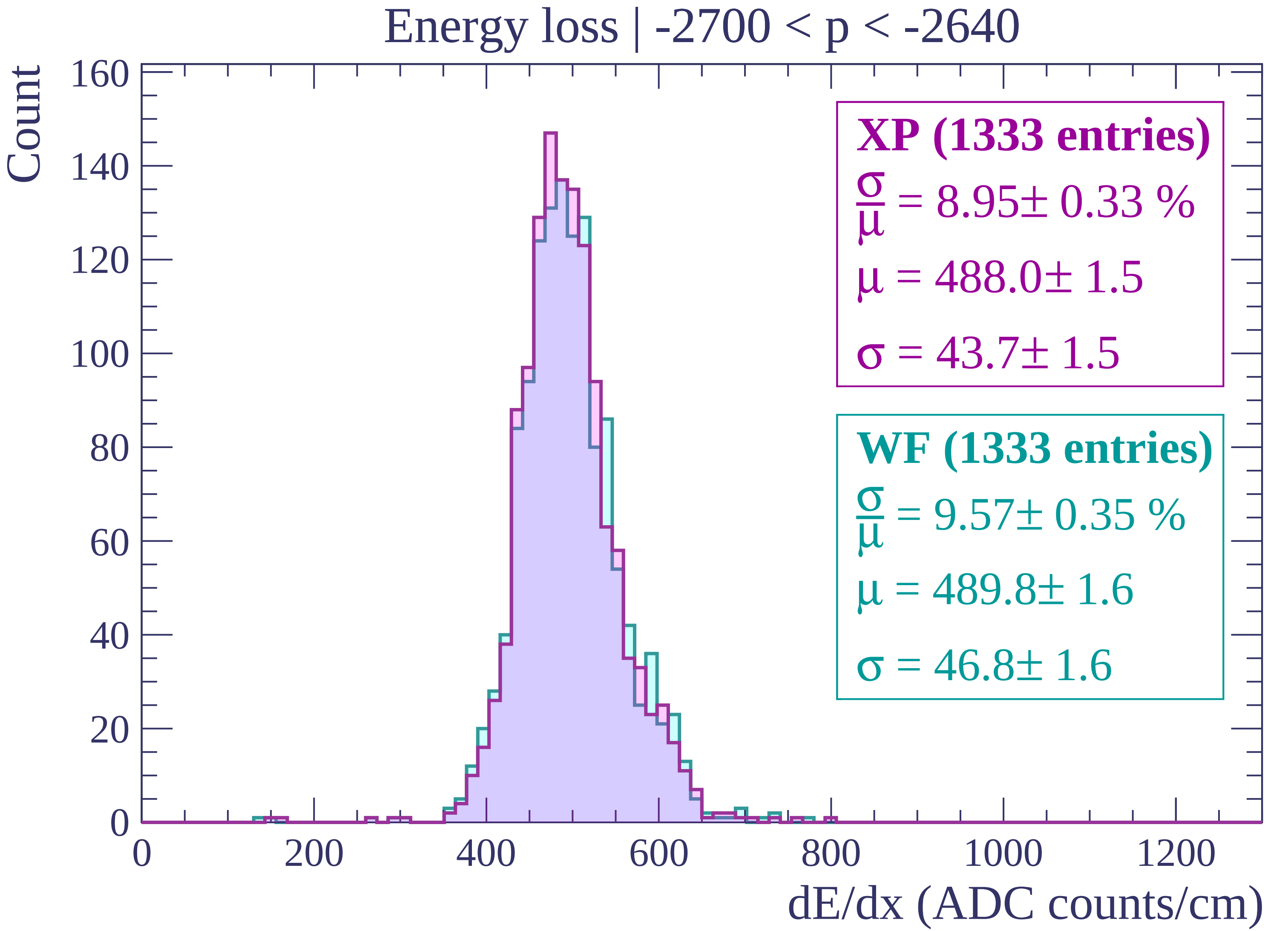

Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
XP (1244 entries)
σ/μ = 9.68± 0.34 %
μ = 492.7± 1.6
σ = 47.7± 1.5
WF (1244 entries)
σ/μ = 10.32± 0.36 %
μ = 496.0± 1.7
σ = 51.2± 1.6
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200


Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
XP (1279 entries)
$\frac{\sigma}{\mu}$ = 10.20± 0.34 %
μ = 494.8± 1.6
σ = 50.5± 1.5
WF (1279 entries)
$\frac{\sigma}{\mu}$ = 10.73± 0.34 %
μ = 497.1± 1.7
σ = 53.3± 1.5
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

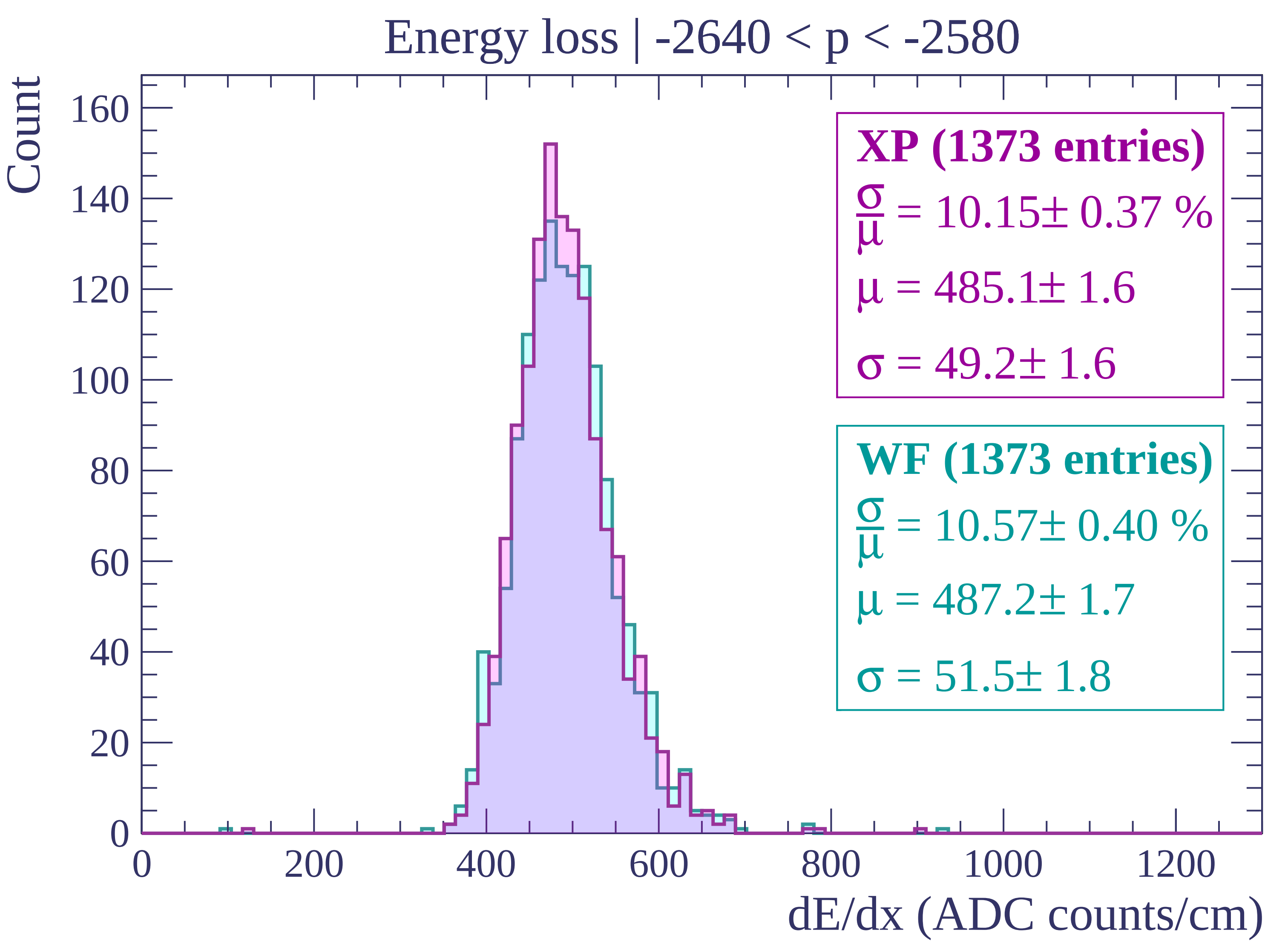


Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1306 entries)
σ/μ = 8.66± 0.31 %
μ = 488.9± 1.5
σ = 42.3± 1.4
WF (1306 entries)
σ/μ = 9.15± 0.37 %
μ = 490.6± 1.6
σ = 44.9± 1.7

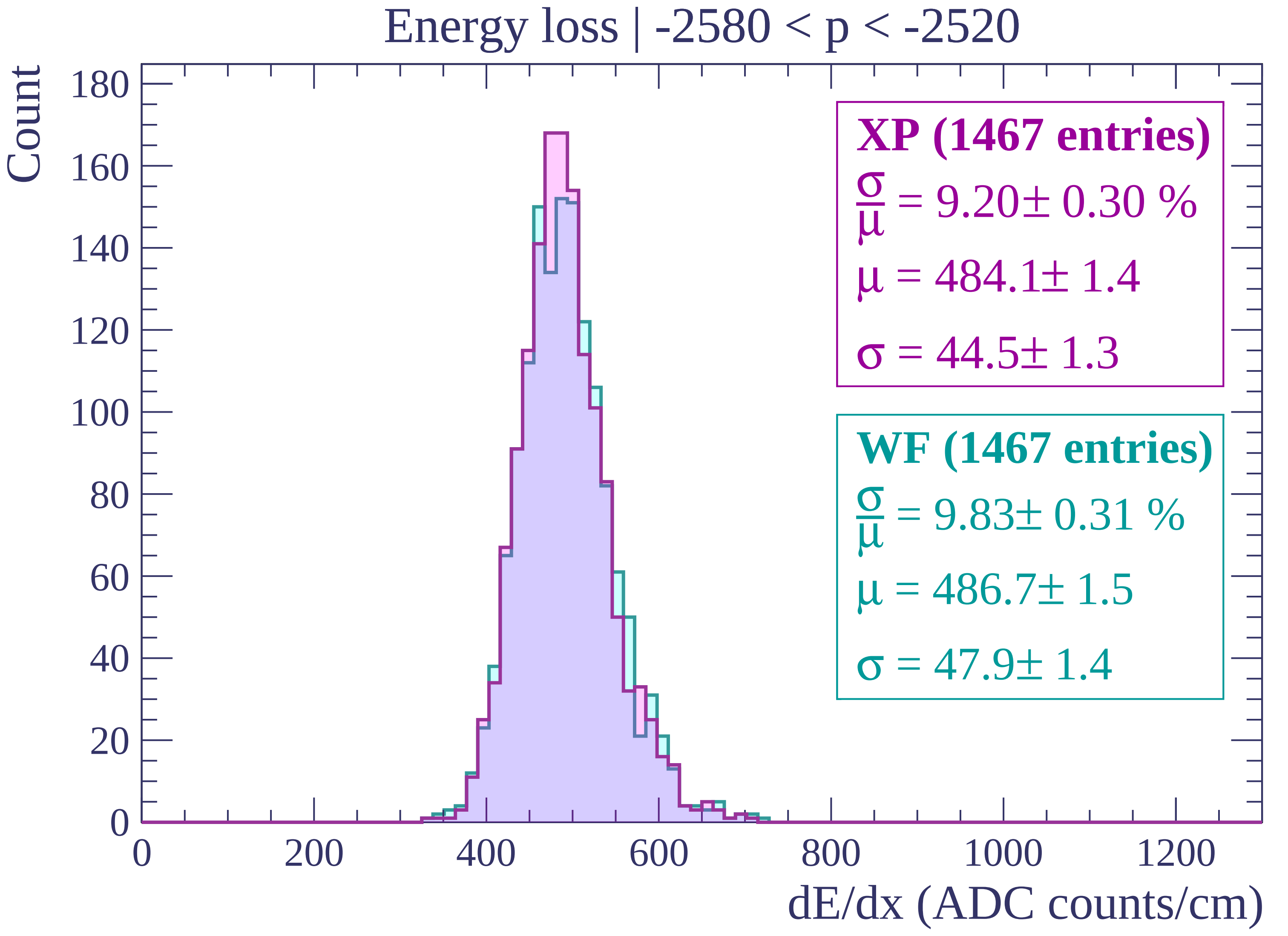

Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1305 entries)
$\frac{\sigma}{\mu}$ = 9.50± 0.35 %
μ = 489.3± 1.5
σ = 46.5± 1.6
WF (1305 entries)
$\frac{\sigma}{\mu}$ = 9.76± 0.34 %
μ = 491.1± 1.6
σ = 47.9± 1.5

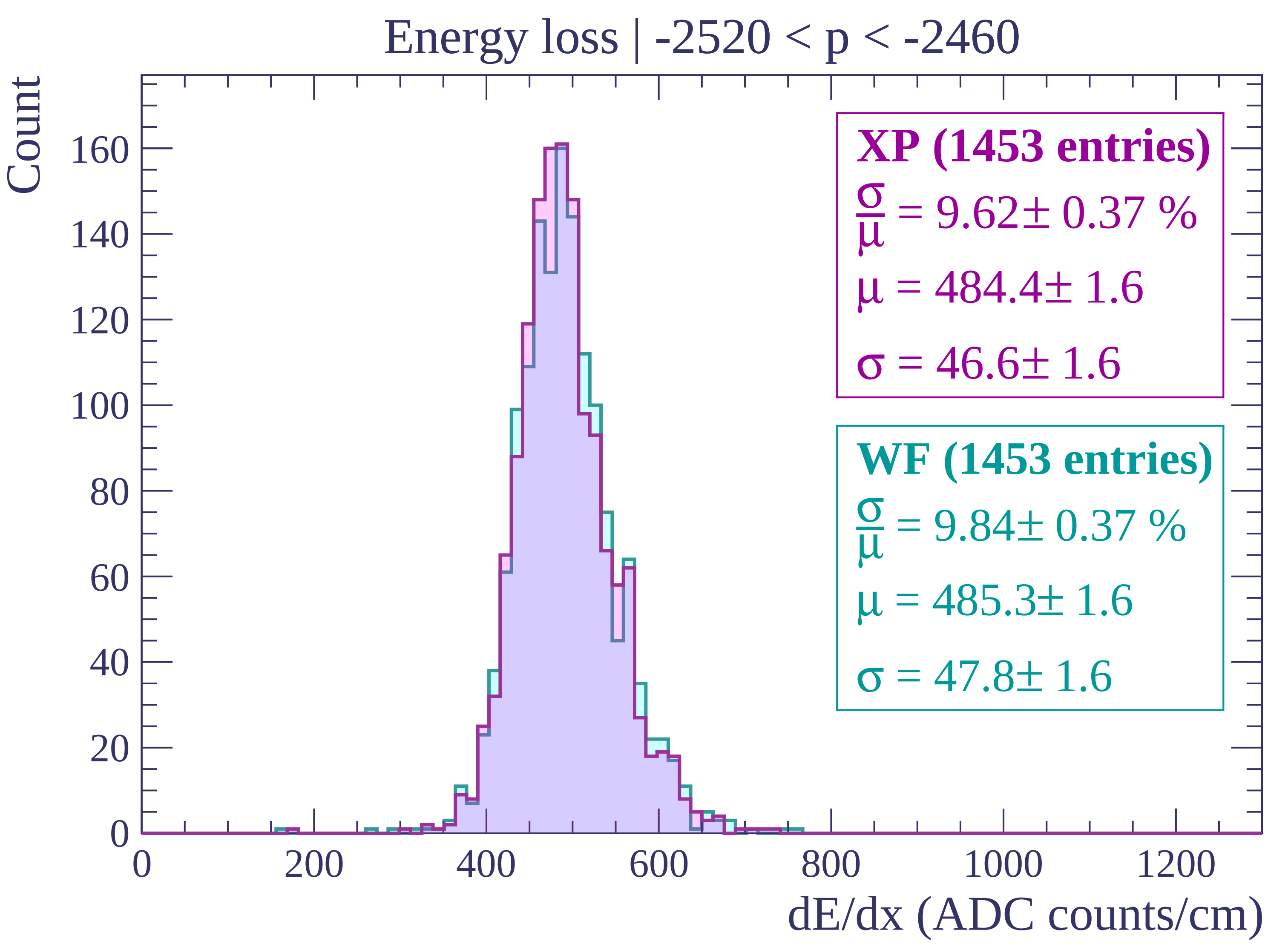


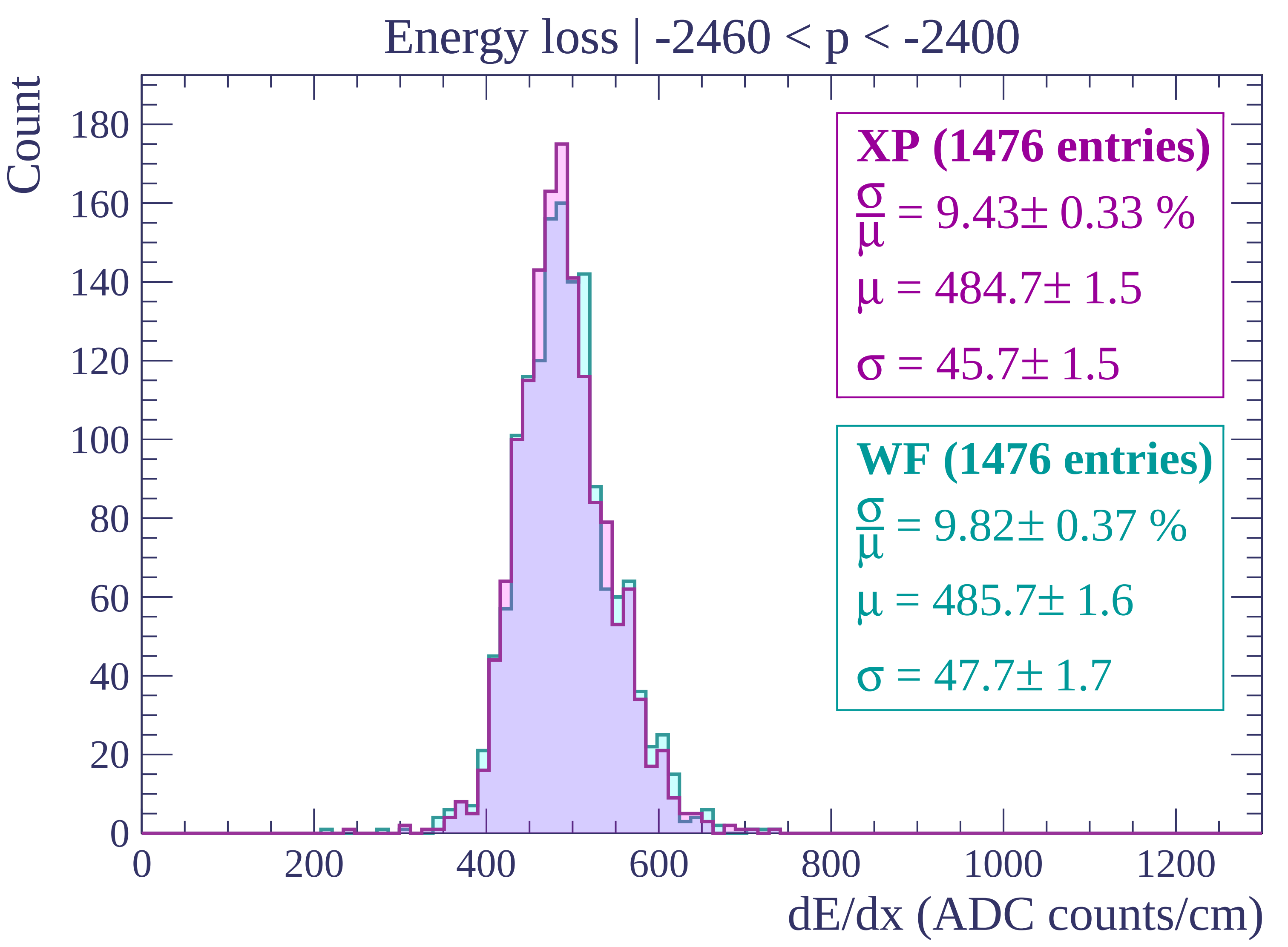
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1333 entries)
σ/μ = 8.95± 0.33 %
μ = 488.0± 1.5
σ = 43.7± 1.5
WF (1333 entries)
σ/μ = 9.57± 0.35 %
μ = 489.8± 1.6
σ = 46.8± 1.6



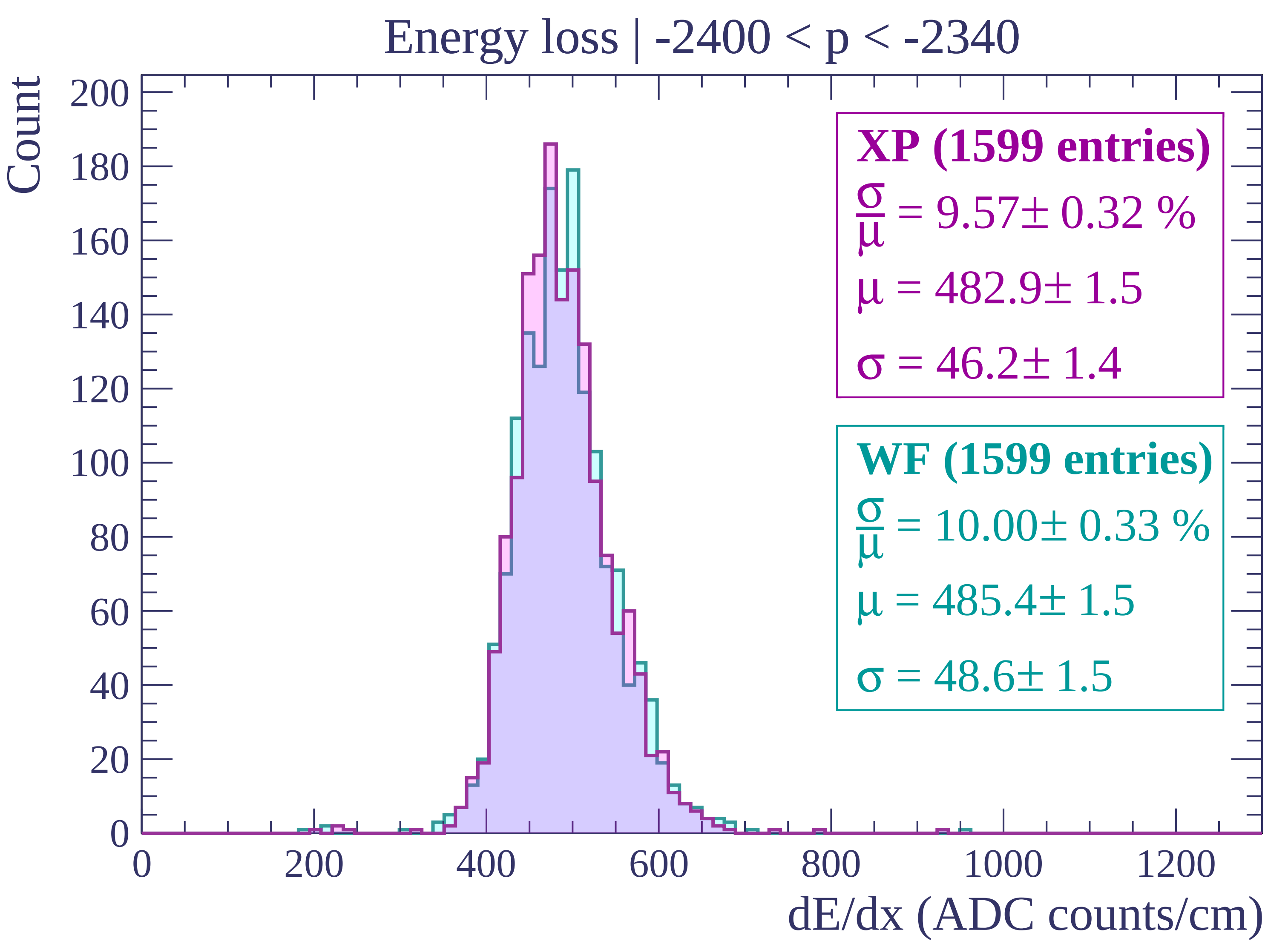
Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1373 entries)
σ/μ = 10.15± 0.37 %
μ = 485.1± 1.6
σ = 49.2± 1.6
WF (1373 entries)
σ/μ = 10.57± 0.40 %
μ = 487.2± 1.7
σ = 51.5± 1.8


Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1467 entries)
σ/μ = 9.20± 0.30 %
μ = 484.1± 1.4
σ = 44.5± 1.3
WF (1467 entries)
σ/μ = 9.83± 0.31 %
μ = 486.7± 1.5
σ = 47.9± 1.4

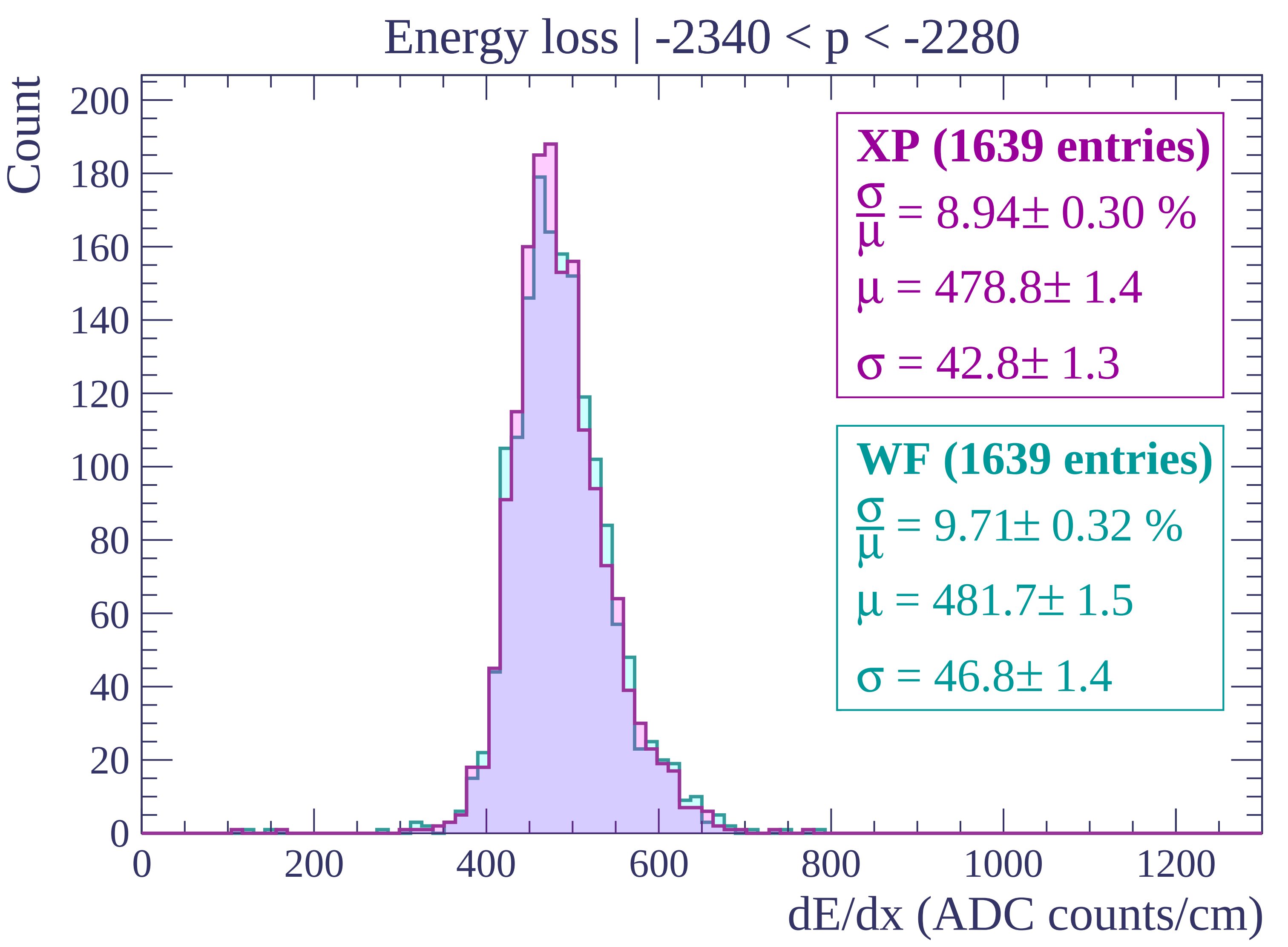

Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1453 entries)
$\frac{\sigma}{\mu}$ = 9.62± 0.37 %
μ = 484.4± 1.6
σ = 46.6± 1.6
WF (1453 entries)
$\frac{\sigma}{\mu}$ = 9.84± 0.37 %
μ = 485.3± 1.6
σ = 47.8± 1.6



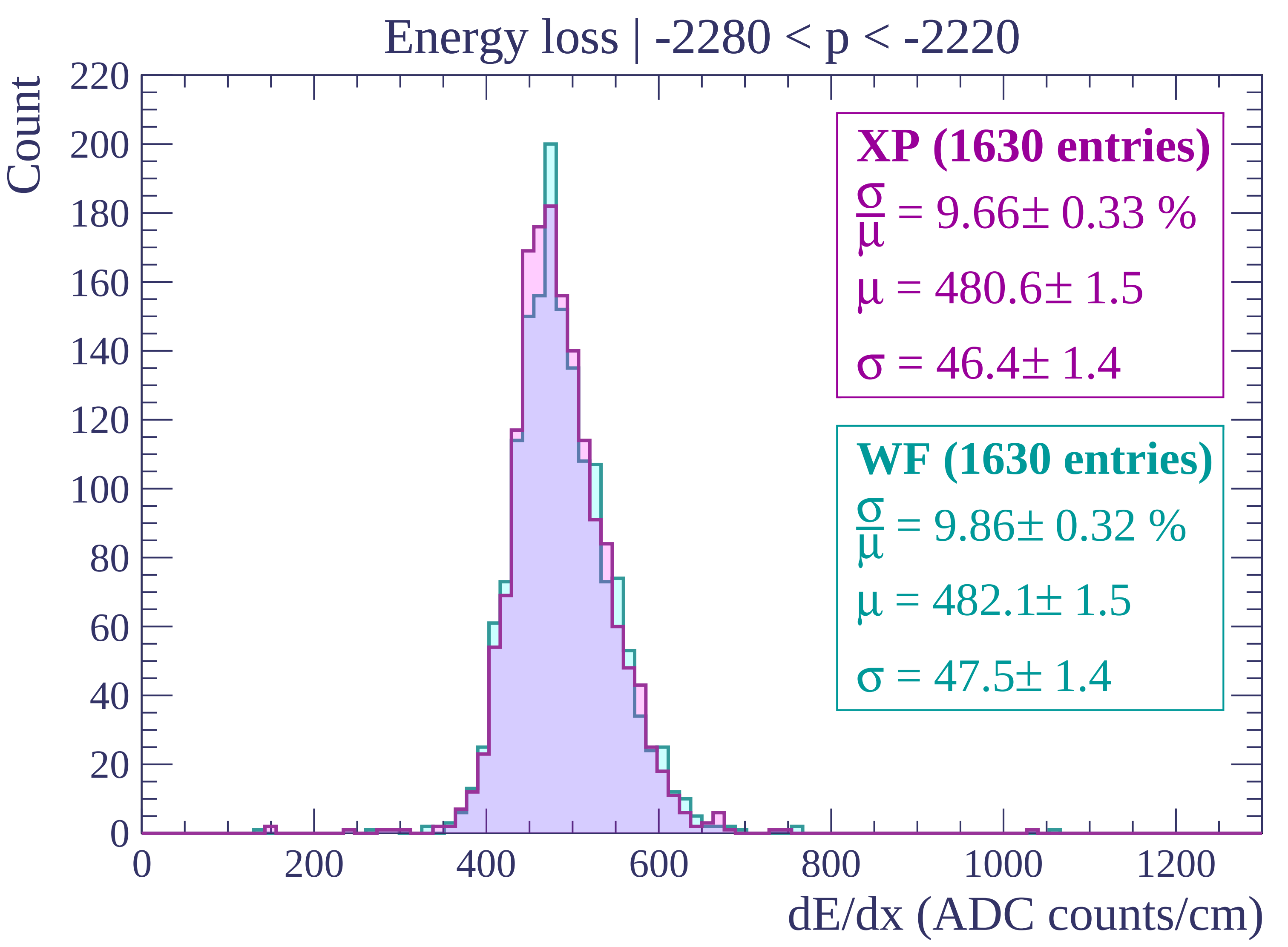
Energy loss | -2460 < p < -2400
Count
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1476 entries)
σ/μ = 9.43± 0.33 %
μ = 484.7± 1.5
σ = 45.7± 1.5
WF (1476 entries)
σ/μ = 9.82± 0.37 %
μ = 485.7± 1.6
σ = 47.7± 1.7



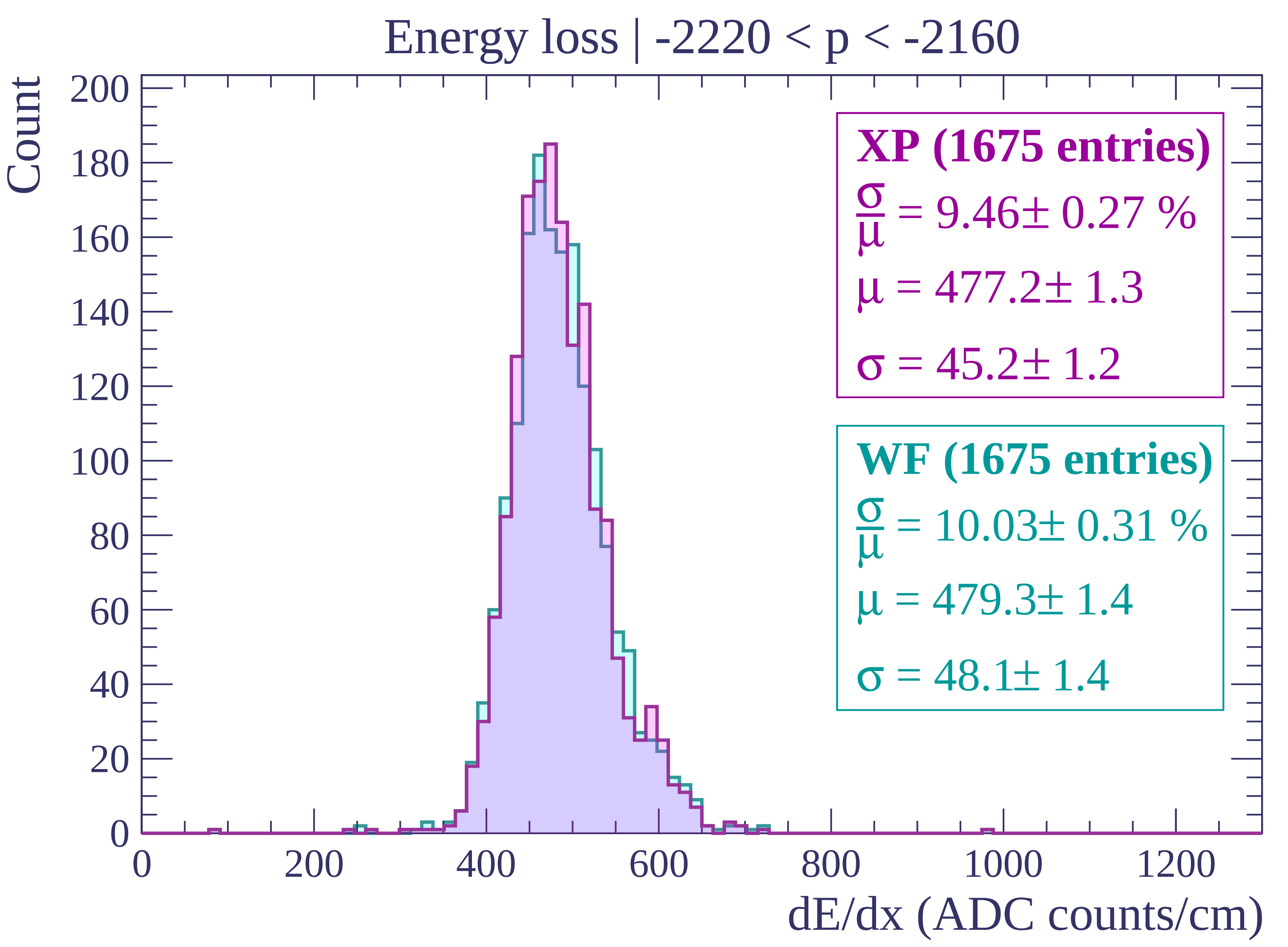
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
XP (1599 entries)
$\frac{\sigma}{\mu}$ = 9.57± 0.32 %
μ = 482.9± 1.5
σ = 46.2± 1.4
WF (1599 entries)
$\frac{\sigma}{\mu}$ = 10.00± 0.33 %
μ = 485.4± 1.5
σ = 48.6± 1.5
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200



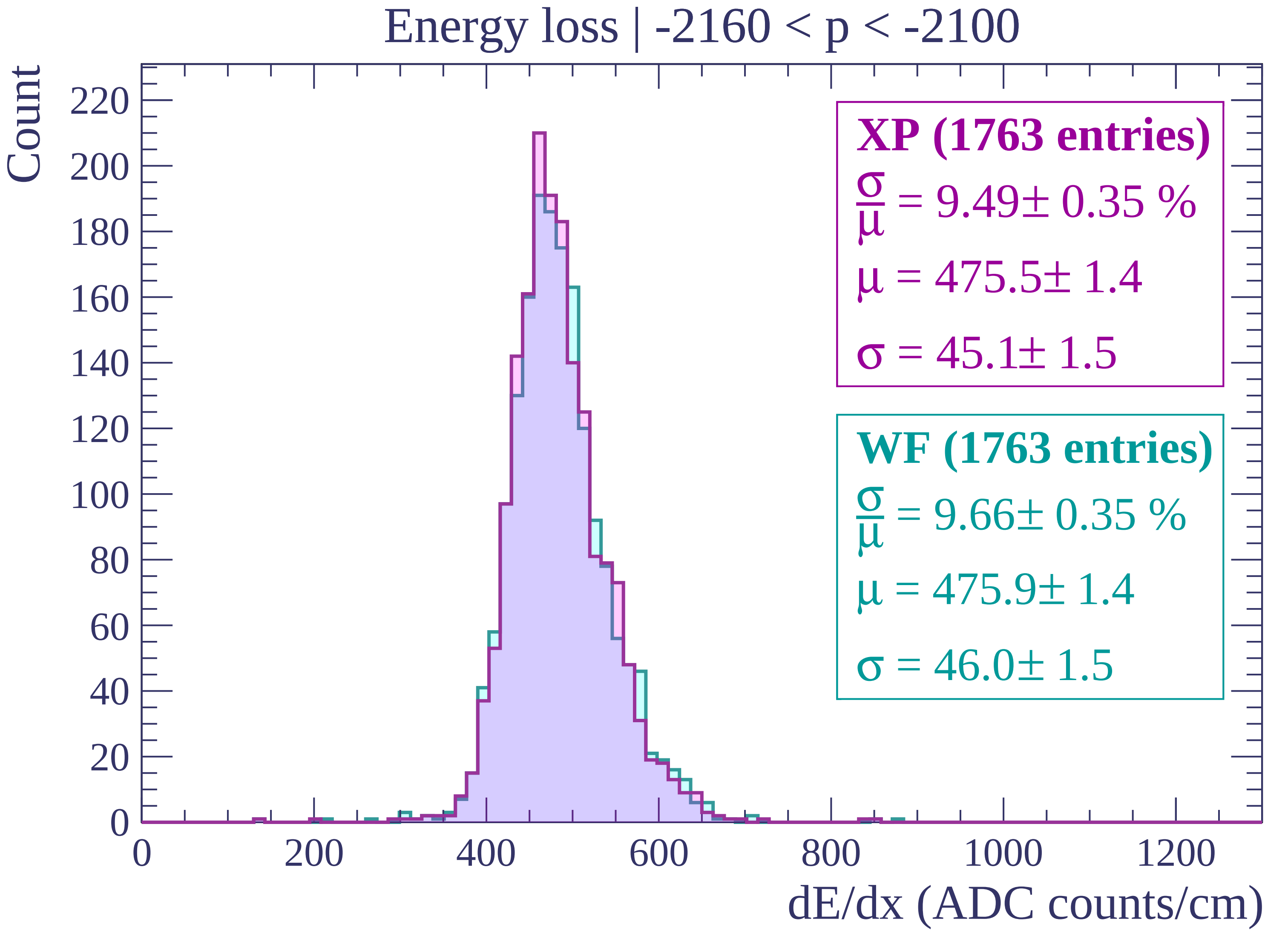
Energy loss | -2340 < p < -2280
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1639 entries)
σ/μ = 8.94± 0.30 %
μ = 478.8± 1.4
σ = 42.8± 1.3
WF (1639 entries)
σ/μ = 9.71± 0.32 %
μ = 481.7± 1.5
σ = 46.8± 1.4



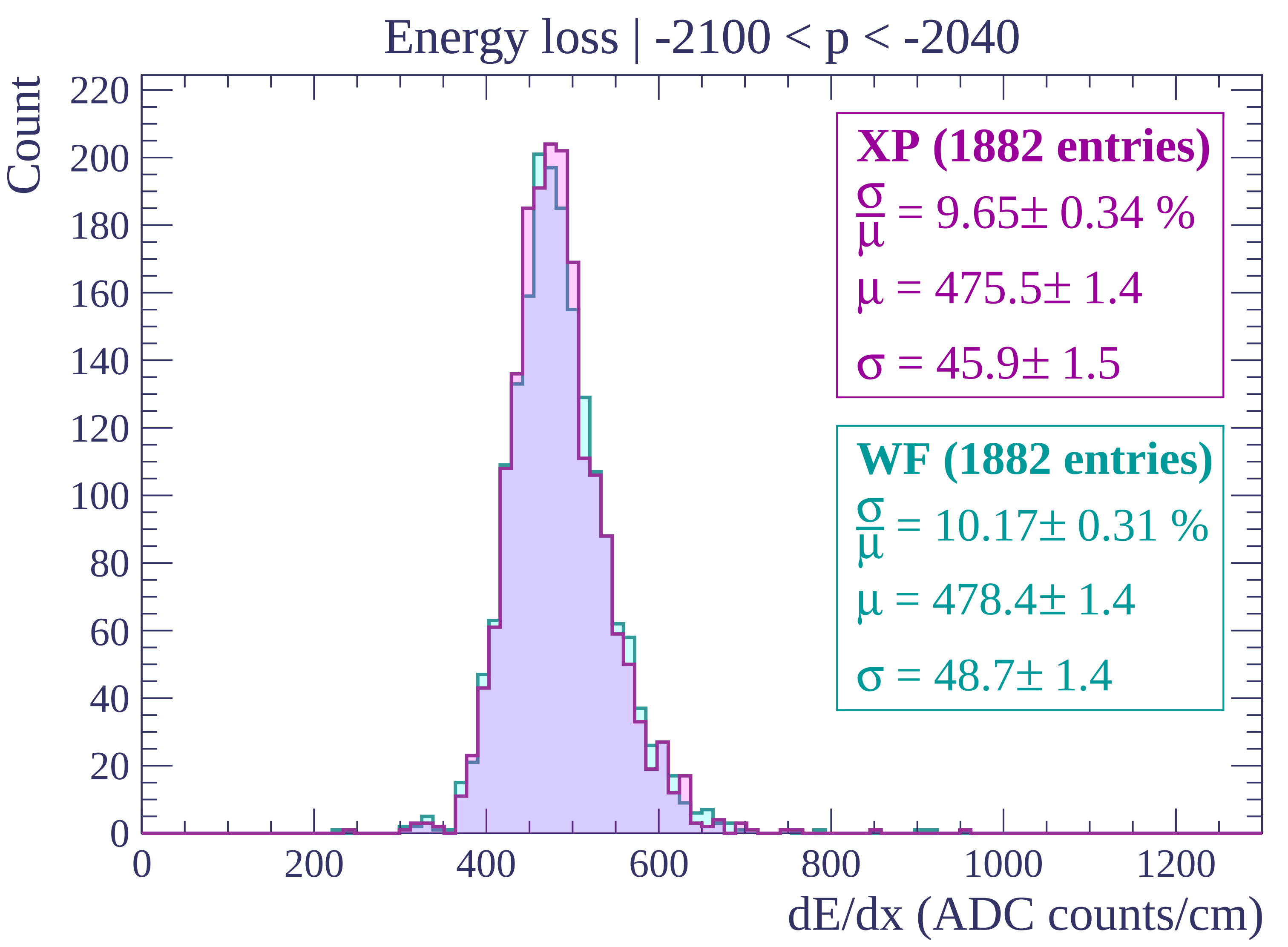
Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
XP (1630 entries)
σ/μ = 9.66± 0.33 %
μ = 480.6± 1.5
σ = 46.4± 1.4
WF (1630 entries)
σ/μ = 9.86± 0.32 %
μ = 482.1± 1.5
σ = 47.5± 1.4
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220

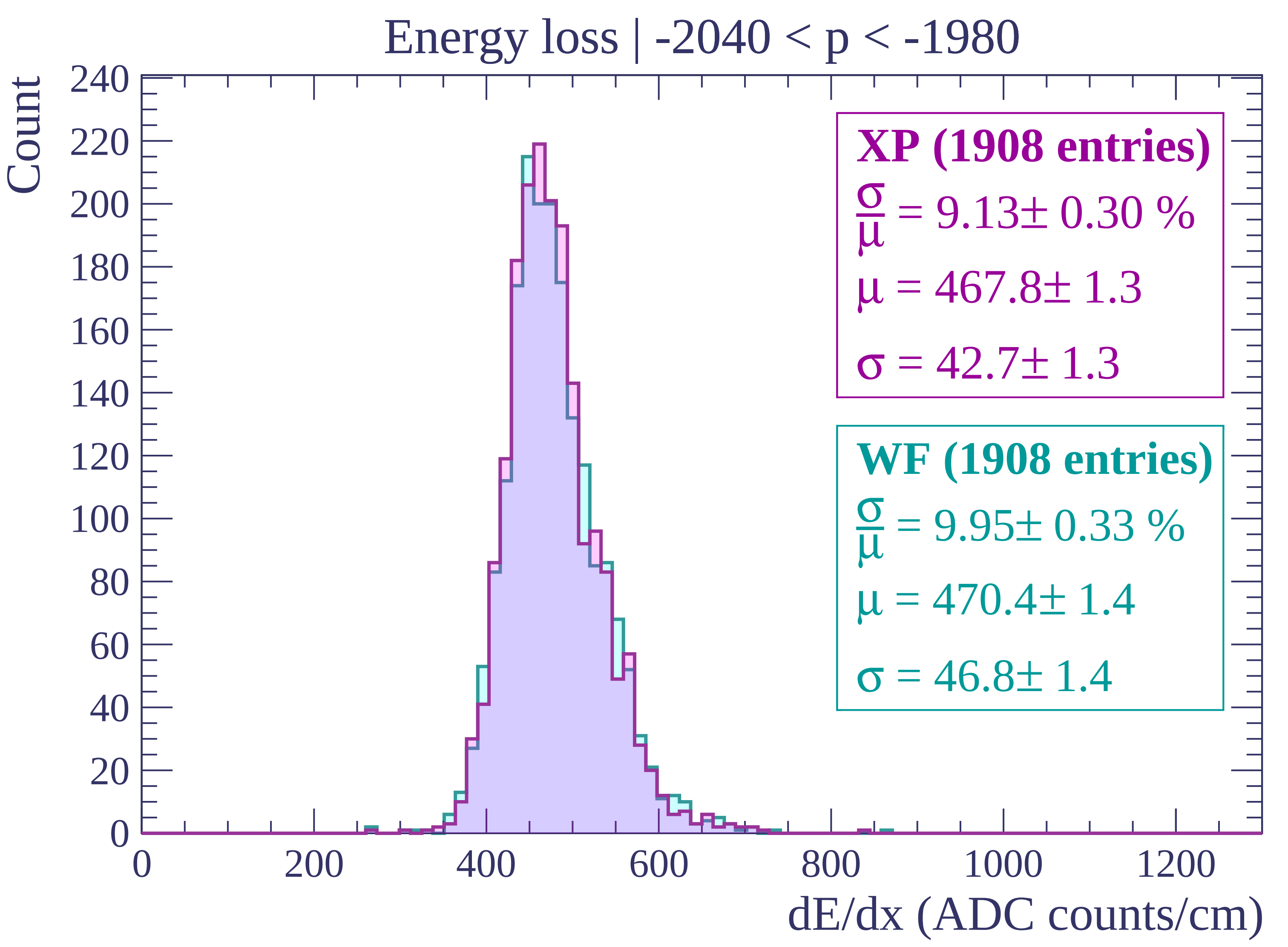


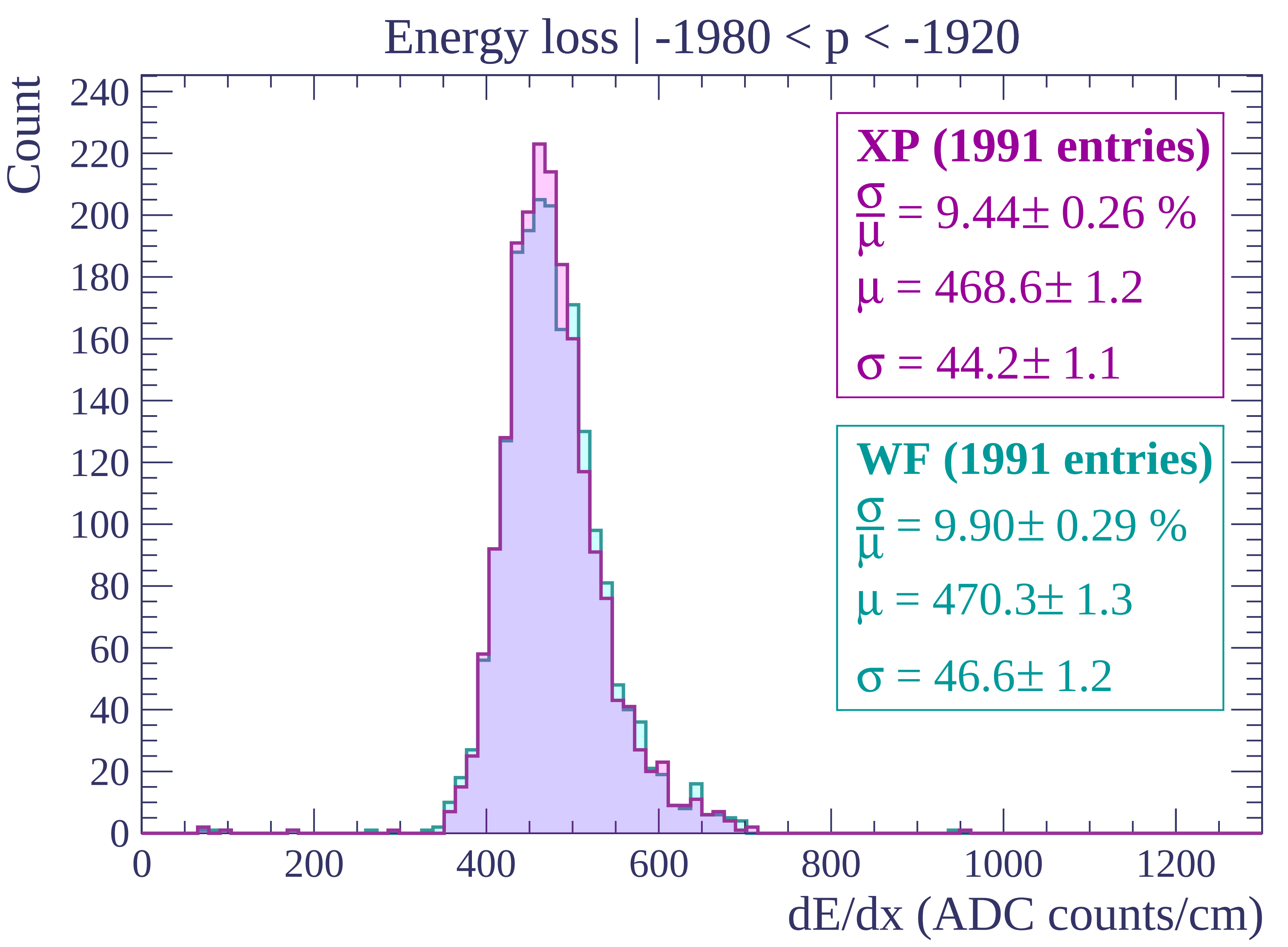
Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP (1675 entries)
$\frac{\sigma}{\mu}$ = 9.46± 0.27 %
μ = 477.2± 1.3
σ = 45.2± 1.2
WF (1675 entries)
$\frac{\sigma}{\mu}$ = 10.03± 0.31 %
μ = 479.3± 1.4
σ = 48.1± 1.4



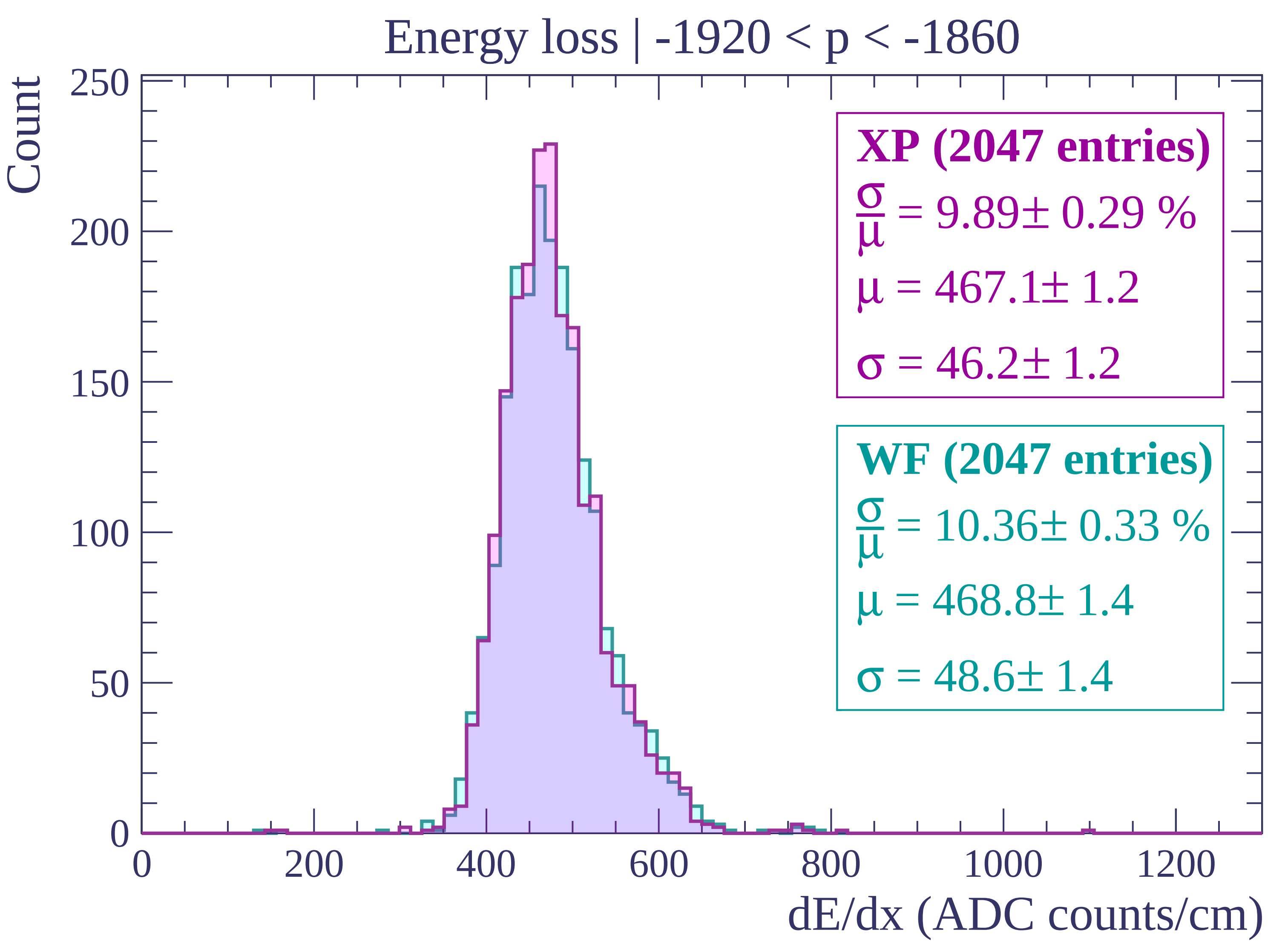
Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (1763 entries)
$\frac{\sigma}{\mu}$ = 9.49± 0.35 %
μ = 475.5± 1.4
σ = 45.1± 1.5
WF (1763 entries)
$\frac{\sigma}{\mu}$ = 9.66± 0.35 %
μ = 475.9± 1.4
σ = 46.0± 1.5


Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
XP (1882 entries)
σ/μ = 9.65± 0.34 %
μ = 475.5± 1.4
σ = 45.9± 1.5
WF (1882 entries)
σ/μ = 10.17± 0.31 %
μ = 478.4± 1.4
σ = 48.7± 1.4

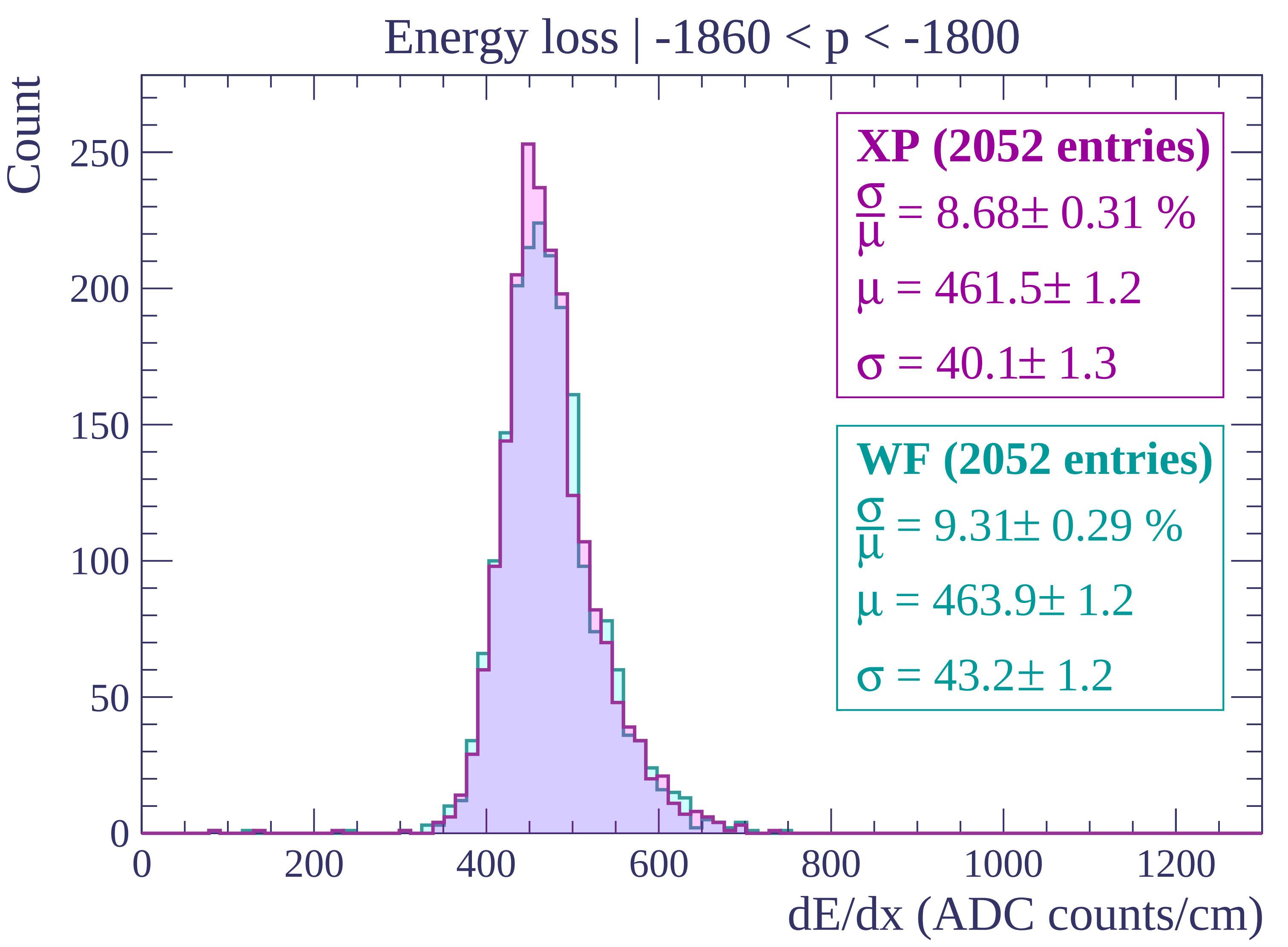

Energy loss | -2040 < p < -1980
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP (1908 entries)
σ/μ = 9.13± 0.30 %
μ = 467.8± 1.3
σ = 42.7± 1.3
WF (1908 entries)
σ/μ = 9.95± 0.33 %
μ = 470.4± 1.4
σ = 46.8± 1.4



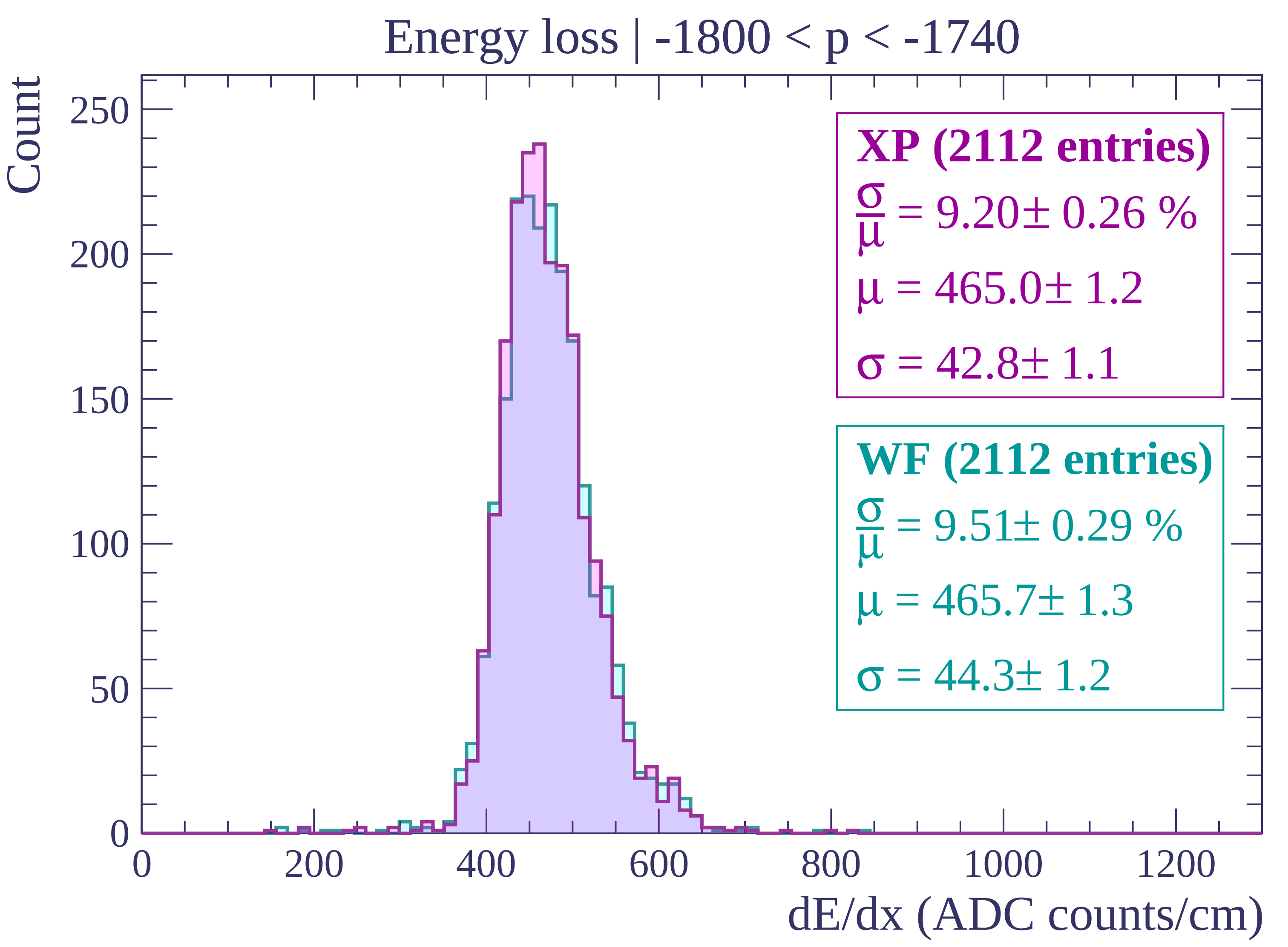
Energy loss | -1980 < p < -1920
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1991 entries)
σ/μ = 9.44± 0.26 %
μ = 468.6± 1.2
σ = 44.2± 1.1
WF (1991 entries)
σ/μ = 9.90± 0.29 %
μ = 470.3± 1.3
σ = 46.6± 1.2



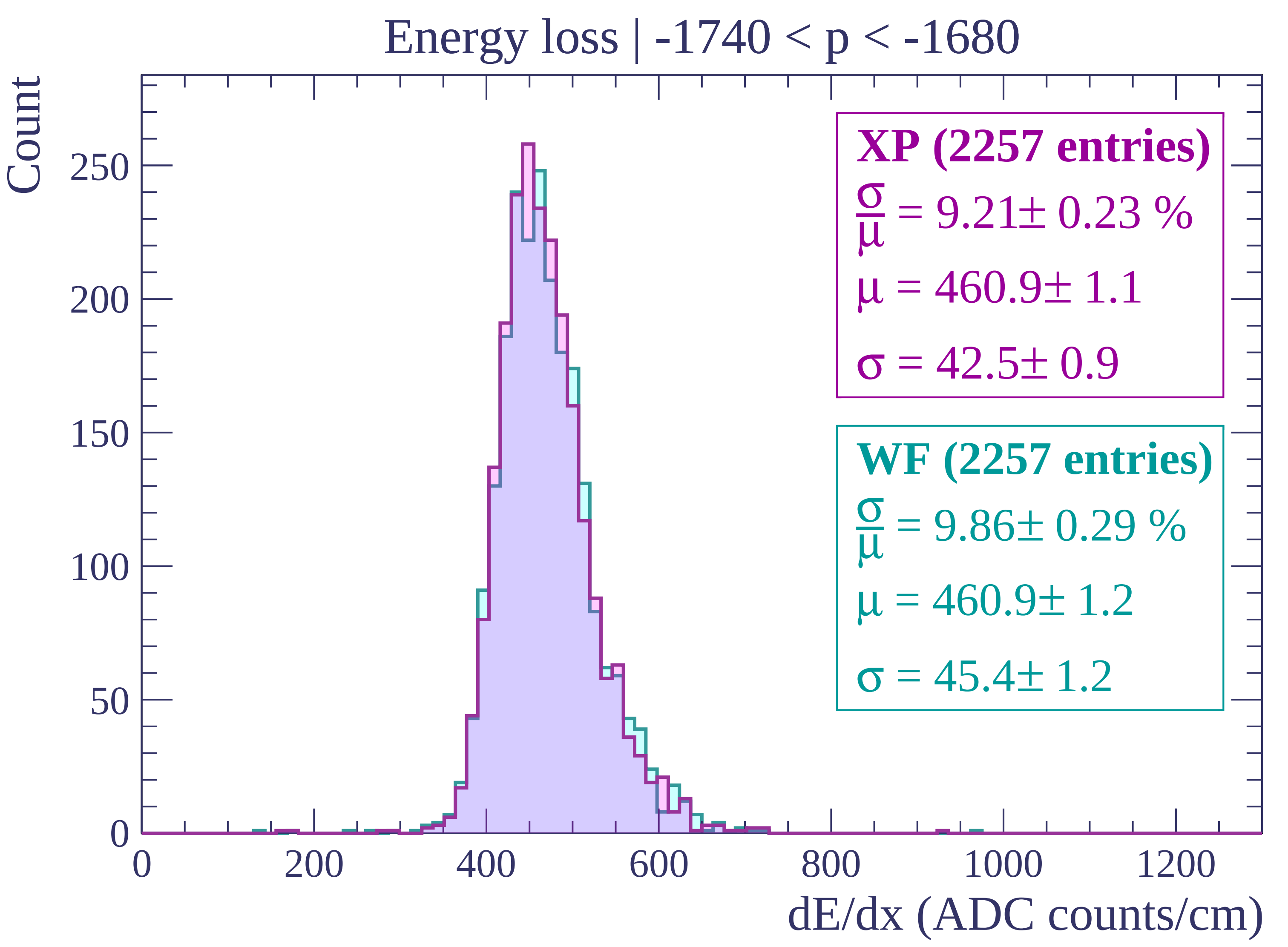
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2047 entries)
σ/μ = 9.89 ± 0.29 %
μ = 467.1 ± 1.2
σ = 46.2 ± 1.2
WF (2047 entries)
σ/μ = 10.36 ± 0.33 %
μ = 468.8 ± 1.4
σ = 48.6 ± 1.4



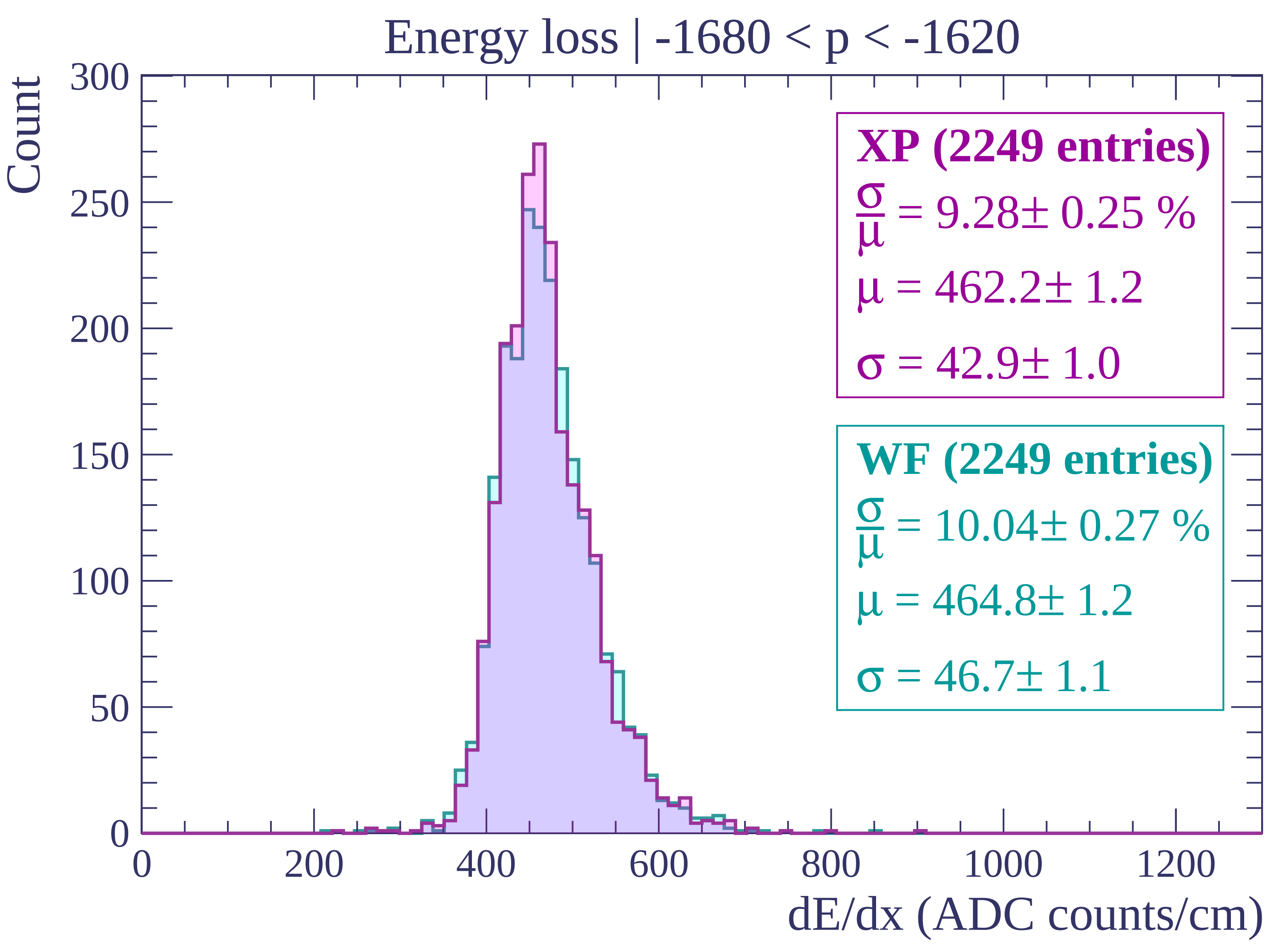
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
XP (2052 entries)
σ/μ = 8.68± 0.31 %
μ = 461.5± 1.2
σ = 40.1± 1.3
WF (2052 entries)
σ/μ = 9.31± 0.29 %
μ = 463.9± 1.2
σ = 43.2± 1.2
0
50
100
150
200
250
0
200
400
600
800
1000
1200



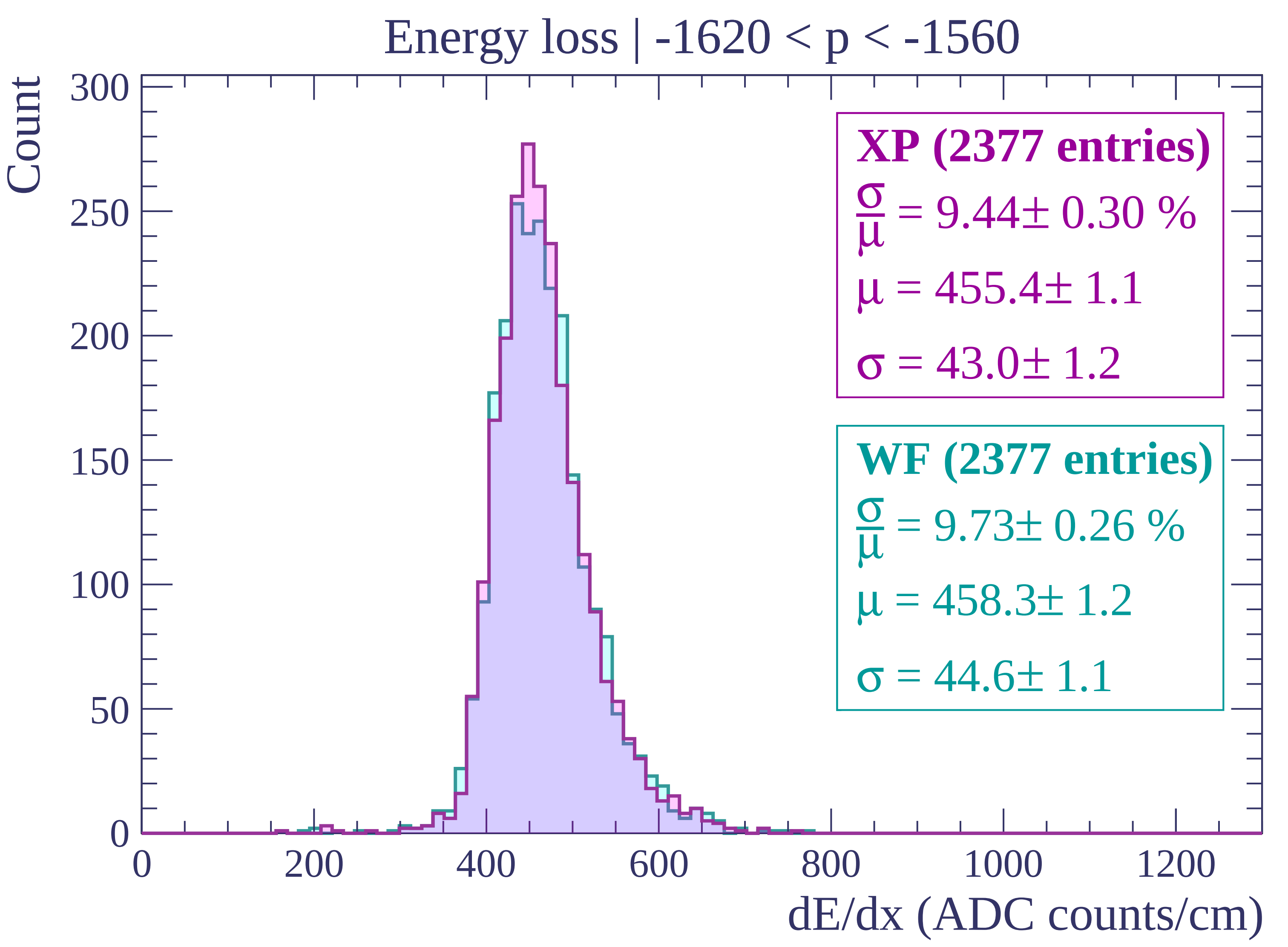
Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2112 entries)
$\frac{\sigma}{\mu}$ = 9.20± 0.26 %
μ = 465.0± 1.2
σ = 42.8± 1.1
WF (2112 entries)
$\frac{\sigma}{\mu}$ = 9.51± 0.29 %
μ = 465.7± 1.3
σ = 44.3± 1.2



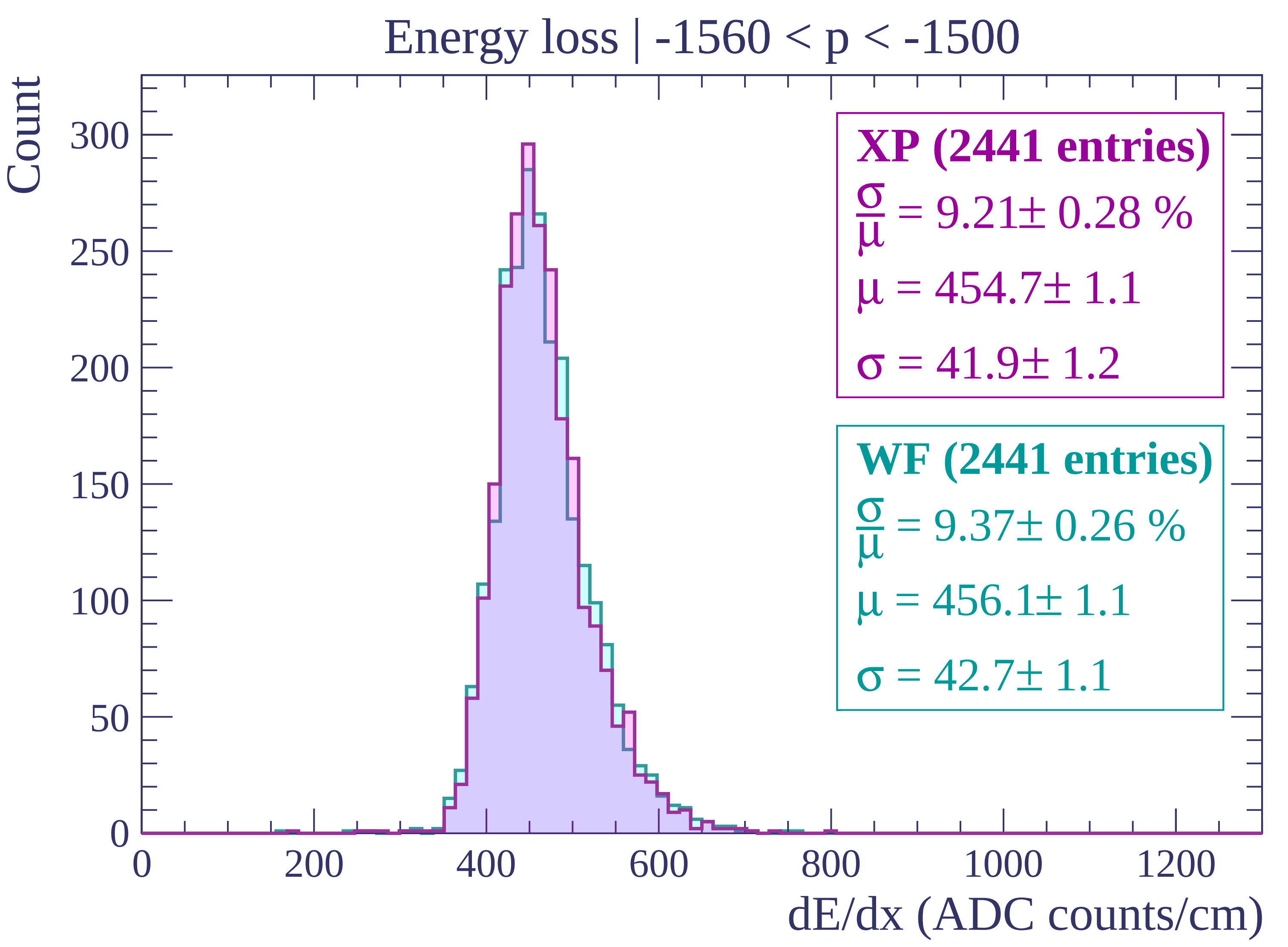
Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2257 entries)
$\frac{\sigma}{\mu}$ = 9.21± 0.23 %
μ = 460.9± 1.1
σ = 42.5± 0.9
WF (2257 entries)
$\frac{\sigma}{\mu}$ = 9.86± 0.29 %
μ = 460.9± 1.2
σ = 45.4± 1.2



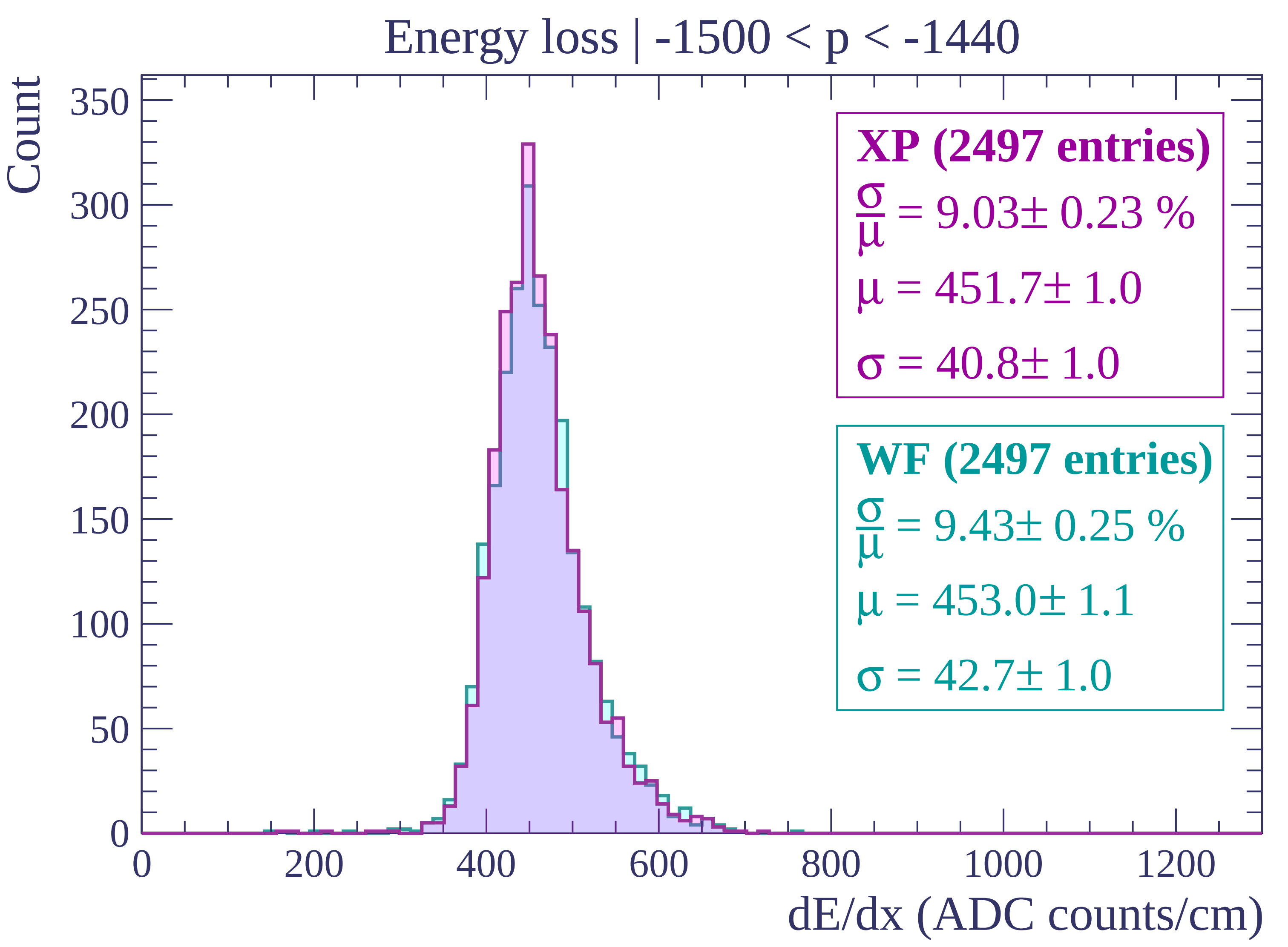
Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2249 entries)
$\frac{\sigma}{\mu}$ = 9.28± 0.25 %
μ = 462.2± 1.2
σ = 42.9± 1.0
WF (2249 entries)
$\frac{\sigma}{\mu}$ = 10.04± 0.27 %
μ = 464.8± 1.2
σ = 46.7± 1.1



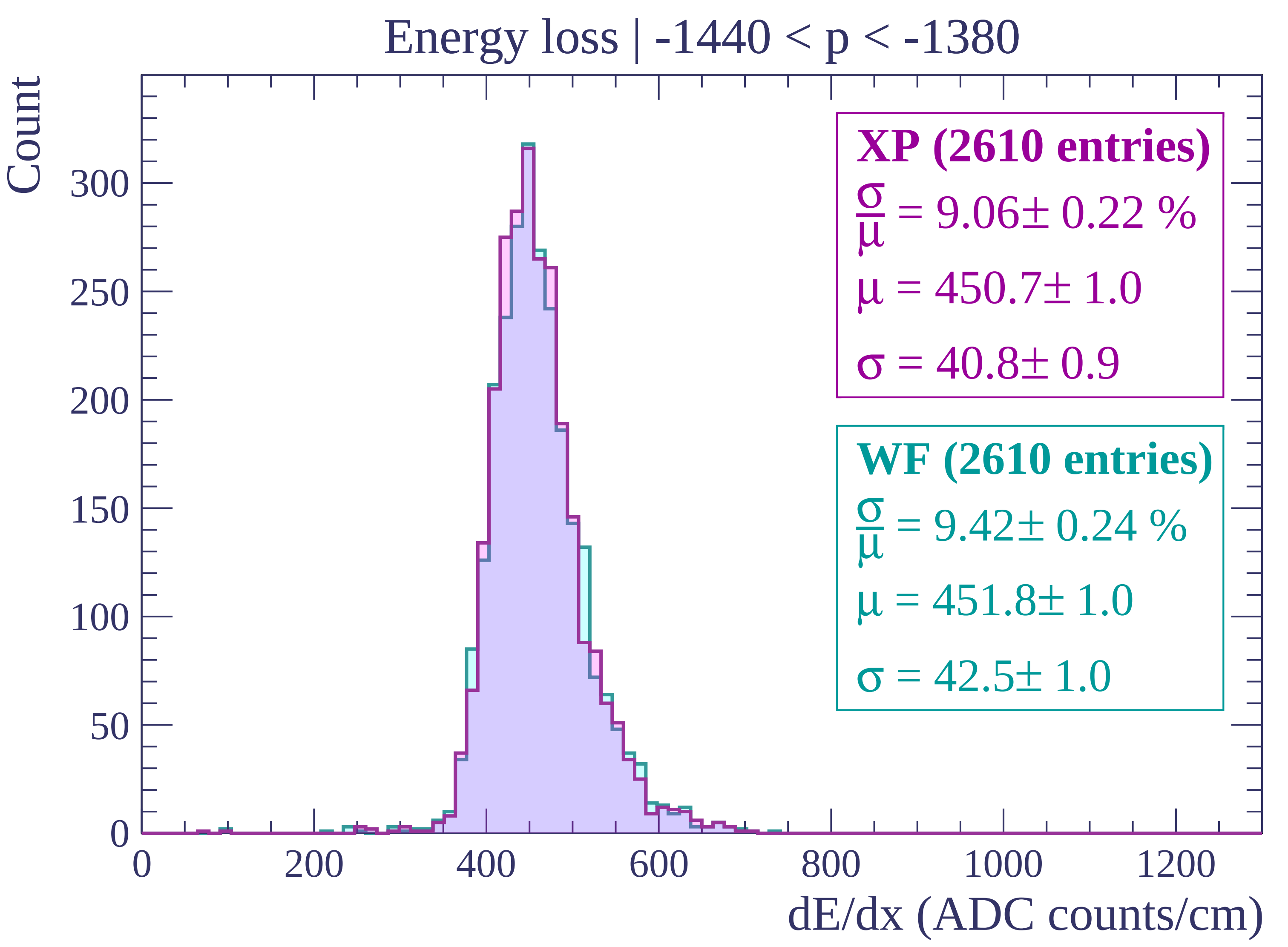
Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2377 entries)
$\frac{\sigma}{\mu}$ = 9.44± 0.30 %
μ = 455.4± 1.1
σ = 43.0± 1.2
WF (2377 entries)
$\frac{\sigma}{\mu}$ = 9.73± 0.26 %
μ = 458.3± 1.2
σ = 44.6± 1.1



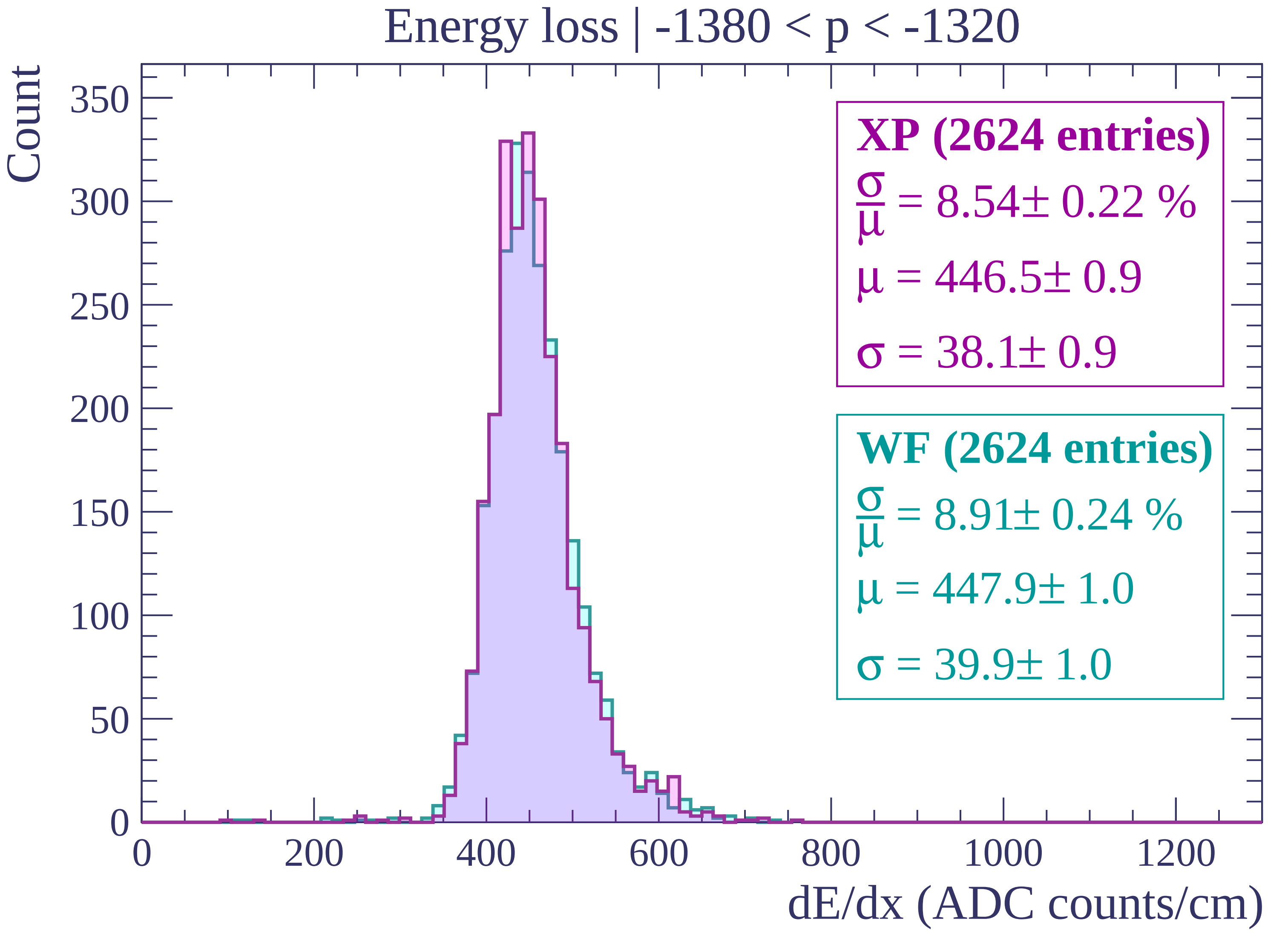
Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2441 entries)
σ/μ = 9.21± 0.28 %
μ = 454.7± 1.1
σ = 41.9± 1.2
WF (2441 entries)
σ/μ = 9.37± 0.26 %
μ = 456.1± 1.1
σ = 42.7± 1.1



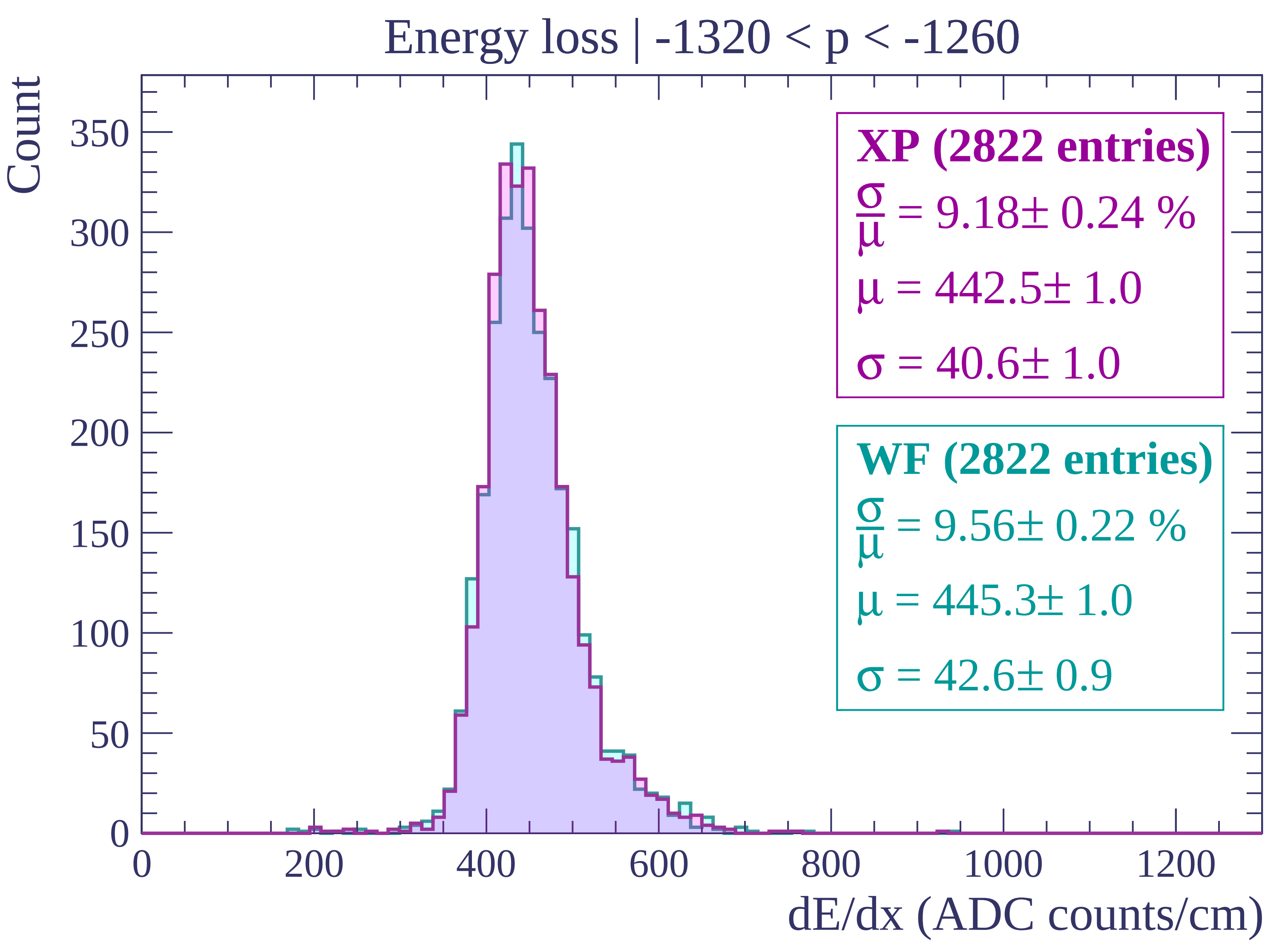
Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2497 entries)
σ/μ = 9.03± 0.23 %
μ = 451.7± 1.0
σ = 40.8± 1.0
WF (2497 entries)
σ/μ = 9.43± 0.25 %
μ = 453.0± 1.1
σ = 42.7± 1.0



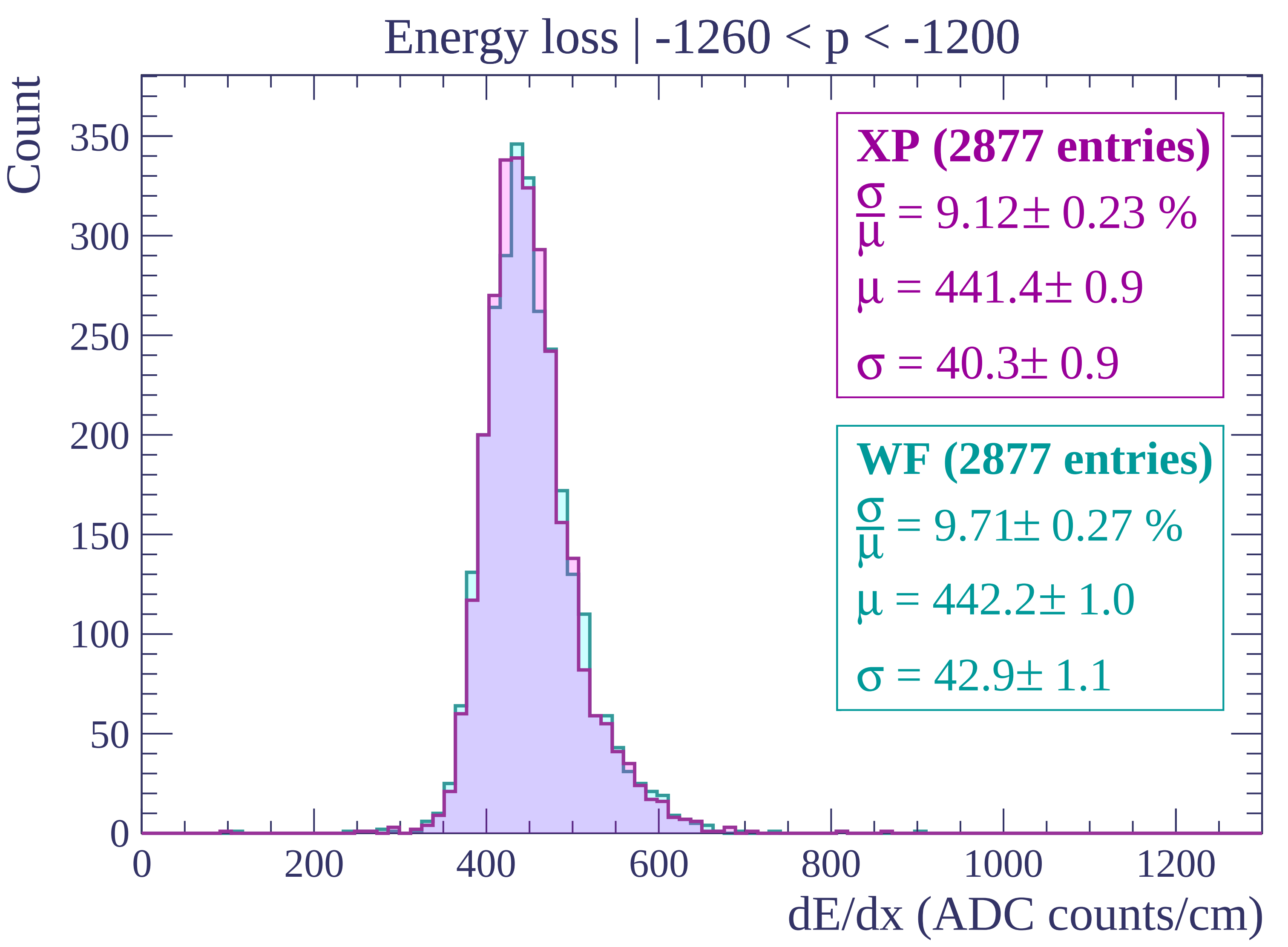
Energy loss | -1440 < p < -1380
Count
dE/dx (ADC counts/cm)
XP (2610 entries)
σ/μ = 9.06± 0.22 %
μ = 450.7± 1.0
σ = 40.8± 0.9
WF (2610 entries)
σ/μ = 9.42± 0.24 %
μ = 451.8± 1.0
σ = 42.5± 1.0
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

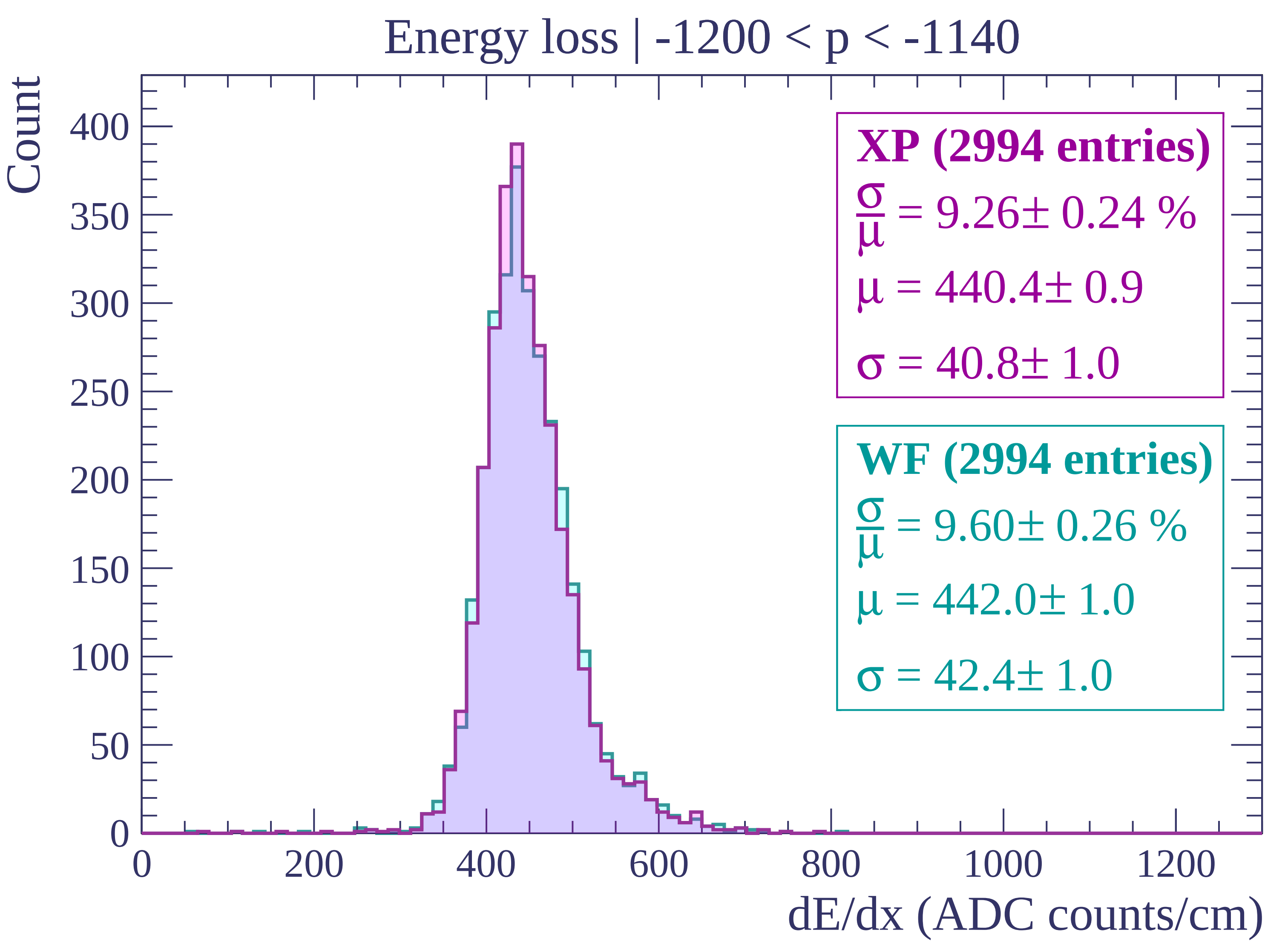


Energy loss | -1380 < p < -1320
Count
dE/dx (ADC counts/cm)
XP (2624 entries)
$\frac{\sigma}{\mu}$ = 8.54± 0.22 %
μ = 446.5± 0.9
σ = 38.1± 0.9
WF (2624 entries)
$\frac{\sigma}{\mu}$ = 8.91± 0.24 %
μ = 447.9± 1.0
σ = 39.9± 1.0

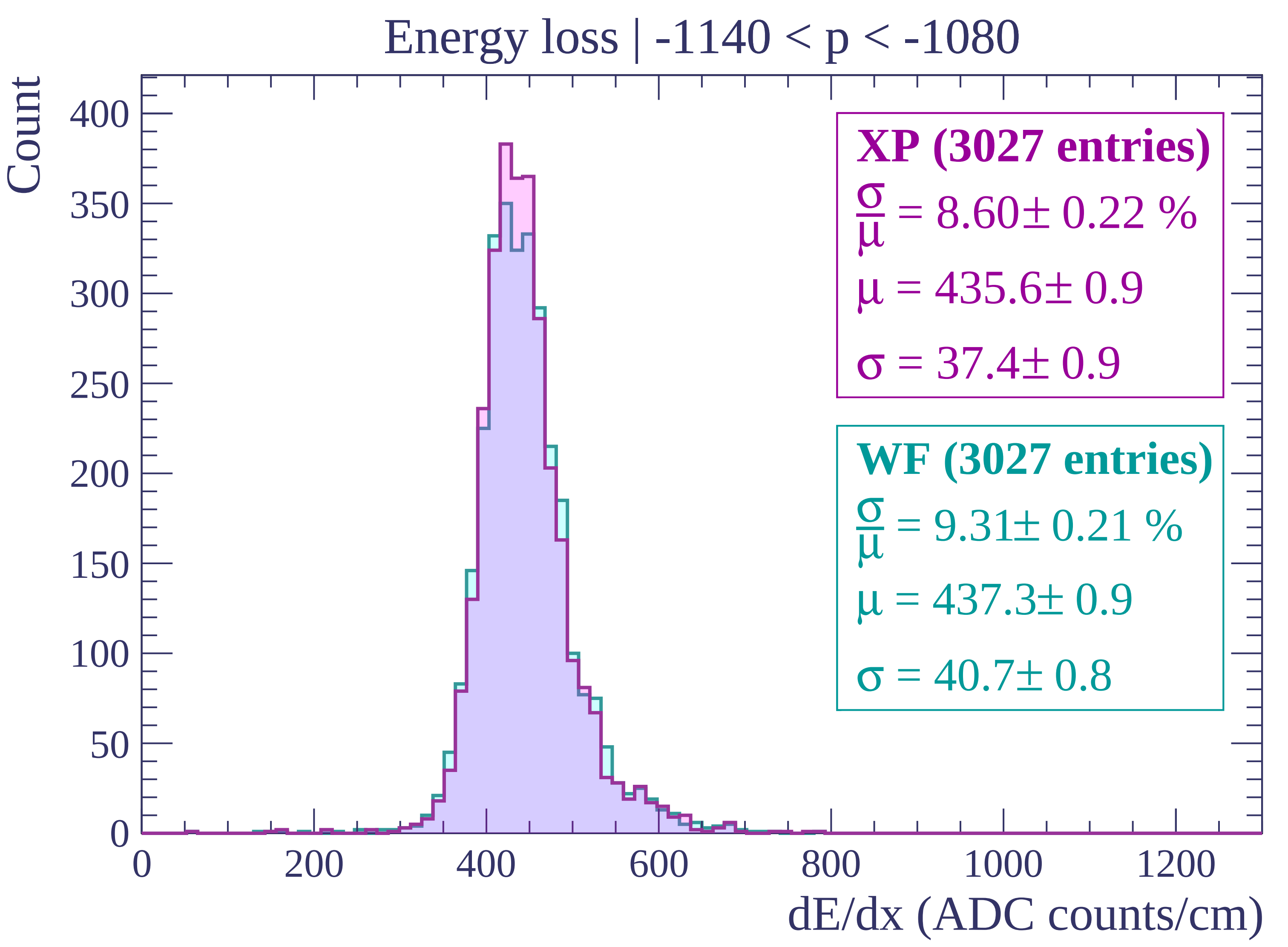


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2822 entries)
σ/μ = 9.18± 0.24 %
μ = 442.5± 1.0
σ = 40.6± 1.0
WF (2822 entries)
σ/μ = 9.56± 0.22 %
μ = 445.3± 1.0
σ = 42.6± 0.9

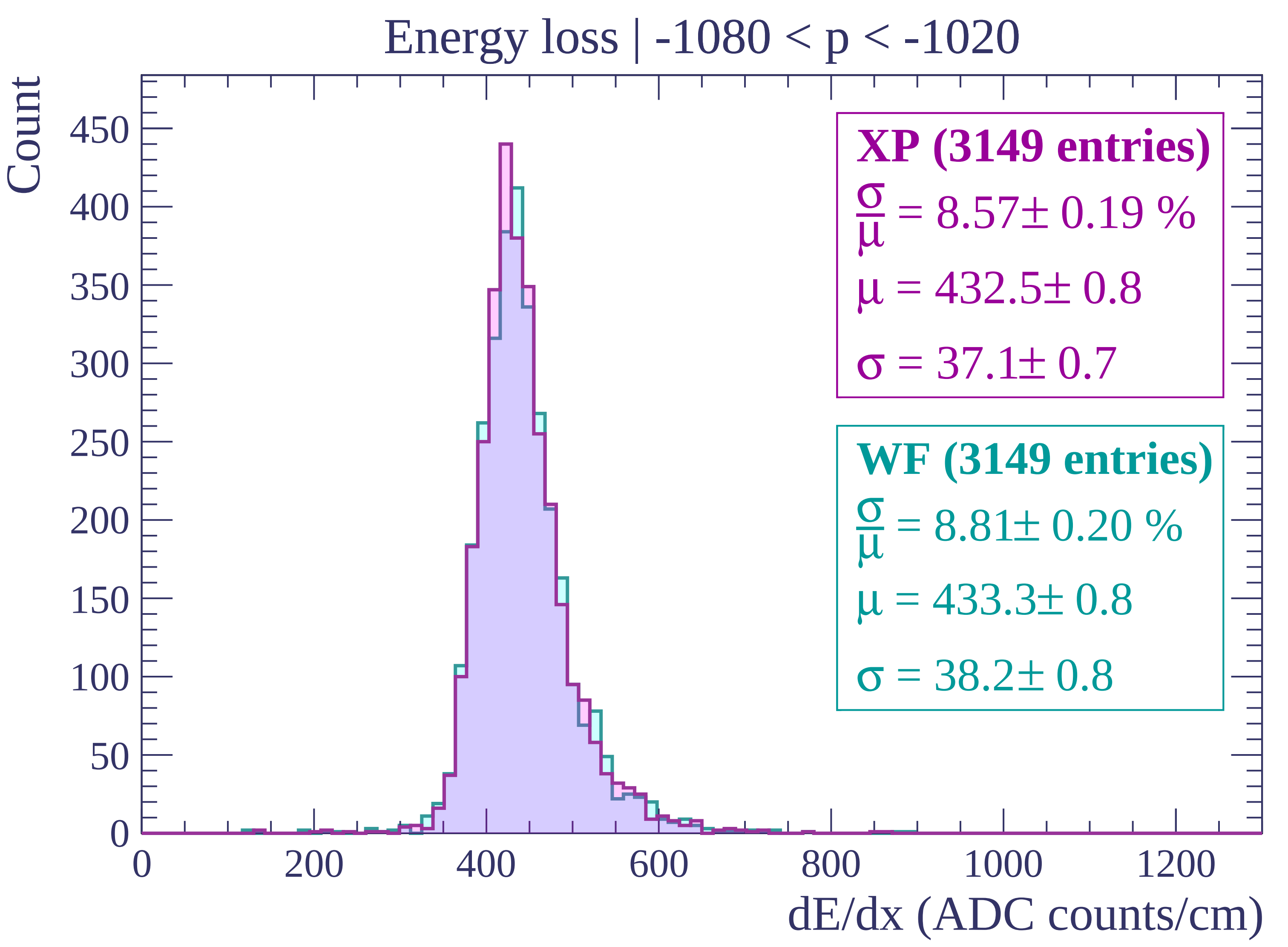

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (2877 entries)
σ/μ = 9.12± 0.23 %
μ = 441.4± 0.9
σ = 40.3± 0.9
WF (2877 entries)
σ/μ = 9.71± 0.27 %
μ = 442.2± 1.0
σ = 42.9± 1.1



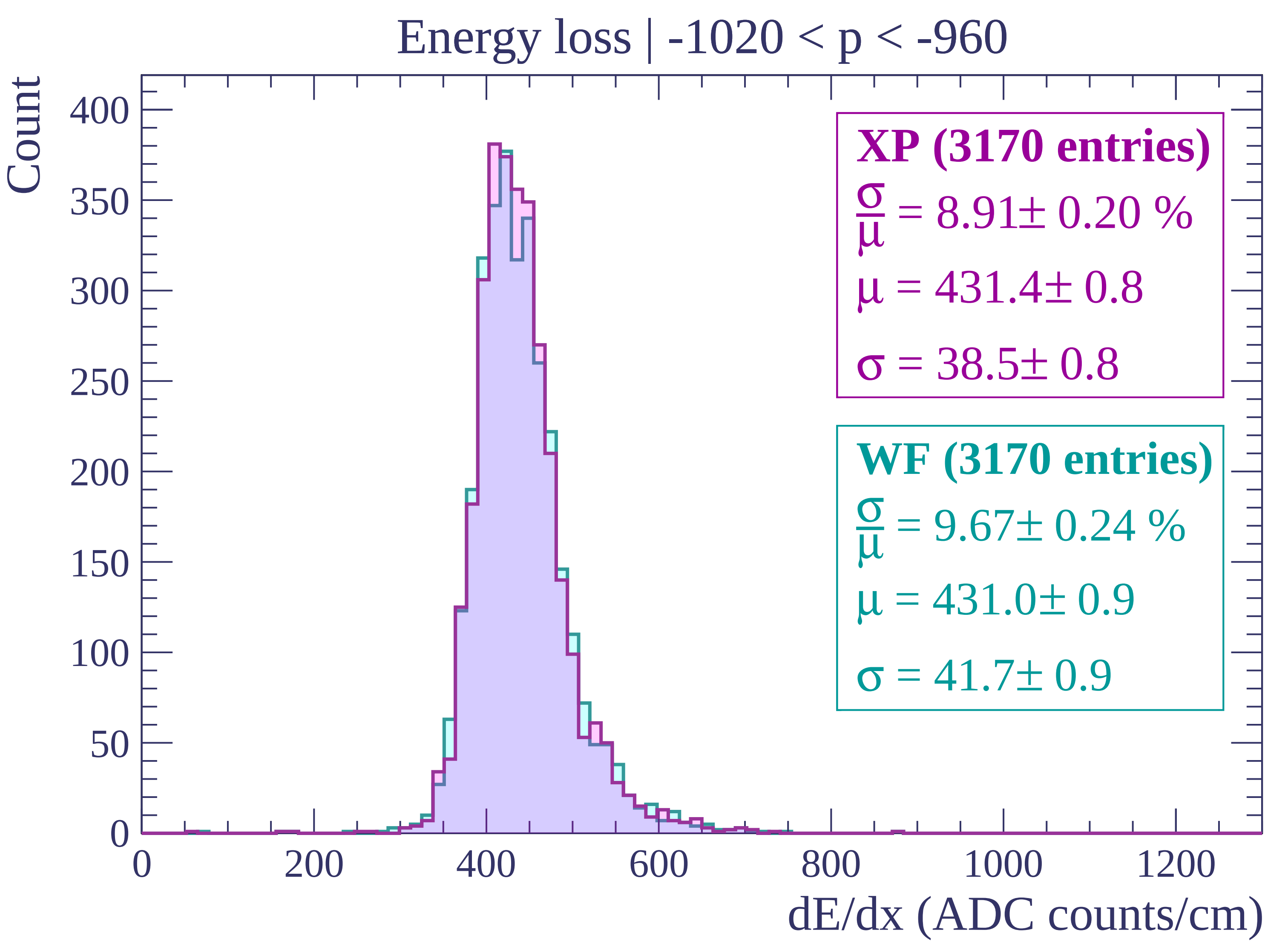
Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (2994 entries)
$\frac{\sigma}{\mu}$ = 9.26± 0.24 %
μ = 440.4± 0.9
σ = 40.8± 1.0
WF (2994 entries)
$\frac{\sigma}{\mu}$ = 9.60± 0.26 %
μ = 442.0± 1.0
σ = 42.4± 1.0



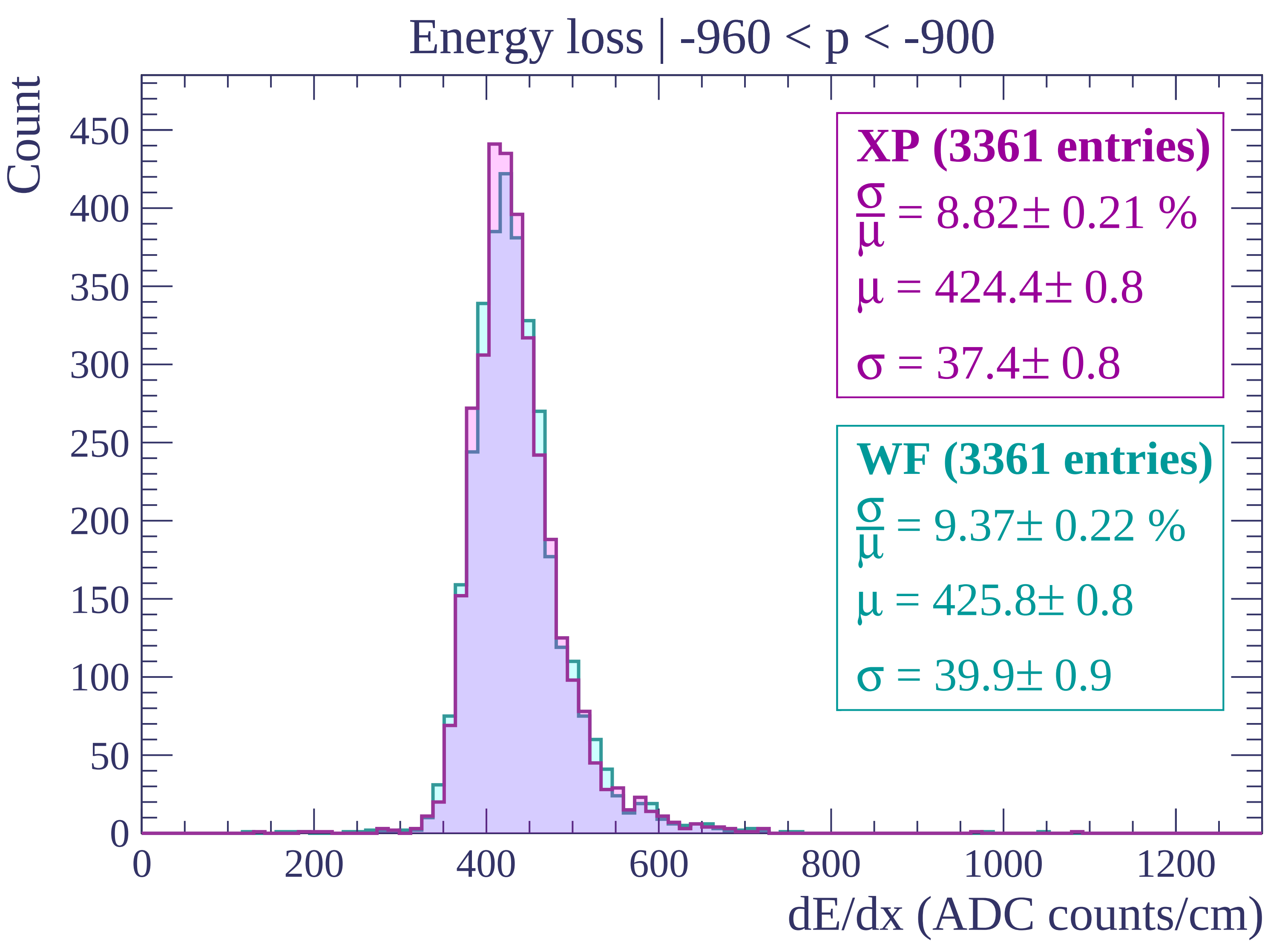
Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3027 entries)
σ/μ = 8.60± 0.22 %
μ = 435.6± 0.9
σ = 37.4± 0.9
WF (3027 entries)
σ/μ = 9.31± 0.21 %
μ = 437.3± 0.9
σ = 40.7± 0.8

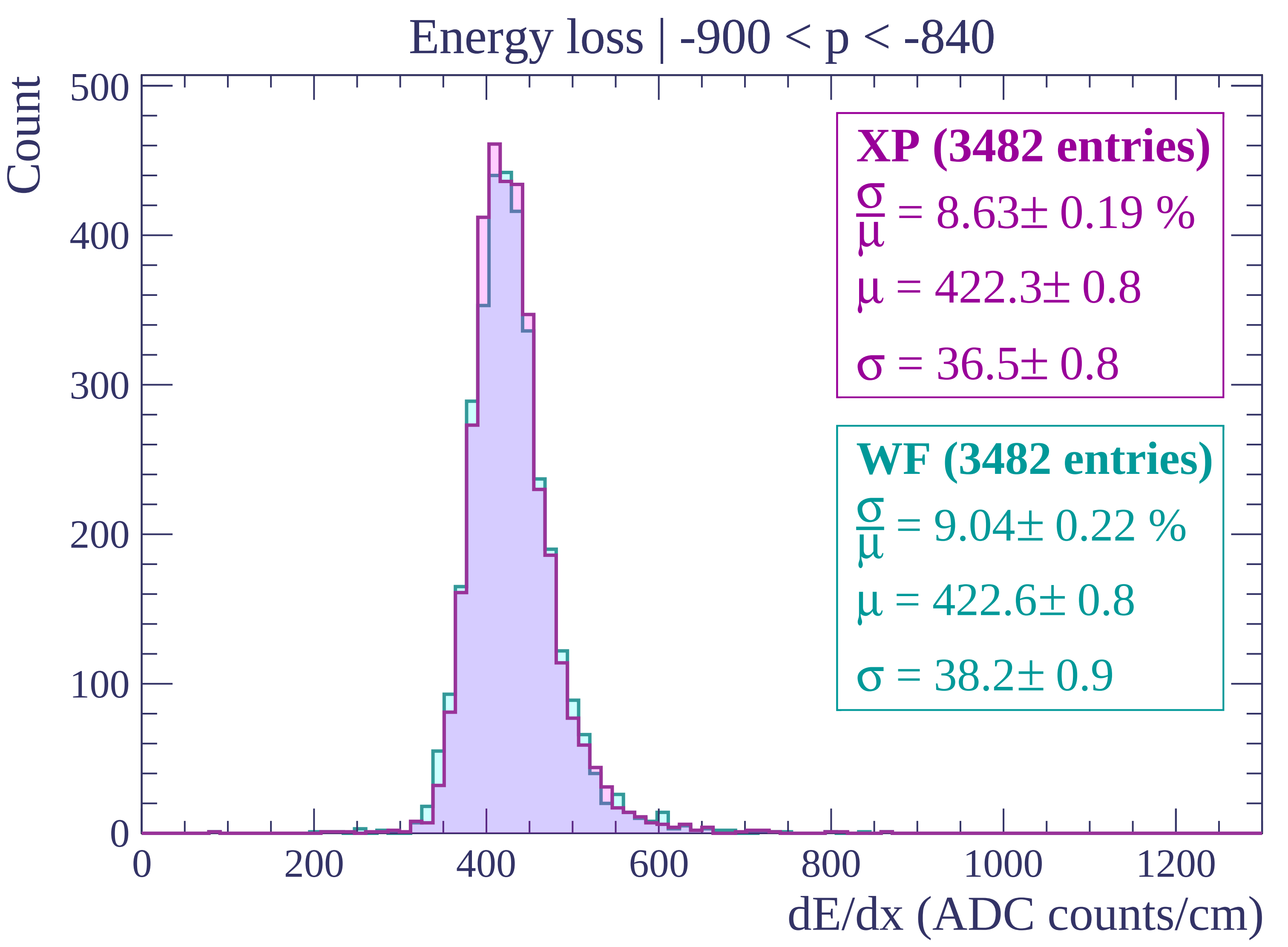


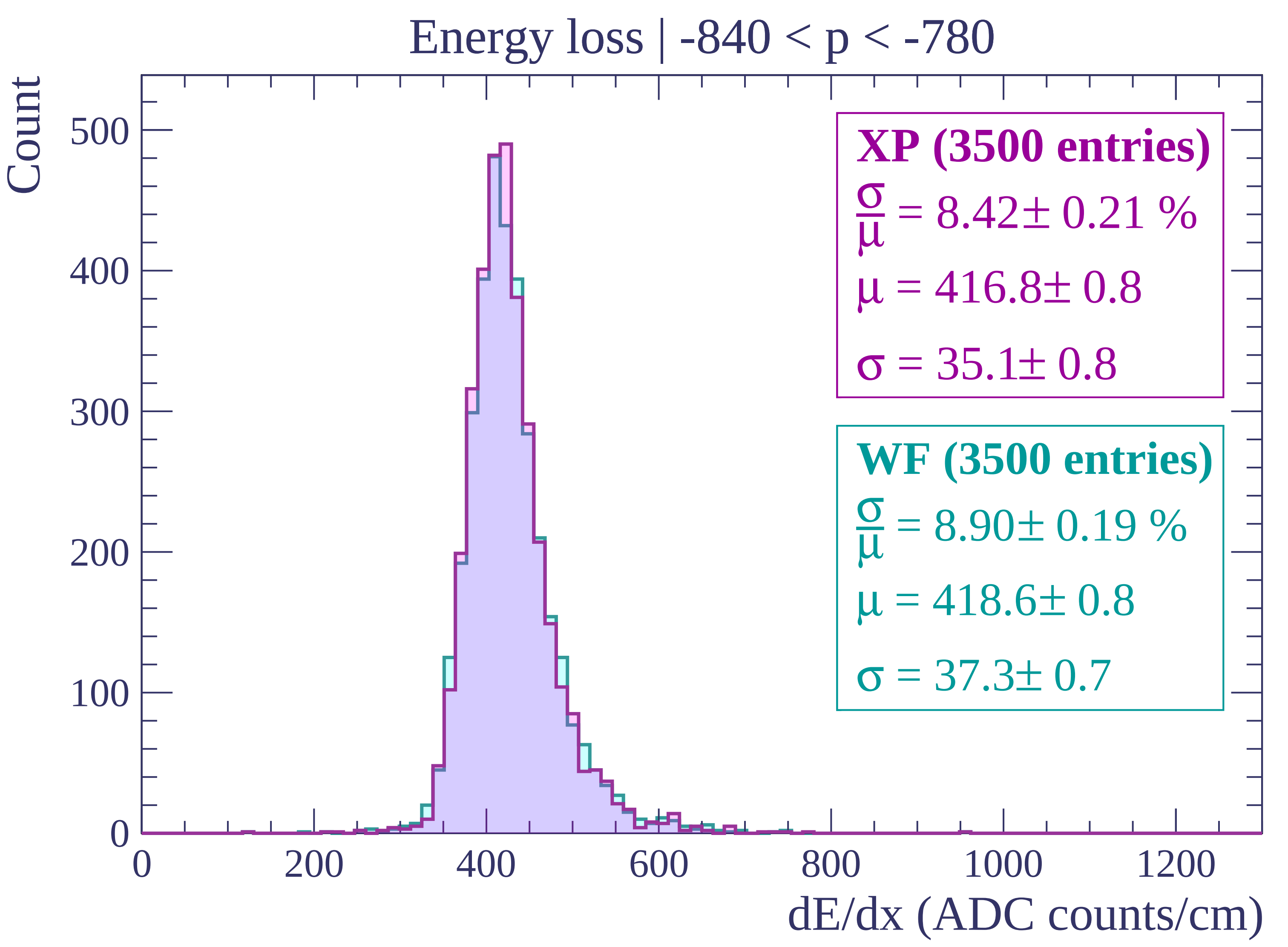
Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (3149 entries)
σ/μ = 8.57± 0.19 %
μ = 432.5± 0.8
σ = 37.1± 0.7
WF (3149 entries)
σ/μ = 8.81± 0.20 %
μ = 433.3± 0.8
σ = 38.2± 0.8



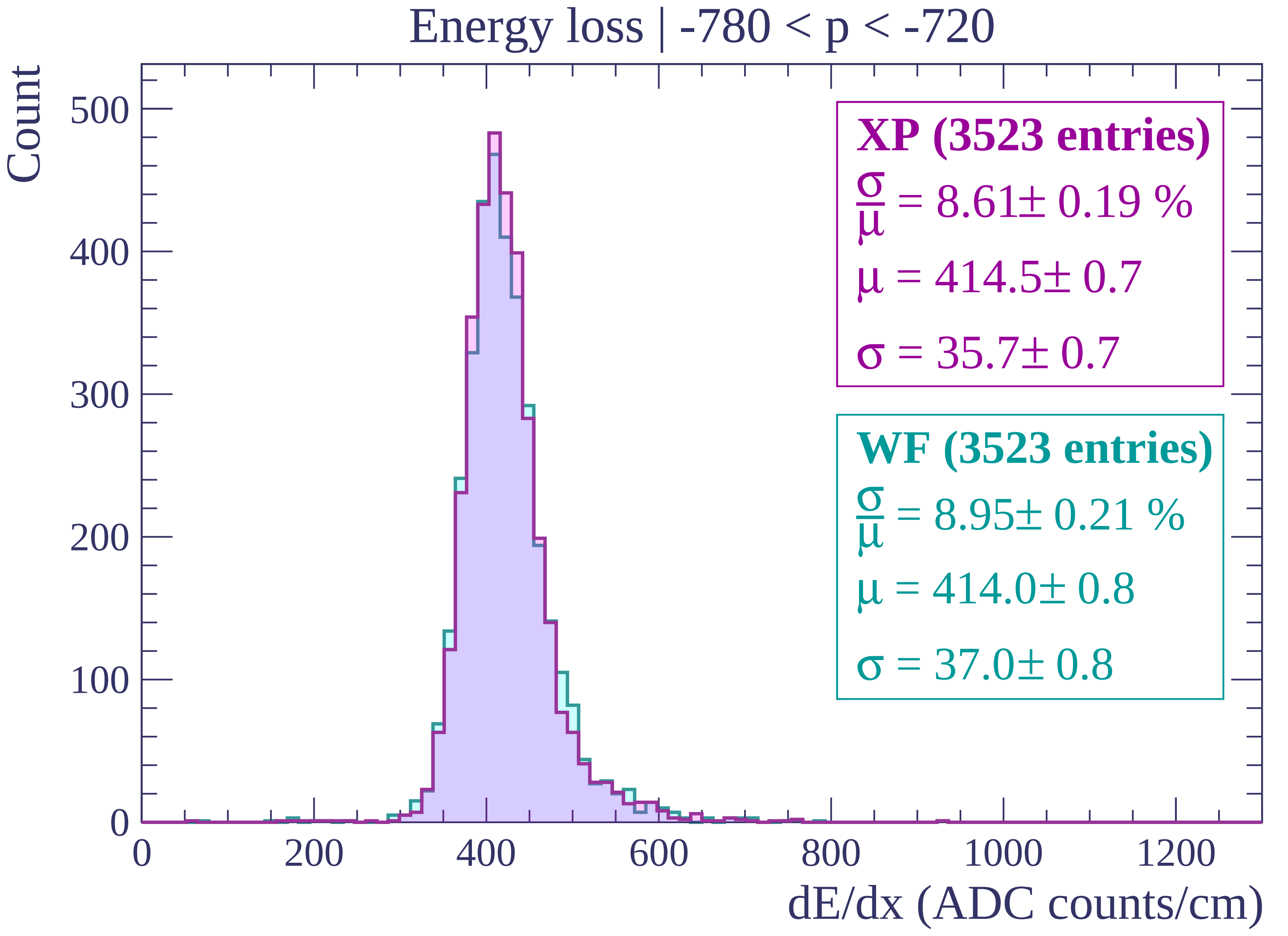
Energy loss | -1020 < p < -960
Count
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3170 entries)
σ/μ = 8.91± 0.20 %
μ = 431.4± 0.8
σ = 38.5± 0.8
WF (3170 entries)
σ/μ = 9.67± 0.24 %
μ = 431.0± 0.9
σ = 41.7± 0.9

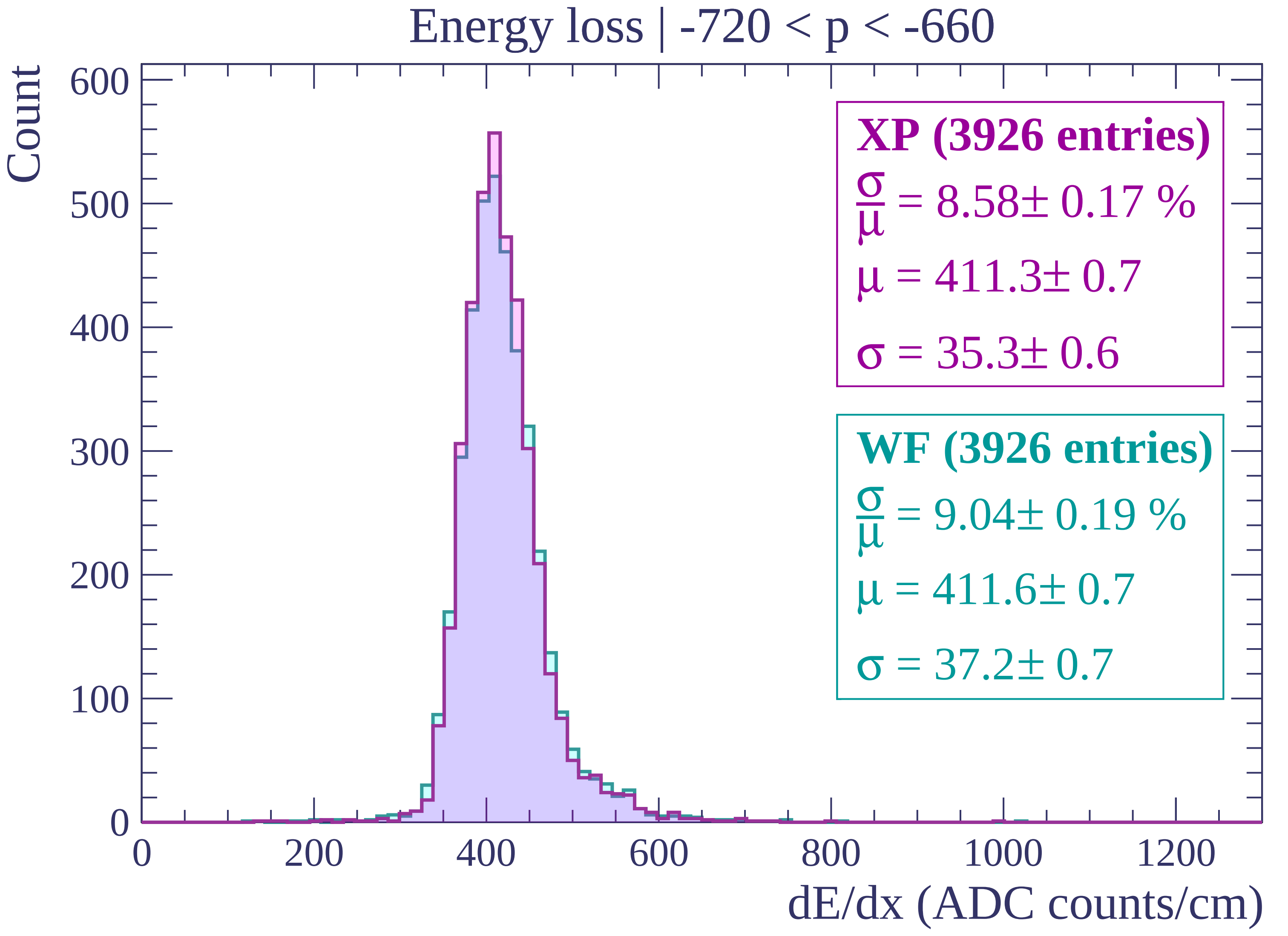

Energy loss | -960 < p < -900
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3361 entries)
σ/μ = 8.82± 0.21 %
μ = 424.4± 0.8
σ = 37.4± 0.8
WF (3361 entries)
σ/μ = 9.37± 0.22 %
μ = 425.8± 0.8
σ = 39.9± 0.9


Energy loss | -900 < p < -840
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3482 entries)
$\frac{\sigma}{\mu}$ = 8.63± 0.19 %
μ = 422.3± 0.8
σ = 36.5± 0.8
WF (3482 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.22 %
μ = 422.6± 0.8
σ = 38.2± 0.9

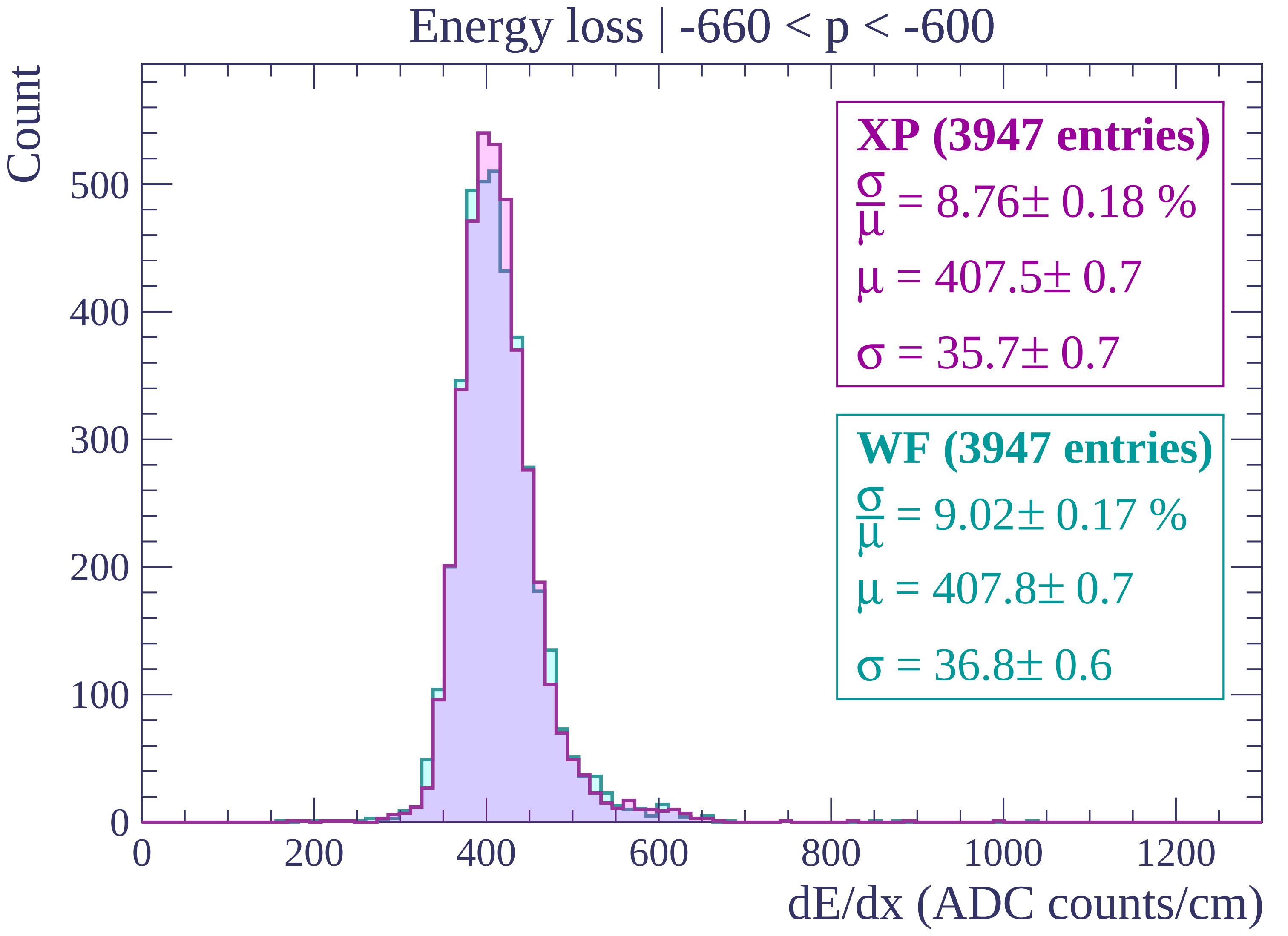


Energy loss | -840 < p < -780
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3500 entries)
$\frac{\sigma}{\mu}$ = 8.42± 0.21 %
μ = 416.8± 0.8
σ = 35.1± 0.8
WF (3500 entries)
$\frac{\sigma}{\mu}$ = 8.90± 0.19 %
μ = 418.6± 0.8
σ = 37.3± 0.7

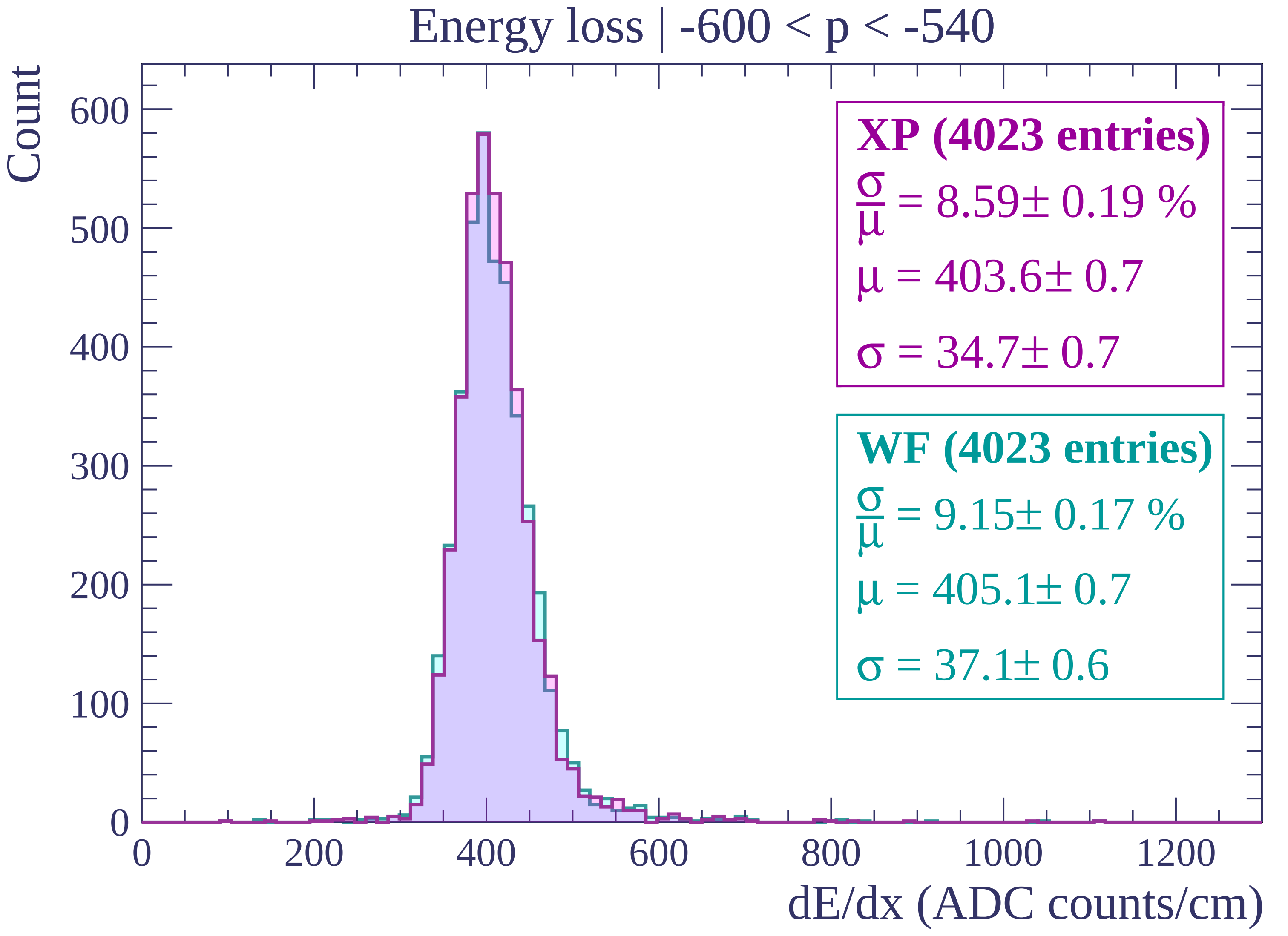

Energy loss | -780 < p < -720
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3523 entries)
σ/μ = 8.61± 0.19 %
μ = 414.5± 0.7
σ = 35.7± 0.7
WF (3523 entries)
σ/μ = 8.95± 0.21 %
μ = 414.0± 0.8
σ = 37.0± 0.8

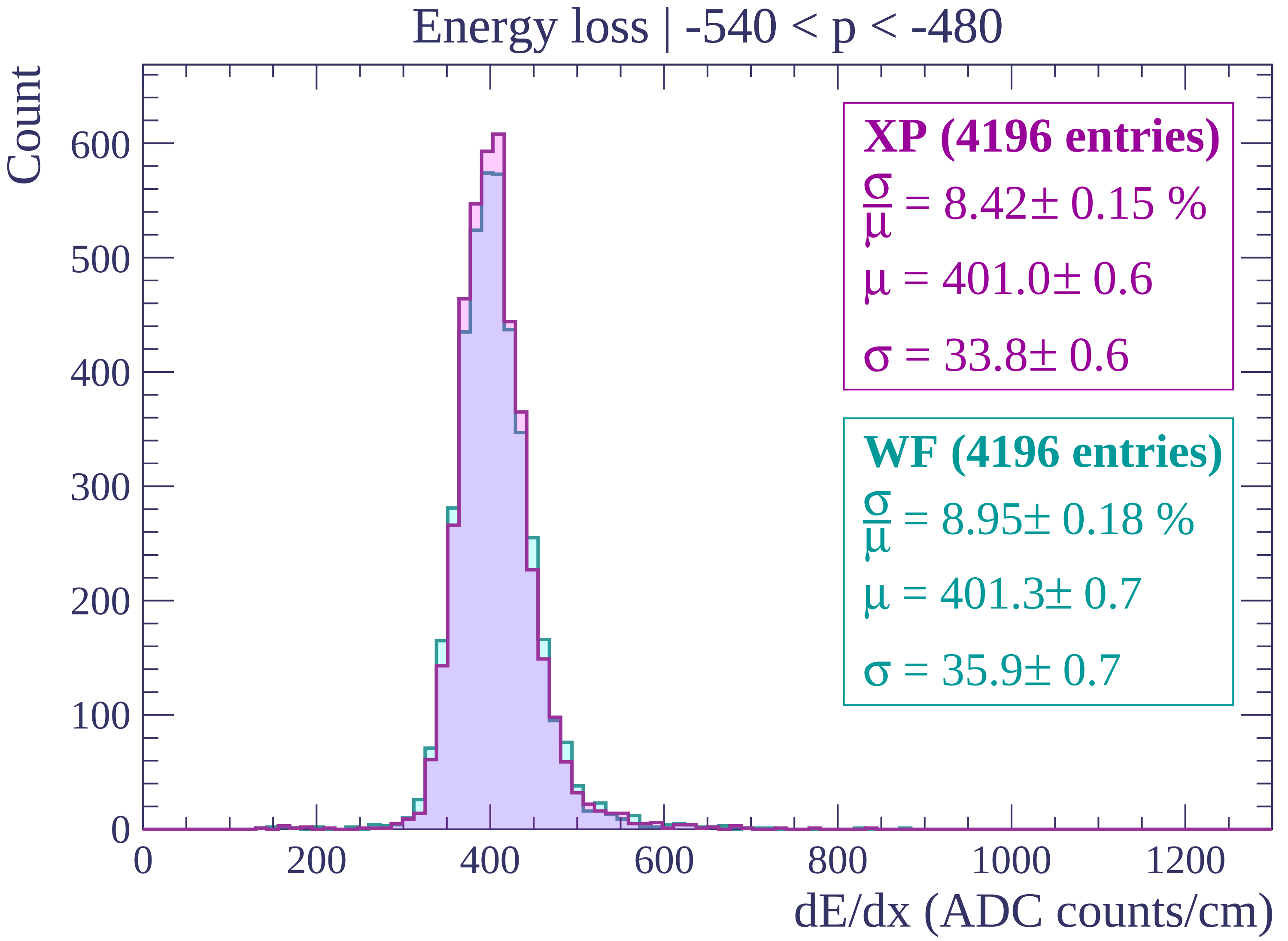


Energy loss | -720 < p < -660
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3926 entries)
$\frac{\sigma}{\mu}$ = 8.58± 0.17 %
μ = 411.3± 0.7
σ = 35.3± 0.6
WF (3926 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.19 %
μ = 411.6± 0.7
σ = 37.2± 0.7

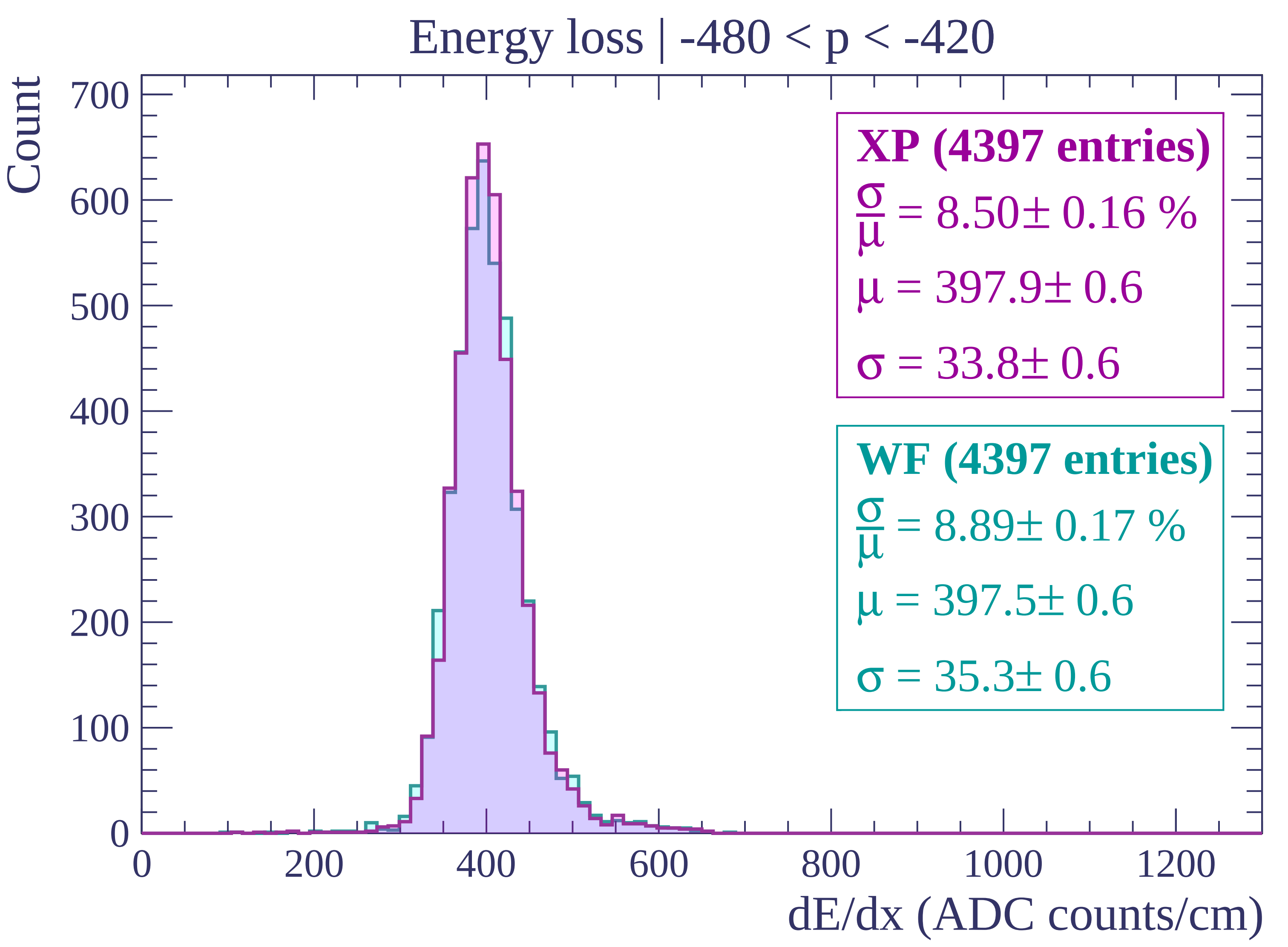


Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (3947 entries)
$\frac{\sigma}{\mu}$ = 8.76± 0.18 %
μ = 407.5± 0.7
σ = 35.7± 0.7
WF (3947 entries)
$\frac{\sigma}{\mu}$ = 9.02± 0.17 %
μ = 407.8± 0.7
σ = 36.8± 0.6

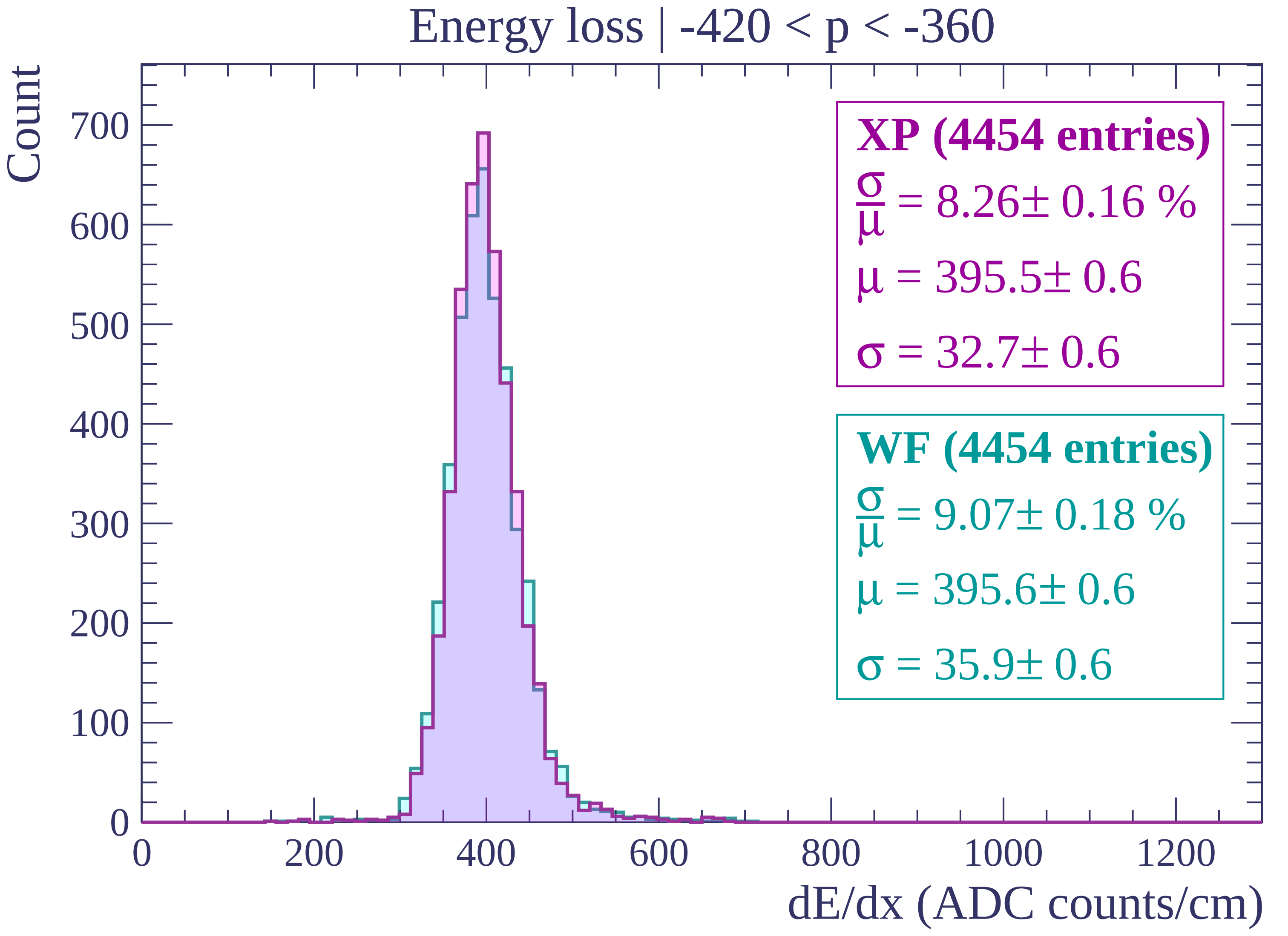

Energy loss | -600 < p < -540
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4023 entries)
σ/μ = 8.59± 0.19 %
μ = 403.6± 0.7
σ = 34.7± 0.7
WF (4023 entries)
σ/μ = 9.15± 0.17 %
μ = 405.1± 0.7
σ = 37.1± 0.6



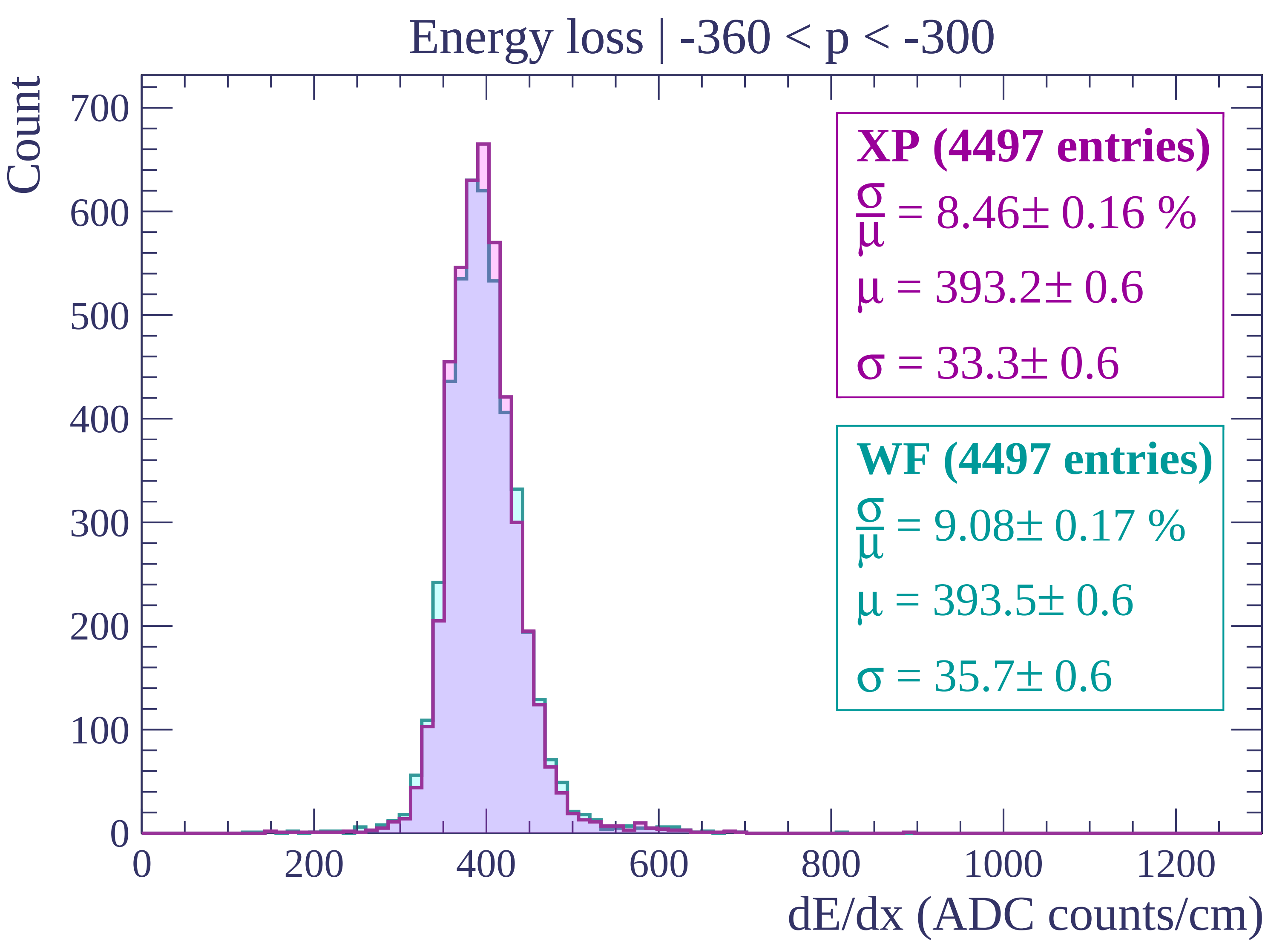
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4196 entries)
σ/μ = 8.42± 0.15 %
μ = 401.0± 0.6
σ = 33.8± 0.6
WF (4196 entries)
σ/μ = 8.95± 0.18 %
μ = 401.3± 0.7
σ = 35.9± 0.7



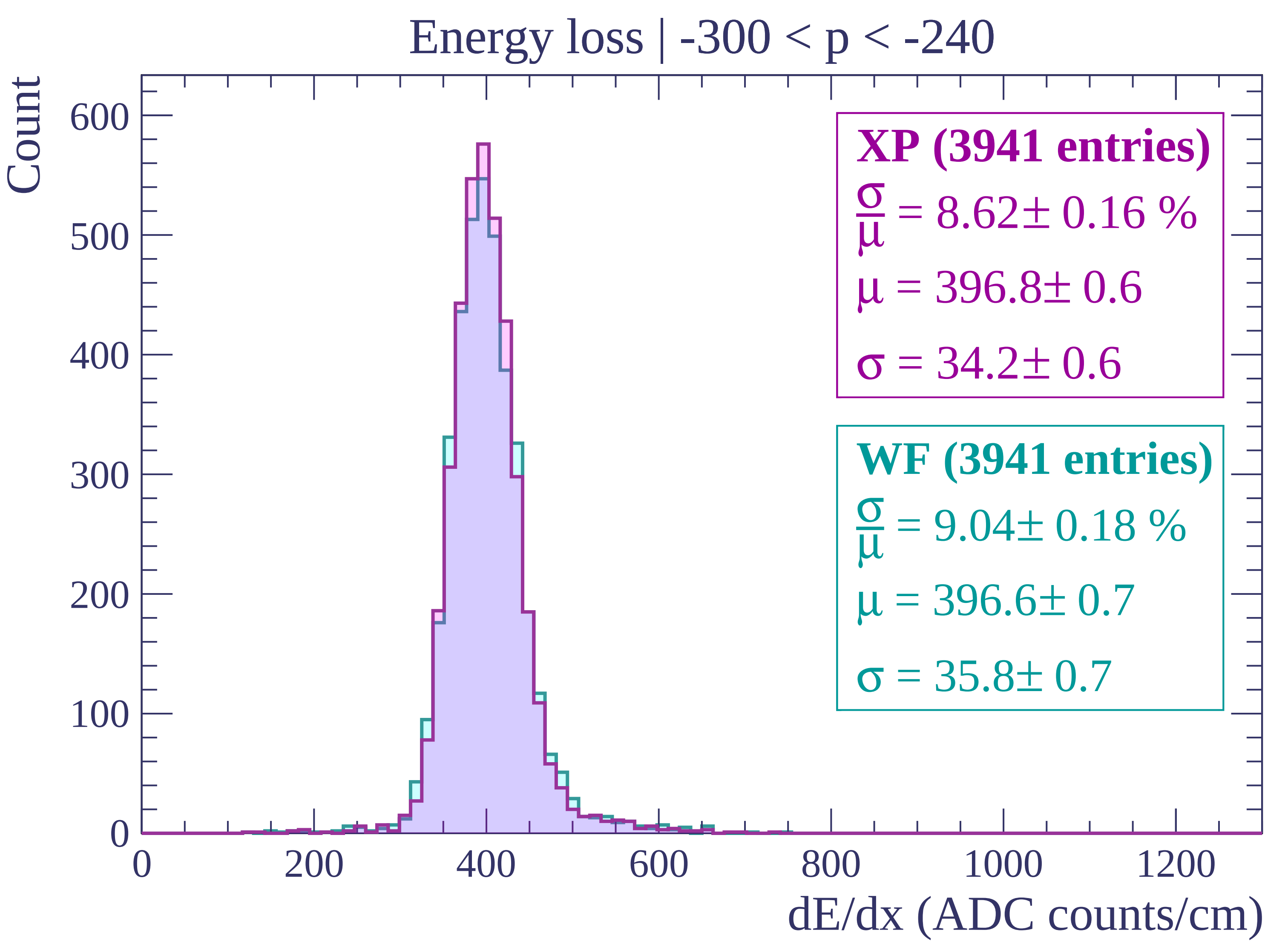
Energy loss | -480 < p < -420
Count
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4397 entries)
$\frac{\sigma}{\mu}$ = 8.50± 0.16 %
μ = 397.9± 0.6
σ = 33.8± 0.6
WF (4397 entries)
$\frac{\sigma}{\mu}$ = 8.89± 0.17 %
μ = 397.5± 0.6
σ = 35.3± 0.6

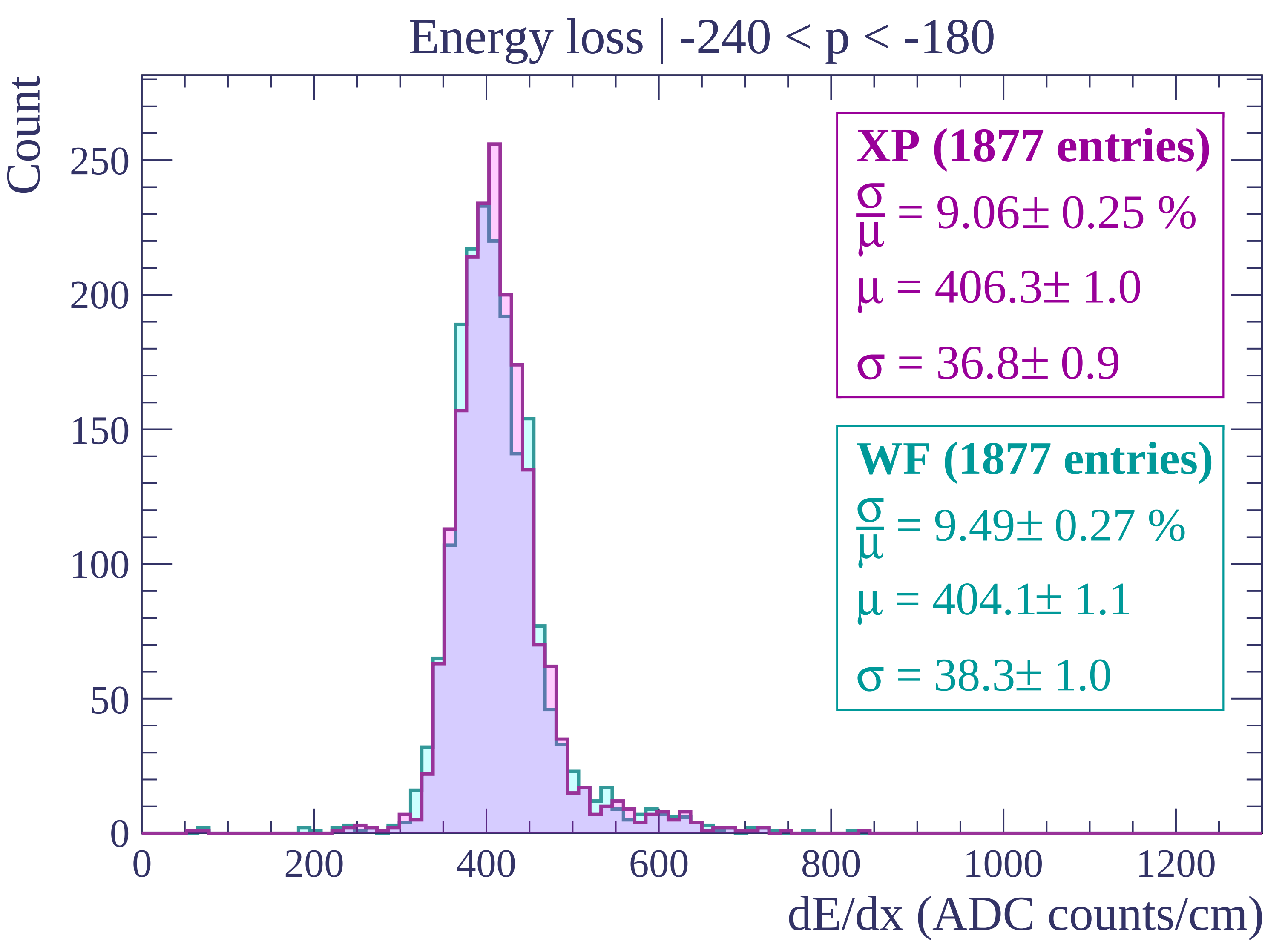


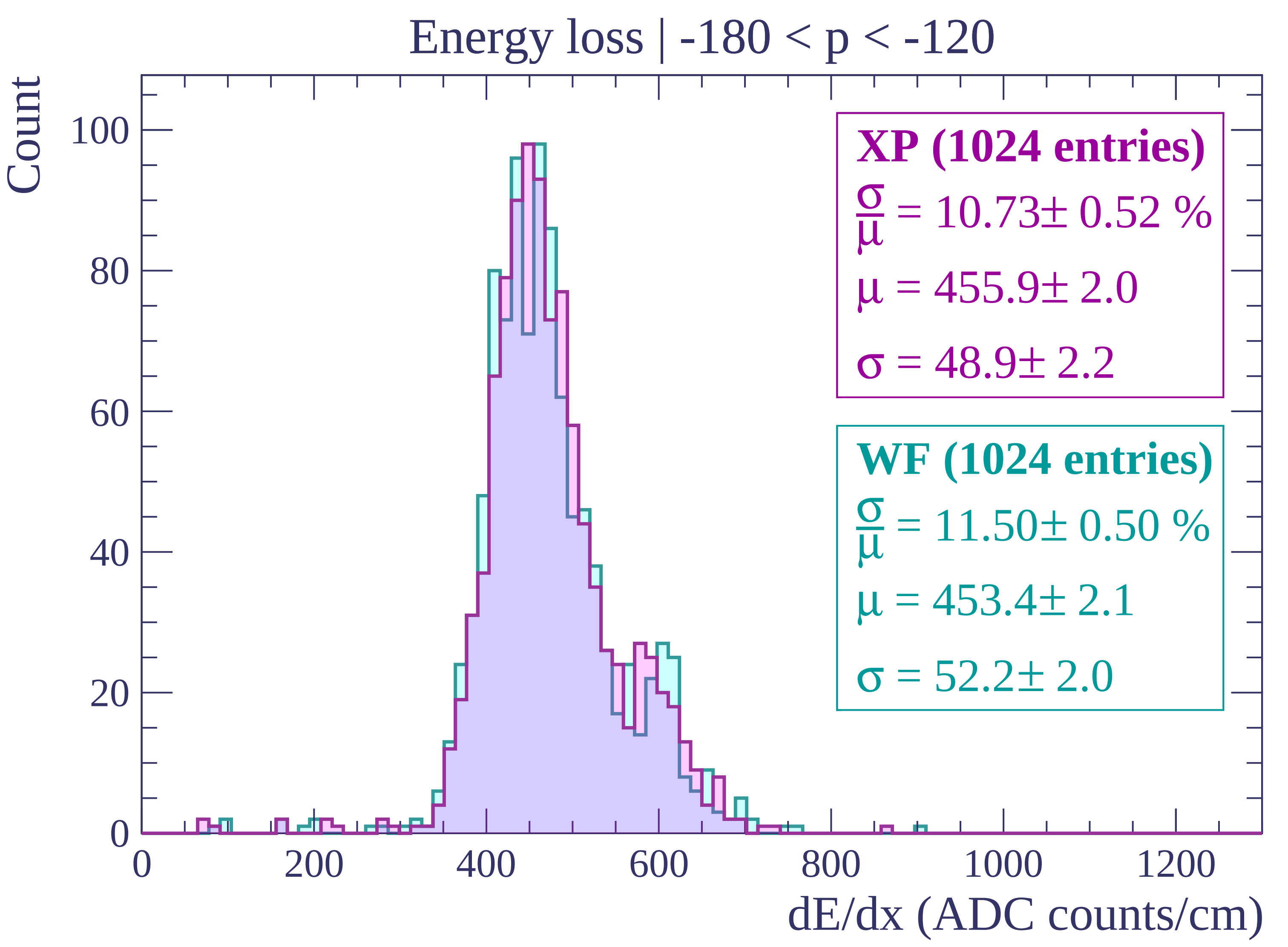
Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4454 entries)
$\frac{\sigma}{\mu}$ = 8.26± 0.16 %
μ = 395.5± 0.6
σ = 32.7± 0.6
WF (4454 entries)
$\frac{\sigma}{\mu}$ = 9.07± 0.18 %
μ = 395.6± 0.6
σ = 35.9± 0.6

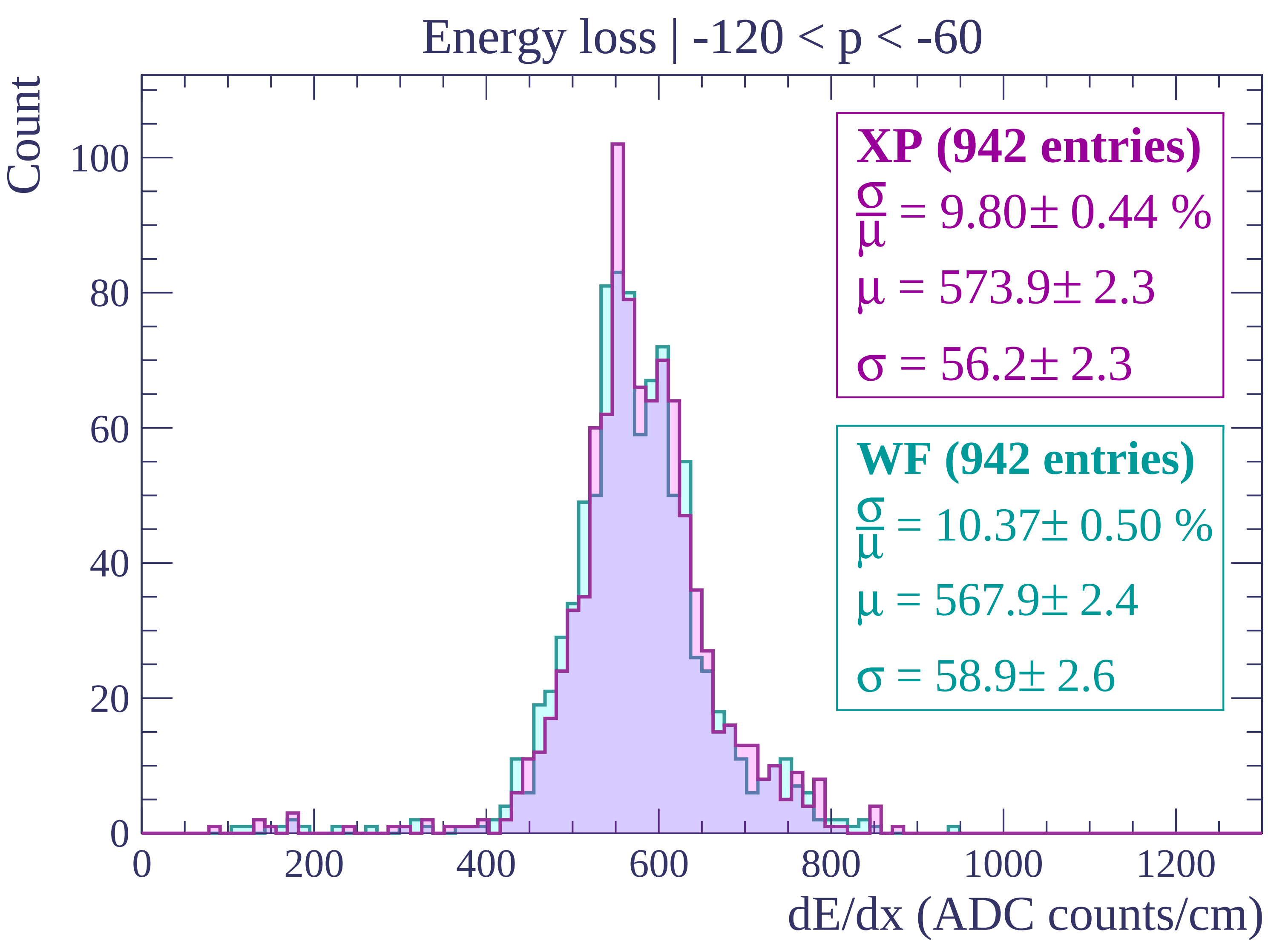


Energy loss | -360 < p < -300
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4497 entries)
$\frac{\sigma}{\mu}$ = 8.46± 0.16 %
μ = 393.2± 0.6
σ = 33.3± 0.6
WF (4497 entries)
$\frac{\sigma}{\mu}$ = 9.08± 0.17 %
μ = 393.5± 0.6
σ = 35.7± 0.6

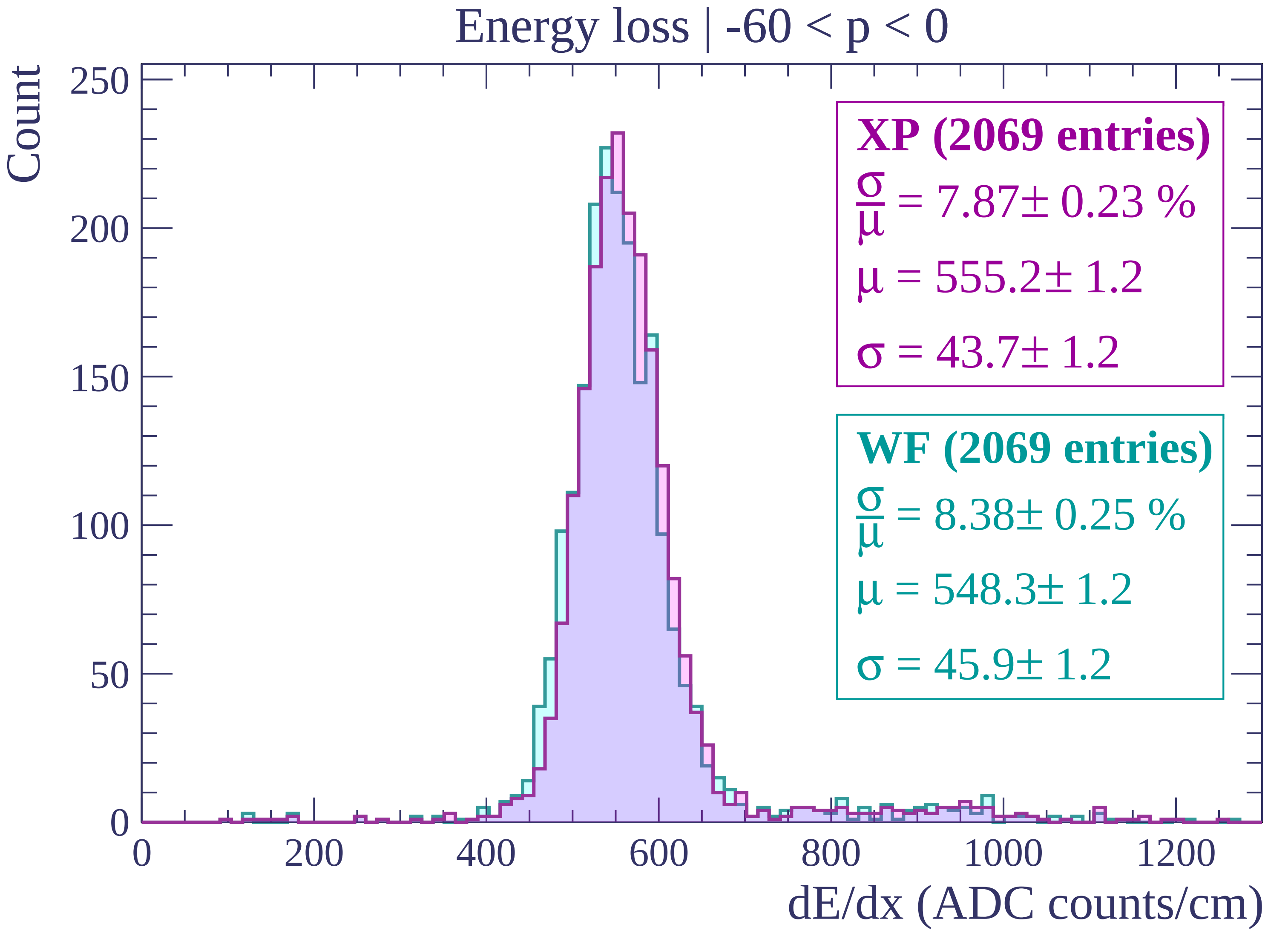

Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (3941 entries)
σ/μ = 8.62± 0.16 %
μ = 396.8± 0.6
σ = 34.2± 0.6
WF (3941 entries)
σ/μ = 9.04± 0.18 %
μ = 396.6± 0.7
σ = 35.8± 0.7


Energy loss | -240 < p < -180
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1877 entries)
$\frac{\sigma}{\mu}$ = 9.06± 0.25 %
μ = 406.3± 1.0
σ = 36.8± 0.9
WF (1877 entries)
$\frac{\sigma}{\mu}$ = 9.49± 0.27 %
μ = 404.1± 1.1
σ = 38.3± 1.0

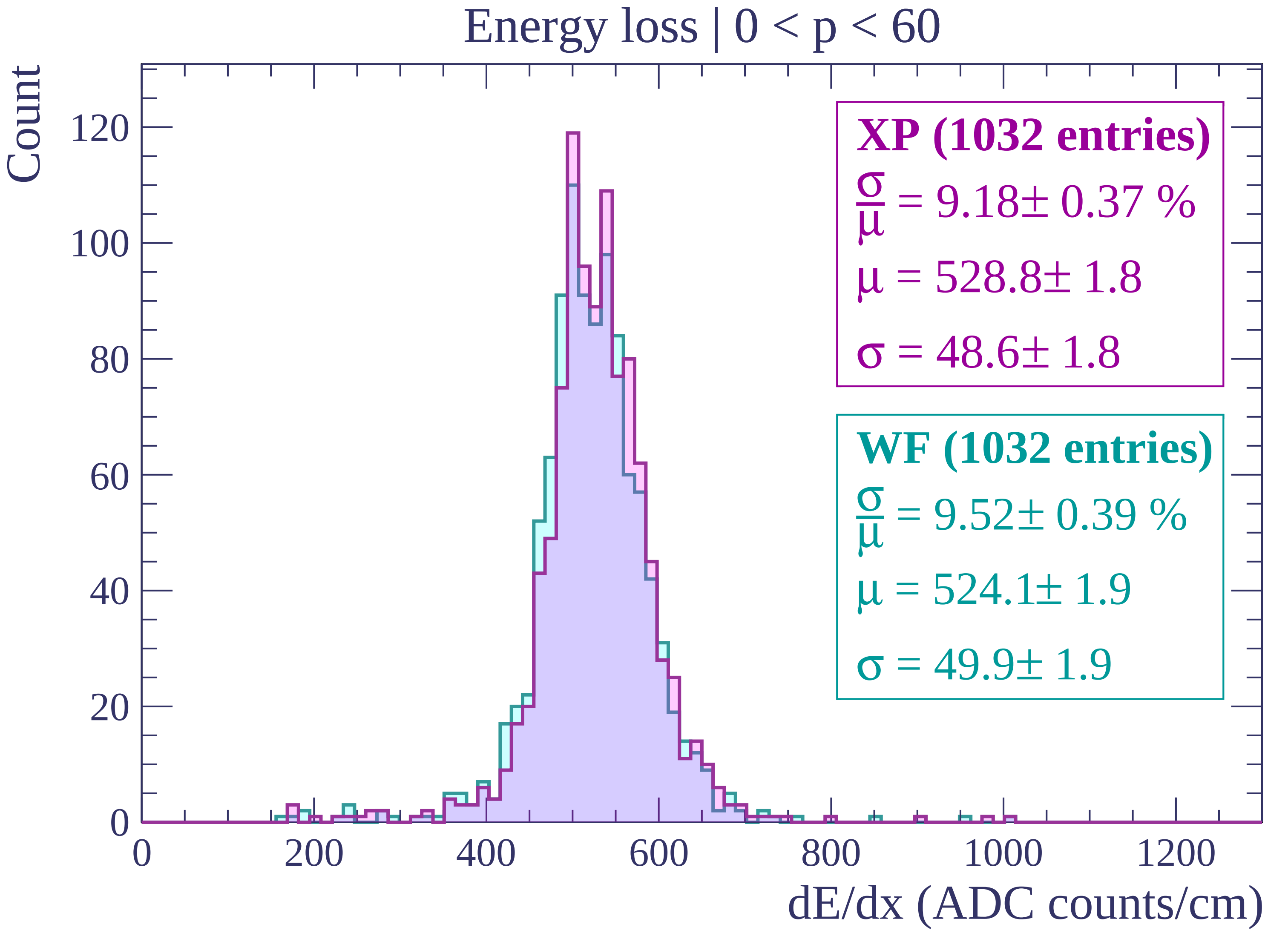

Energy loss | -180 < p < -120
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1024 entries)
σ/μ = 10.73± 0.52 %
μ = 455.9± 2.0
σ = 48.9± 2.2
WF (1024 entries)
σ/μ = 11.50± 0.50 %
μ = 453.4± 2.1
σ = 52.2± 2.0

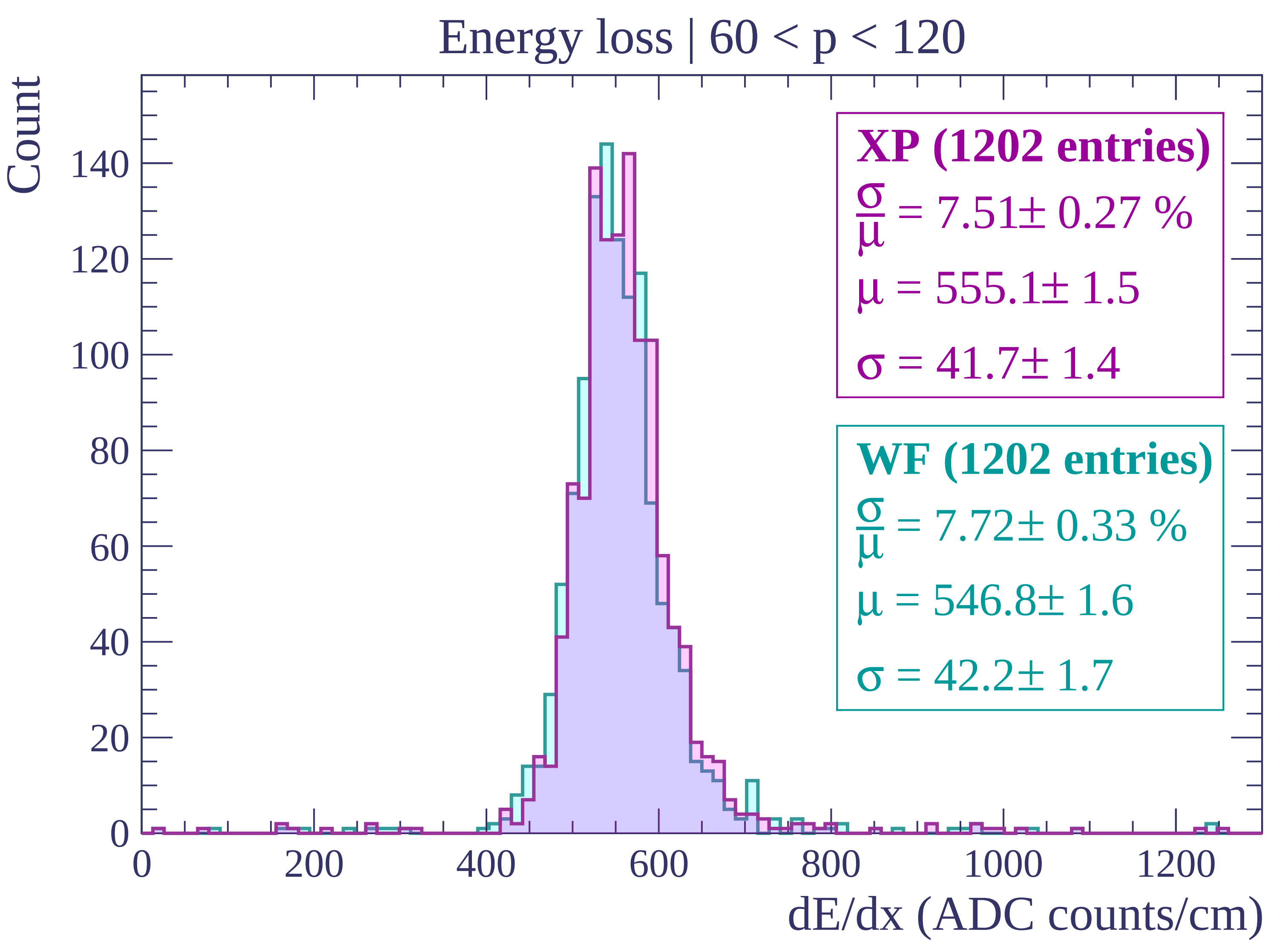


Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP (942 entries)
σ/μ = 9.80± 0.44 %
μ = 573.9± 2.3
σ = 56.2± 2.3
WF (942 entries)
σ/μ = 10.37± 0.50 %
μ = 567.9± 2.4
σ = 58.9± 2.6

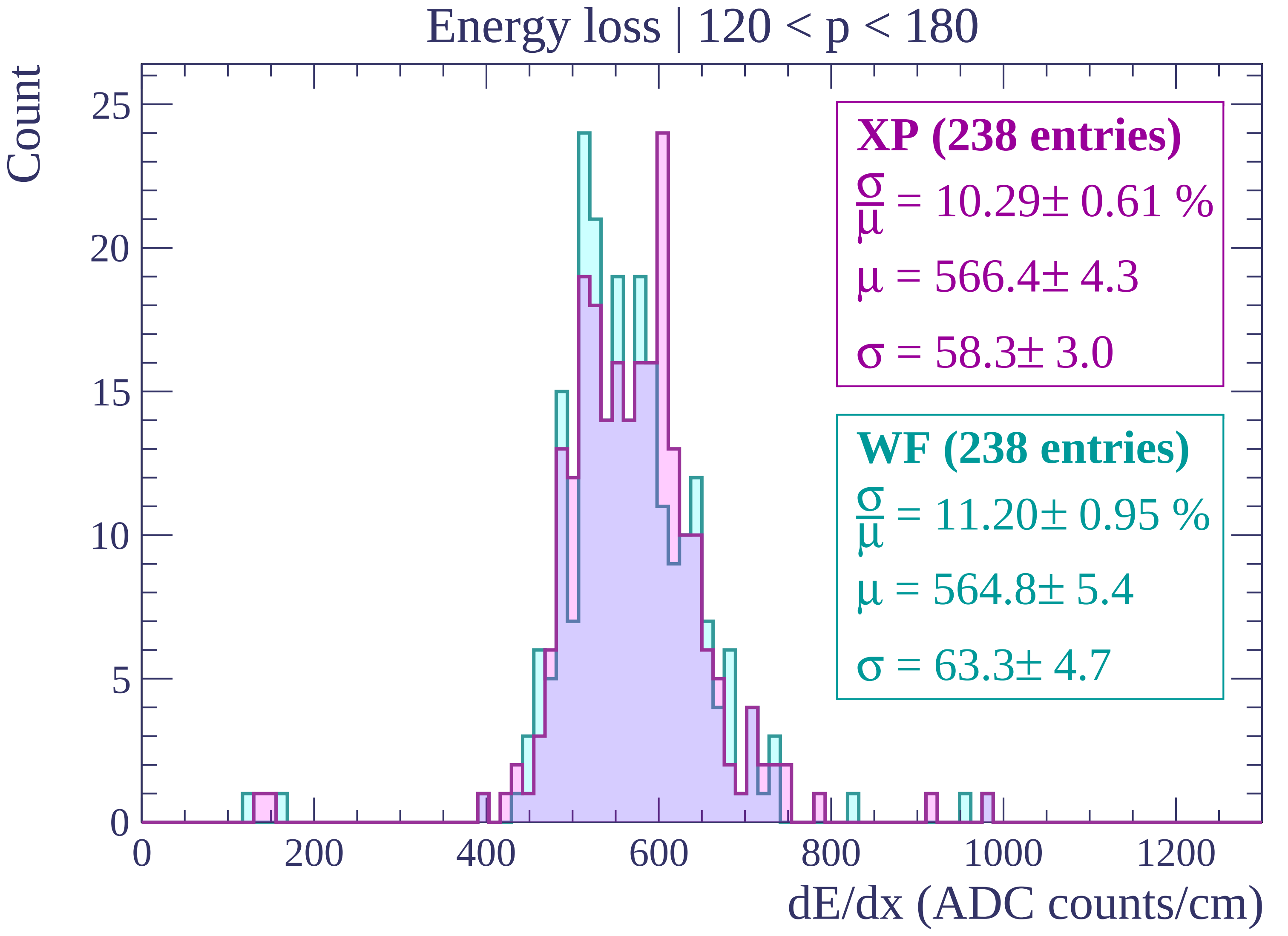


Energy loss | -60 < p < 0
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2069 entries)
$\frac{\sigma}{\mu}$ = 7.87± 0.23 %
μ = 555.2± 1.2
σ = 43.7± 1.2
WF (2069 entries)
$\frac{\sigma}{\mu}$ = 8.38± 0.25 %
μ = 548.3± 1.2
σ = 45.9± 1.2

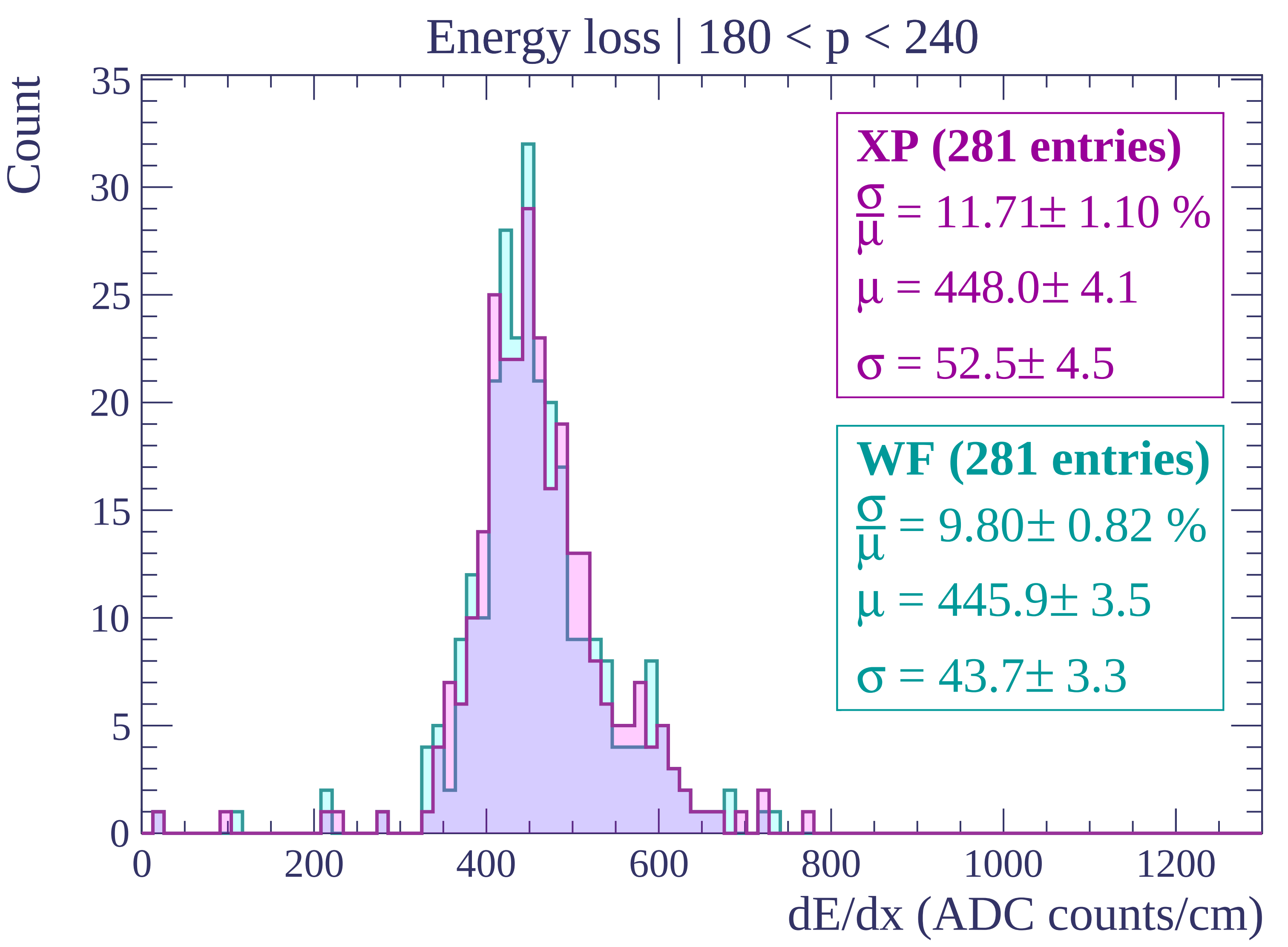

Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP (1032 entries)
$\frac{\sigma}{\mu}$ = 9.18± 0.37 %
μ = 528.8± 1.8
σ = 48.6± 1.8
WF (1032 entries)
$\frac{\sigma}{\mu}$ = 9.52± 0.39 %
μ = 524.1± 1.9
σ = 49.9± 1.9
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

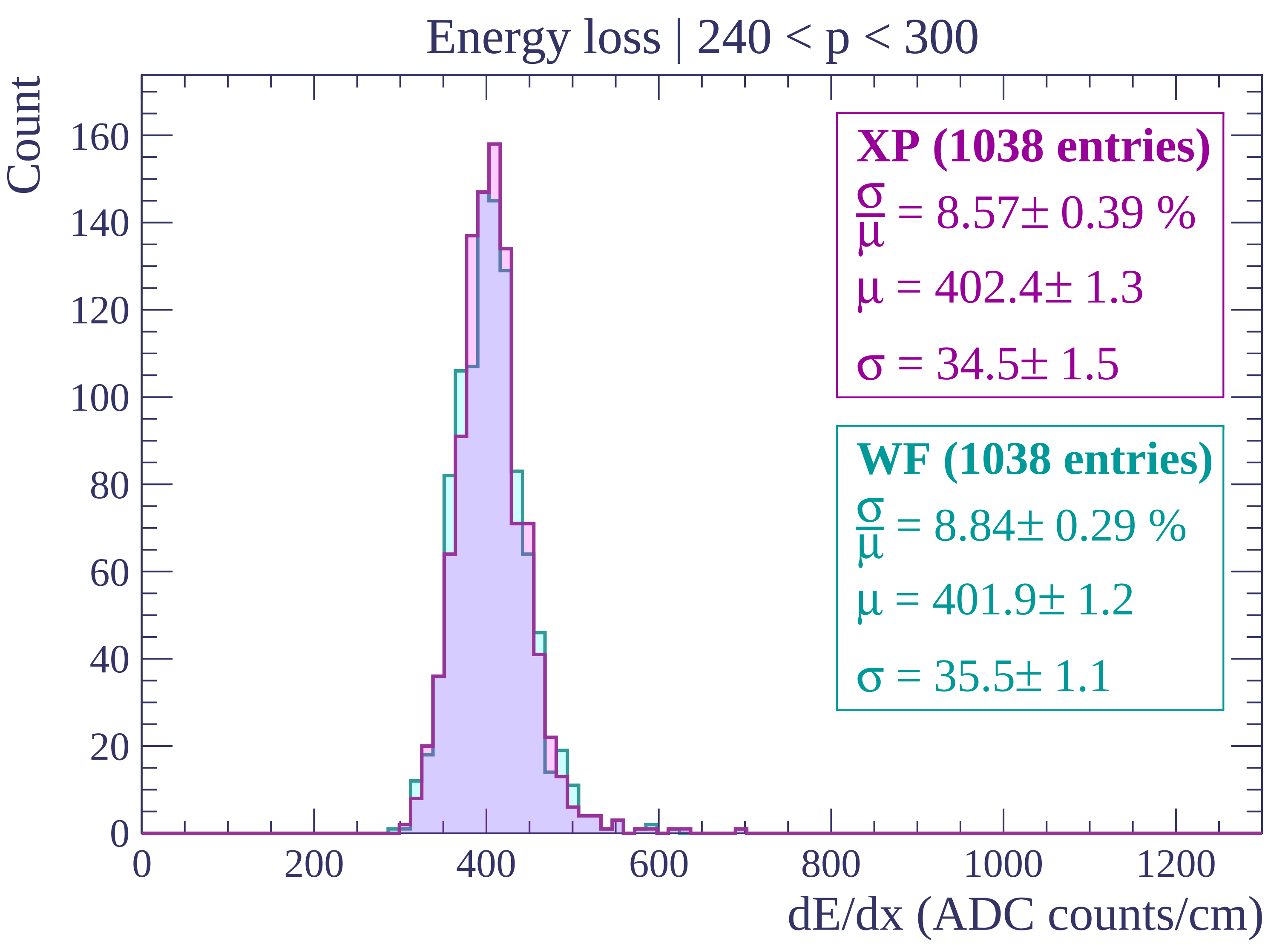


Energy loss | 60 < p < 120
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1202 entries)
σ/μ = 7.51± 0.27 %
μ = 555.1± 1.5
σ = 41.7± 1.4
WF (1202 entries)
σ/μ = 7.72± 0.33 %
μ = 546.8± 1.6
σ = 42.2± 1.7

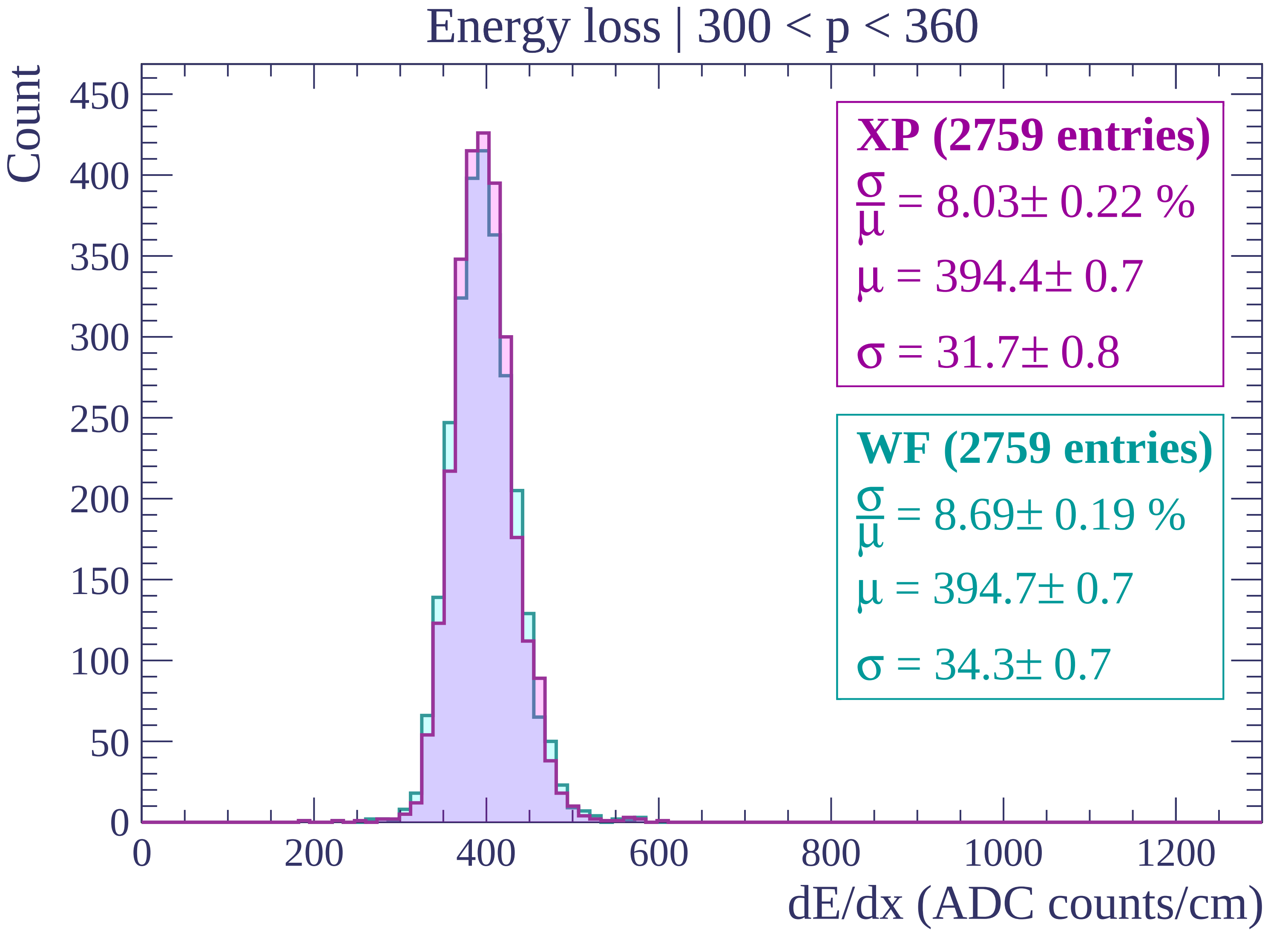


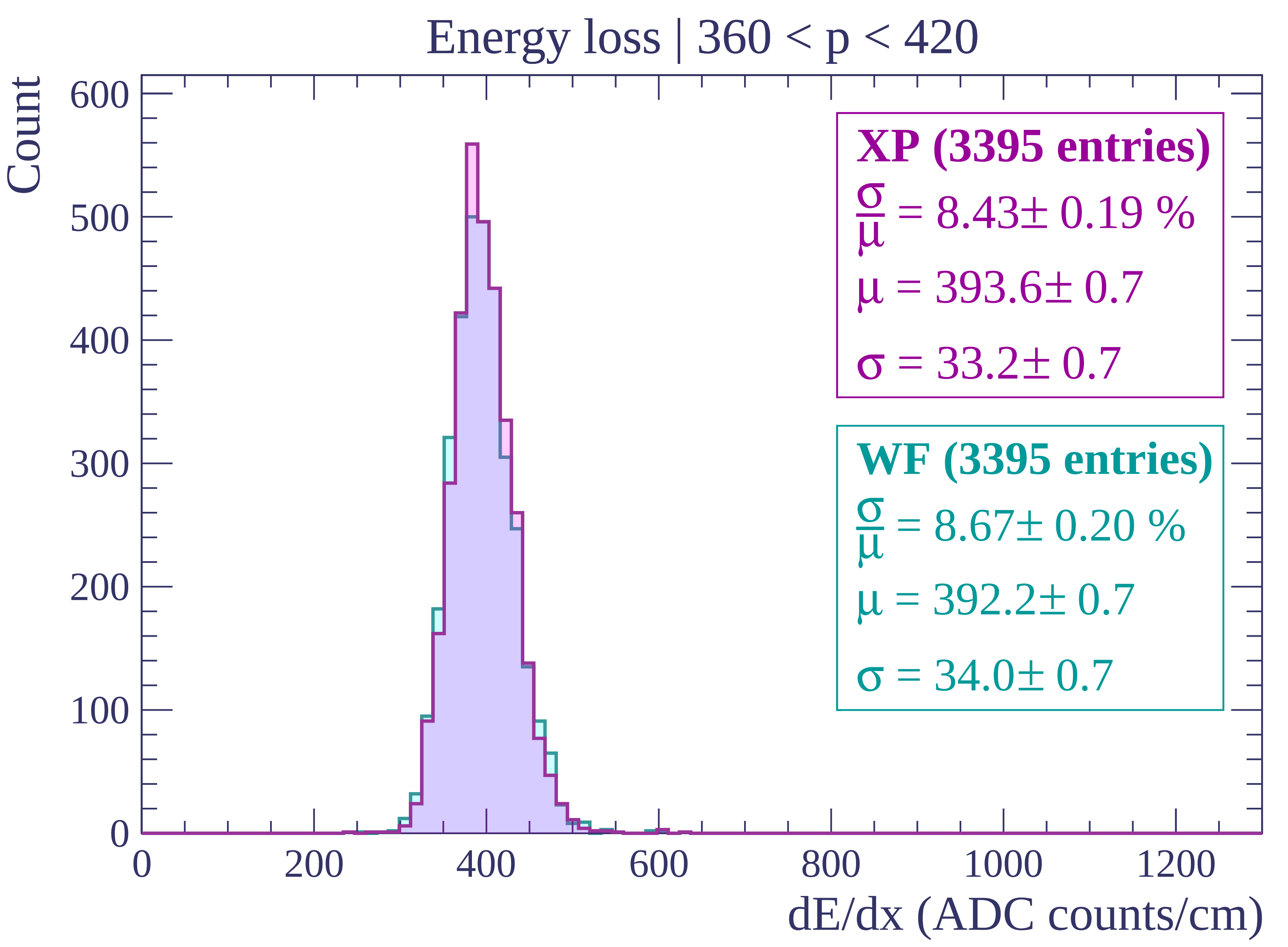
Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP (238 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.61 %
μ = 566.4± 4.3
σ = 58.3± 3.0
WF (238 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.95 %
μ = 564.8± 5.4
σ = 63.3± 4.7

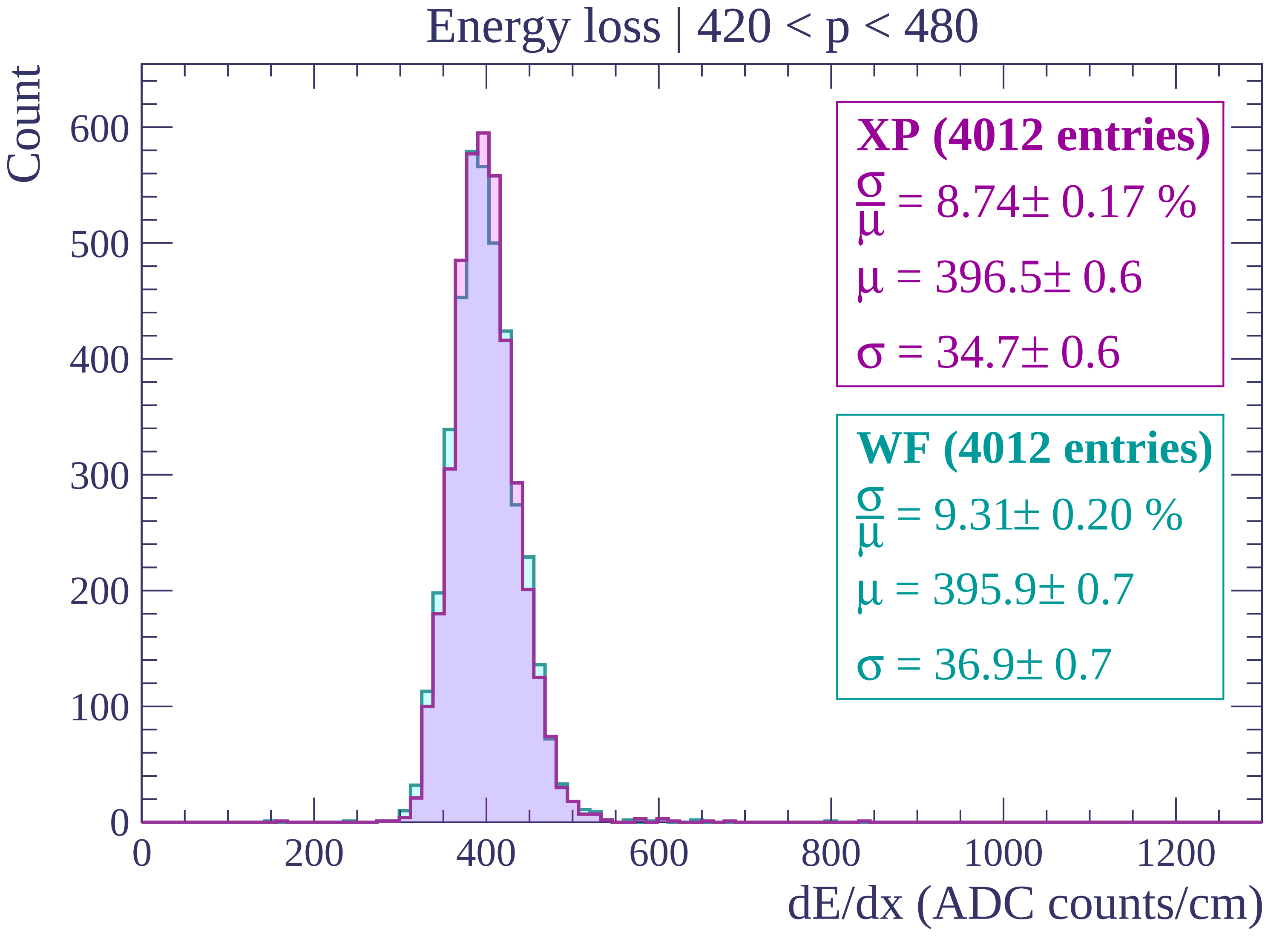

Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP (281 entries)
$\frac{\sigma}{\mu}$ = 11.71± 1.10 %
μ = 448.0± 4.1
σ = 52.5± 4.5
WF (281 entries)
$\frac{\sigma}{\mu}$ = 9.80± 0.82 %
μ = 445.9± 3.5
σ = 43.7± 3.3



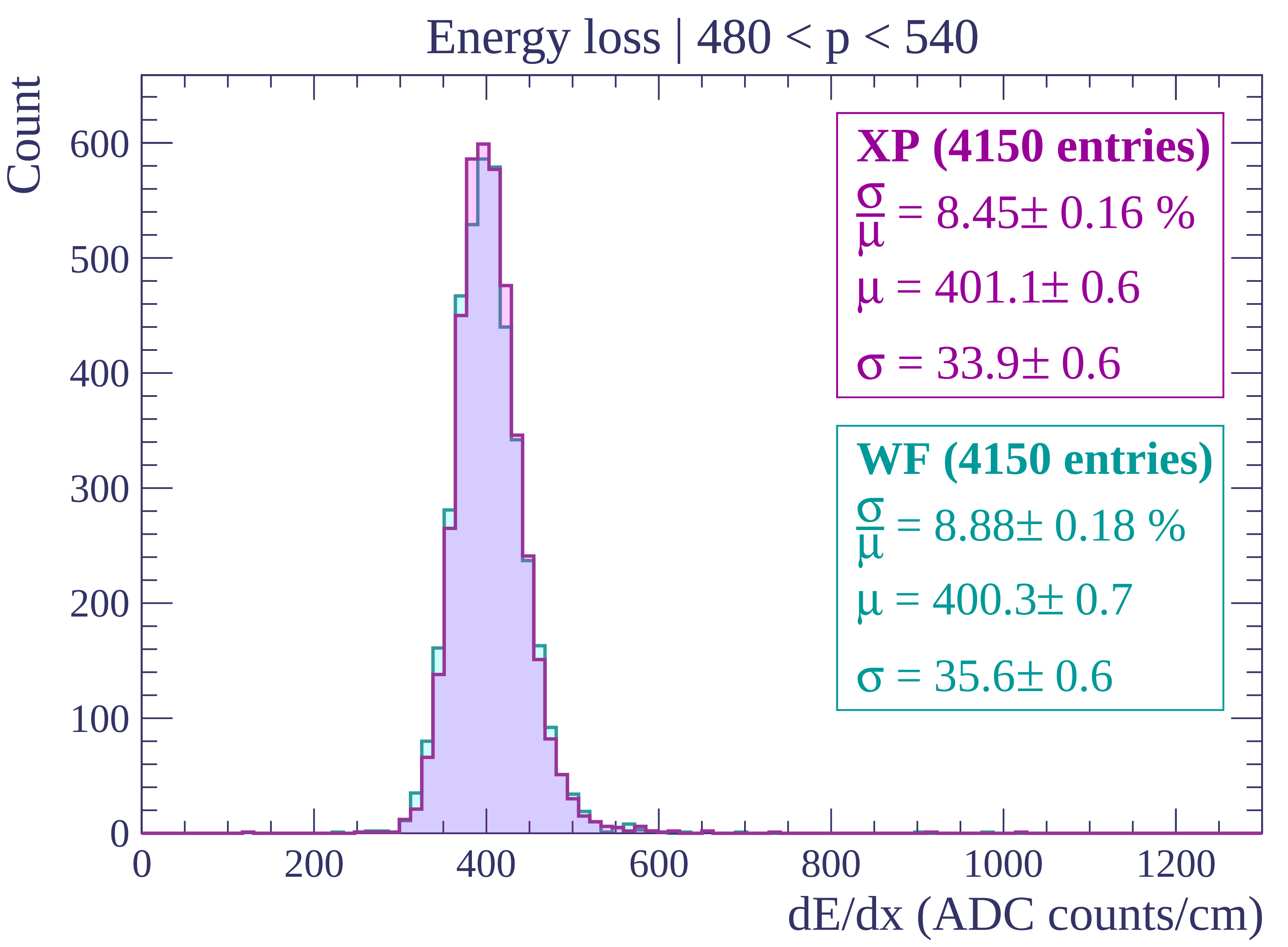
Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP (1038 entries)
$\frac{\sigma}{\mu}$ = 8.57± 0.39 %
μ = 402.4± 1.3
σ = 34.5± 1.5
WF (1038 entries)
$\frac{\sigma}{\mu}$ = 8.84± 0.29 %
μ = 401.9± 1.2
σ = 35.5± 1.1


Energy loss | 300 < p < 360
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2759 entries)
σ/μ = 8.03± 0.22 %
μ = 394.4± 0.7
σ = 31.7± 0.8
WF (2759 entries)
σ/μ = 8.69± 0.19 %
μ = 394.7± 0.7
σ = 34.3± 0.7

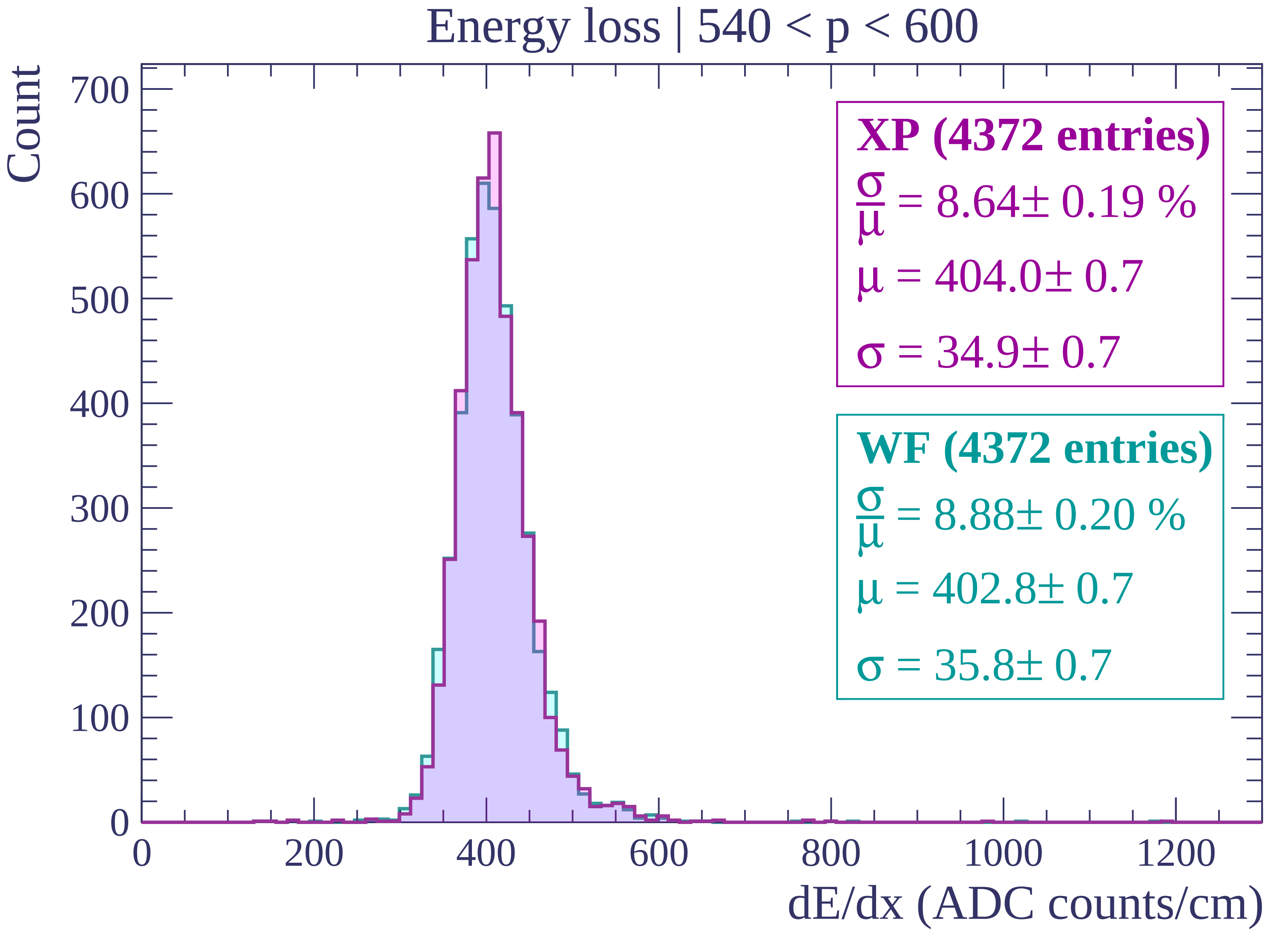

Energy loss | 360 < p < 420
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3395 entries)
σ/μ = 8.43± 0.19 %
μ = 393.6± 0.7
σ = 33.2± 0.7
WF (3395 entries)
σ/μ = 8.67± 0.20 %
μ = 392.2± 0.7
σ = 34.0± 0.7



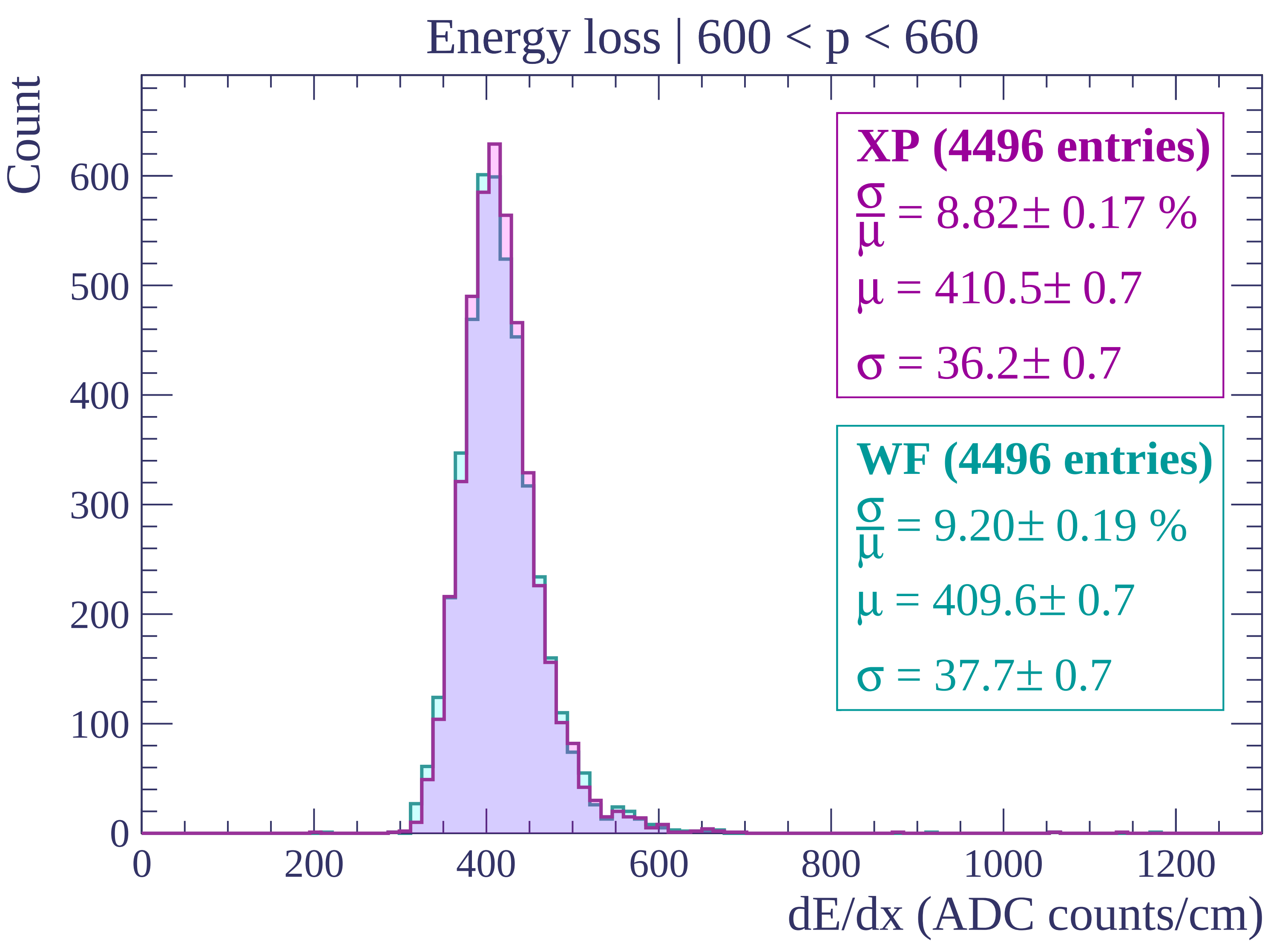
Energy loss | 420 < p < 480
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4012 entries)
σ/μ = 8.74± 0.17 %
μ = 396.5± 0.6
σ = 34.7± 0.6
WF (4012 entries)
σ/μ = 9.31± 0.20 %
μ = 395.9± 0.7
σ = 36.9± 0.7

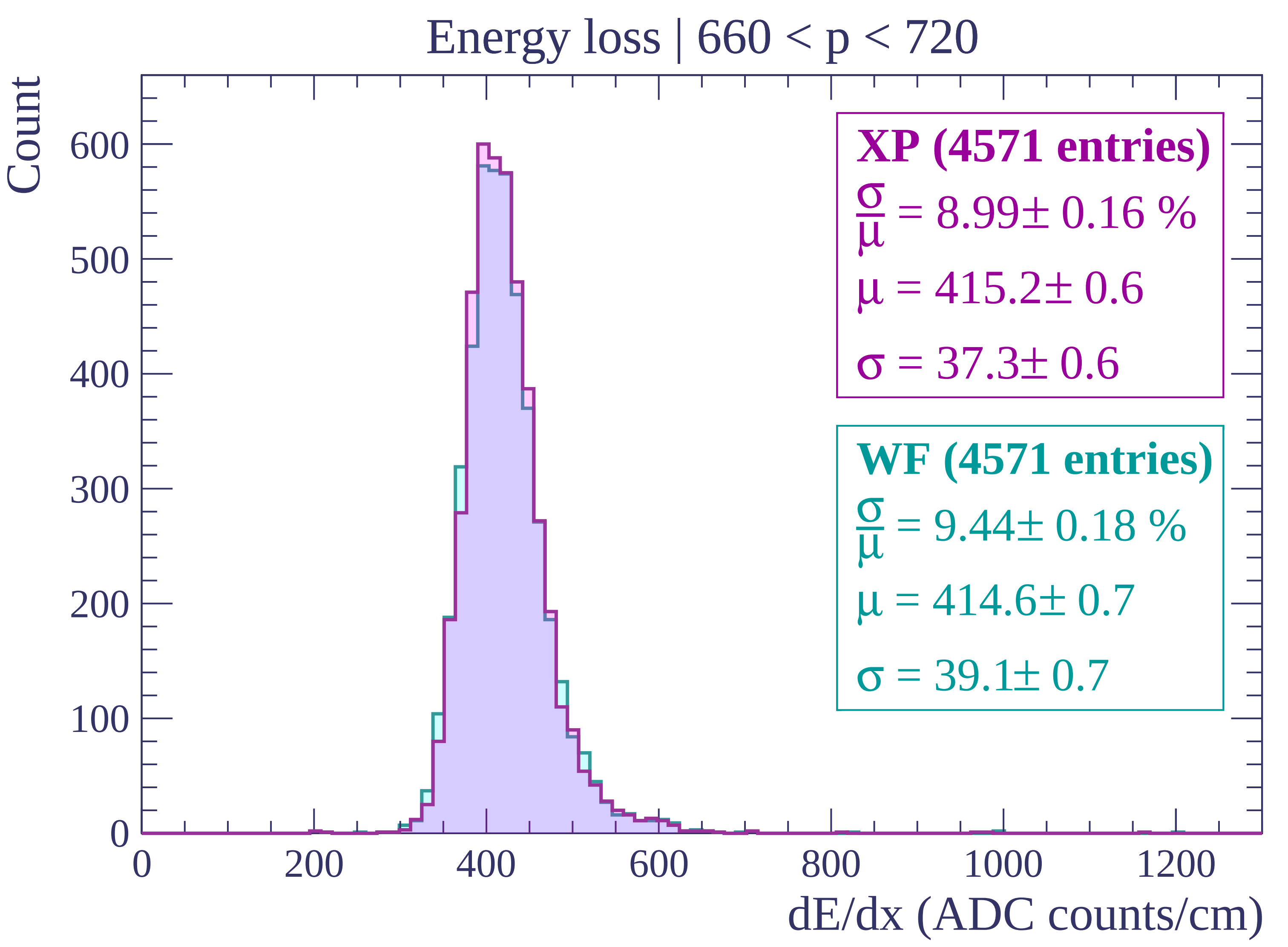


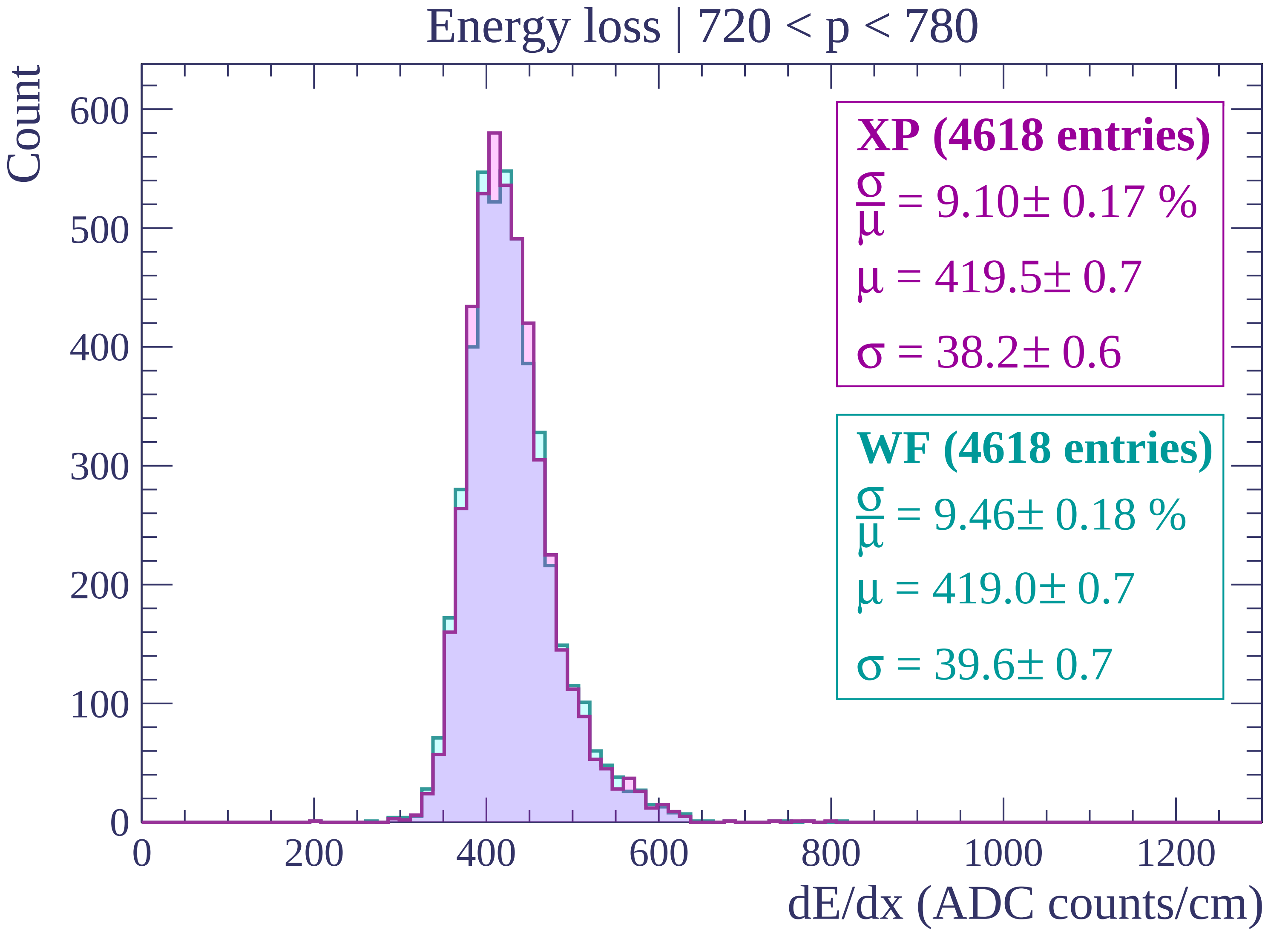
Energy loss | 480 < p < 540
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4150 entries)
$\frac{\sigma}{\mu}$ = 8.45± 0.16 %
μ = 401.1± 0.6
σ = 33.9± 0.6
WF (4150 entries)
$\frac{\sigma}{\mu}$ = 8.88± 0.18 %
μ = 400.3± 0.7
σ = 35.6± 0.6

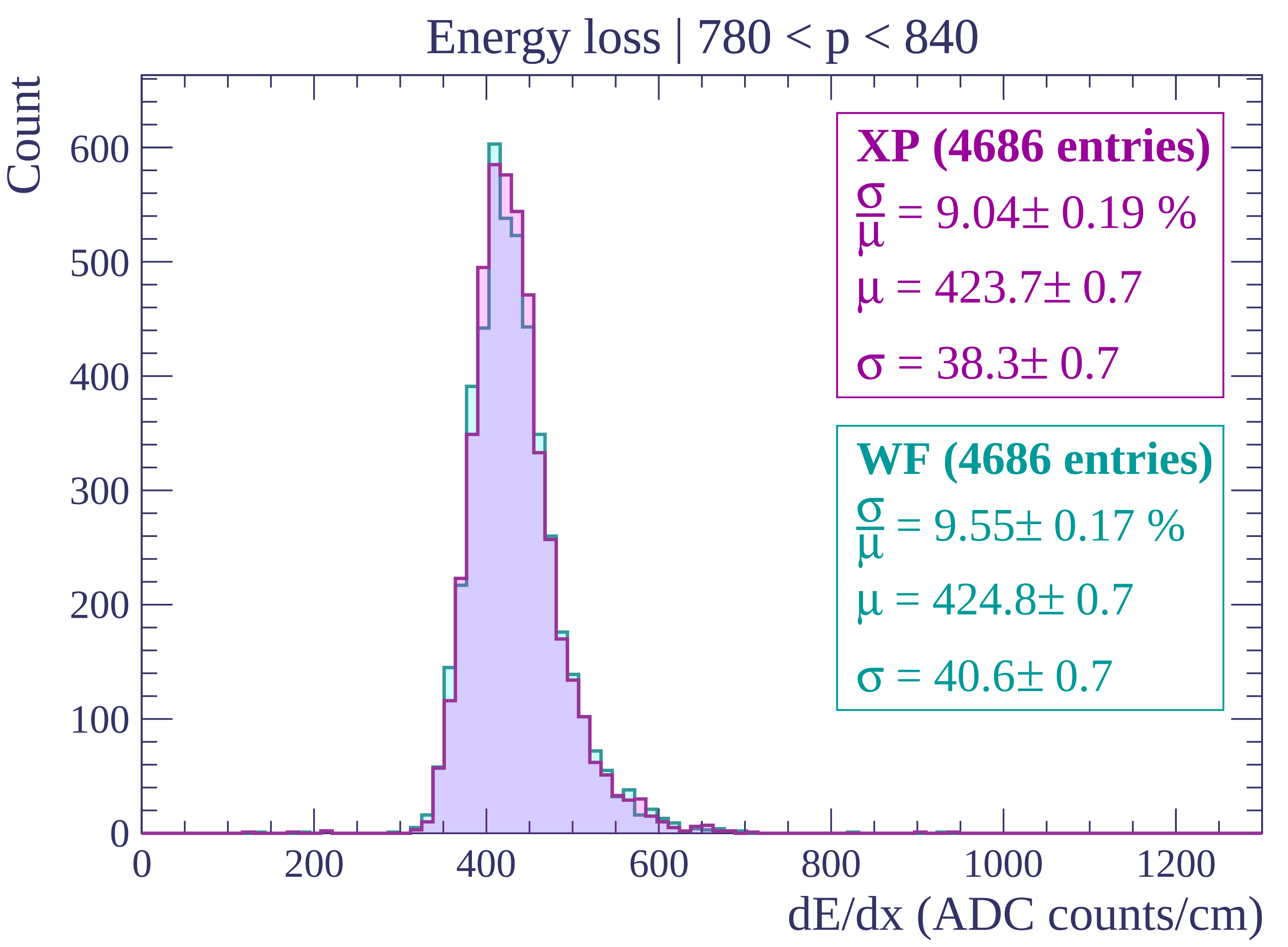


Energy loss | 540 < p < 600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP (4372 entries)
σ/μ = 8.64± 0.19 %
μ = 404.0± 0.7
σ = 34.9± 0.7
WF (4372 entries)
σ/μ = 8.88± 0.20 %
μ = 402.8± 0.7
σ = 35.8± 0.7

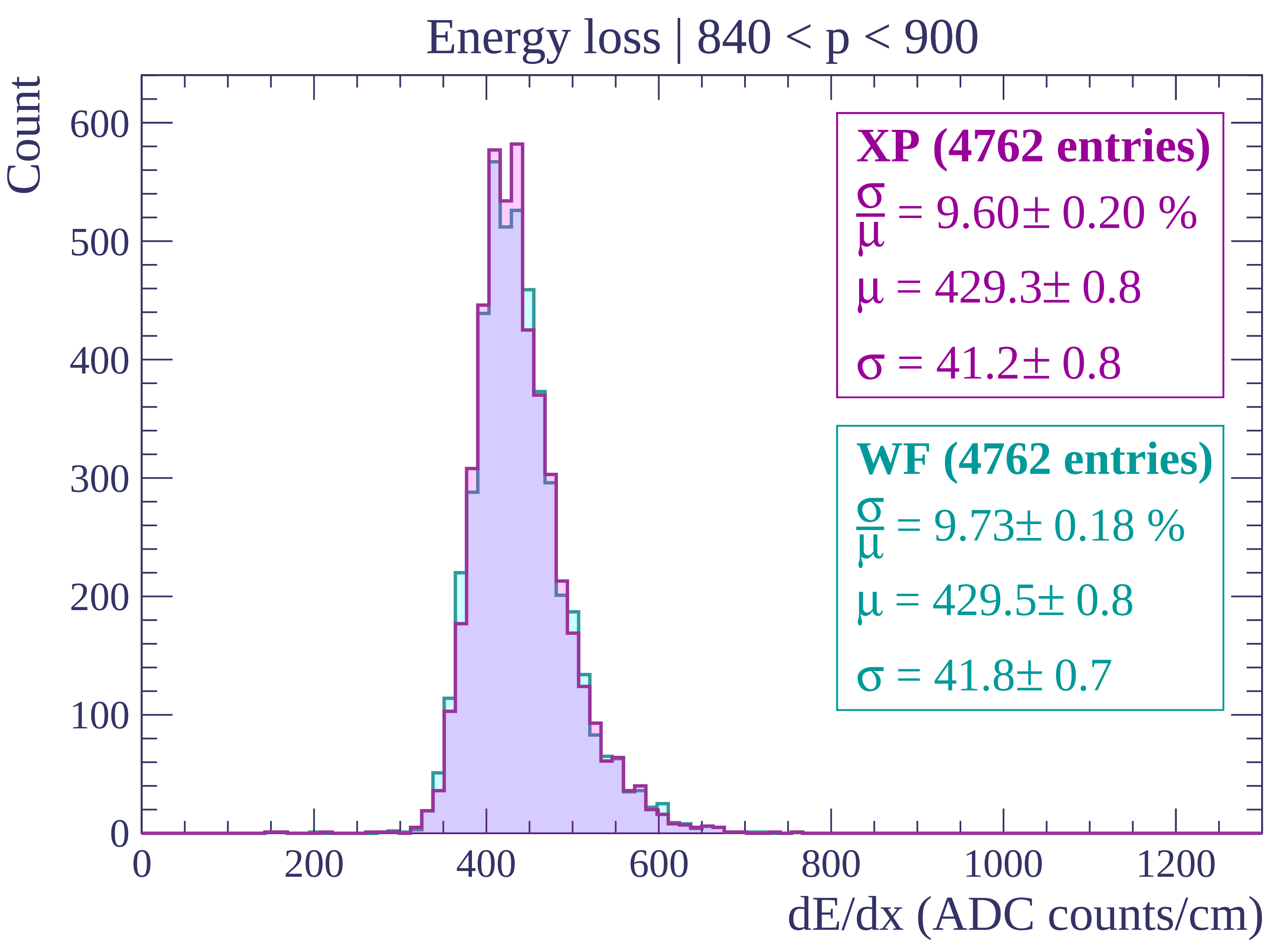

Energy loss | 600 < p < 660
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP (4496 entries)
σ/μ = 8.82± 0.17 %
μ = 410.5± 0.7
σ = 36.2± 0.7
WF (4496 entries)
σ/μ = 9.20± 0.19 %
μ = 409.6± 0.7
σ = 37.7± 0.7

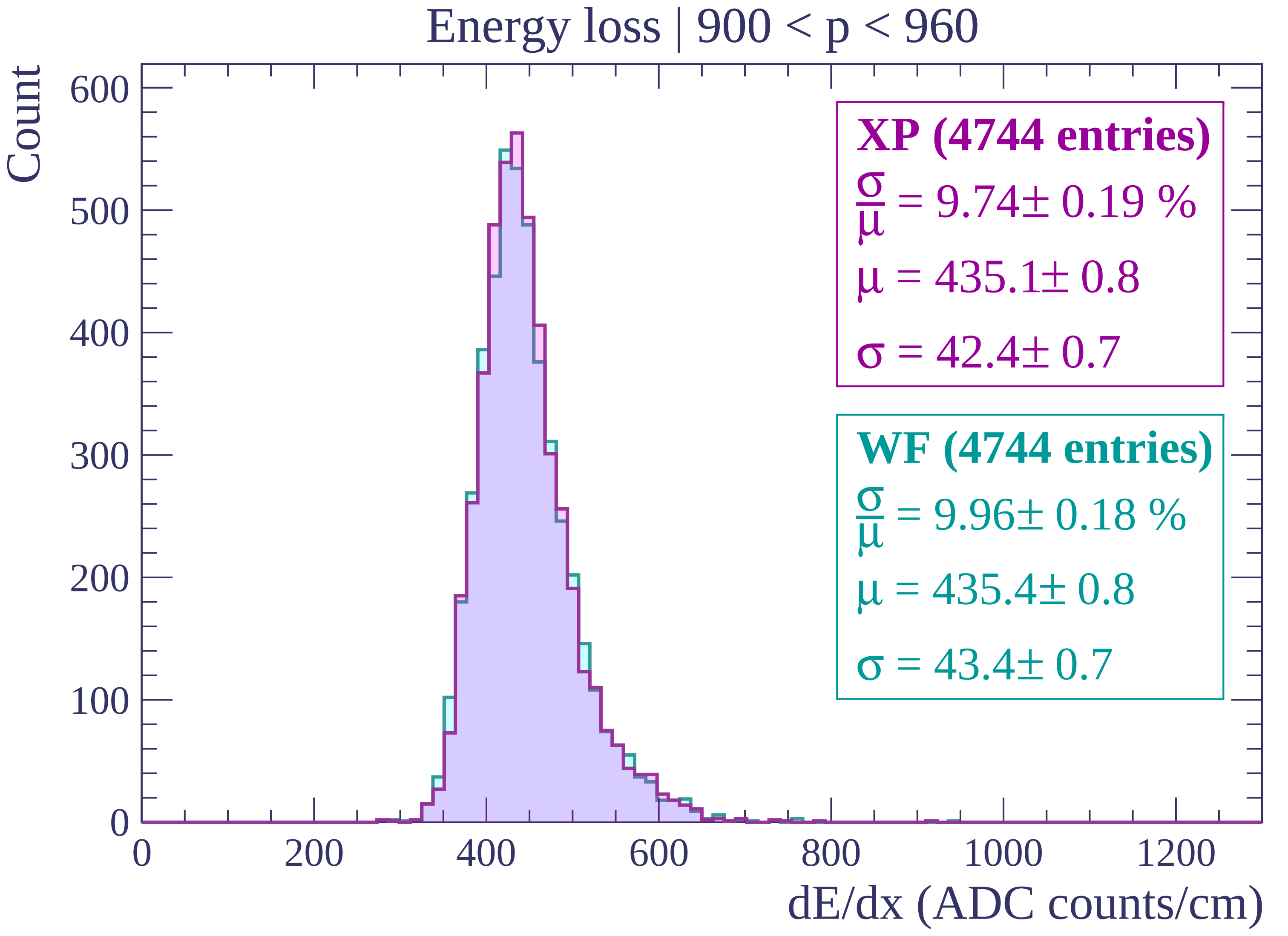

Energy loss | 660 < p < 720
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4571 entries)
σ/μ = 8.99± 0.16 %
μ = 415.2± 0.6
σ = 37.3± 0.6
WF (4571 entries)
σ/μ = 9.44± 0.18 %
μ = 414.6± 0.7
σ = 39.1± 0.7


Energy loss | 720 < p < 780
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4618 entries)
σ/μ = 9.10± 0.17 %
μ = 419.5± 0.7
σ = 38.2± 0.6
WF (4618 entries)
σ/μ = 9.46± 0.18 %
μ = 419.0± 0.7
σ = 39.6± 0.7

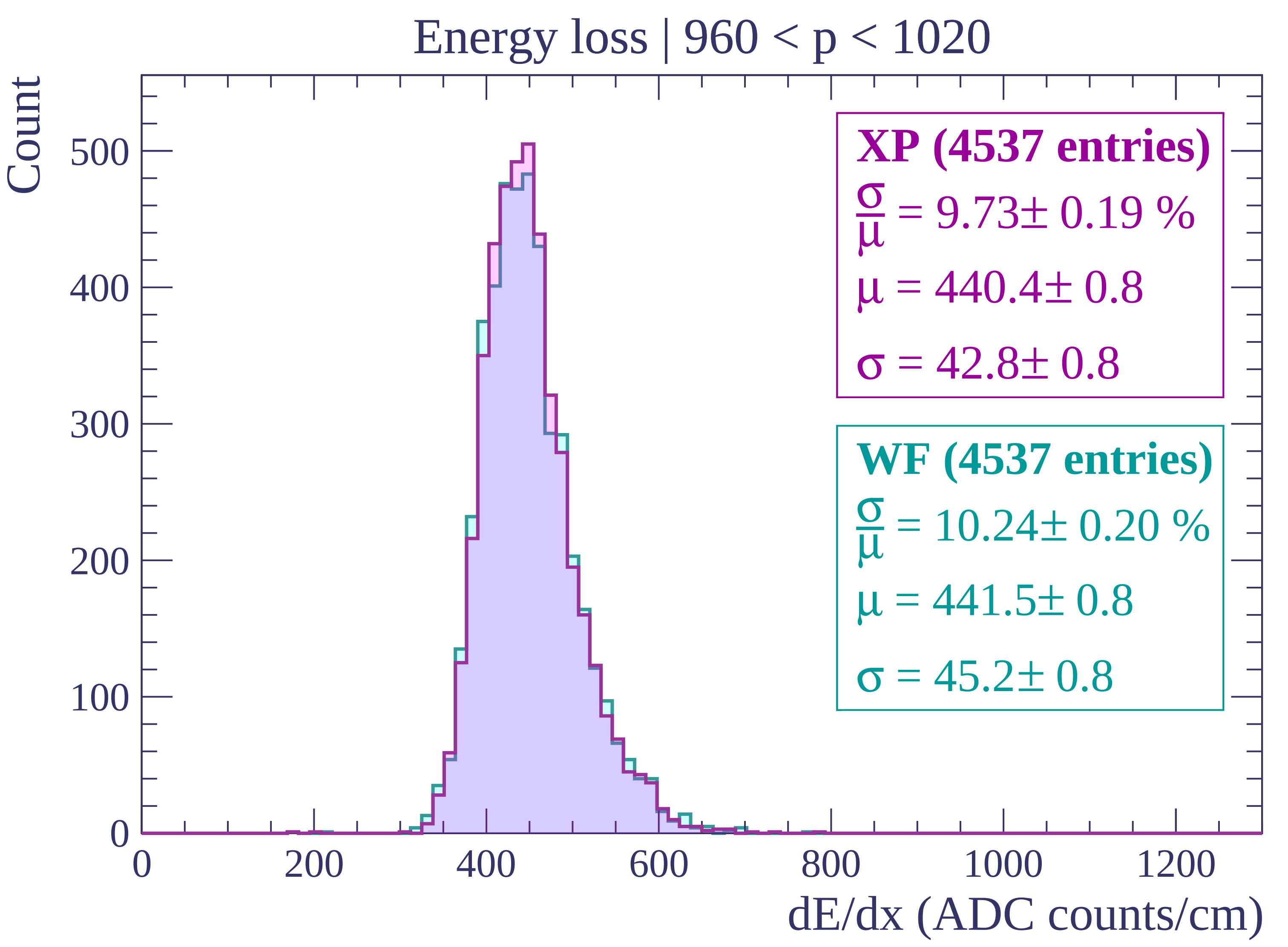


Energy loss | 780 < p < 840
Count
dE/dx (ADC counts/cm)
XP (4686 entries)
$\frac{\sigma}{\mu}$ = 9.04± 0.19 %
μ = 423.7± 0.7
σ = 38.3± 0.7
WF (4686 entries)
$\frac{\sigma}{\mu}$ = 9.55± 0.17 %
μ = 424.8± 0.7
σ = 40.6± 0.7

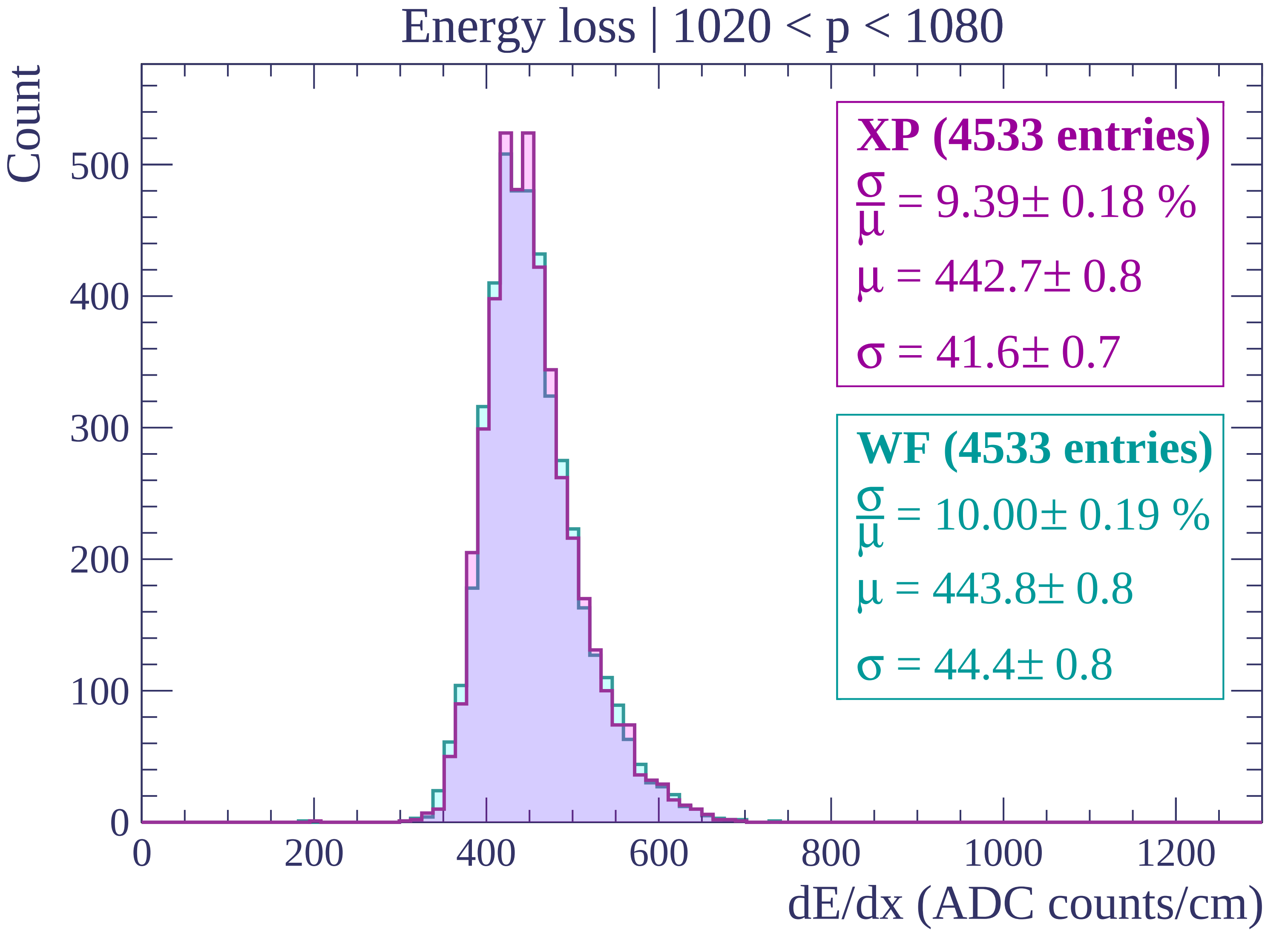

Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
XP (4762 entries)
σ/μ = 9.60± 0.20 %
μ = 429.3± 0.8
σ = 41.2± 0.8
WF (4762 entries)
σ/μ = 9.73± 0.18 %
μ = 429.5± 0.8
σ = 41.8± 0.7
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



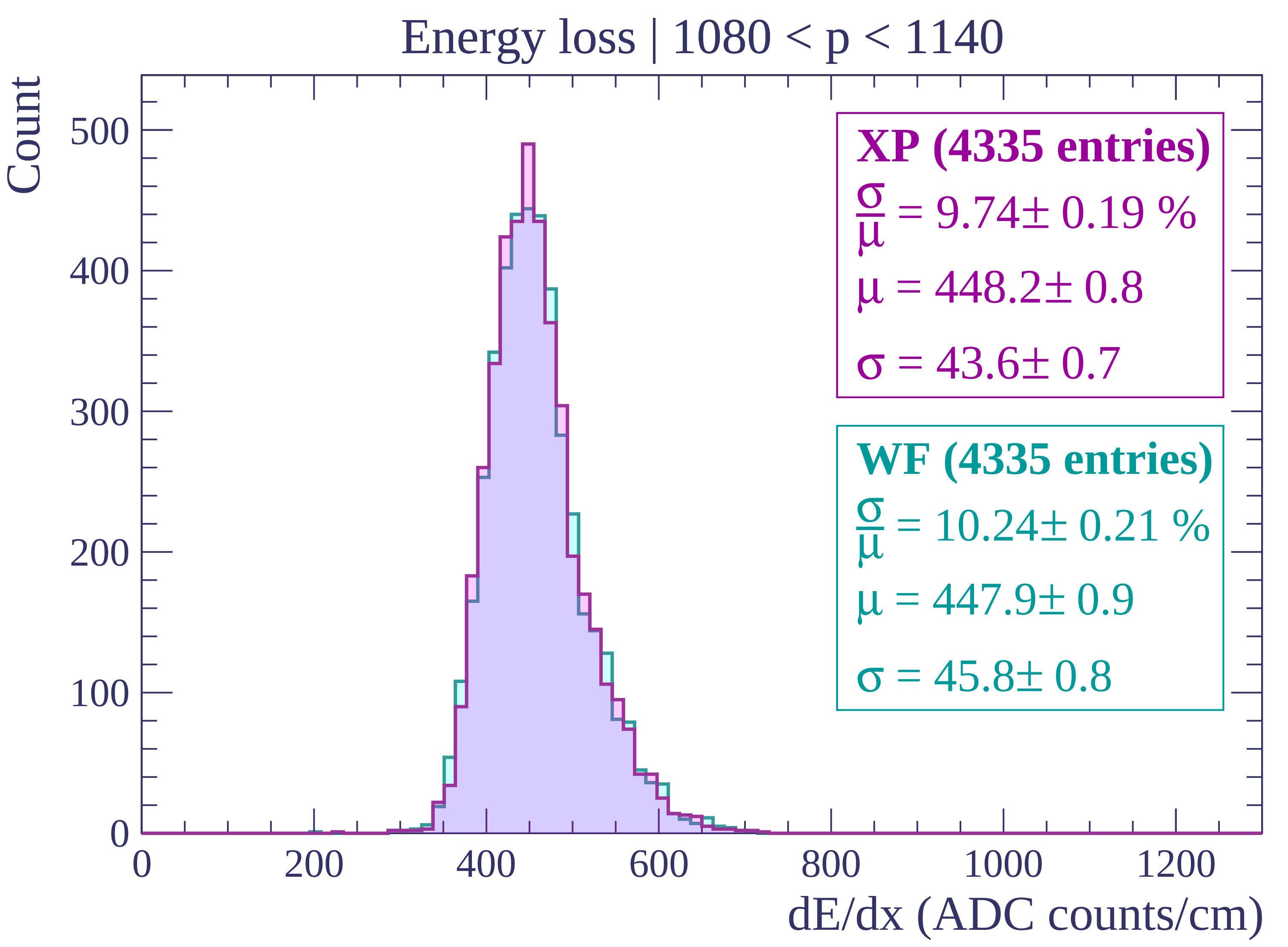
Energy loss | 900 < p < 960
Count
XP (4744 entries)
σ/μ = 9.74± 0.19 %
μ = 435.1± 0.8
σ = 42.4± 0.7
WF (4744 entries)
σ/μ = 9.96± 0.18 %
μ = 435.4± 0.8
σ = 43.4± 0.7
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

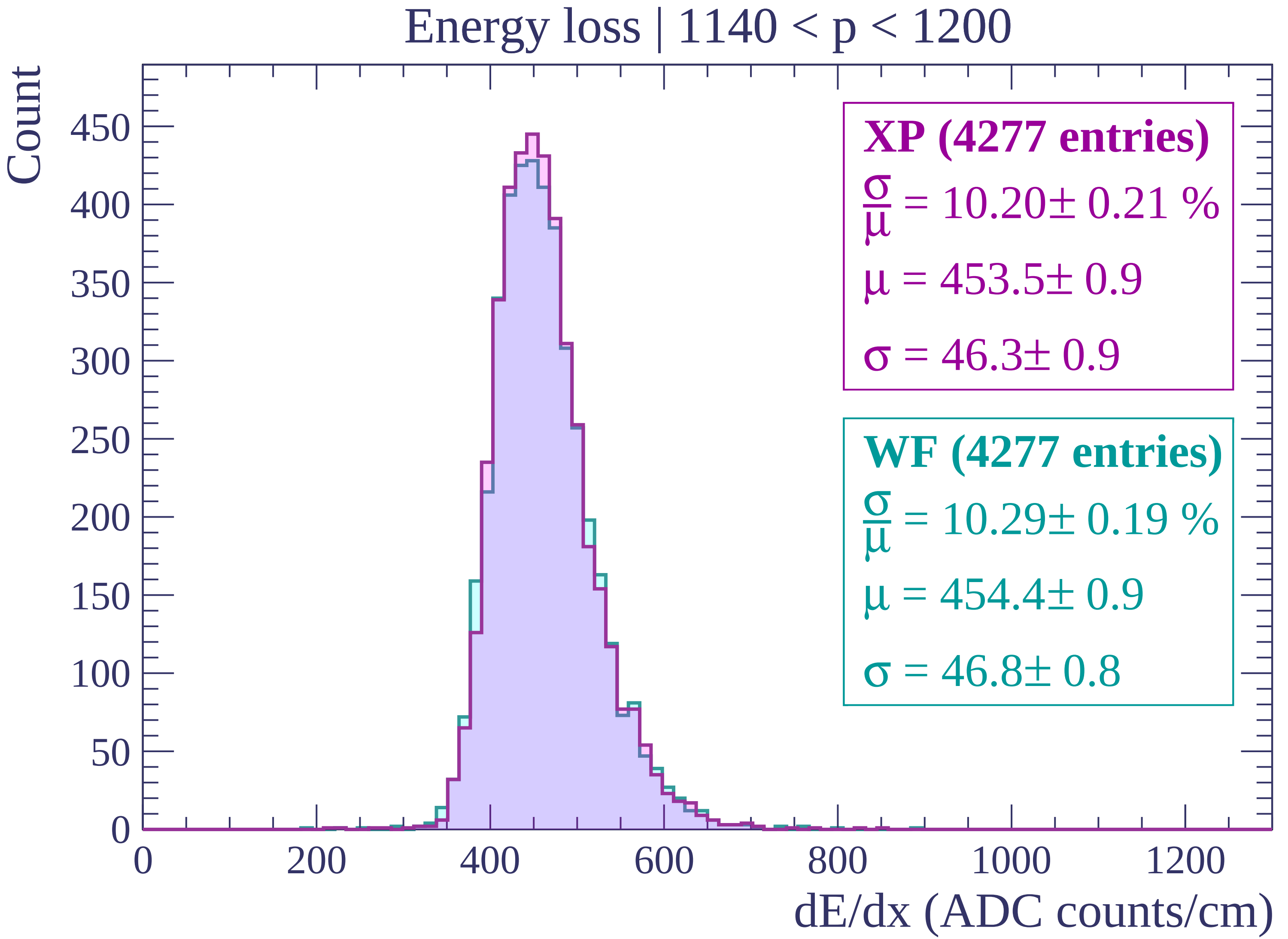


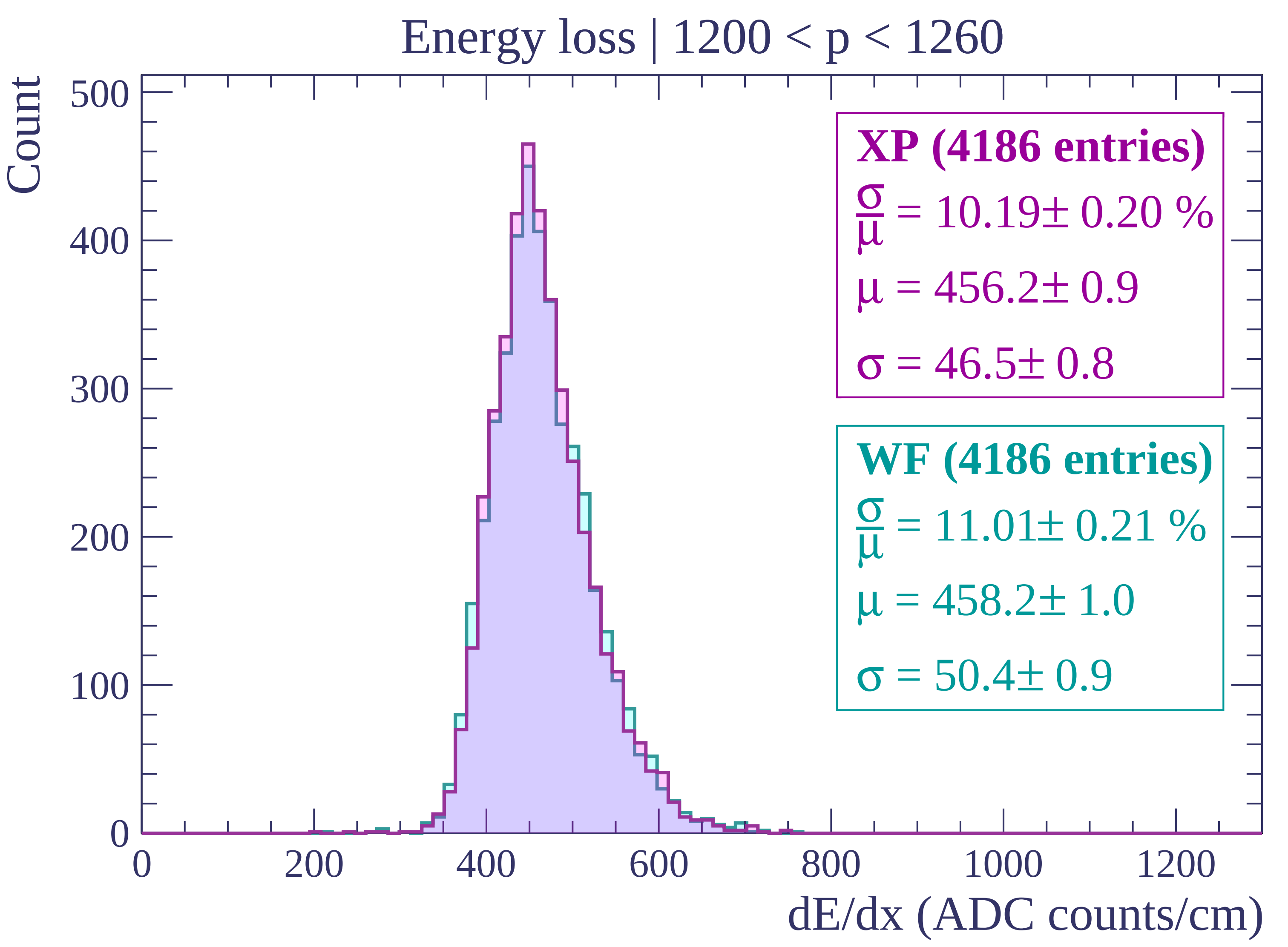
Energy loss | 960 < p < 1020
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (4537 entries)
σ/μ = 9.73± 0.19 %
μ = 440.4± 0.8
σ = 42.8± 0.8
WF (4537 entries)
σ/μ = 10.24± 0.20 %
μ = 441.5± 0.8
σ = 45.2± 0.8



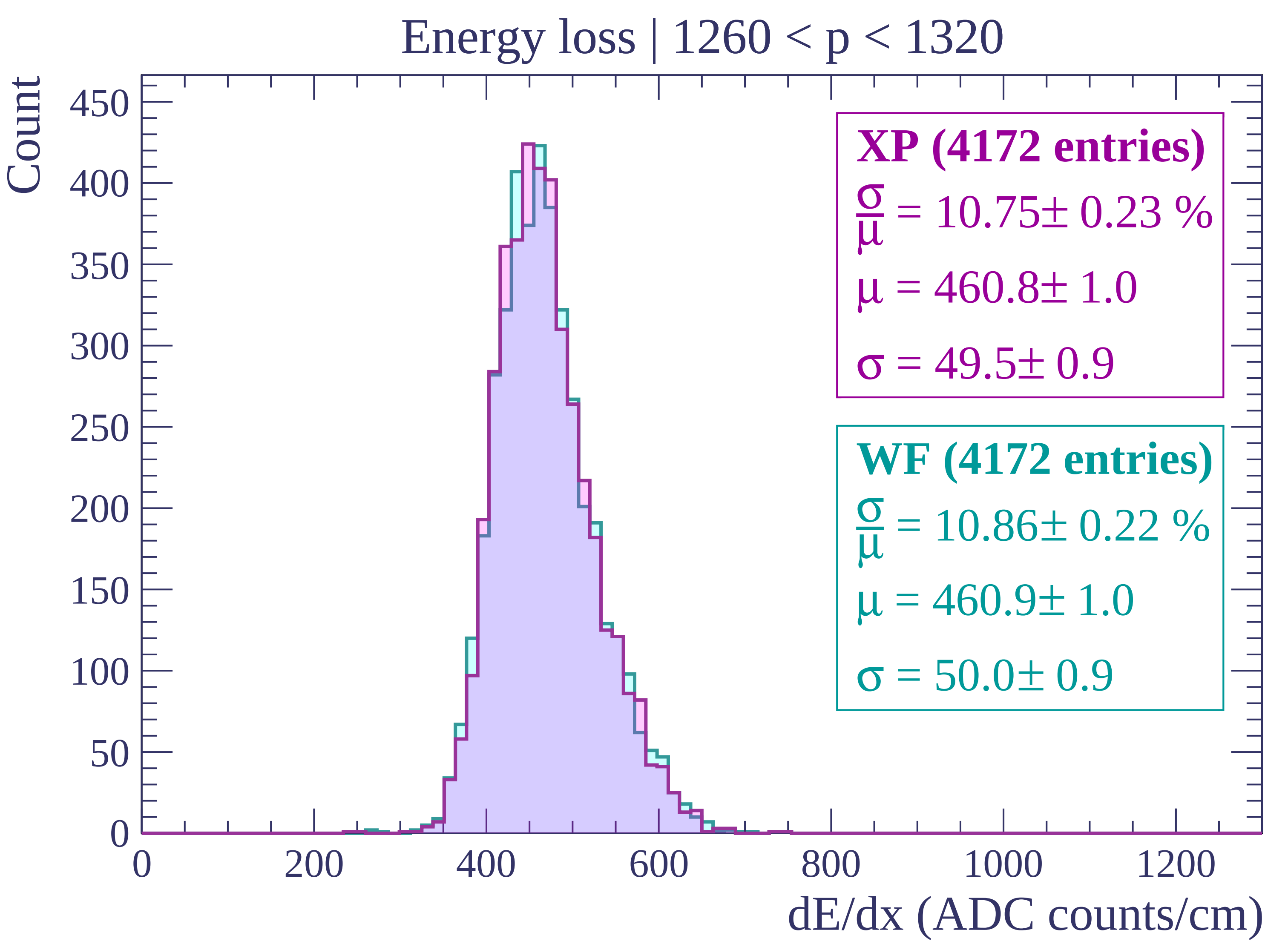
Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4533 entries)
σ/μ = 9.39± 0.18 %
μ = 442.7± 0.8
σ = 41.6± 0.7
WF (4533 entries)
σ/μ = 10.00± 0.19 %
μ = 443.8± 0.8
σ = 44.4± 0.8


Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4335 entries)
σ/μ = 9.74± 0.19 %
μ = 448.2± 0.8
σ = 43.6± 0.7
WF (4335 entries)
σ/μ = 10.24± 0.21 %
μ = 447.9± 0.9
σ = 45.8± 0.8

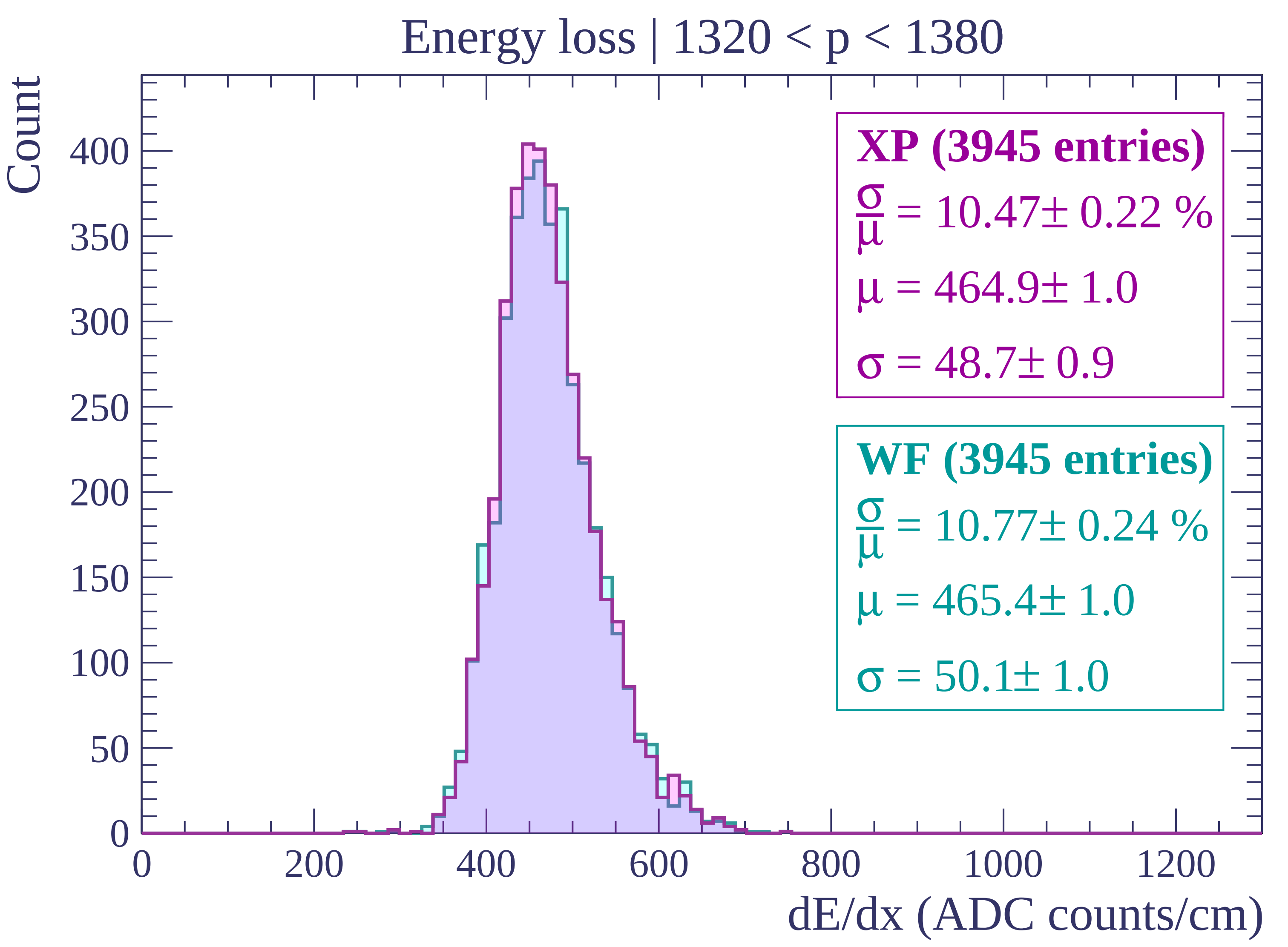

Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
XP (4277 entries)
$\frac{\sigma}{\mu}$ = 10.20± 0.21 %
μ = 453.5± 0.9
σ = 46.3± 0.9
WF (4277 entries)
$\frac{\sigma}{\mu}$ = 10.29± 0.19 %
μ = 454.4± 0.9
σ = 46.8± 0.8
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



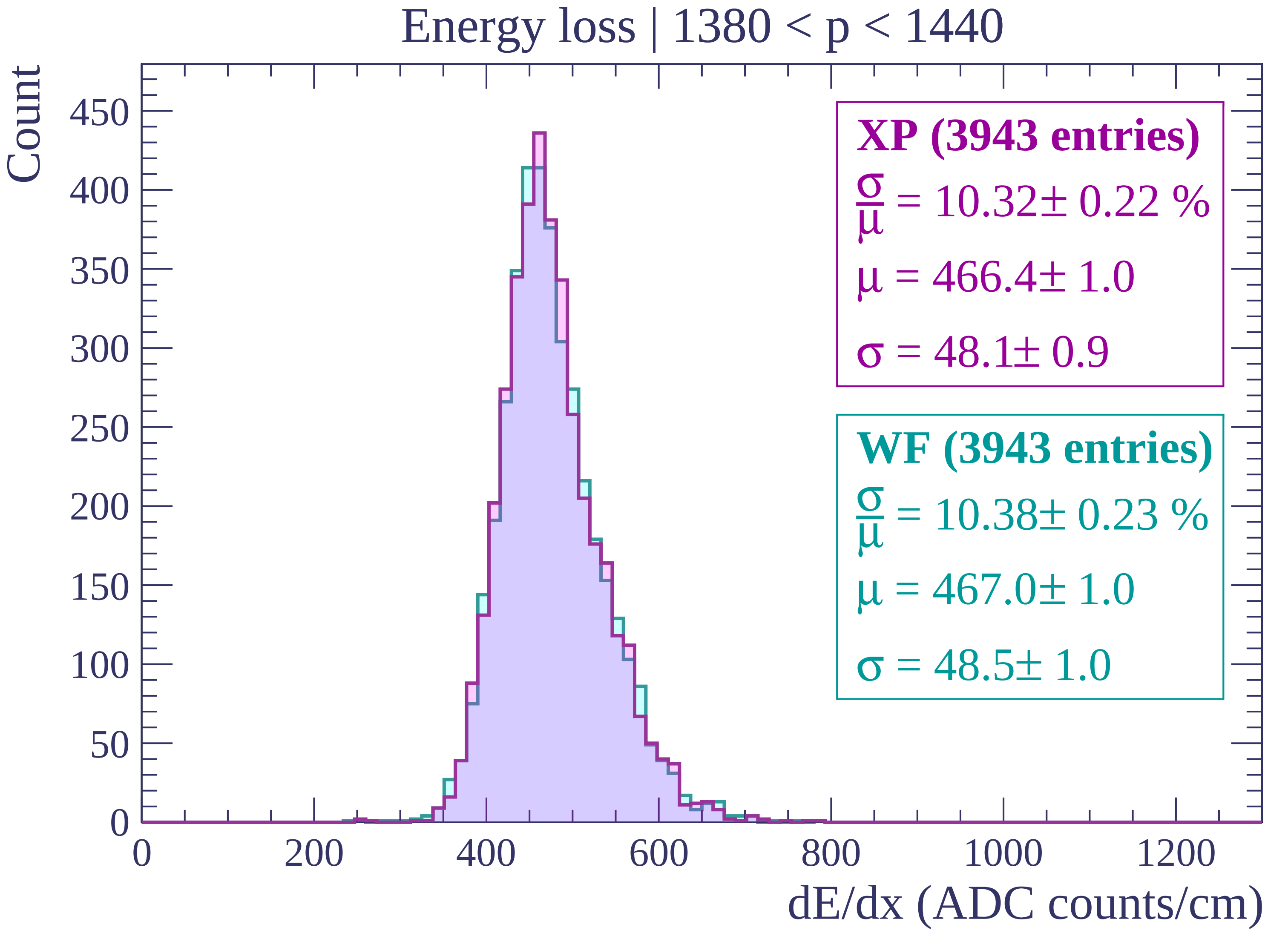
Energy loss | 1200 < p < 1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP (4186 entries)
$\frac{\sigma}{\mu}$ = 10.19± 0.20 %
μ = 456.2± 0.9
σ = 46.5± 0.8
WF (4186 entries)
$\frac{\sigma}{\mu}$ = 11.01± 0.21 %
μ = 458.2± 1.0
σ = 50.4± 0.9

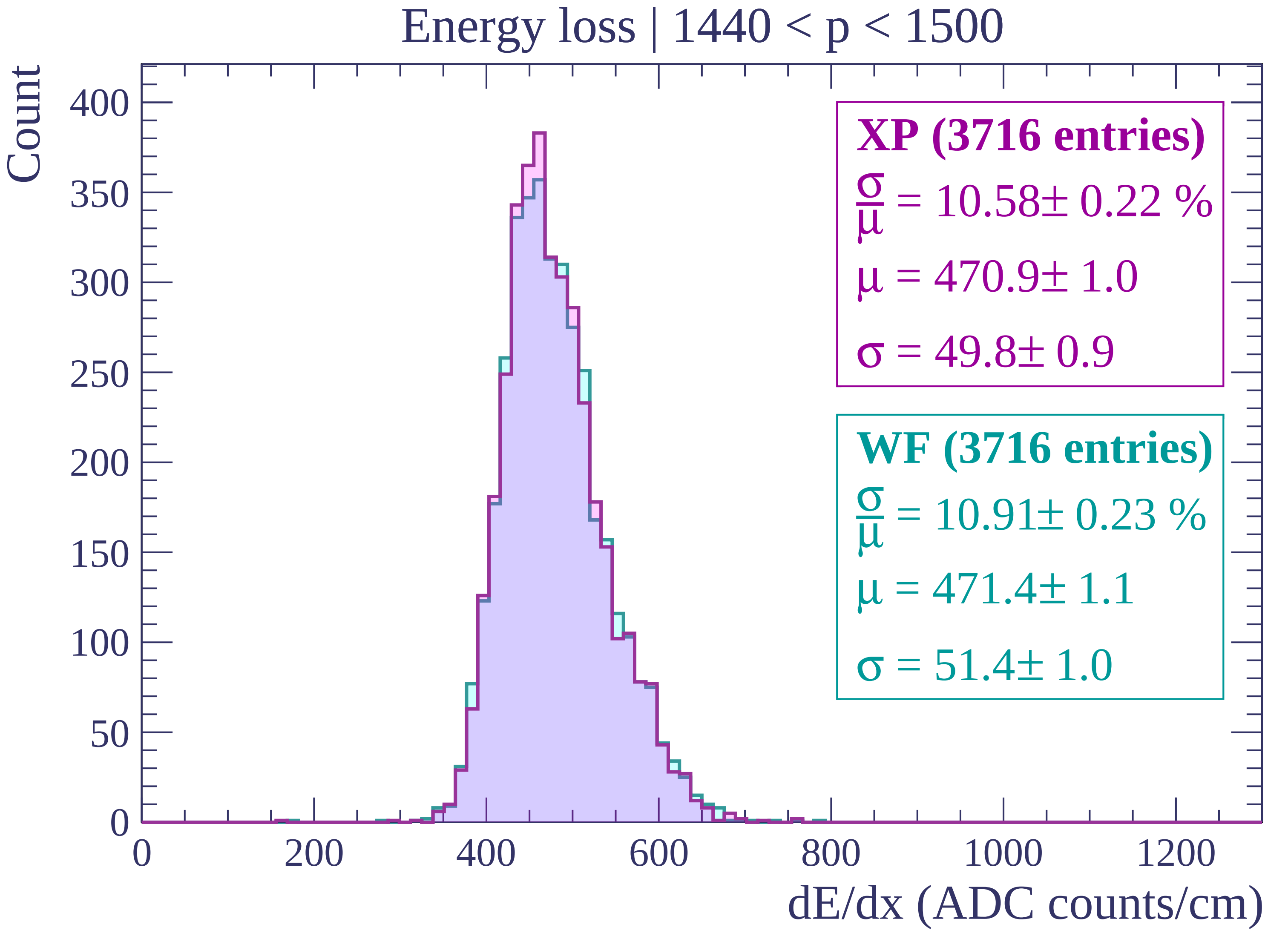


Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP (4172 entries)
$\frac{\sigma}{\mu}$ = 10.75± 0.23 %
μ = 460.8± 1.0
σ = 49.5± 0.9
WF (4172 entries)
$\frac{\sigma}{\mu}$ = 10.86± 0.22 %
μ = 460.9± 1.0
σ = 50.0± 0.9

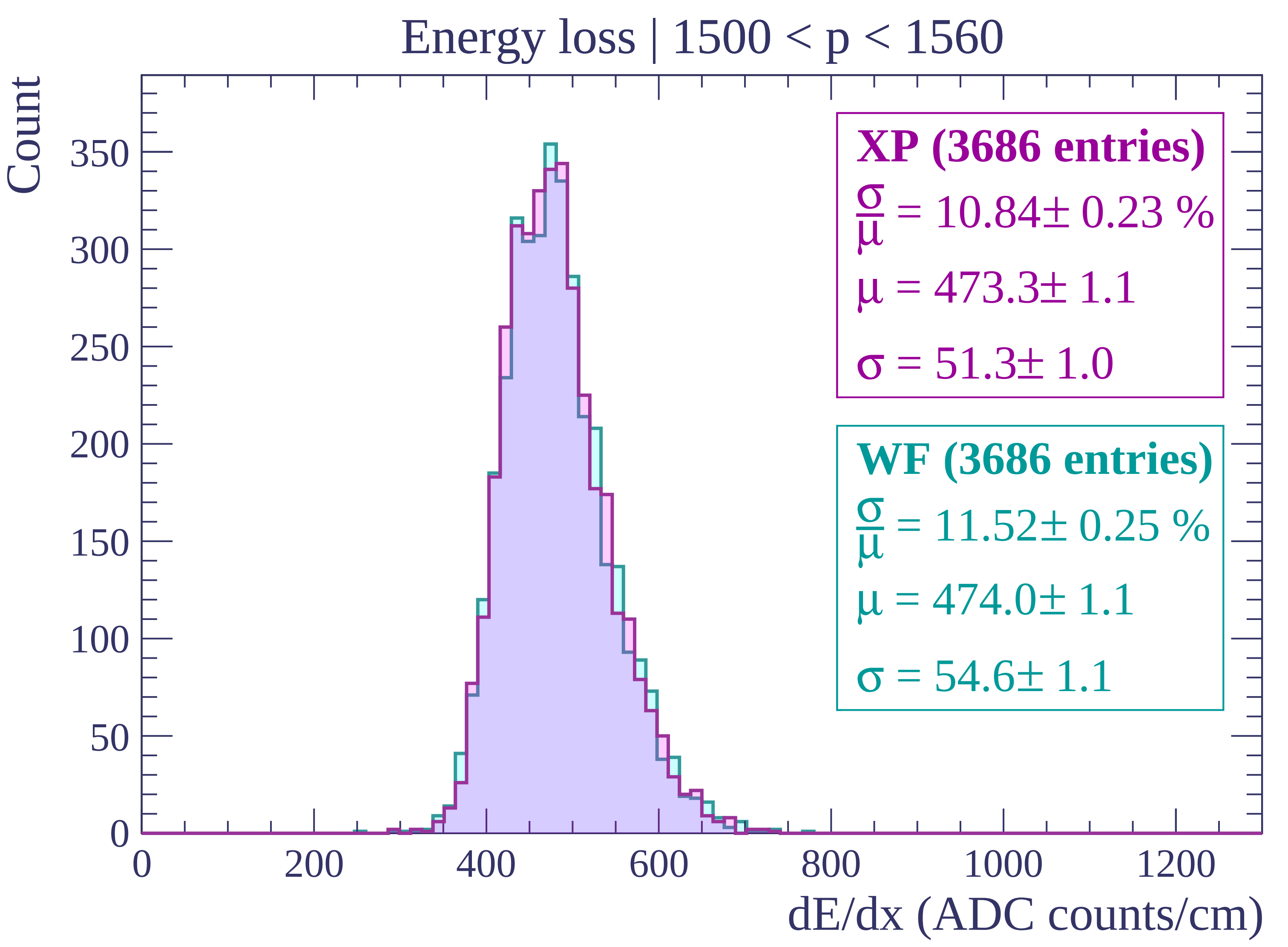


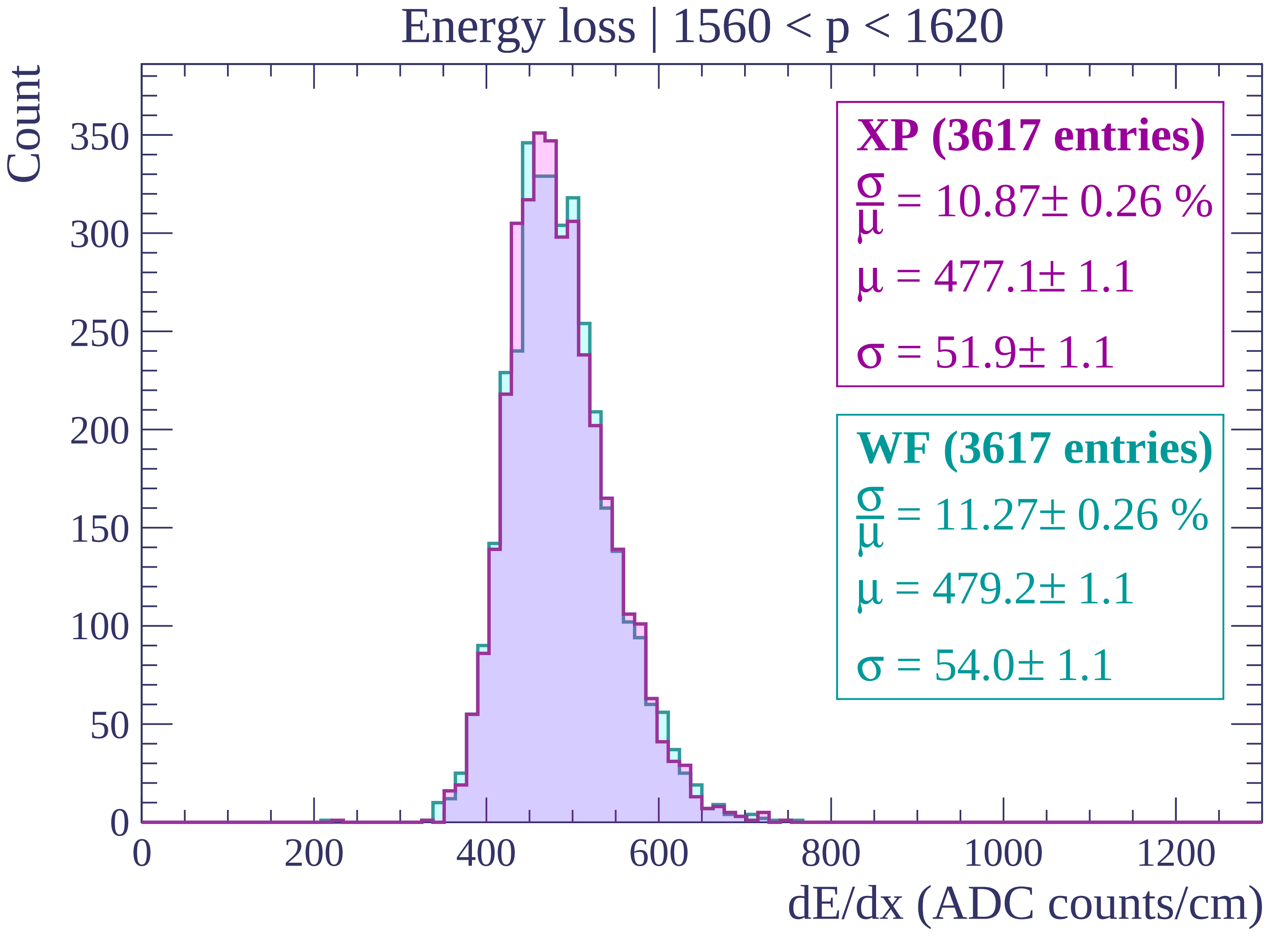
Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP (3945 entries)
σ/μ = 10.47± 0.22 %
μ = 464.9± 1.0
σ = 48.7± 0.9
WF (3945 entries)
σ/μ = 10.77± 0.24 %
μ = 465.4± 1.0
σ = 50.1± 1.0

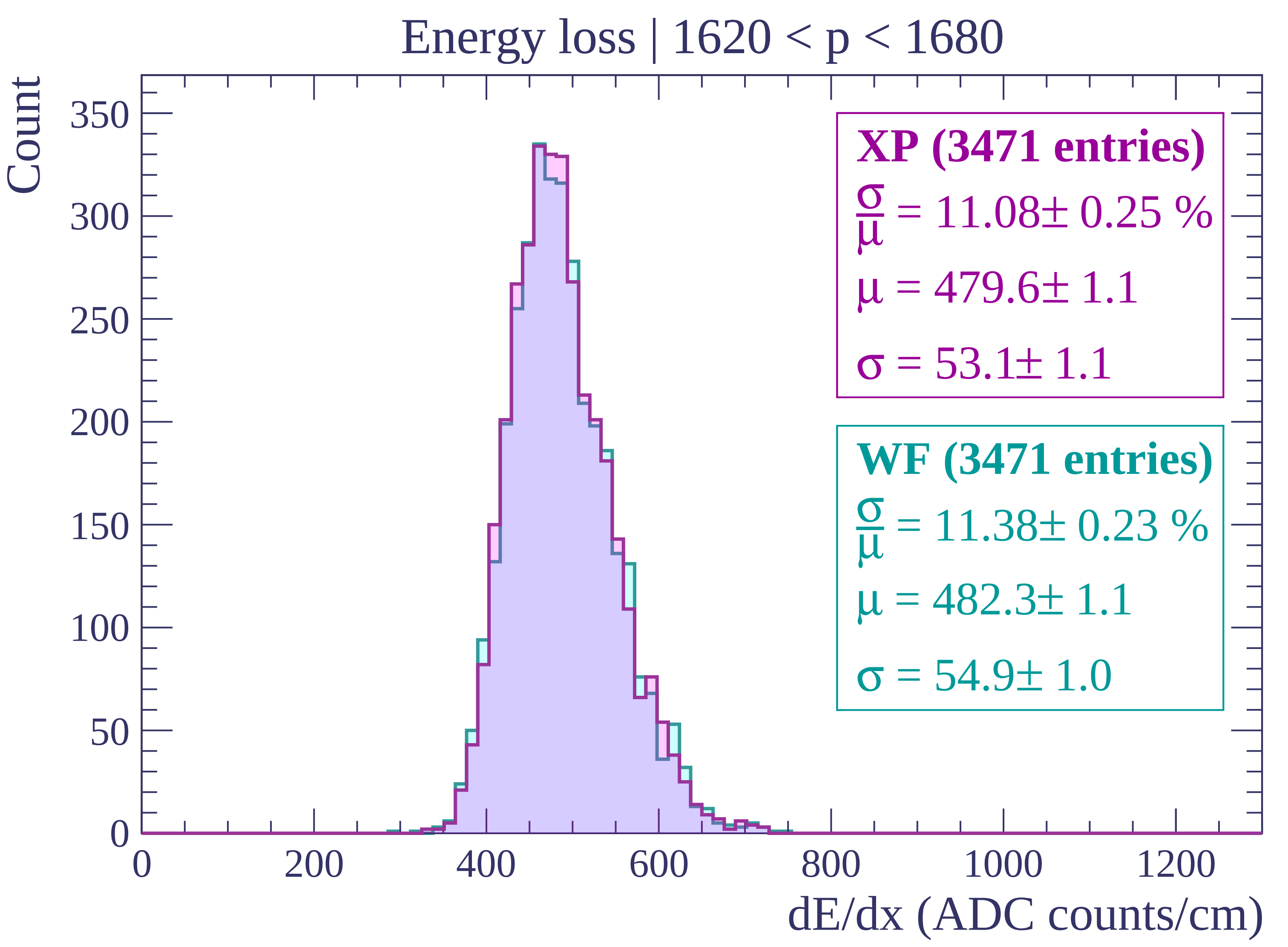

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
450
XP (3943 entries)
$\frac{\sigma}{\mu}$ = 10.32± 0.22 %
μ = 466.4± 1.0
σ = 48.1± 0.9
WF (3943 entries)
$\frac{\sigma}{\mu}$ = 10.38± 0.23 %
μ = 467.0± 1.0
σ = 48.5± 1.0

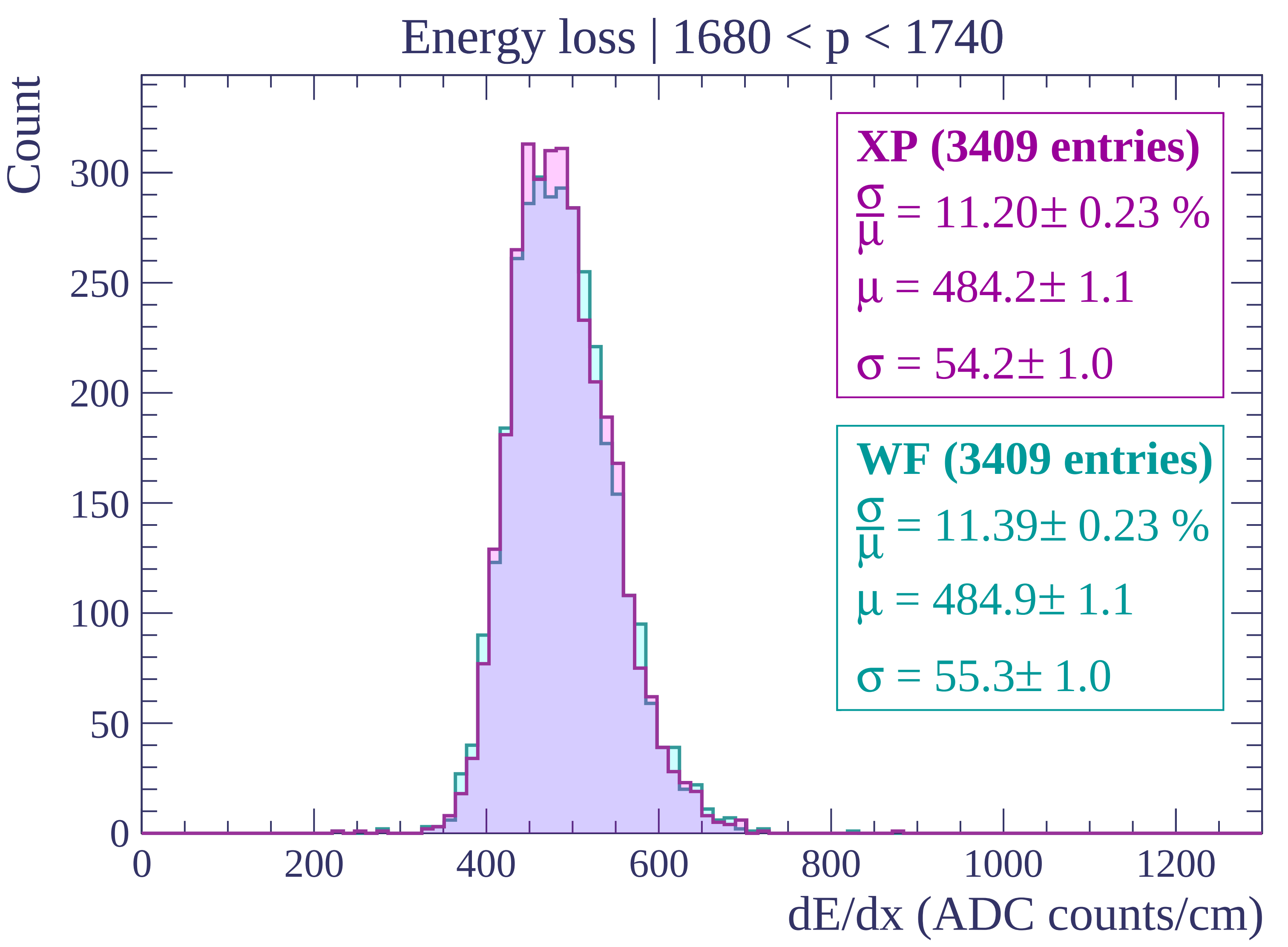


Energy loss | 1440 < p < 1500
Count
dE/dx (ADC counts/cm)
XP (3716 entries)
σ/μ = 10.58± 0.22 %
μ = 470.9± 1.0
σ = 49.8± 0.9
WF (3716 entries)
σ/μ = 10.91± 0.23 %
μ = 471.4± 1.1
σ = 51.4± 1.0

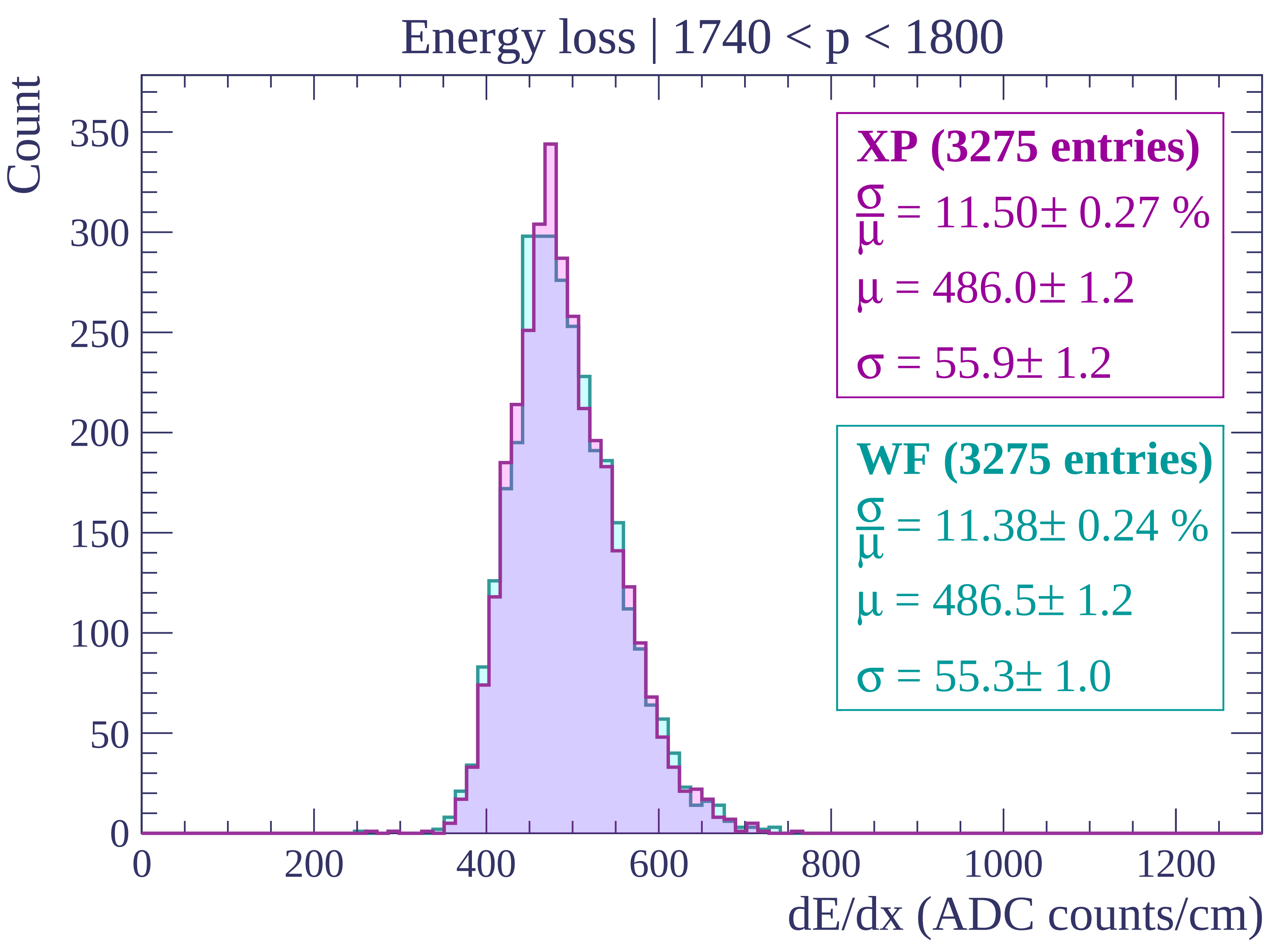

Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
XP (3686 entries)
σ/μ = 10.84± 0.23 %
μ = 473.3± 1.1
σ = 51.3± 1.0
WF (3686 entries)
σ/μ = 11.52± 0.25 %
μ = 474.0± 1.1
σ = 54.6± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | 1560 < p < 1620
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP (3617 entries)
σ/μ = 10.87± 0.26 %
μ = 477.1± 1.1
σ = 51.9± 1.1
WF (3617 entries)
σ/μ = 11.27± 0.26 %
μ = 479.2± 1.1
σ = 54.0± 1.1

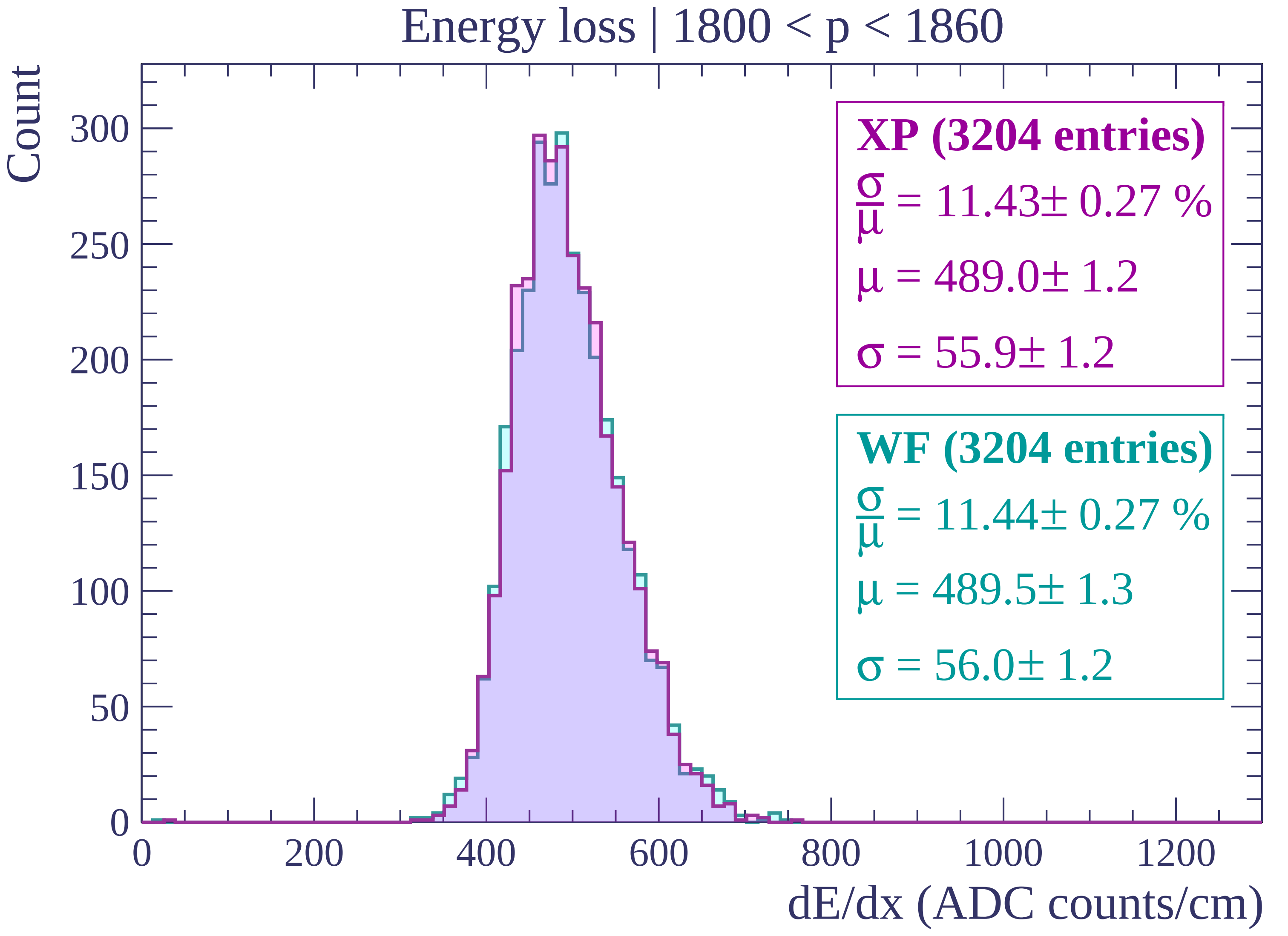

Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP (3471 entries)
σ/μ = 11.08± 0.25 %
μ = 479.6± 1.1
σ = 53.1± 1.1
WF (3471 entries)
σ/μ = 11.38± 0.23 %
μ = 482.3± 1.1
σ = 54.9± 1.0

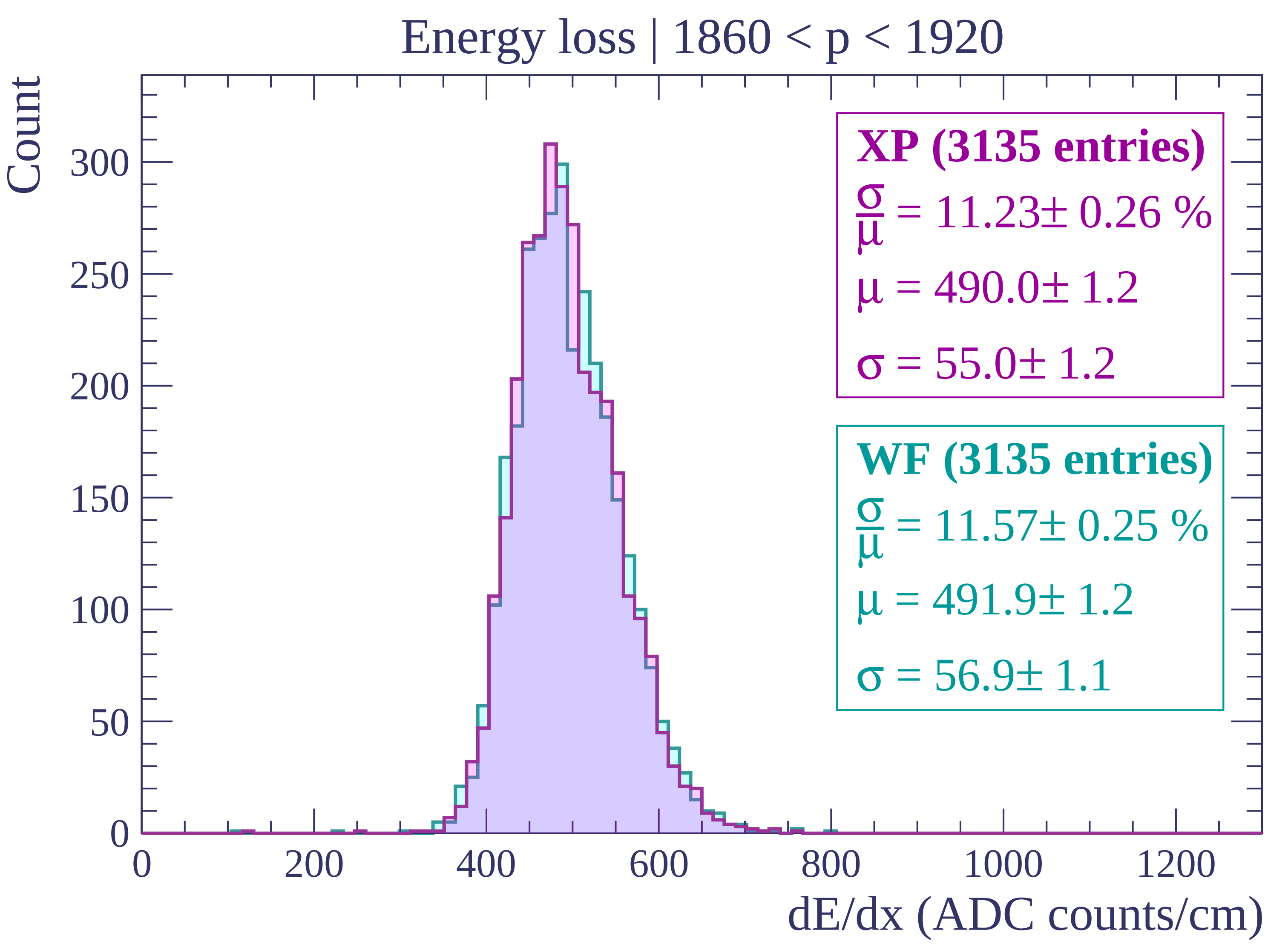


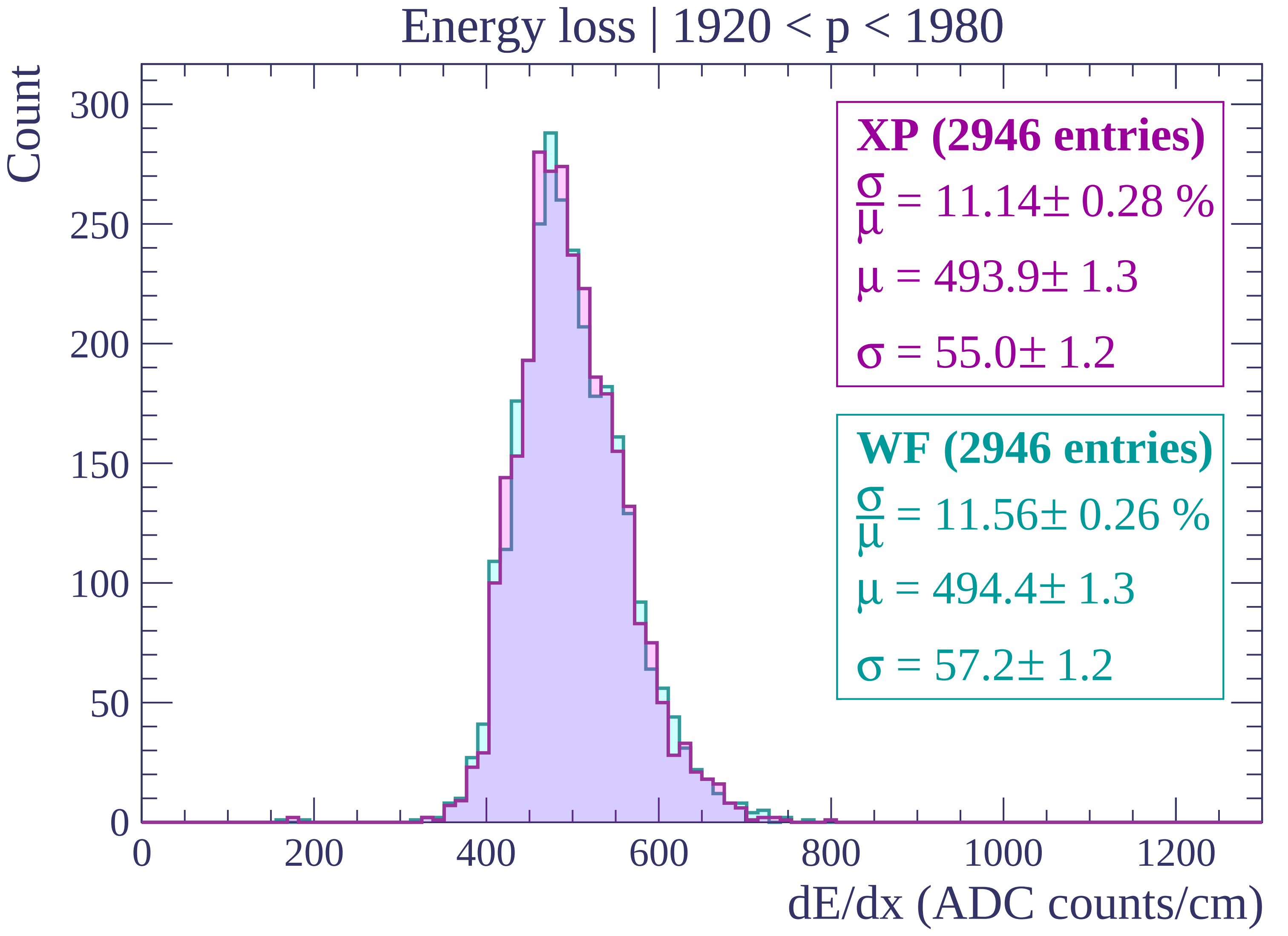
Energy loss | 1680 < p < 1740
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (3409 entries)
$\frac{\sigma}{\mu}$ = 11.20± 0.23 %
μ = 484.2± 1.1
σ = 54.2± 1.0
WF (3409 entries)
$\frac{\sigma}{\mu}$ = 11.39± 0.23 %
μ = 484.9± 1.1
σ = 55.3± 1.0



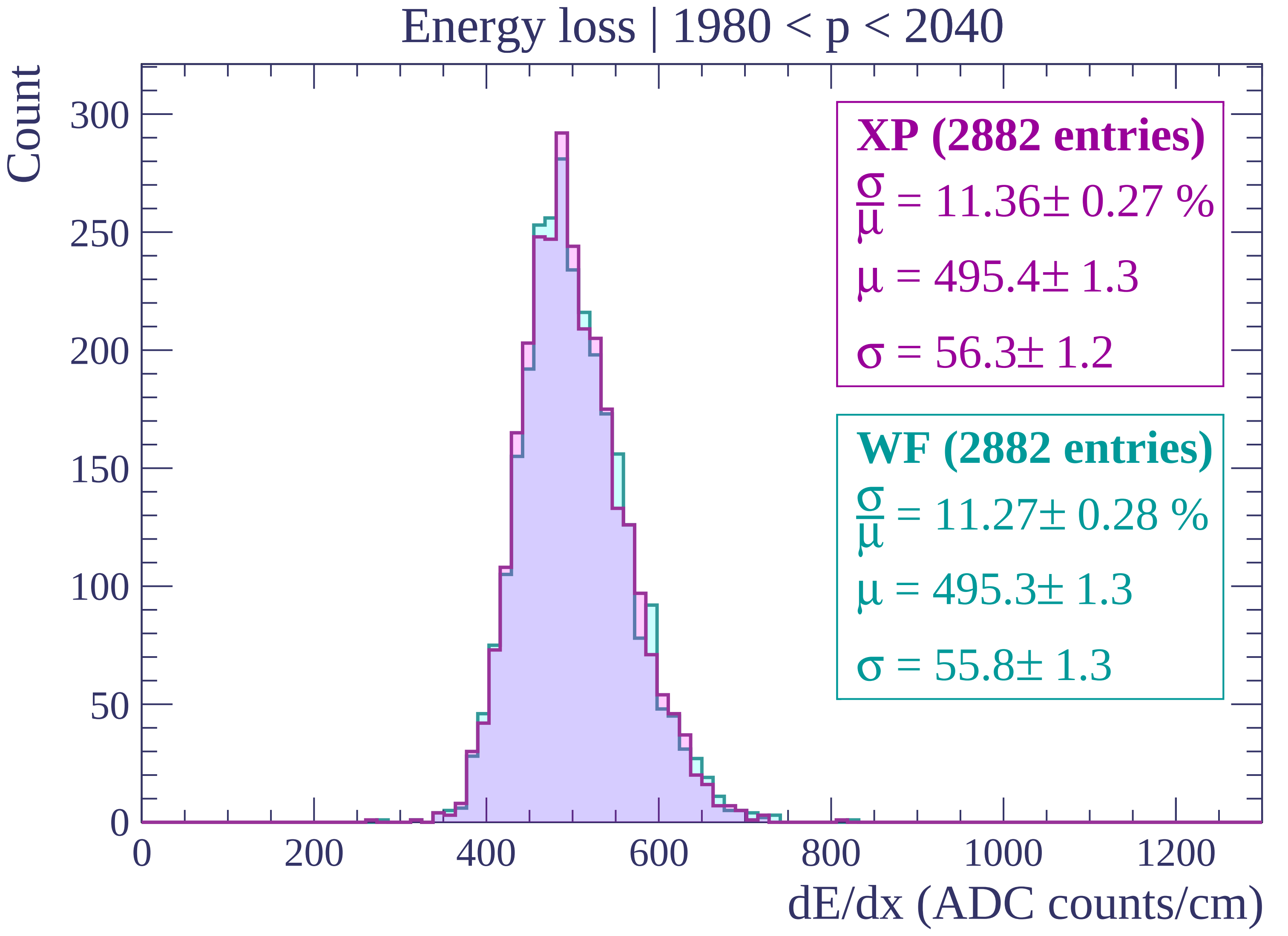
Energy loss | 1740 < p < 1800
Count
dE/dx (ADC counts/cm)
XP (3275 entries)
$\frac{\sigma}{\mu}$ = 11.50± 0.27 %
μ = 486.0± 1.2
σ = 55.9± 1.2
WF (3275 entries)
$\frac{\sigma}{\mu}$ = 11.38± 0.24 %
μ = 486.5± 1.2
σ = 55.3± 1.0
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200


Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3204 entries)
σ/μ = 11.43± 0.27 %
μ = 489.0± 1.2
σ = 55.9± 1.2
WF (3204 entries)
σ/μ = 11.44± 0.27 %
μ = 489.5± 1.3
σ = 56.0± 1.2

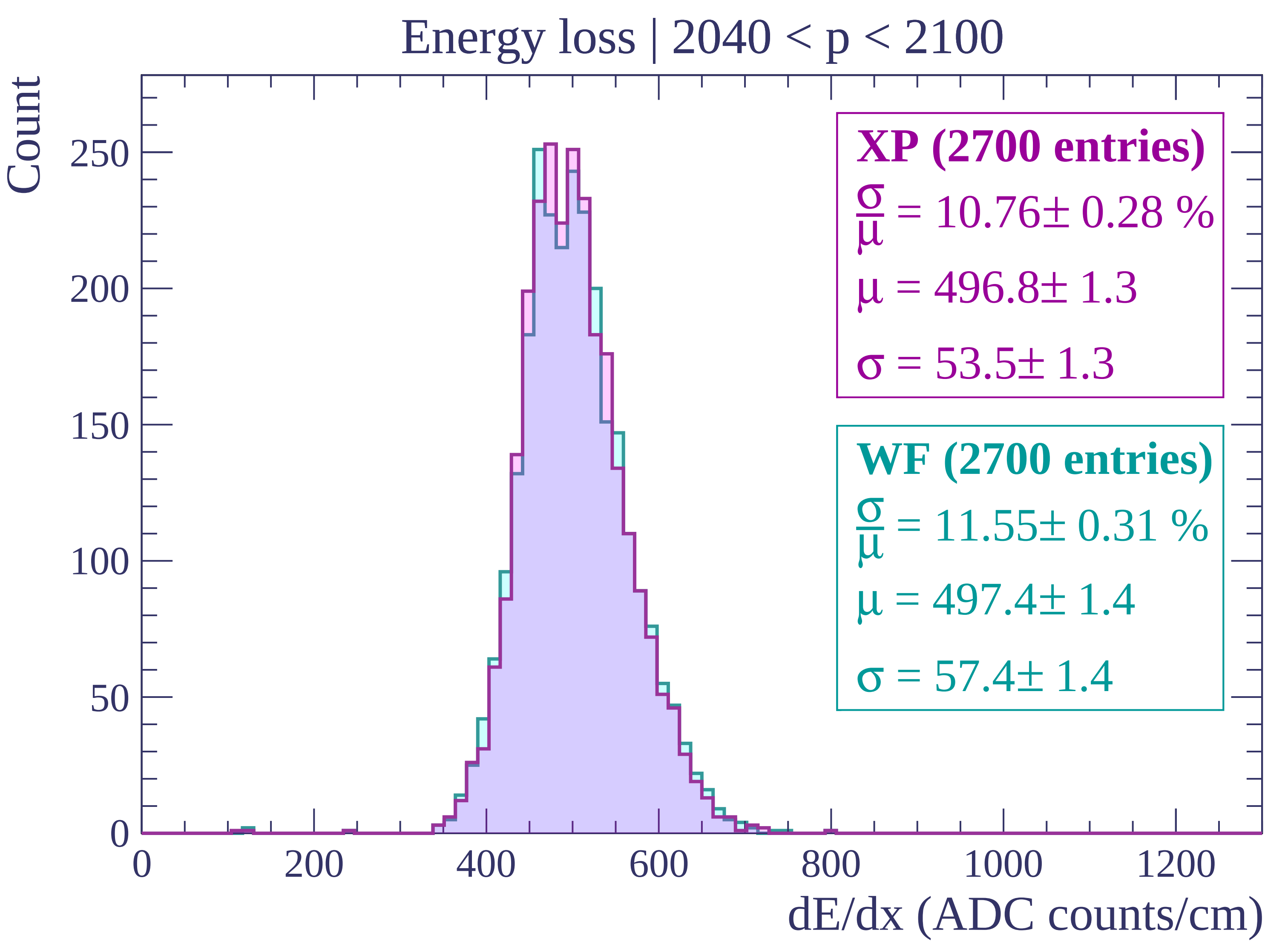

Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (3135 entries)
$\frac{\sigma}{\mu}$ = 11.23± 0.26 %
μ = 490.0± 1.2
σ = 55.0± 1.2
WF (3135 entries)
$\frac{\sigma}{\mu}$ = 11.57± 0.25 %
μ = 491.9± 1.2
σ = 56.9± 1.1

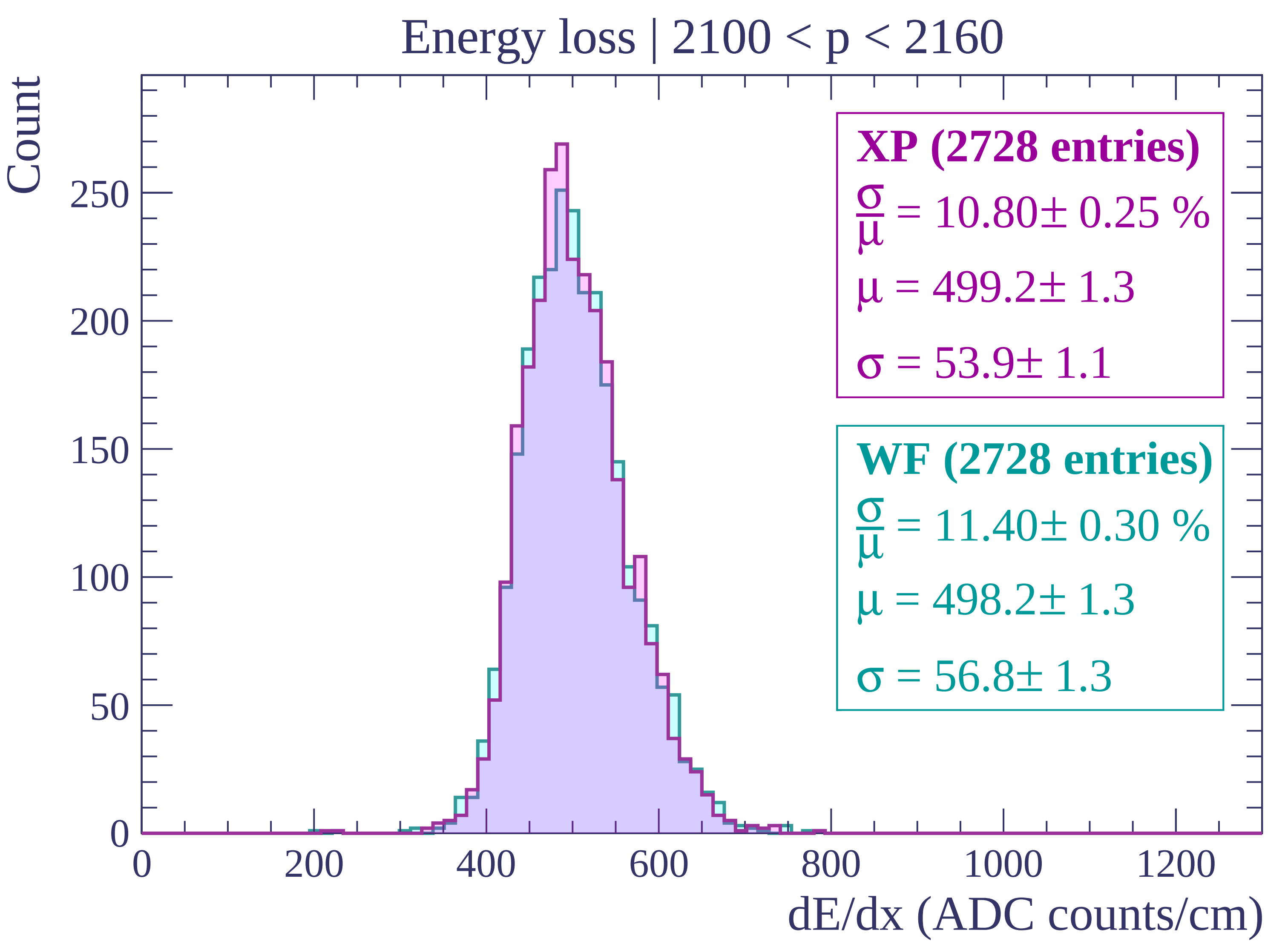


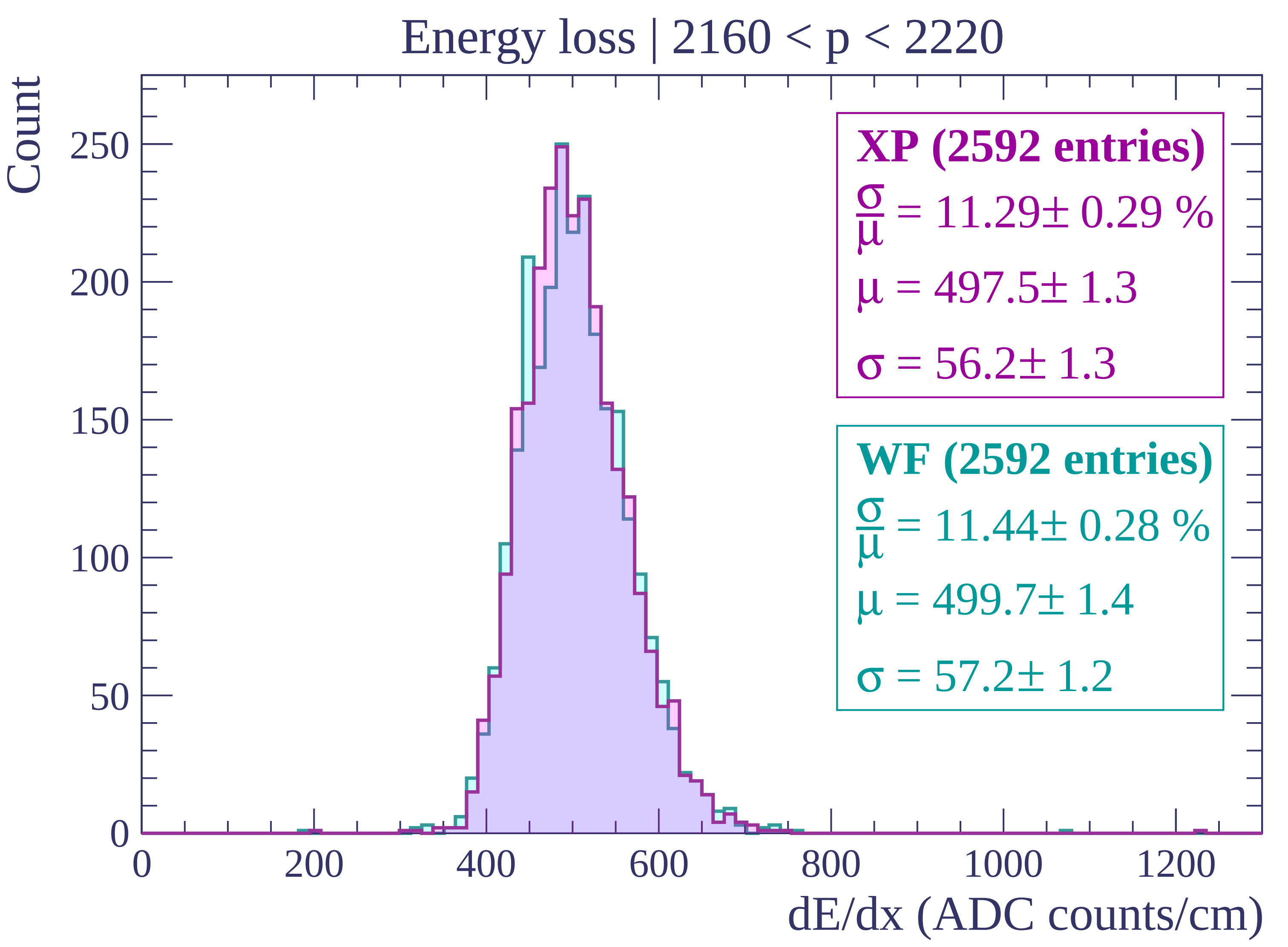
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2946 entries)
σ/μ = 11.14± 0.28 %
μ = 493.9± 1.3
σ = 55.0± 1.2
WF (2946 entries)
σ/μ = 11.56± 0.26 %
μ = 494.4± 1.3
σ = 57.2± 1.2

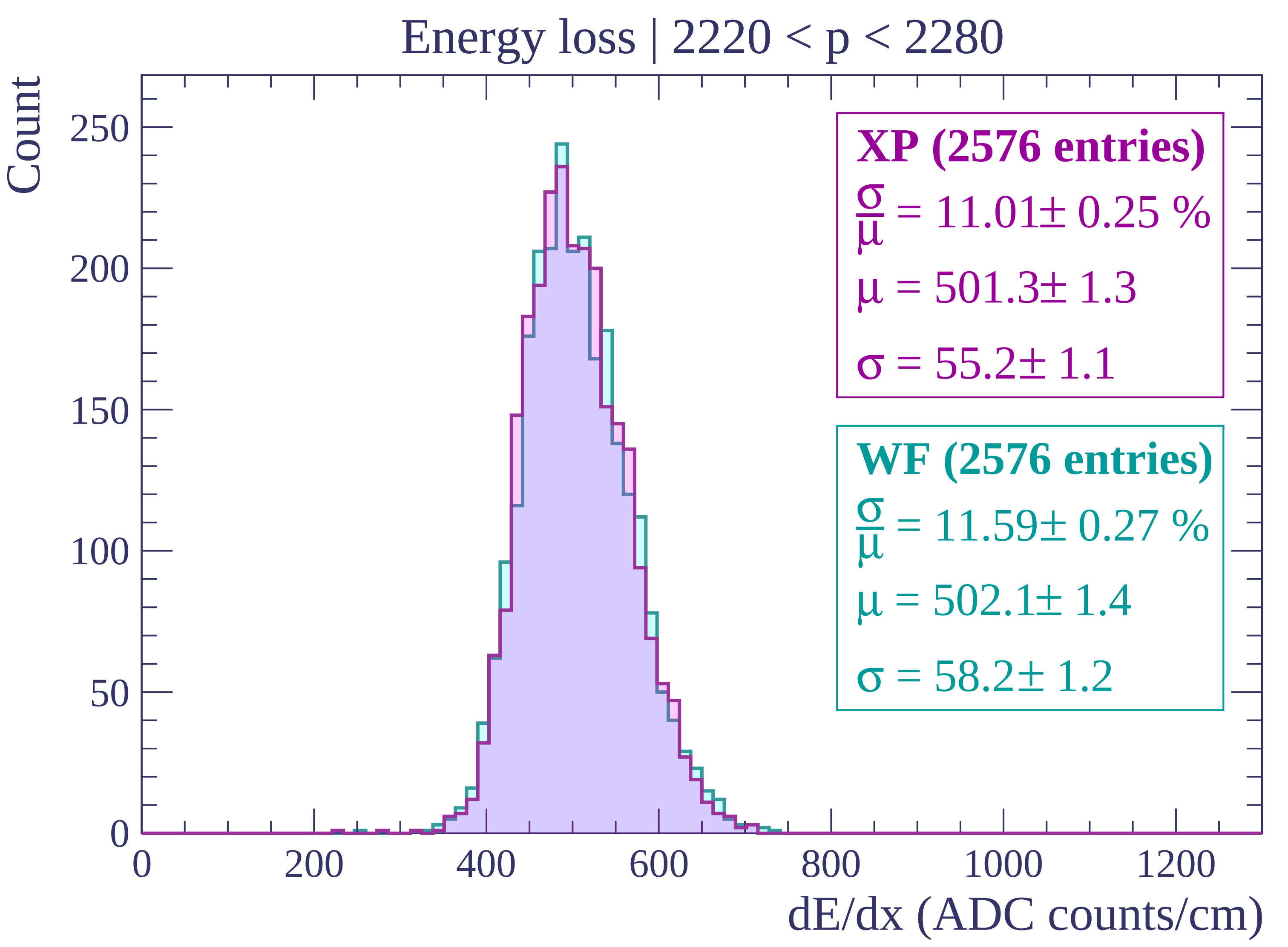


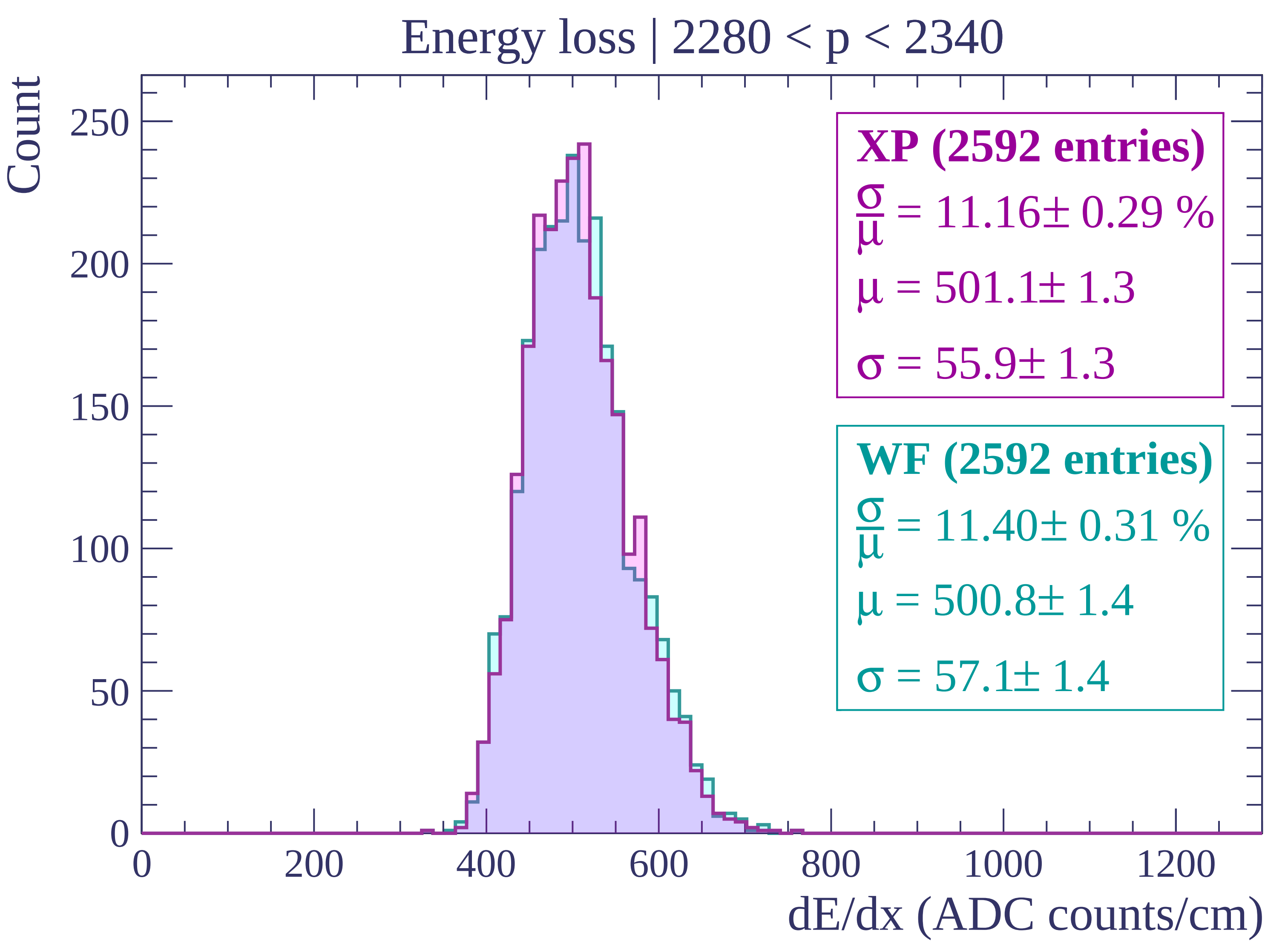
Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP (2882 entries)
σ/μ = 11.36± 0.27 %
μ = 495.4± 1.3
σ = 56.3± 1.2
WF (2882 entries)
σ/μ = 11.27± 0.28 %
μ = 495.3± 1.3
σ = 55.8± 1.3

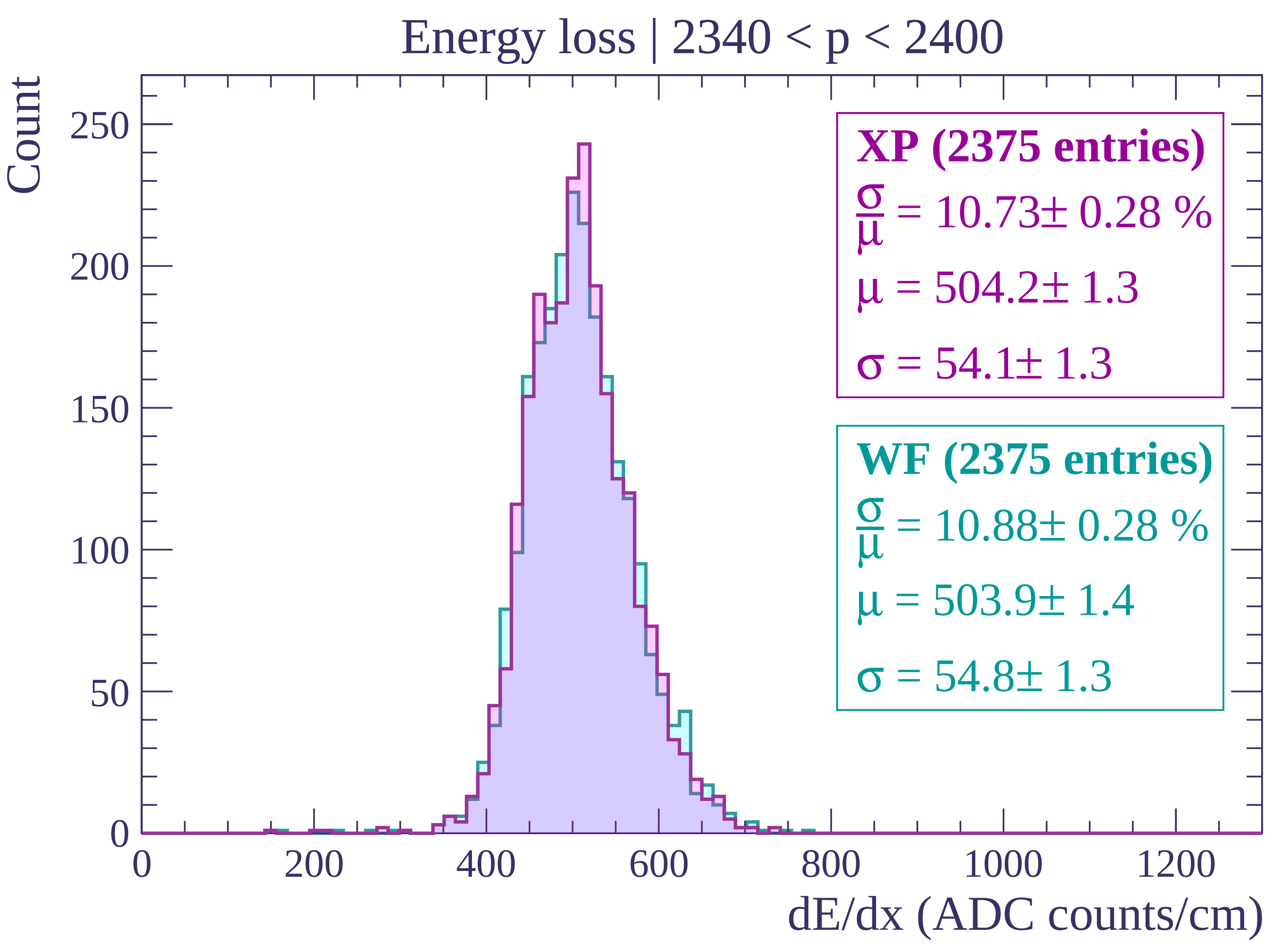

Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2700 entries)
$\frac{\sigma}{\mu}$ = 10.76± 0.28 %
μ = 496.8± 1.3
σ = 53.5± 1.3
WF (2700 entries)
$\frac{\sigma}{\mu}$ = 11.55± 0.31 %
μ = 497.4± 1.4
σ = 57.4± 1.4

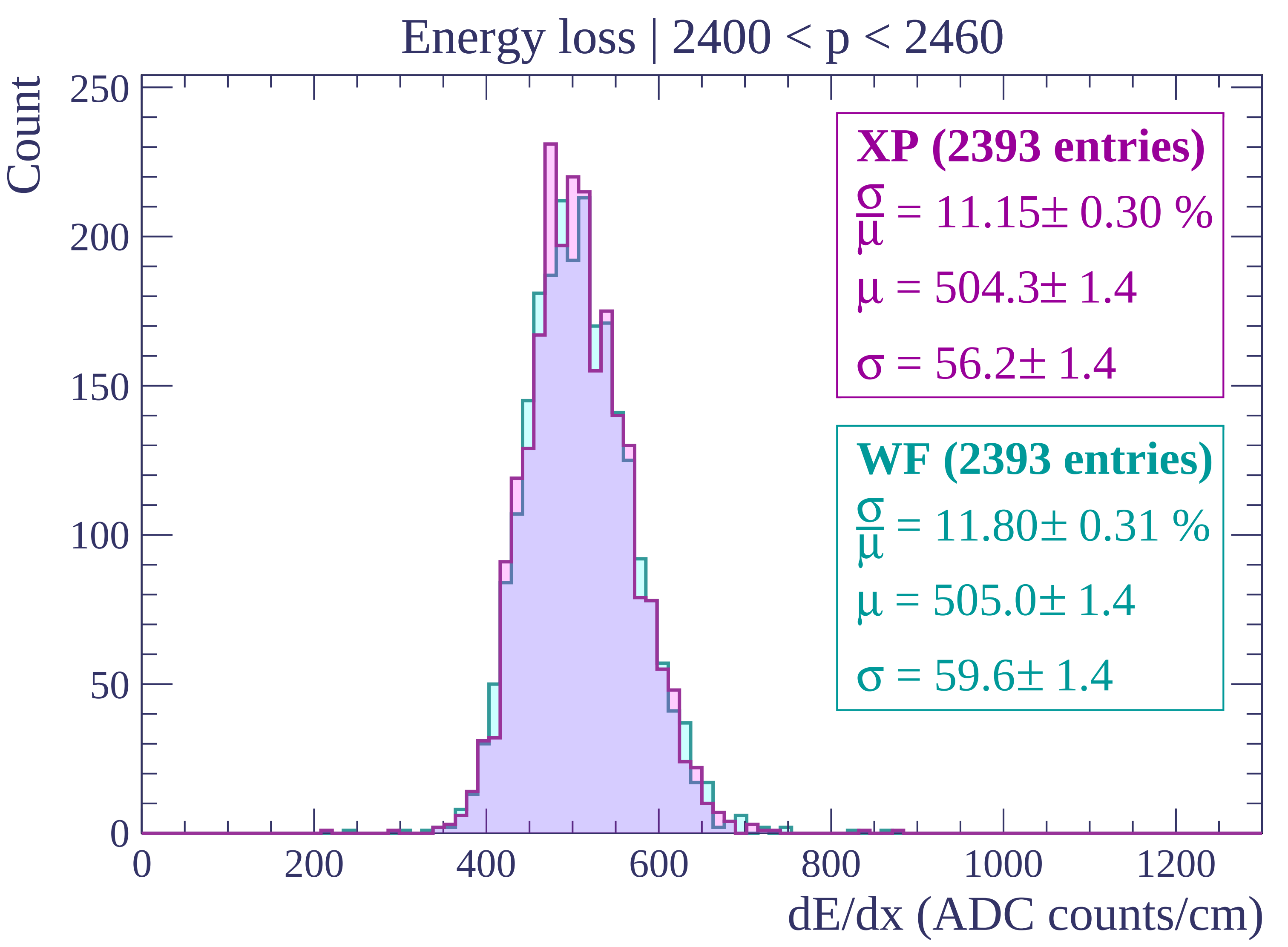

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (2728 entries)
$\frac{\sigma}{\mu}$ = 10.80± 0.25 %
μ = 499.2± 1.3
σ = 53.9± 1.1
WF (2728 entries)
$\frac{\sigma}{\mu}$ = 11.40± 0.30 %
μ = 498.2± 1.3
σ = 56.8± 1.3

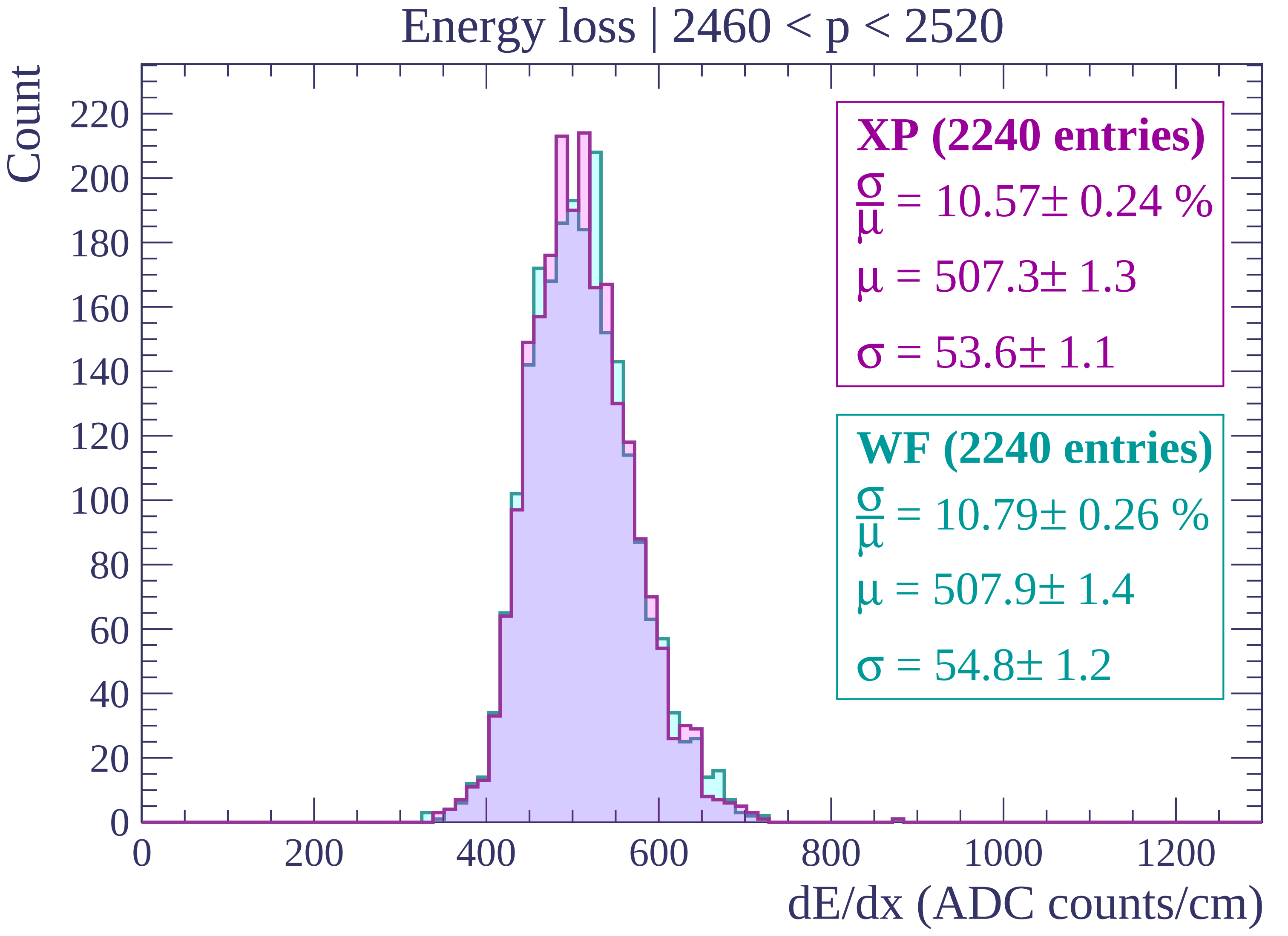

Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2592 entries)
$\frac{\sigma}{\mu}$ = 11.29± 0.29 %
μ = 497.5± 1.3
σ = 56.2± 1.3
WF (2592 entries)
$\frac{\sigma}{\mu}$ = 11.44± 0.28 %
μ = 499.7± 1.4
σ = 57.2± 1.2


Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP (2576 entries)
σ/μ = 11.01± 0.25 %
μ = 501.3± 1.3
σ = 55.2± 1.1
WF (2576 entries)
σ/μ = 11.59± 0.27 %
μ = 502.1± 1.4
σ = 58.2± 1.2

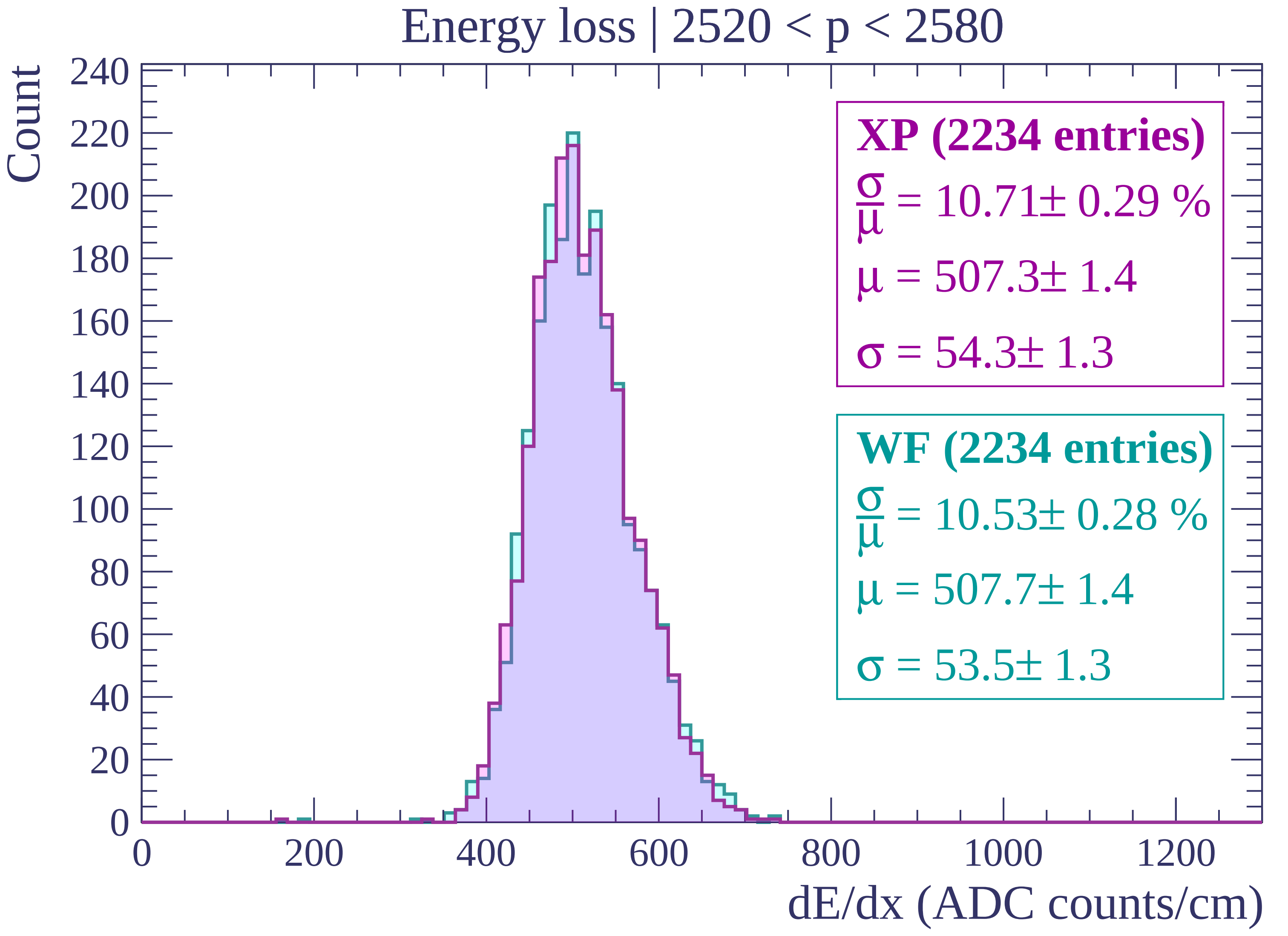

Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2592 entries)
σ/μ = 11.16± 0.29 %
μ = 501.1± 1.3
σ = 55.9± 1.3
WF (2592 entries)
σ/μ = 11.40± 0.31 %
μ = 500.8± 1.4
σ = 57.1± 1.4


Energy loss | 2340 < p < 2400
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2375 entries)
σ/μ = 10.73± 0.28 %
μ = 504.2± 1.3
σ = 54.1± 1.3
WF (2375 entries)
σ/μ = 10.88± 0.28 %
μ = 503.9± 1.4
σ = 54.8± 1.3

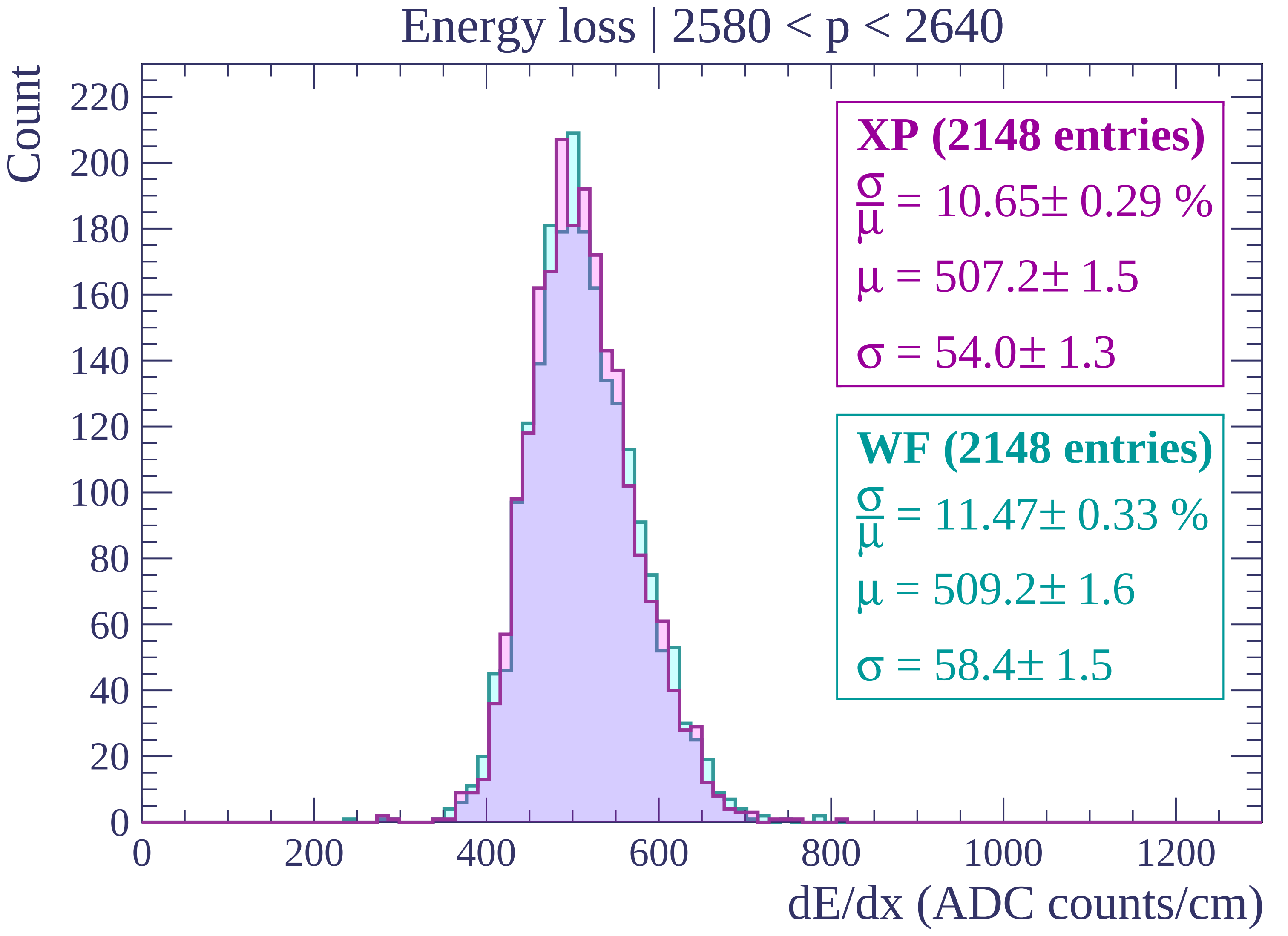

Energy loss | 2400 < p < 2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP (2393 entries)
σ/μ = 11.15± 0.30 %
μ = 504.3± 1.4
σ = 56.2± 1.4
WF (2393 entries)
σ/μ = 11.80± 0.31 %
μ = 505.0± 1.4
σ = 59.6± 1.4



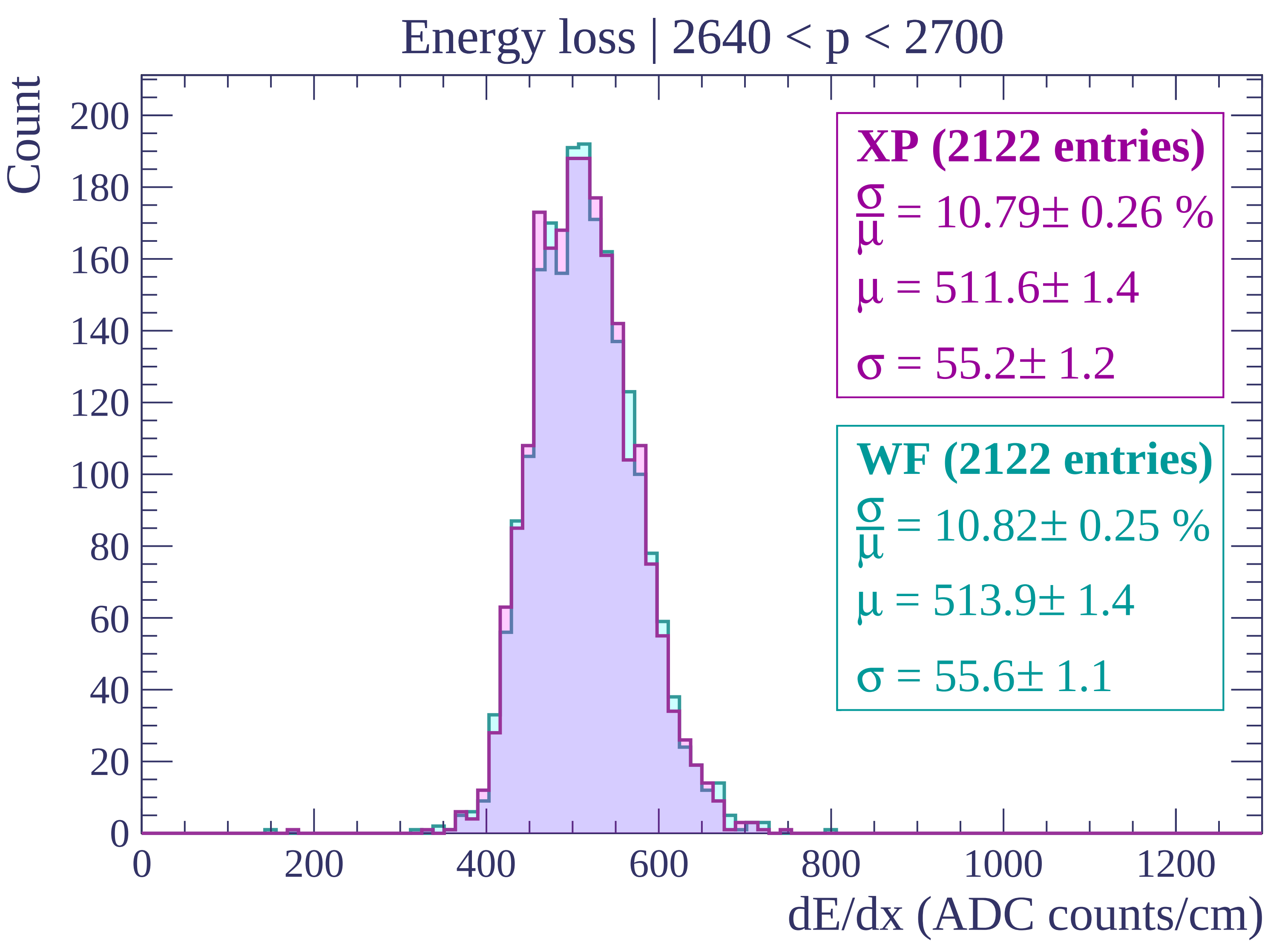
Energy loss | 2460 < p < 2520
Count
dE/dx (ADC counts/cm)
XP (2240 entries)
$\frac{\sigma}{\mu}$ = 10.57± 0.24 %
μ = 507.3± 1.3
σ = 53.6± 1.1
WF (2240 entries)
$\frac{\sigma}{\mu}$ = 10.79± 0.26 %
μ = 507.9± 1.4
σ = 54.8± 1.2
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200



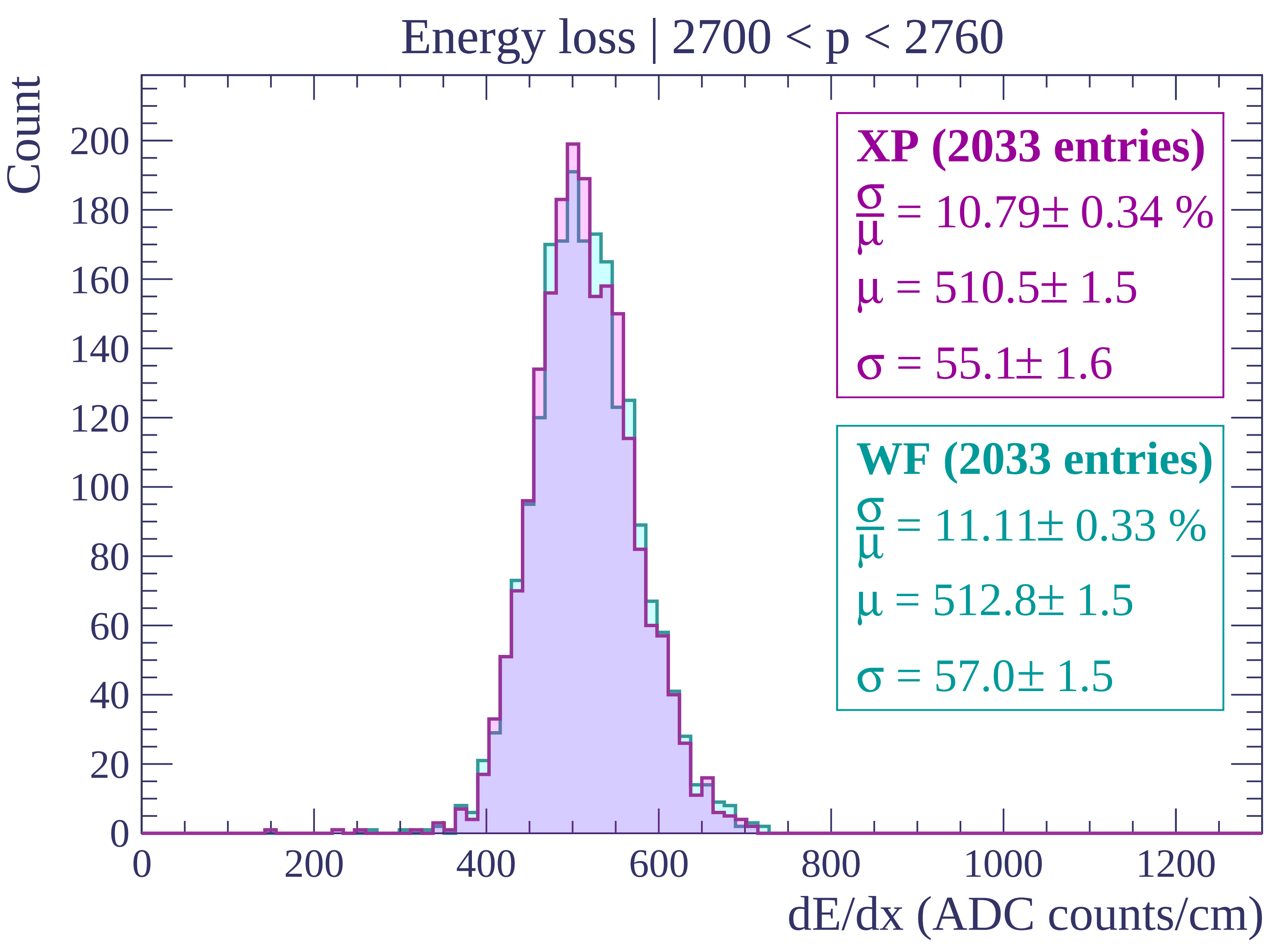
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
240
XP (2234 entries)
σ/μ = 10.71± 0.29 %
μ = 507.3± 1.4
σ = 54.3± 1.3
WF (2234 entries)
σ/μ = 10.53± 0.28 %
μ = 507.7± 1.4
σ = 53.5± 1.3



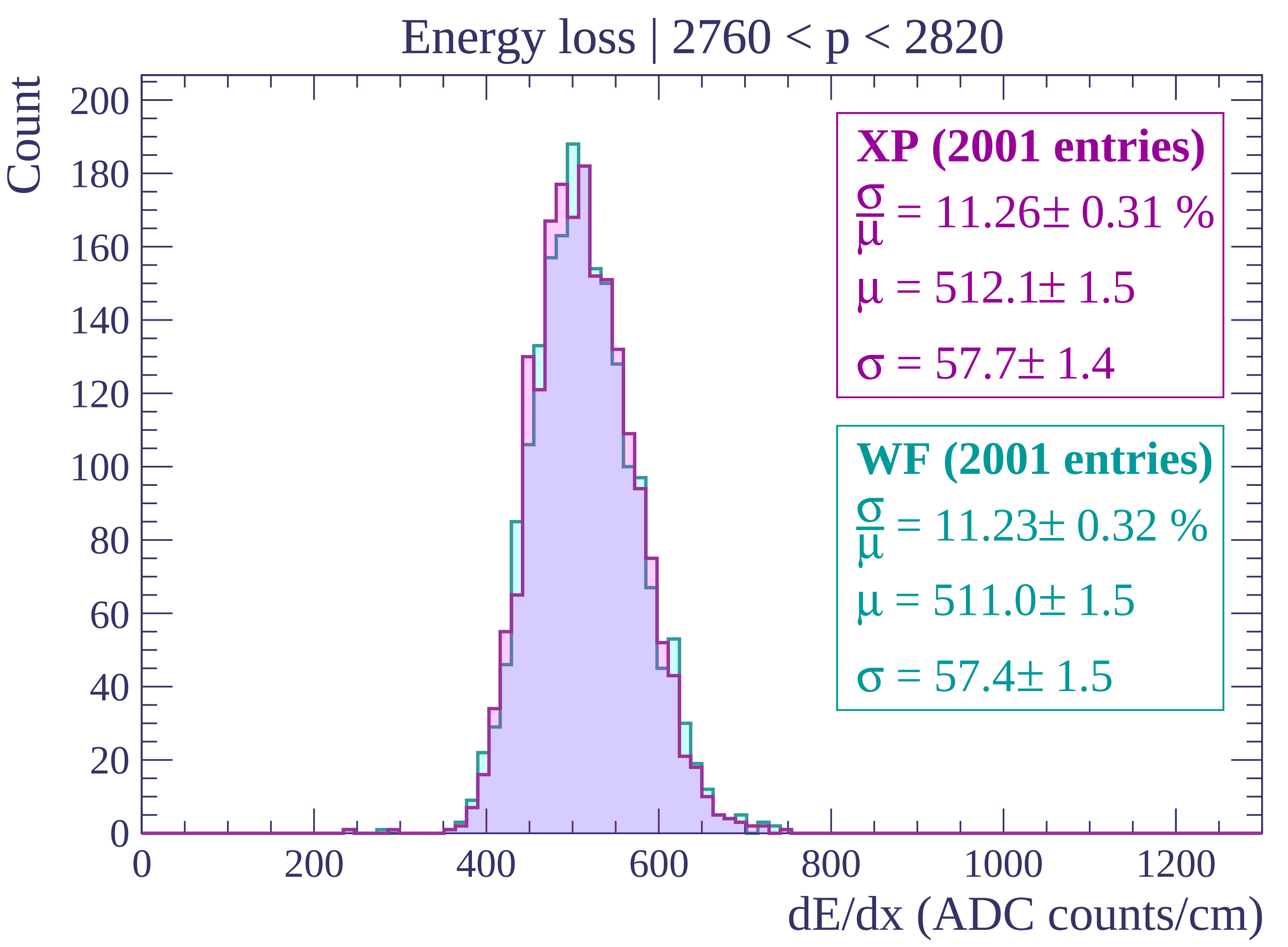
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP (2148 entries)
σ/μ = 10.65± 0.29 %
μ = 507.2± 1.5
σ = 54.0± 1.3
WF (2148 entries)
σ/μ = 11.47± 0.33 %
μ = 509.2± 1.6
σ = 58.4± 1.5

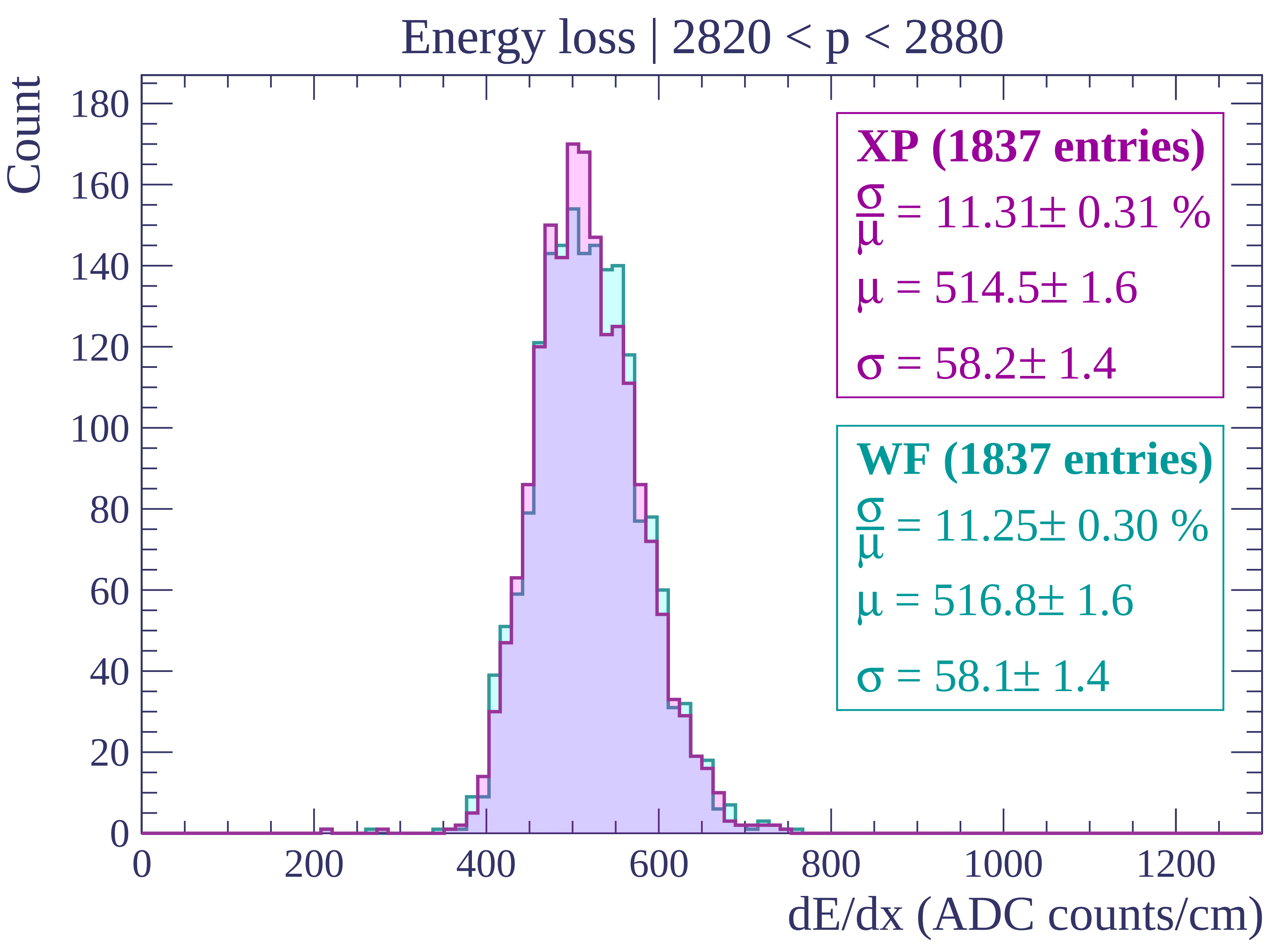


Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP (2122 entries)
σ/μ = 10.79± 0.26 %
μ = 511.6± 1.4
σ = 55.2± 1.2
WF (2122 entries)
σ/μ = 10.82± 0.25 %
μ = 513.9± 1.4
σ = 55.6± 1.1

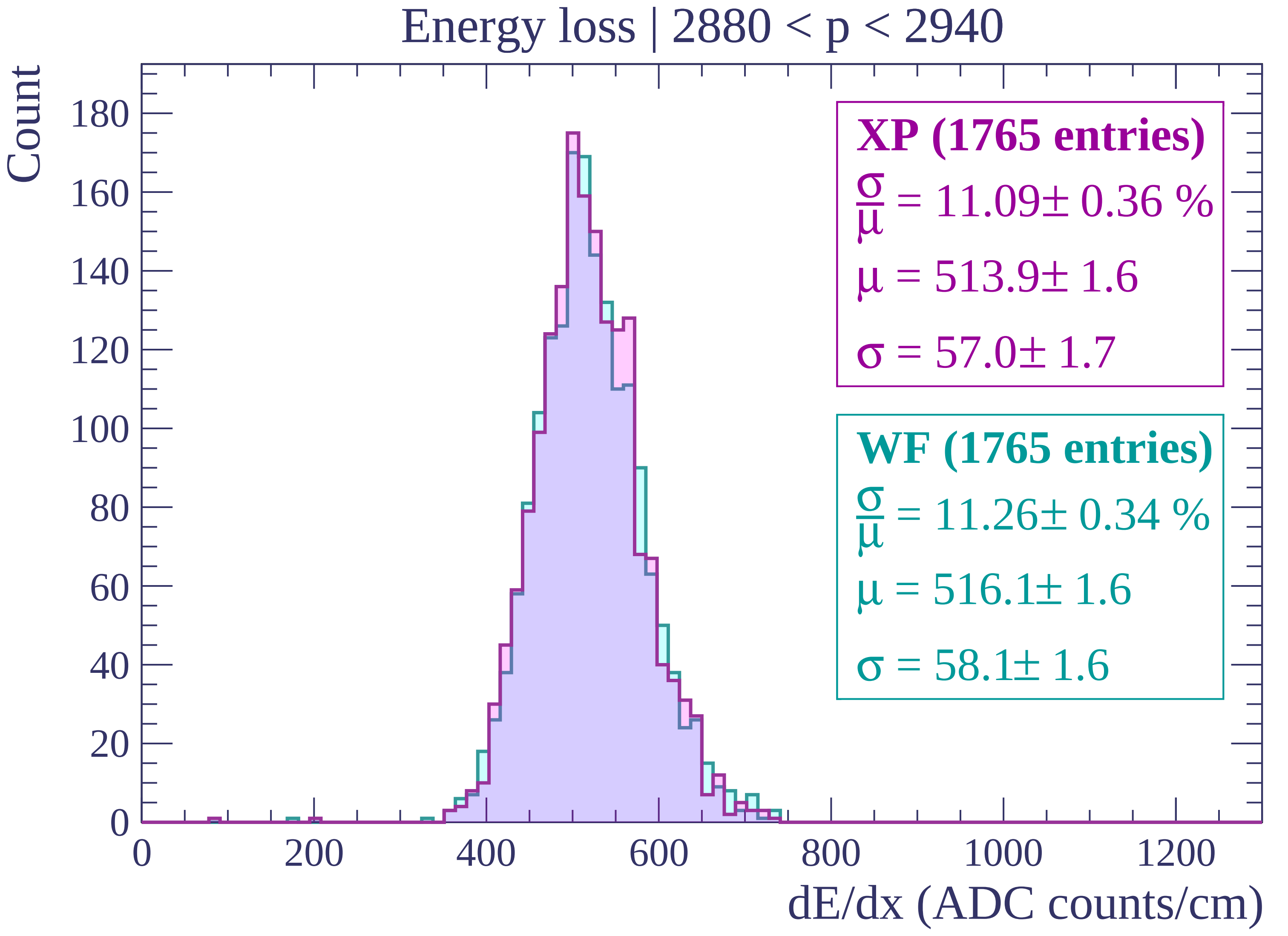


Energy loss | 2700 < p < 2760
Count
dE/dx (ADC counts/cm)
XP (2033 entries)
σ/μ = 10.79± 0.34 %
μ = 510.5± 1.5
σ = 55.1± 1.6
WF (2033 entries)
σ/μ = 11.11± 0.33 %
μ = 512.8± 1.5
σ = 57.0± 1.5

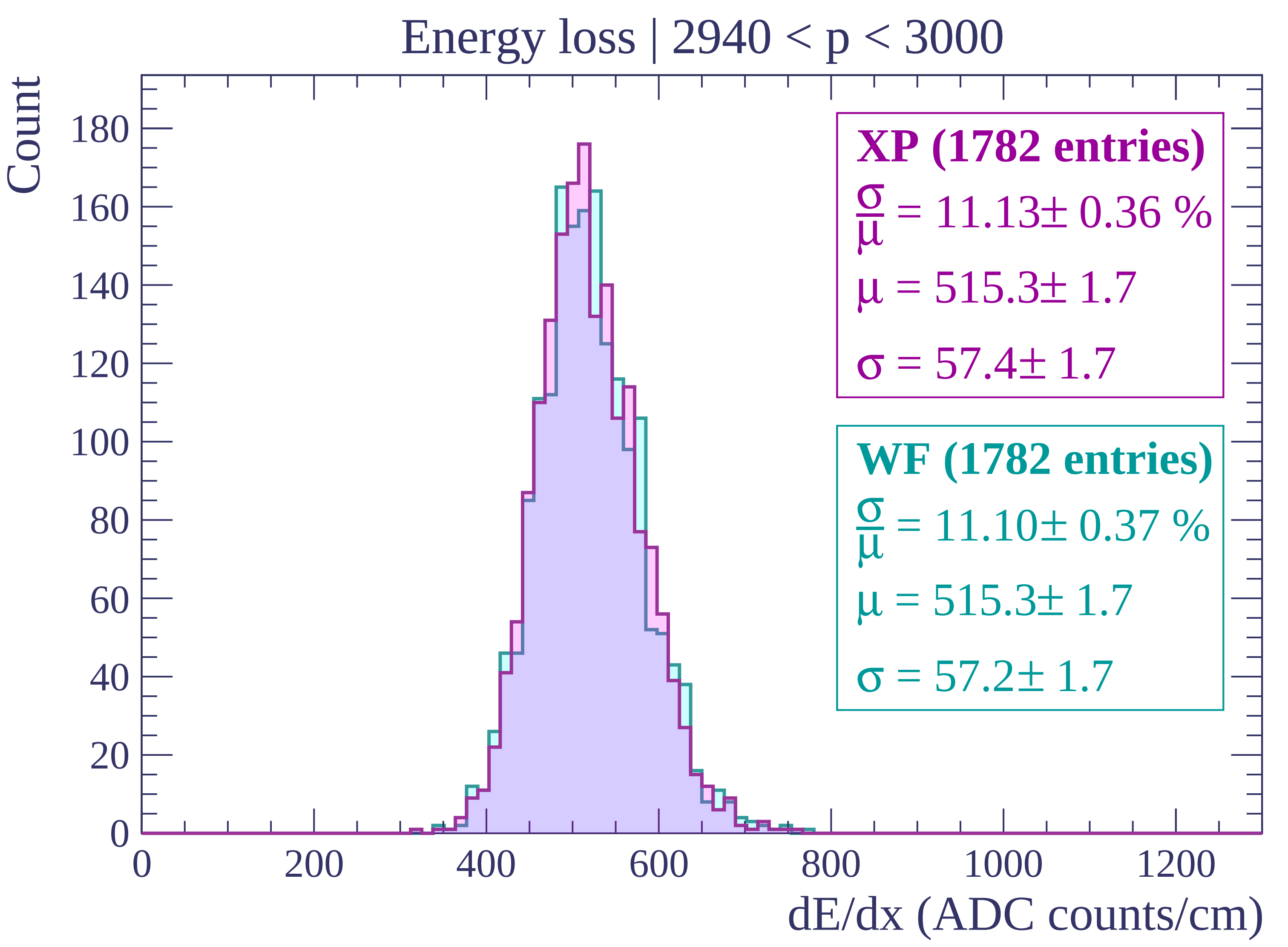

Energy loss | 2760 < p < 2820
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (2001 entries)
σ/μ = 11.26± 0.31 %
μ = 512.1± 1.5
σ = 57.7± 1.4
WF (2001 entries)
σ/μ = 11.23± 0.32 %
μ = 511.0± 1.5
σ = 57.4± 1.5



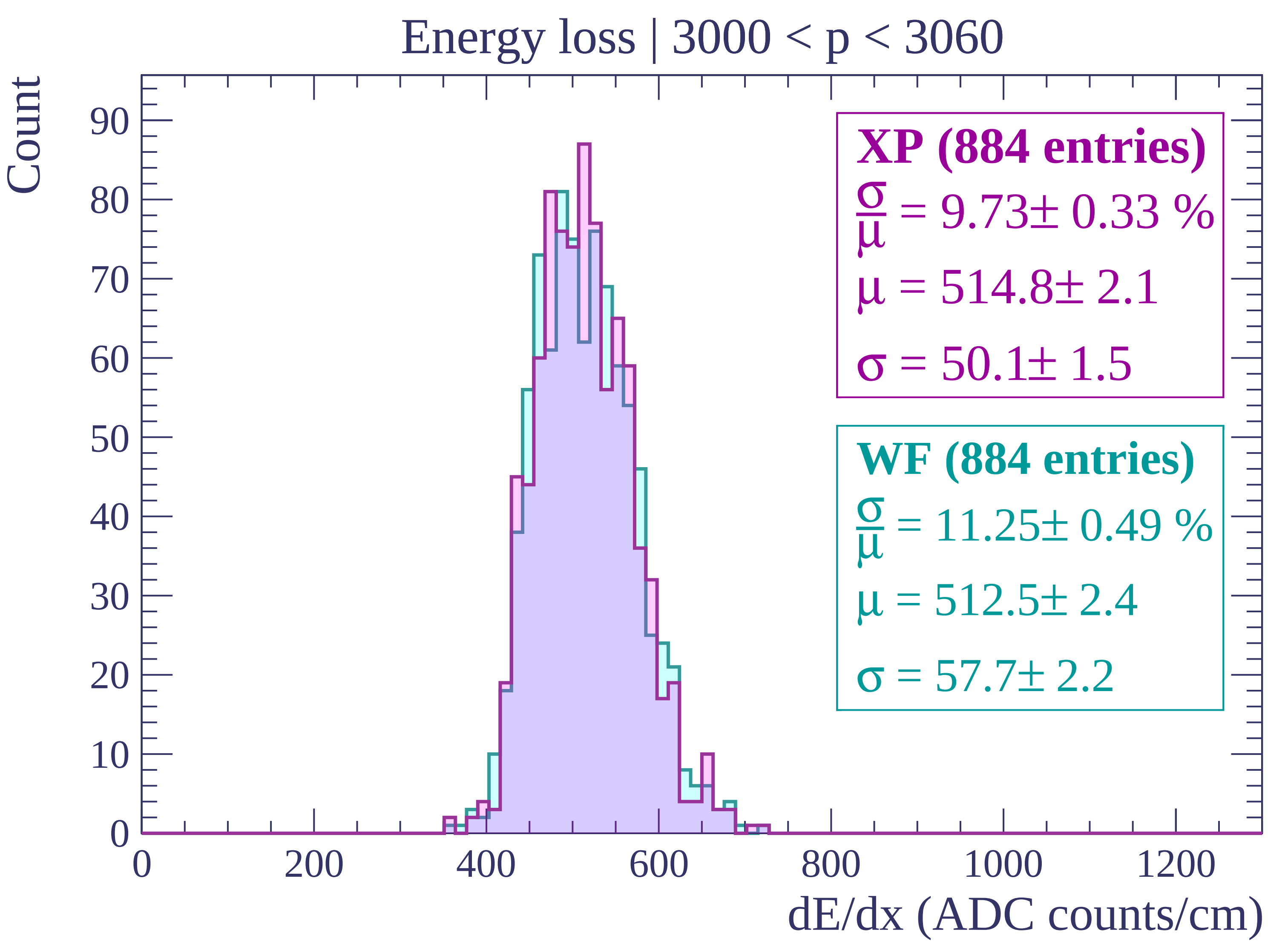
Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP (1837 entries)
$\frac{\sigma}{\mu}$ = 11.31± 0.31 %
μ = 514.5± 1.6
σ = 58.2± 1.4
WF (1837 entries)
$\frac{\sigma}{\mu}$ = 11.25± 0.30 %
μ = 516.8± 1.6
σ = 58.1± 1.4



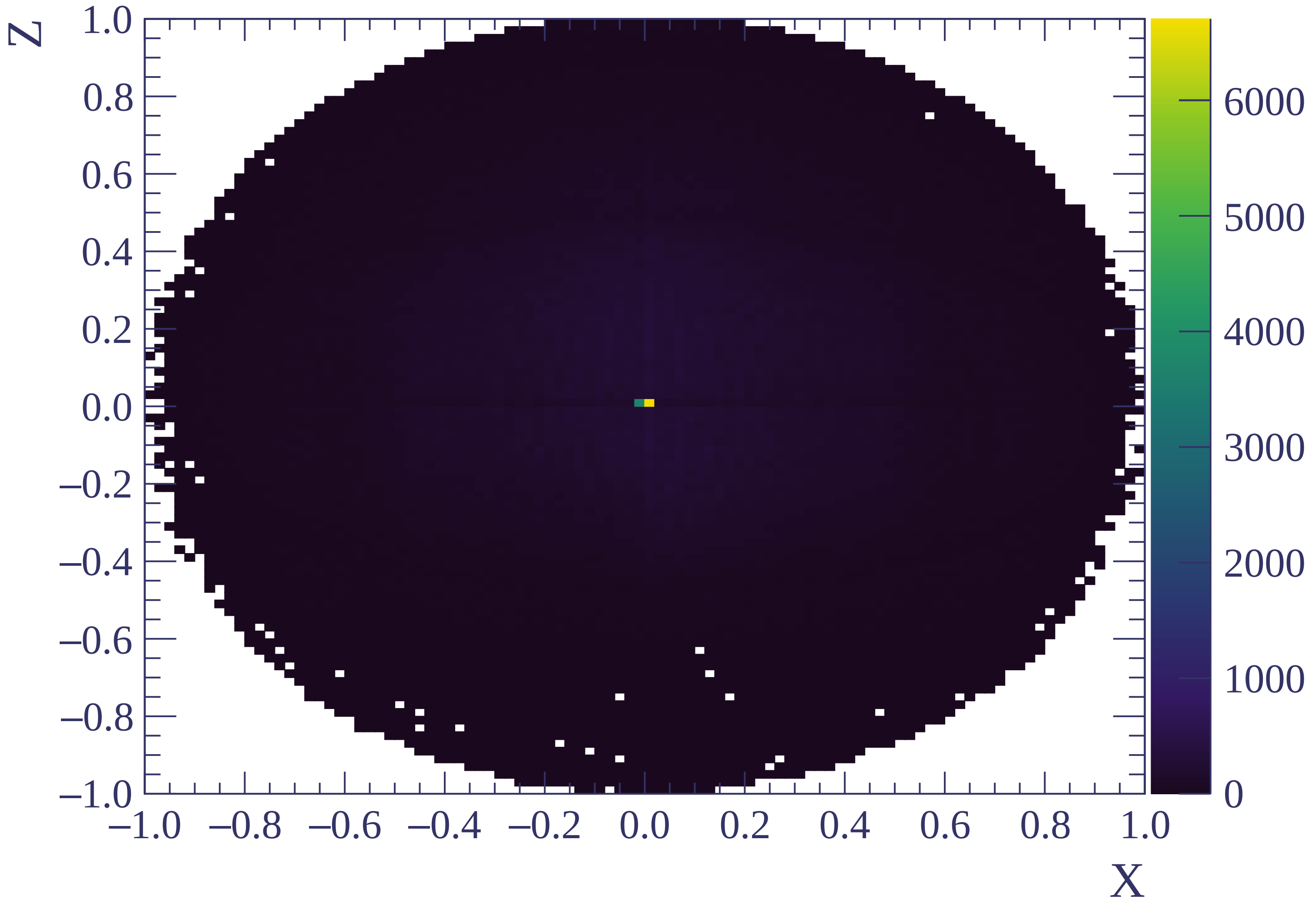
Energy loss | 2880 < p < 2940
Count
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP (1765 entries)
σ/μ = 11.09± 0.36 %
μ = 513.9± 1.6
σ = 57.0± 1.7
WF (1765 entries)
σ/μ = 11.26± 0.34 %
μ = 516.1± 1.6
σ = 58.1± 1.6



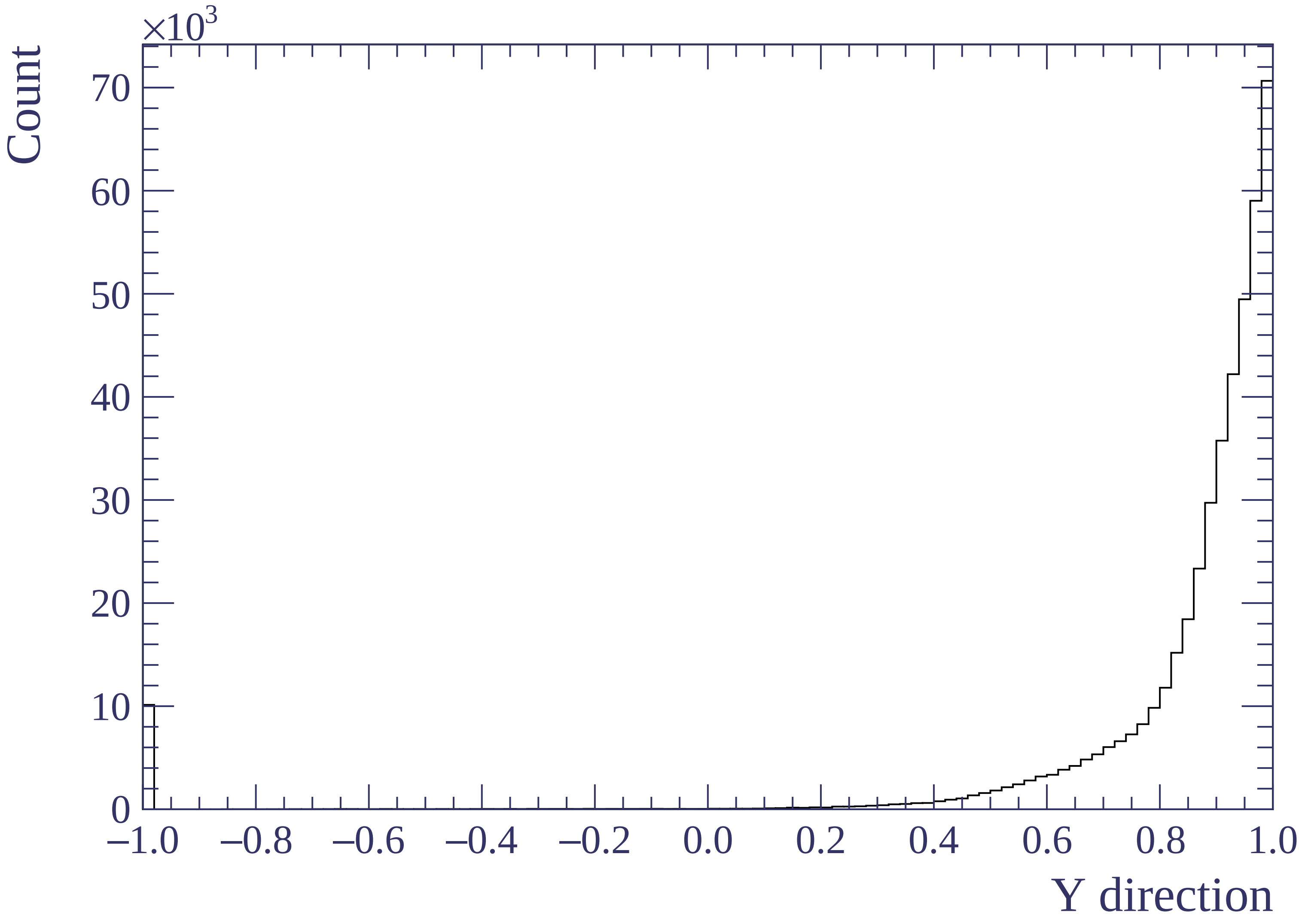
Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
XP (1782 entries)
σ/μ = 11.13± 0.36 %
μ = 515.3± 1.7
σ = 57.4± 1.7
WF (1782 entries)
σ/μ = 11.10± 0.37 %
μ = 515.3± 1.7
σ = 57.2± 1.7

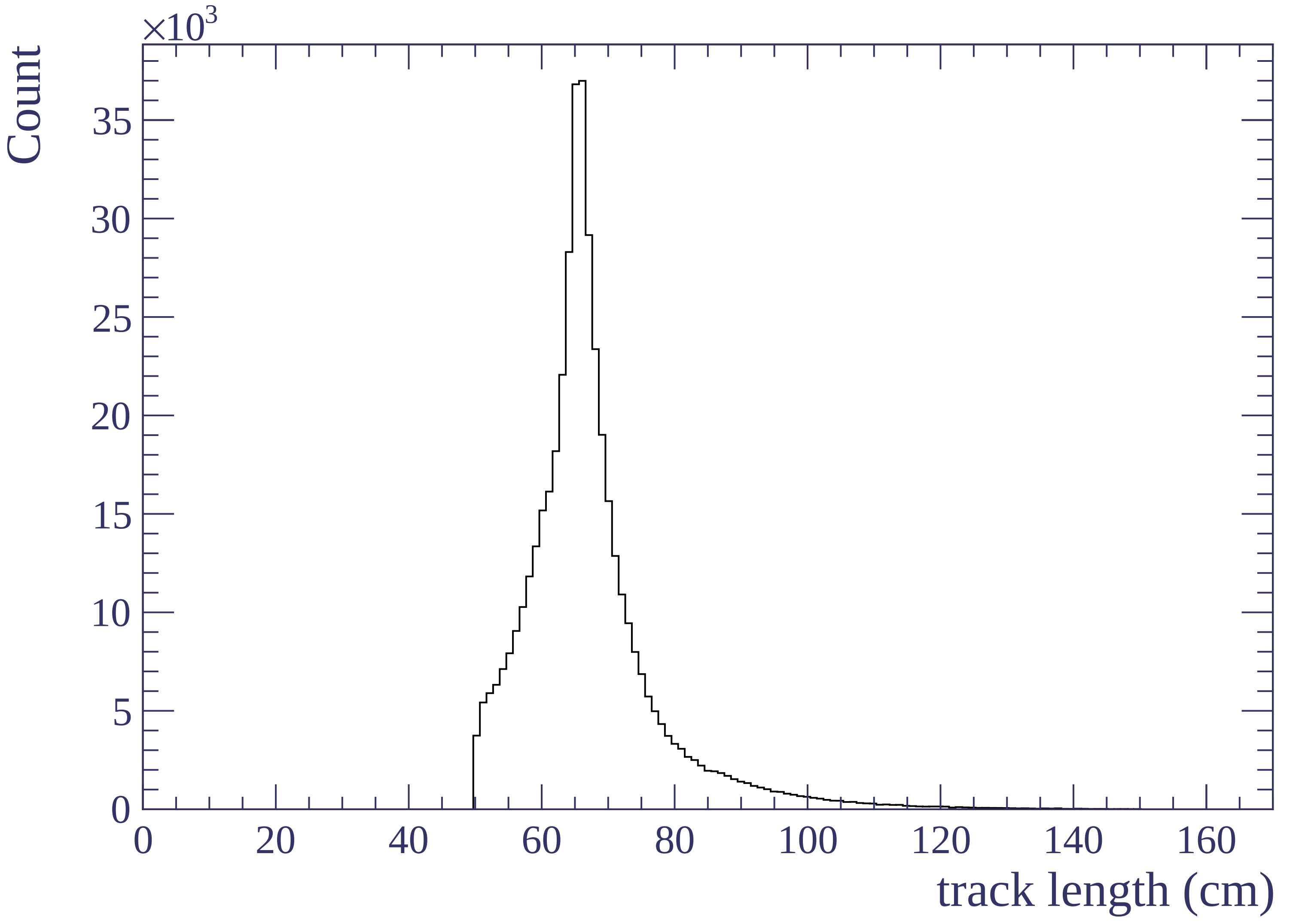


Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP (884 entries)
σ/μ = 9.73± 0.33 %
μ = 514.8± 2.1
σ = 50.1± 1.5
WF (884 entries)
σ/μ = 11.25± 0.49 %
μ = 512.5± 2.4
σ = 57.7± 2.2

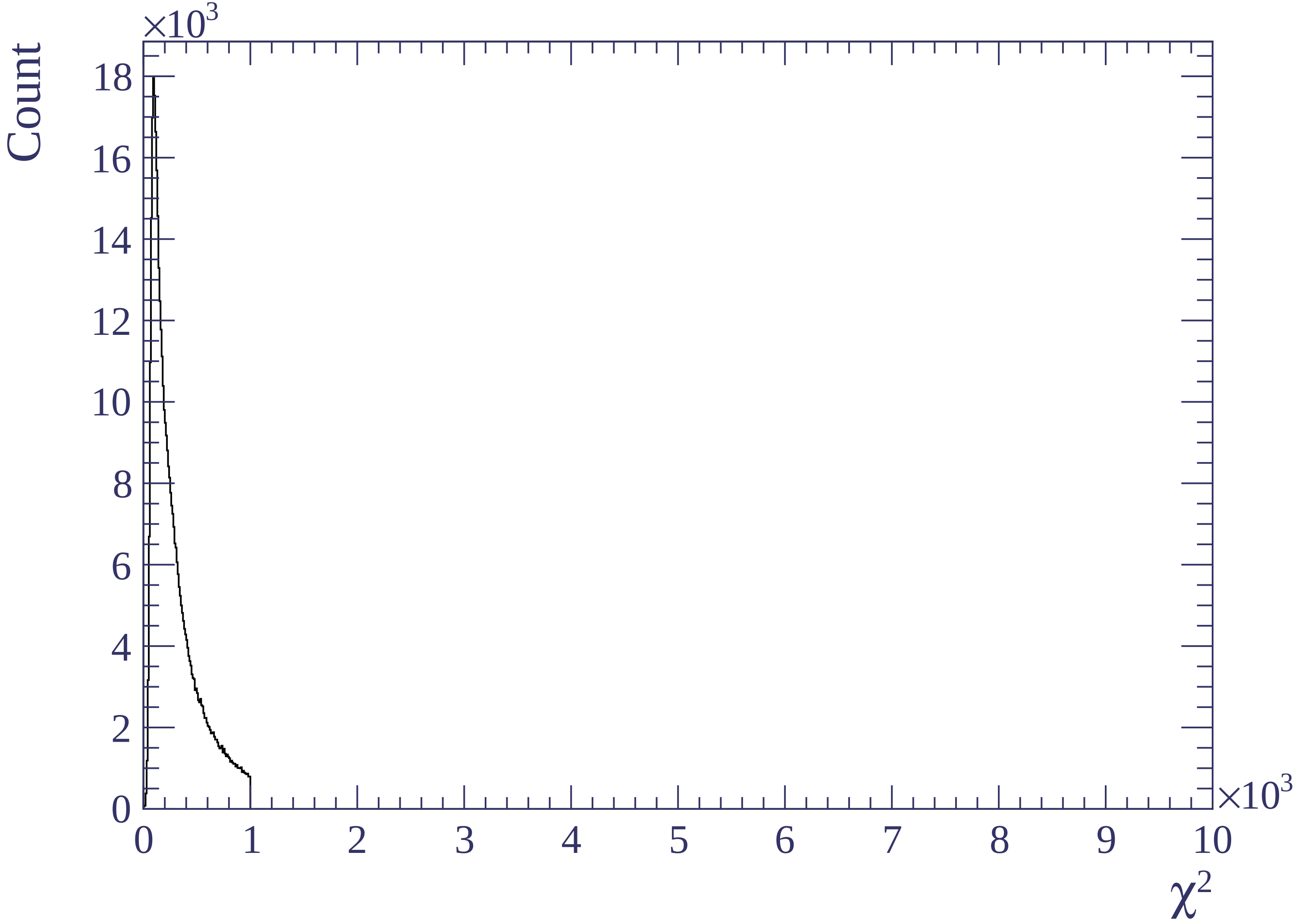


Z
X
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
6000
5000
4000
3000
2000
1000
0

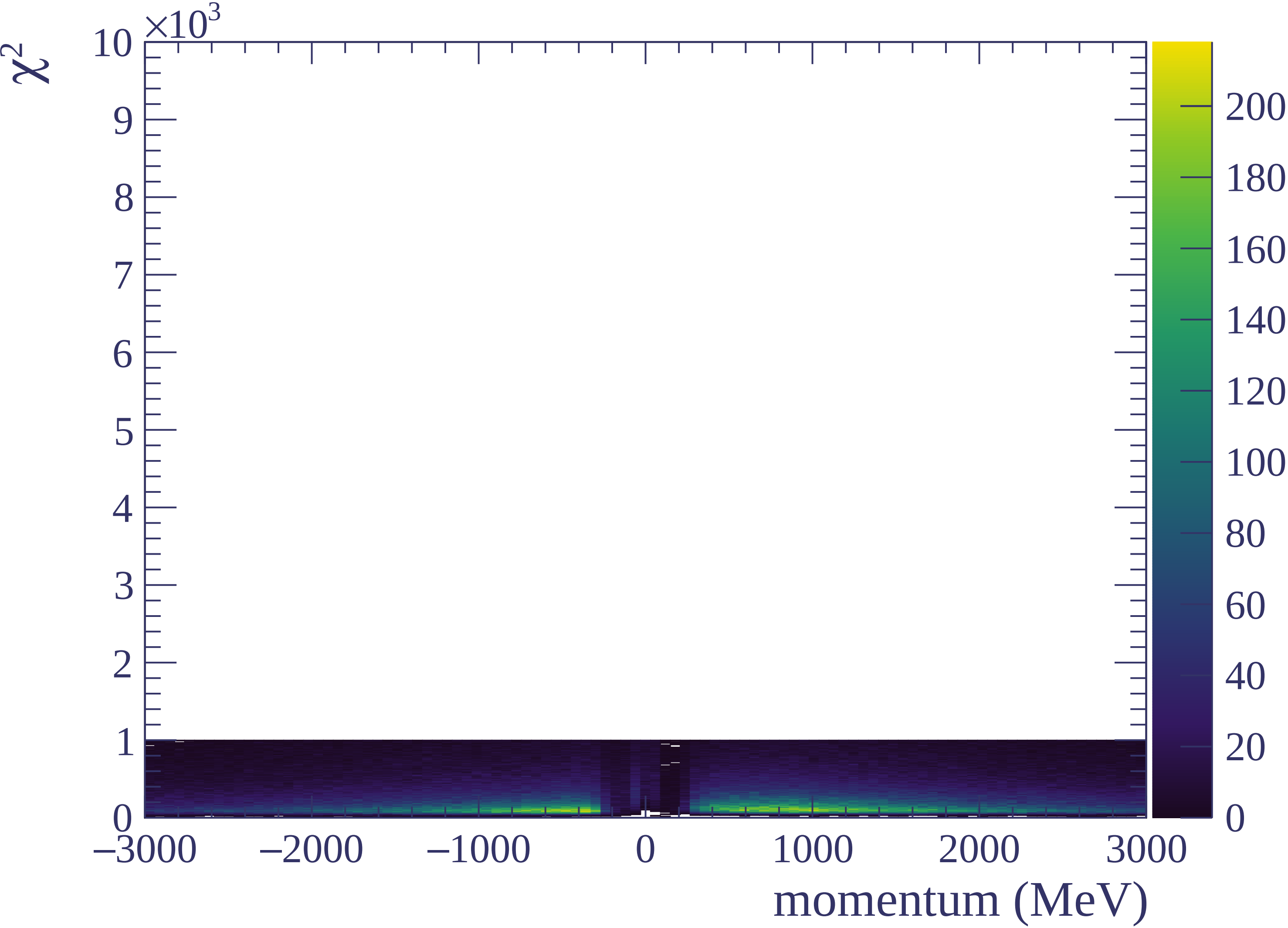

×10^3
Count
70
60
50
40
30
20
10
0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
Y direction



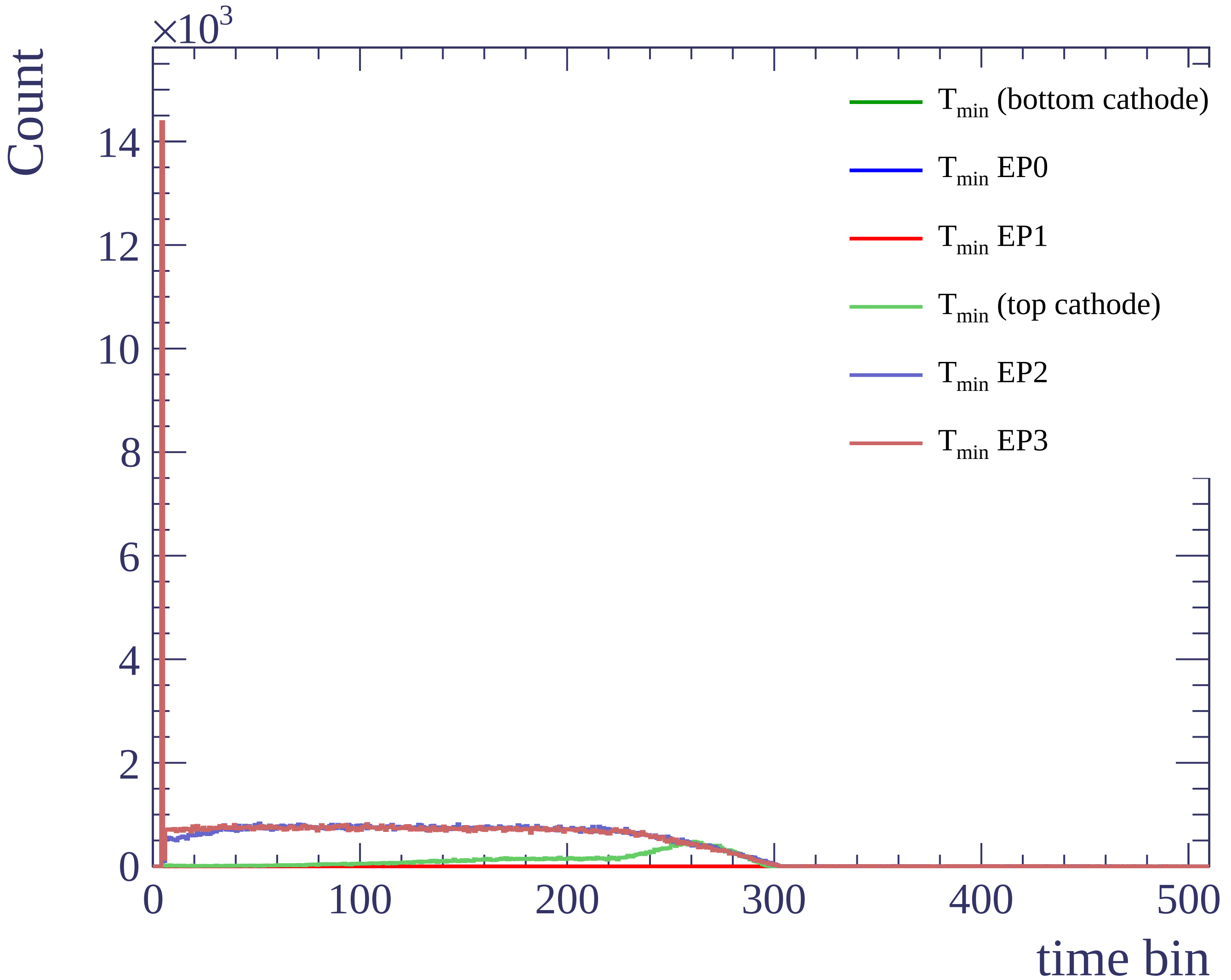
×10^3
Count
35
30
25
20
15
10
5
0
0
20
40
60
80
100
120
140
160
track length (cm)



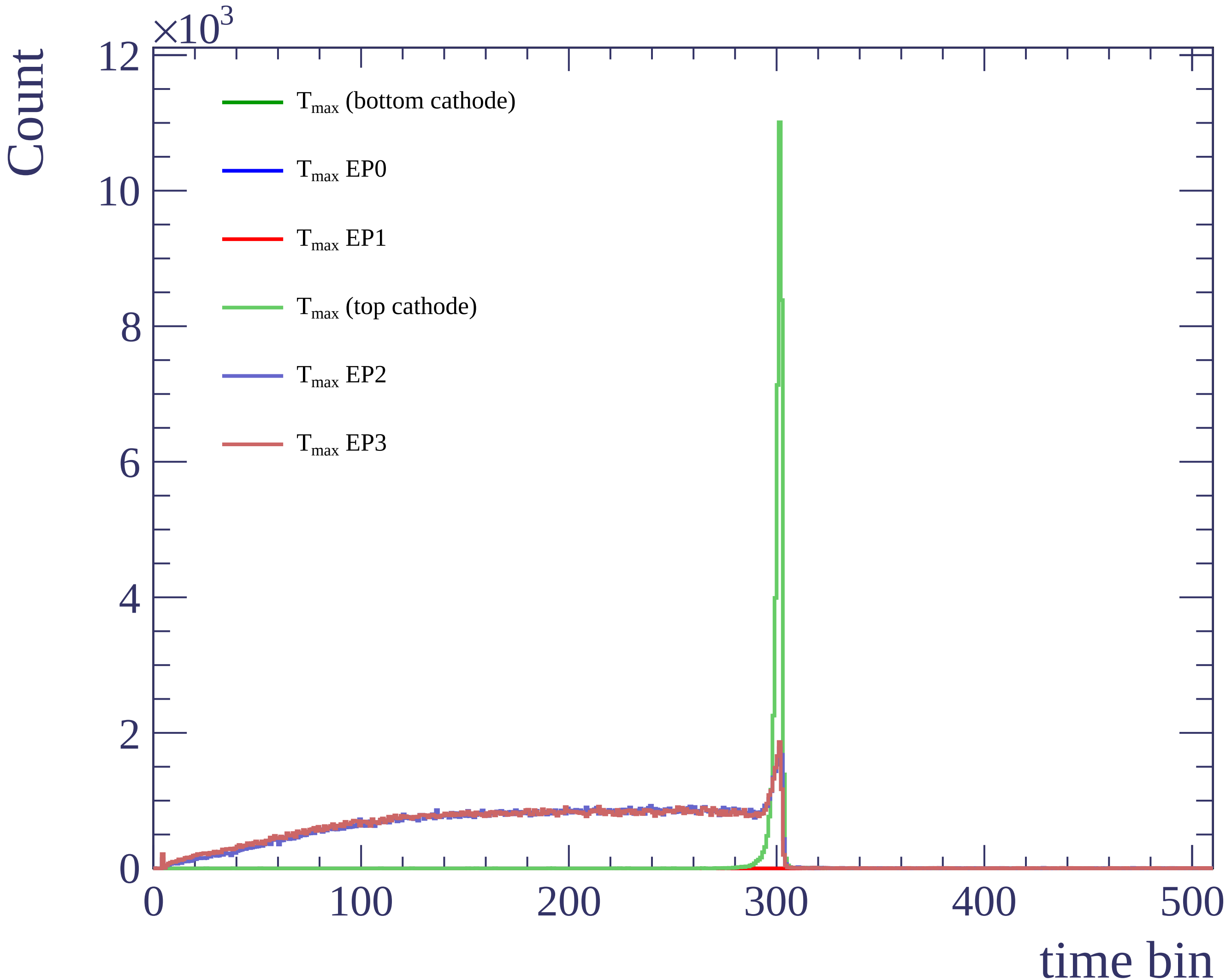
Count
×10^3
18
16
14
12
10
8
6
4
2
0
0
1
2
3
4
5
6
7
8
9
10
×10^3
χ^2



χ^2
$\times 10^3$
10
9
8
7
6
5
4
3
2
1
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
200
180
160
140
120
100
80
60
40
20
0



$\times 10^3$
Count
14
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
$\times 10^3$
12
10
8
6
4
2
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3